

An underwater photograph of a person in a swimming pool. The person is wearing a green swimsuit and is positioned vertically, with their head near the surface and legs pointing downwards. They are surrounded by a large cloud of bubbles, suggesting they have just jumped or are moving through the water. The water is a deep blue color.

ProMinent®

Swimming pool technology

2015

The ultimate in water treatment

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Swimming pool technology

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1 DULCODOS® Pool Swimming Pool Metering Systems

1.1

DULCODOS® Pool Swimming Pool Metering Systems

The metering systems DULCODOS® Pool ensure the best water quality. The systems are available in four different designs. It is easy to work out which type is best suited to your requirements.

Chlorine or active oxygen?

Historically, swimming pool water has always been treated with chlorine. Because it is an effective disinfectant and is highly oxidising, chlorine is also the chemical of choice for public pools. Clear and hygienically safe water is guaranteed.

The metering systems DULCODOS® Pool reliably keep the operating parameters in an optimum range and unpleasant side-effects, such as the smell of chlorine or burning eyes, are very rare.

Active oxygen is less effective than chlorine. It can be used for very gentle and environmentally-sound water treatment in pools with fewer users.

■ Soft

DULCODOS® Pool Soft is especially suited to private pools used by a small number of people. It works with active oxygen substances, which are less effective than chlorine. Water treatment with active oxygen is a good alternative for ecologically-minded pool owners or if users are allergic to chlorine. DULCODOS® Pool Soft uses no chlorine chemicals.

■ Basic

DULCODOS® Pool Basic regulates the pH and chlorine content using the redox potential. This is the direct measurement of effective oxidation in the water and is therefore an indication of the disinfectant effect and concentration of the metered chlorine. The concentration of chlorine cannot be determined with accuracy with this process. ORP measurements allow a particular range of chlorine to be set. DULCODOS® Pool Basic is robust and requires little maintenance.

■ Comfort

DULCODOS® Pool Comfort uses highly specific chlorine sensors to measure the chlorine content. The concentration of chlorine in the water can be determined and set with accuracy. The effectiveness of the pool filter is boosted by an integrated feeder assembly for flocculant, resulting in crystal-clear water! Numerous features to enhance operating convenience, such as measured values being mapped by a screen plotter or remote control from your PC, iPad or other tablet device using an integrated web server, make the metering system very popular with customers.

■ Professional

In addition to the features described above, DULCODOS® Pool Professional also measures the combined chlorine. This is an important parameter in public pools. It can be incorporated in the building management system via OPC and KNX and alarm messages can be sent by text or e-mail. Eco!Mode operating mode reduces the energy consumption of the filter pumps. The integrated soft PLC control can be used to operate several peripheral devices and functions. The swimming pool controller becomes the central control unit for all the swimming pool technology.



1 DULCODOS® Pool Swimming Pool Metering Systems

■ Choice of pumps

The metering systems DULCODOS®Pool allow you to choose which metering pump to fit on your complete system. The choice of pump depends entirely on the size of your pool and how often it is used.

- **Peristaltic pumps DULCO®flex** are suited to applications requiring few chemicals, such as small pools or those used infrequently. The pump reliably eliminates bubbles of gas formed during periods of non-use. Depending on the amount of use, the metering hose has to be replaced once or twice a year.
- **Motor-driven metering pumps alpha** have a higher capacity and longer maintenance intervals. Like peristaltic pumps, they are silent.
- **Solenoid metering pumps Beta®** are not controlled by switching them on and off, like DULCO®flex or alpha, instead, their metering frequency is adjusted continuously, enabling the pump to have an extremely precise control action.
- Pumps with **CAN bus system** can be used in the DULCODOS® Pool Professional series. They communicate all operating messages, such as two-stage monitoring of the chemical reservoir, to the control.

■ Accessories

Whether you are looking for collecting pans for chemical tanks or portable test devices for measurement parameters – or even software for digital control: optional accessories make it even easier for you to operate the system.

■ Service

Installation, commissioning, training in how the system works, operation and maintenance: When you buy a DULCODOS®Pool system, it comes with service you can rely on – even if your pool is already quite old.



1 DULCODOS® Pool Swimming Pool Metering Systems

1.2 Metering System DULCODOS® Pool Soft

Ecologically convincing: chlorine-free water treatment with active oxygen in private swimming pools – fully automatically and correctly.

For swimming pools with volumes up to 100 m³

Chlorine-free water treatment system for environmentally operated private pools. Safe water disinfection with active oxygen as a turnkey complete solution.

Complete system DULCODOS® Pool Soft for pH adjustment and chlorine-free disinfection with active oxygen. To prevent any germs in the pool from building up resistance to active oxygen, it is not metered continuously, but injected at intervals controlled by a timer.

Peristaltic pumps of the product range DULCO®flex, motor-driven metering pumps type alpha or solenoid metering pumps type Beta® are used, depending on demand and the circulation volume.

Sensors, controllers and metering pumps form a unit with the chemical storage tanks, which can get to work without a lot of installation effort on your part.

The control device performs numerous functions to enhance operating convenience, such as mapping measured values using a screen plotter and an SD memory card or remote access via the integral WEB server and LAN interface (optional).

Your benefits

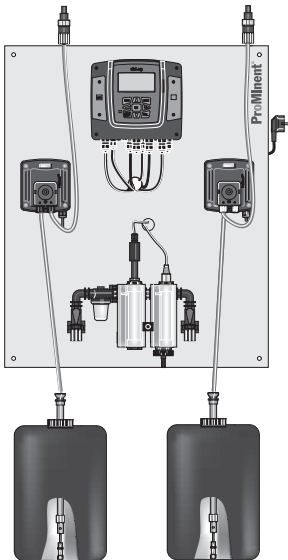
- Simple, quick assembly
- Simple, menu-driven operation
- Chlorine-free
- Constantly good water quality
- Versatile monitoring functions

Technical details

- 2-channel controller DULCOMETER® diaLog DACa with measurement/control of the pH value and metering of active oxygen using an integrated timer function, mounted on a wall plate ready for use.
- In-line probe housing with sample water monitoring, sample water filter and measuring probe for pH value
- Monitoring of the chemical reservoir
- Dosing monitor to protect against over-metering
- Screen plotter for graphic mapping of measured values, data logger with SD card
- Embedded web server with LAN interface (optional)
- Metering pumps alpha, DULCO®flex or Beta® to control the pH value and active oxygen content.
- Connector for point of injection: Injection valves with 1/2" screw thread
- Connectors for metering pumps/points of injection: PVC hose 10 x 4 mm
- Sample water connector: PE hose 8 x 5 mm
- Digital pause input
- Alarm relay output
- Electrical connection: 230 VAC, 50/60 Hz
- Dimensions with metering pumps alpha or Beta®:
 - 595 x 745 x 150 mm (W x H x D) mounting plate for measuring technology
 - 595 x 400 x 150 mm (W x H x D) mounting plate for pumps
- Dimensions with metering pumps DULCO®flex: 595 x 745 x 150 mm (W x H x D)
- Weight: approx. 10 kg or 6 kg (without pumps)

Field of application

- Private swimming pool



P_DD_0042_SW1



1 DULCODOS® Pool Swimming Pool Metering Systems

Identity Code Ordering System for DULCODOS® Pool Soft

DSPa	Measured variable
DO2	pH / Timer control H ₂ O ₂
Hardware-additional functions	
0	Standard
Software-additional functions	
1	Screen plotter with measured data backup including SD card
Communication interfaces	
0	none
5	Embedded web server, LAN (available from mid-2015)
Electrical connection	
A	230 V, 50/60 Hz, European standard plug
B	230 V, 50/60 Hz, Swiss plug
Sensor equipment	
0	with sensors
1	without sensors
Version	
0	with ProMinent® logo
1	without ProMinent® Logo
Language	
A	Swedish
D	German
E	English
F	French
G	Czech
I	Italian
N	Dutch
P	Polish
R	Russian
S	Spanish
Metering pumps for acids/alkalis	
0	without metering pumps
1	0.8 l/h (DULCO®flex DF2a 0208)
2	1.6 l/h (DULCO®flex DF2a 0216)
3	2.4 l/h (DULCO®flex DF2a 0224)
4	1.8 l/h (alpha ALPc 1002 PVT)
5	3.5 l/h (alpha ALPc 1004 PVT)
6	1.5 l/h (Beta® BT4b 0401 PVT)
7	2.8 l/h (Beta® BT4b 0402 PVT)
8	4.5 l/h (Beta® BT4b 0404 PVT)
Multifunctional valve for acid/alkali pump	
0	without
1	with MFV (only for alpha and Beta®)
Metering pumps for disinfection	
0	without metering pumps
1	0.8 l/h (DULCO®flex DF2a for 0208) for pools up to a volume of 20 m ³
2	1.6 l/h (DULCO®flex DF2a for 0216) for pools up to a volume of 40 m ³
3	2.4 l/h (DULCO®flex DF2a for 0224) for pools up to a volume of 60 m ³
4	1.8 l/h (alpha ALPc 1002 PVT) for pools up to a volume of 45 m ³
5	3.5 l/h (alpha ALPc 1004 PVT) for pools up to a volume of 90 m ³
6	1.5 l/h (Beta® BT4b 0401 PVT) for pools up to a volume of 25 m ³
7	2.8 l/h (Beta® BT4b 0402 PVT) for pools up to a volume of 50 m ³
8	4.5 l/h (Beta® BT4b 0404 PVT) for pools up to a volume of 100 m ³
Multifunctional valve for disinfection pump	
0	without
1	with MFV (only for alpha and Beta®)
Installation	
0	supplied loose without mounting plate
1	assembled on a base plate
Approvals	
0	with CE certification

DSPa DO2 0 1 0 A 0 0 D 2 0 2 0 1 0 Identity code as a representative example



1 DULCODOS® Pool Swimming Pool Metering Systems

1.3 Metering System DULCODOS® Pool Basic

Convenient and simple: pure water in private swimming pools – fully automatically and correctly.

For swimming pools with a circulation capacity of up to 200 m³/h

The chlorine metering system DULCODOS® Pool Basic is a complete solution for private swimming pools where the chlorine content is controlled using the low-maintenance measurement of the redox potential.

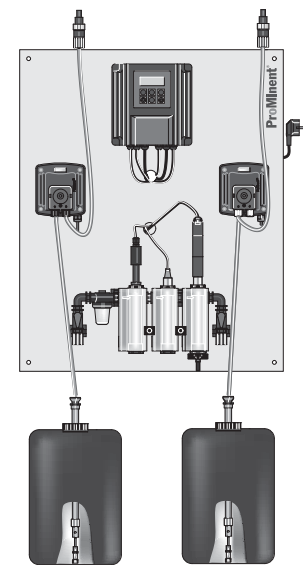
Complete system for the fully automatic adjustment of pH and chlorine content (using the measured variable redox potential) in swimming pool water. Peristaltic pumps of the product range DULCO®flex or motor-driven metering pumps type alpha are used, depending on demand and the circulation volume. Sensors, controllers and metering pumps form a single perfectly coordinated unit with the chemical storage tanks, which can reliably get to work without a lot of installation effort on your part.

Your benefits

- Simple, quick assembly
- Simple, menu-driven operation
- Constantly good water quality
- Versatile monitoring functions

Technical details

- 2-channel swimming pool controller Splash Control PPCa with measurement, control and metering functions for pH and redox potential (chlorine metering)
- In-line probe housing with sample water monitoring, sample water filter and measuring probe for pH value and redox potential, fitted on a wall panel.
- 2 metering pumps alpha or DULCO®flex
- Monitoring of the chemical reservoir
- Dosing monitor to protect against over-metering
- Connectors for points of injection: Injection valves with 1/2" screw thread
- Connectors for metering pumps/points of injection: PVC hose 10 x 4 mm
- Sample water connector: PE hose 8 x 5 mm
- Digital pause input
- Alarm relay output
- Electrical connection: 230 VAC, 50/60 Hz
- Dimensions: 595 x 745 x 150 mm (W x H x D)
- Weight: approx. 10 kg or 6 kg (without pumps)



pk_7_100_SW1

Field of application

- Private swimming pool



1 DULCODOS® Pool Swimming Pool Metering Systems

Identity Code Ordering System for DULCODOS® Pool Basic

DSPa	Measured variable														
	PRO	pH / ORP													
		Hardware-additional functions													
		0	Standard												
			Software-additional functions												
			0	none											
				Communication interfaces											
				0	none										
					Electrical connection										
					A	230 V, 50/60 Hz, European standard plug									
					B	230 V, 50/60 Hz, Swiss plug									
						Sensor equipment									
						0	with sensors								
						A	without sensors								
							Version								
							0	with ProMinent® logo							
							1	without ProMinent® Logo							
								Language							
								D	German						
								E	English						
								F	French						
								G	Czech						
								I	Italian						
								N	Dutch						
								R	Russian						
								S	Spanish						
									Metering pumps for acids/alkalis						
									0	without metering pumps					
									1	0,8 l/h (DULCO®flex DF2a 0208)					
									2	1,6 l/h (DULCO®flex DF2a 0216)					
									3	2,4 l/h (DULCO®flex DF2a 0224)					
									4	1,8 l/h (alpha ALPc 1002 PVT)					
									5	3,5 l/h (alpha ALPc 1004 PVT)					
										Multifunctional valve for acid/alkali pump					
										0	without				
										1	with MFV (alpha only)				
											Metering pumps for disinfection				
											0	without metering pumps			
											1	0.8 l/h DULCO®flex for up to 45/10 m³/h circulation HB/FB*			
											2	1.6 l/h DULCO®flex for up to 90/20 m³/h circulation HB/FB*			
											3	2.4 l/h DULCO®flex for up to 140/30 m³/h circulation HB/FB*			
											4	1.8 l/h alpha for up to 100/20 m³/h circulation HB/FB*			
											5	3.5 l/h alpha for up to 200/40 m³/h circulation HB/FB*			
												Multifunctional valve for disinfection pump			
											0	without			
											0	with MFV (alpha only)			
												Installation			
												0	supplied loose without mounting plate		
												1	assembled on a base plate		
													Approvals		
													0	with CE certification	
DSPa	PRO	0	0	0	A	0	0	D	2	0	2	0	1	0	Identity code as a representative example

* Calculated for 12 percent sodium-calcium hypochlorite
 HB = Indoor swimming pool
 FB = Outdoor swimming pool

1 DULCODOS® Pool Swimming Pool Metering Systems

1.4

Metering System DULCODOS® Pool Comfort

Convenient and simple: crystal-clear water in private swimming pools – fully automatically and correctly.

For swimming pools with a circulation capacity of up to 225 m³/h

The chlorine metering system DULCODOS® Pool Comfort is the convenient solution for pH adjustment and disinfection of swimming pools with liquid chlorine products. Remote access is possible via LAN interface.

Your benefits

Complete system DULCODOS® Pool Comfort for pH adjustment and disinfection with liquid chlorine products. Peristaltic pumps of the product range DULCO®flex, motor-driven metering pumps type alpha or solenoid metering pumps type Beta® are used, depending on demand and the circulation volume.

An integrated flocculant metering station (optional) ensures crystal-clear water.

Sensors, controllers and metering pumps form a unit with the chemical storage tanks, which can get to work without a lot of installation effort on your part.

The control device performs numerous functions to enhance operating convenience, such as mapping measured values using a screen plotter and an SD memory card or remote access via the integral WEB server and LAN interface (optional).

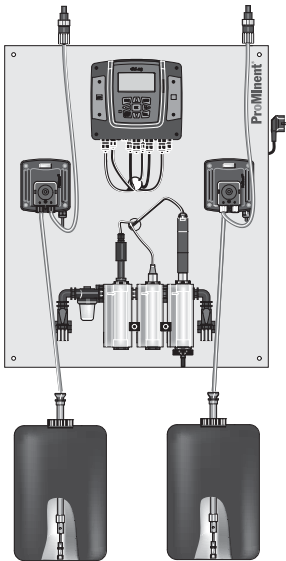
- Simple, quick assembly
- Simple, menu-driven operation
- Brilliant water quality
- Versatile monitoring functions

Technical details

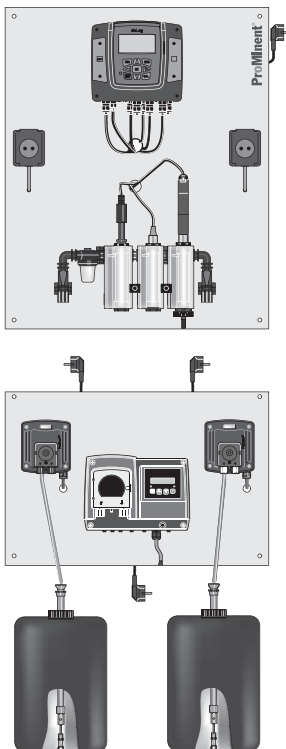
- 2-channel controller DULCOMETER® diaLog DACa with measurement/control and metering functions for pH and chlorine concentration, mounted on a wall plate ready for use
- Integrated flocculant metering station (optional)
- In-line probe housing with sample water monitoring, sample water filter and measuring probes for pH and chlorine content (DC2 for free chlorine, DC4 for free chlorine in the presence of isocyanuric acid stabiliser)
- Monitoring of the chemical reservoir
- Dosing monitor to protect against over-metering
- Screen plotter for graphic mapping of measured values, data logger with SD card
- Embedded web server with LAN interface (optional)
- Metering pumps alpha, DULCO®flex or Beta® to control the pH value and chlorine content, DULCO®flex for metering flocculant (optional).
- Connector for point of injection: Injection valves with 1/2" screw thread
- Connectors for metering pumps/points of injection: PVC hose 10 x 4 mm
- Sample water connector: PE hose 8 x 5 mm
- Digital pause input
- Alarm relay output
- Electrical connection: 230 VAC, 50/60 Hz
- Dimensions with metering pumps alpha or Beta® and/or with "flocculant metering" option:
 - 595 x 745 x 150 mm (W x H x D) mounting plate for measuring technology
 - 595 x 400 x 150 mm (W x H x D) mounting plate for pumps
- Dimensions with metering pumps DULCO®flex: 595 x 745 x 150 mm (W x H x D)
- Weight: approx. 10 kg or 6 kg (without pumps)

Field of application

- High-end private pool



P_DD_0037_SW1



P_DD_0045_SW1



1 DULCODOS® Pool Swimming Pool Metering Systems

Identity Code Ordering System for DULCODOS® Pool Comfort

DSPa	Measured variable
DC2	pH / free chlorine (chlorine sensor CLE 3-mA-2ppm)
DC4	pH / free chlorine in the presence of the stabiliser isocyanuric acid (chlorine sensor CGE 2-mA-2ppm)
Hardware-additional functions	
0	Standard
Software-additional functions	
1	Screen plotter with measured data backup including SD card
Communication interfaces	
0	none
5	Embedded web server, LAN (available from mid-2015)
Electrical connection	
A	230 V, 50/60 Hz, European standard plug
B	230 V, 50/60 Hz, Swiss plug
Sensor equipment	
0	with sensors
B	Measured variable DC2 without sensors
C	Measured variable DC4 without sensors
Version	
0	with ProMinent® logo
1	without ProMinent® Logo
Language	
A	Swedish
D	German
E	English
F	French
G	Czech
I	Italian
N	Dutch
P	Polish
R	Russian
S	Spanish
Metering pumps for acids/alkalis	
0	without metering pumps
1	0,8 l/h (DULCO®flex DF2a 0208)
2	1,6 l/h (DULCO®flex DF2a 0216)
3	2,4 l/h (DULCO®flex DF2a 0224)
4	1,8 l/h (alpha ALPc 1002 PVT)
5	3,5 l/h (alpha ALPc 1004 PVT)
6	1,5 l/h (Beta® BT4b 0401 PVT)
7	2,8 l/h (Beta® BT4b 0402 PVT)
8	4,5 l/h (Beta® BT4b 0404 PVT)
Multifunctional valve for acid/alkali pump	
0	without
1	with MFV (only for alpha and Beta®)
Metering pumps for disinfection	
0	without metering pumps
1	0.8 l/h DULCO®flex for up to 45/10 m³/h circulation HB/FB*
2	1.6 l/h DULCO®flex for up to 90/20 m³/h circulation HB/FB*
3	2.4 l/h DULCO®flex for up to 140/30 m³/h circulation HB/FB*
4	1.8 l/h alpha for up to 100/20 m³/h circulation HB/FB*
5	3.5 l/h alpha for up to 200/40 m³/h circulation HB/FB*
6	1.5 l/h Beta® for up to 85/20 m³/h circulation HB/FB*
7	2.8 l/h Beta® for up to 160/35 m³/h circulation HB/FB*
8	4.5 l/h Beta® for up to 260/55 m³/h circulation HB/FB*
Multifunctional valve for disinfection pump	
0	without
1	with MFV (only for alpha and Beta®)
Installation	
0	supplied loose without mounting plate
1	assembled on a base plate
B	Base plate with flocculant pump DF4a fitted
Approvals	
0	with CE certification

DSPa DC4 0 1 0 A 0 0 D 2 0 2 0 1 0 Identity code as a representative example

* Calculated for 12 percent sodium-calcium hypochlorite
 HB = Indoor swimming pool
 FB = Outdoor swimming pool



1 DULCODOS® Pool Swimming Pool Metering Systems

1.5

Metering System DULCODOS® Pool Professional

Professional and demanding: crystal-clear water in public swimming pools – fully automatically and correctly, with minimal energy consumption, thanks to Eco!Mode.

For swimming pools with a circulation capacity of up to 1,130 m³/h

Chlorine metering system for individual adjustment and monitoring of all common hygiene auxiliary parameters in public pools. DULCODOS® Pool Professional ensures crystal-clear water quality and lowers operating costs thanks to Eco!Mode.

Complete system DULCODOS® Pool Professional for individual adjustment and monitoring of all common hygiene auxiliary parameters in public pools, such as pH, redox potential and free and combined chlorine. Peristaltic pumps of the product range DULCO®flex, motor-driven metering pumps type alpha or solenoid metering pumps type Beta® are used, depending on demand and the circulation volume.

An integrated flocculant metering station (optional) ensures crystal-clear water.

In Eco!Mode, the circulating volume of the swimming pool pumps is optimised depending on the water quality, enabling you to efficiently save energy.

The integrated SoftPLC allows the system to control optional peripheral devices and functions, such as UV systems, water attractions, lighting, heating and water top-up.

The system has many different communication interfaces that enable it to be integrated in networks or a building management system and it can also be remotely monitored and controlled with an Apple® iPad or tablet PC.

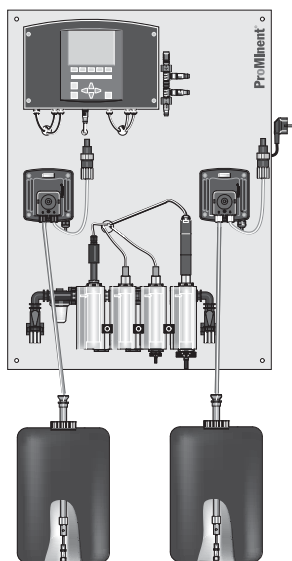
Sensors, controllers, metering pumps and the process chemical storage tanks form a single unit with the other peripheral swimming pool technology used, which can handle your work without a lot of installation effort on your part.

Your benefits

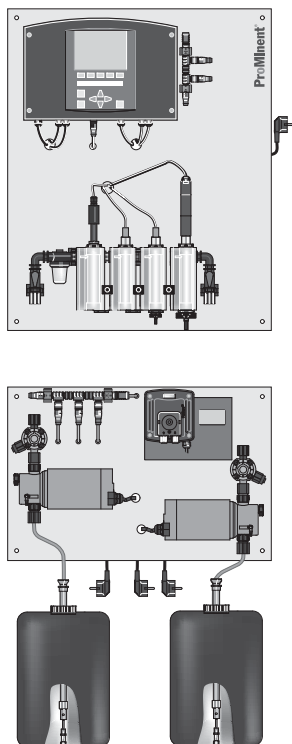
- Simple, quick assembly
- Brilliant water quality
- Eco!Mode helps cut operating costs
- Versatile communication interfaces
- Central control of peripheral devices and functions too

Technical details

- Multi-channel, multi-parameter controller DULCOMETER® DULCOMARIN® II with measuring, control and metering functions for pH, redox potential, free and combined chlorine in various combinations depending on the type, ready-wired for use and mounted on a wall panel
- Optional integrated flocculant metering station
- In-line probe housing with sample water monitoring, sample water filter and all sensors
- Monitoring of the chemical reservoir with a pre-alarm (options A and F)
- Dosing monitor to protect against over-metering
- Screen plotter for graphic mapping of measured values, data logger with SD card
- Embedded web server with LAN interface (optional)
- OPC and KNX for integration in building management systems, alarm function by text or e-mail (optional)
- Integrated SoftPLC for control of peripheral devices (option F)
- Metering pumps alpha, DULCO®flex or Beta® to control the pH value and chlorine content, DULCO®flex for metering flocculant (optional)
- Connector for point of injection: Injection valves with 1/2" screw thread
- Connectors for metering pumps/points of injection: PVC hose 10 x 4 mm
- Sample water connector: PE hose 8 x 5 mm.
- Digital pause input
- 3 contact inputs, freely configurable (option A)
- 5 contact inputs, freely configurable (option F)
- CAN bus for connection of chlorine measuring cells and metering pumps Beta® and DULCO®flex DF4a
- Temperature measuring input Pt 100/Pt 1000
- Alarm relay output
- 3 output relay outputs, freely configurable
- 3 relay outputs for the control of metering pumps (option A)
- 6 output relay outputs, freely configurable (option F)
- 4 analogue outputs 0/4-20 mA, freely configurable (option A)
- 2 analogue outputs 0/4-20 mA, freely configurable (option F)
- Electrical connection: 230 VAC, 50/60 Hz.



P_DD_0035_SW1



pk_7_105_SW1



1 DULCODOS® Pool Swimming Pool Metering Systems

- Dimensions with metering pumps alpha, Beta® or DULCO®flex DF4a and/or with "floculant metering" option:
 - 595 x 745 x 150 mm (W x H x D) mounting plate for measuring technology
 - 595 x 400 x 150 mm (W x H x D) mounting plate for pumps
- Dimensions with metering pumps DULCO®flex DF2a: 595 x 745 x 150 mm (W x H x D)
- Weight: approx. 12 kg or 7 kg (without pumps)

Field of application

- High-end private pool
- Public swimming pool
- Therapy pool

Type	pH	ORP	Measured variables:		
			Chlorine	Chlorine/isocyanuric acid	Combined chlorine
PC5	x	x			
PC6	x		x		
PC7	x	x	x		
PC8	x	x	x		x
PC9	x			x	
PCA	x	x		x	
PCD	x	x		x	x

Type PC5: pH and redox potential (chlorine metering)

Type PC6: pH and free chlorine

Type PC7: pH, redox potential and free chlorine

Type PC8: pH, redox potential, free chlorine and combined chlorine

Type PC9: pH and free chlorine in the presence of the stabiliser isocyanuric acid

Type PCA: pH, redox potential and free chlorine in the presence of the stabiliser isocyanuric acid

Type PCD: pH, redox potential, free chlorine in the presence of the stabiliser isocyanuric acid and combined chlorine



1 DULCODOS® Pool Swimming Pool Metering Systems

Identity Code Ordering System for DULCODOS® Pool Professional

DSPa	Measured variable
PC5	pH / ORP
PC6	pH / free chlorine (chlorine sensor CLE 3.1-CAN)
PC7	pH / ORP / free chlorine (chlorine sensor CLE 3.1-CAN)
PC8	pH / ORP / free chlorine / total chlorine (chlorine sensors CLE 3.1-CAN and CTE 1-CAN)
PC9	pH / total chlorine (chlorine sensor CGE 2-CAN)
PCA	pH / ORP / total chlorine (chlorine sensor CGE 2-CAN)
PCD	pH / ORP / free chlorine / total chlorine (chlorine sensors CGE 2-CAN and CTE 1-CAN)
Hardware-additional functions	
0	Standard
A	4 standard signal outputs, 0/4-20 mA measured value (A module)
F	Functional module (F-module)
Software-additional functions	
1	Screen plotter with measured data backup including SD card
Communication interfaces	
0	none
5	Embedded web server, LAN
6	OPC server + Embedded web server + KNX function + alarm by text or e-mail
Electrical connection	
A	230 V, 50/60 Hz, European standard plug
B	230 V, 50/60 Hz, Swiss plug
Sensor equipment	
0	with sensors
A	Measured variable PC5 without sensors
E	Measured variable PC6 without sensors
F	Measured variable PC7 without sensors
G	Measured variable PC8 without sensors
H	Measured variable PC9 without sensors
I	Measured variable PCA without sensors
L	Measured variable PCD without sensors
Version	
0	with ProMinent® logo
1	without ProMinent® Logo
Language	
D	German
E	English
F	French
I	Italian
P	Polish
S	Spanish
Metering pumps for acids/alkalis	
0	without metering pumps
1	0,8 l/h (DULCO®flex DF2a 0208)
2	1,6 l/h (DULCO®flex DF2a 0216)
3	2,4 l/h (DULCO®flex DF2a 0224)
4	1,8 l/h (alpha ALPc 1002 PVT)
5	3,5 l/h (alpha ALPc 1004 PVT)
A	1.5 l/h (Beta® CANopen BT4a 0401 PVT)
B	2.8 l/h (Beta® CANopen BT4a 0402 PVT)
C	5.3 l/h (Beta® CANopen BT4a 0405 PVT)
D	1.5 l/h (DULCO®flex DF4a 04015 CAN Bus)
E	6.0 l/h (DULCO®flex DF4a 03060 CAN Bus)
F	8.3 l/h (Beta® CANopen BT4a 0408 PVT)
Multifunctional valve for acid/alkali pump	
0	without
1	with MFV (only for alpha and Beta®)
Metering pumps for disinfection	
0	without metering pumps
1	0.8 l/h DULCO®flex DF2a for up to 45/10 m³/h circulation HB/FB*
2	1.6 l/h DULCO®flex DF2a for up to 90/20 m³/h circulation HB/FB*
3	2.4 l/h DULCO®flex DF2a for up to 140/30 m³/h circulation HB/FB*
4	1.8 l/h alpha for up to 100/20 m³/h circulation HB/FB*
5	3.5 l/h alpha for up to 200/40 m³/h circulation HB/FB*
A	1.5 l/h Beta® CANopen for up to 85/20 m³/h circulation HB/FB*
B	2.8 l/h Beta® CANopen for up to 160/35 m³/h circulation HB/FB*
C	5.3 l/h Beta® CANopen for up to 300/65 m³/h circulation HB/FB*
D	1.5 l/h DULCO®flex DF4a CANopen for up to 85/20 m³/h circulation HB/FB*
E	6.0 l/h DULCO®flex DF4a CANopen for up to 340/70 m³/h circulation HB/FB*
F	19.5 l/h Beta® CANopen for up to 1050/225 m³/h circulation HB/FB*
Multifunctional valve for disinfection pump	
0	without
1	with MFV (only for alpha and Beta®)
Installation	
0	supplied loose without mounting plate
1	assembled on a base plate
B	Base plate with flocculant pump DF4a fitted
Approvals	
0	with CE certification

DSPa PC7 A 1 5 A 0 0 D B 0 B 0 1 0 Identity code as a representative example

* Calculated for 12 % sodium-calcium hypochlorite

HB = Indoor swimming pool FB = Outdoor swimming pool



1 DULCODOS® Pool Swimming Pool Metering Systems

1.6 Maintenance Kits

The following are needed for the maintenance of a measuring, control and metering system DULCODOS® Pool:

- 2 no. maintenance kits for metering pumps
- 1 no. maintenance kit for the measured variable

1.6.1 Maintenance Kits for Metering Pumps

The following table shows the assignment of the maintenance kits to the types of metering pumps used.

	Product range	Type	Order no.
Hose, complete 4.8 x 8.0 PharMed	DF2a	0208, 0216, 0224	1009480
Hose, complete 1.6 x 4.8 PharMed	DF4a	04015	1030722
Hose, complete 3.2 x 6.4 PharMed	DF4a	03060	1030723
Spare parts kit 1005-2/1605-2 PVT	ALPc, BT4a	1002PVT/1004PVT (ALPc), 0405PVT (BT4a)	1023110
Spare parts kits 1601 – 2 PVT, PPT, NPT	BT4a, BT4b	0401PVT (BT4a), 0401PVT (BT4b)	1023108
Spare parts kits 1602 – 2 PVT, PPT, NPT	BT4a, BT4b	0402PVT (BT4a), 0402PVT (BT4b)	1023109
Spare parts kits 0708 – 2/1008 – 2 PVT, PPT, NPT	BT4a	0408PVT	1023111
Spare parts kit 9.2/33.5/12 x 9 PVT	BT4a	0220PVT	1023113
Spare parts kits 1604 – 2 PVT, PPT, NPT	BT4b	0404PVT	1035332

1.6.2 Maintenance Kits for Measured Variables

Maintenance kits are put together for the measured variables of the DSPa. Depending on the measured variable, the maintenance kits consist of:

- Buffer solutions
- Electrolytes
- Diaphragm caps
- 1 stainless steel screen 300 µm for the water filter
- 1 NBR flat seal for the water filter

The following table shows the assignment of the maintenance kits to the types of DULCODOS® Pool.

	Type	Order no.
DSPA maintenance kit PR0, PC5, 333, 335, 735, 736	Basic, Professional PC5	1050631
DSPA maintenance kit DO2	Soft	1050632
DSPA maintenance kit DC2, PC6, 640, 645, 745	Comfort DC2, Professional PC6	1050633
DSPA maintenance kit DC4, PC9	Comfort DC4, Professional PC9	1050644
DSPA maintenance kit PC7, PCB, 781, 785, 786	Professional PC7	1050645
DSPA maintenance kit PC8	Professional PC8	1050646
DSPA maintenance kit PCA	Professional PCA	1050647
DSPA maintenance kit PCD	Professional PCD	1050648

1.6.3 Buffer Solutions

Quality buffer solutions are provided for calibration of pH and ORP sensors.

The following table shows the assignment of the buffer solutions to the sensors.

	Measured variable	Order no.
Buffer solution pH 4, 50 ml, red	pH	506251
Buffer solution pH 7, 50 ml, green	pH	506253
Buffer solution ORP 465 mV, 50 ml	ORP	506240



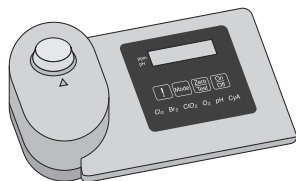
1 DULCODOS® Pool Swimming Pool Metering Systems

1.7 Test Equipment

1.7.1 Photometer

Precise measurement results through high-quality interference filters

Photometers measure nearly all disinfectants and the pH value based on the photometric principle. They are portable, compact and make safe, simple measurement possible.



P_DT_0074_SW
Photometer

The photometers DT1B, DT2C, DT3B and DT4B are used, among other things, as a reference method for calibrating the electrochemical sensors for chlorine, chlorine dioxide, fluoride, chlorite, H_2O_2 , bromine and ozone. They have been adapted to today's requirements and can be used in almost all areas of water analysis. High-quality interference filters and long-term stable LEDs are used as the light source in the high-precision optics. The entire measuring unit is maintenance-free. Precise and reproducible analysis results are achieved with minimum time and effort. The units are winning customers over with their excellent operating convenience, ergonomic design, compact dimensions and ease of use.

Your benefits

- Portable and compact
- Simple to operate with text support
- Safe, simple measurement of chlorine, chlorine dioxide, fluoride, chlorite, H_2O_2 , bromine, ozone, pH and trichloroisocyanuric acid
- Can be calibrated
- Memory function for the last measurements
- Backlit display
- Real-time clock
- Countdown
- Watertight, degree of protection IP 68

Technical details

Measuring ranges of the DT1B:

- 0.05... 6.0 mg/l free chlorine (DPD1) + total chlorine (DPD1+3)
- 5 ... 200 mg/l free chlorine (high range)
- 0.1 ... 13.0 mg/l bromine (DPD1)
- 0.05 ... 11 mg/l chlorine dioxide (DPD1)
- 0.03 ... 4.0 mg/l ozone (DPD4)
- 6.5 ... 8.4 pH (phenol red)
- 1 ... 80 mg/l cyanuric acid

Measuring ranges of the DT2C:

- 0.05 ... 2.0 mg/l fluoride
- 0.05... 6.0 mg/l free chlorine and total chlorine
- 0.05 ... 11.0 mg/l chlorine dioxide

Measuring ranges of the DT3B:

- 1 ... 50 / 40 ... 500 mg/l hydrogen peroxide (H_2O_2)

Measuring ranges of the DT4B:

- 0.03 ... 2.5 mg/l chlorite
- 0.05 ... 11 mg/l chlorine dioxide
- 0.05 ... 6 mg/l chlorine

Measuring tolerance: depending on the measured value and measuring method

Battery: 4 no. AA/LR6 batteries

Permissible ambient temperature range: 5...40 °C

Relative humidity: 30 ... 90 % (non-condensing)

Degree of protection: IP 68

Housing material: ABS

Keypad: polycarbonate film

Dimensions: 190 x 110 x 55 mm (L x W x H)

Weight: 0.4 kg



1 DULCODOS® Pool Swimming Pool Metering Systems

	Order no.
Photometer DT1B	1039315
Photometer DT2C	1039316
Photometer DT3B	1039317
Photometer DT4B	1039318

Photometers supplied with accessories, container vessels and reagents.

Consumable items

	Order no.
DPD 1 buffer, 15 ml	1002857
DPD 1 reagent, 15 ml	1002858
DPD 3 solution, 15 ml	1002859
Phenol red tablets R 175 (100 in each)	305532
Cyanuric acid tablets (100 in each)	1039744
SPADNS reagent, 250 ml for fluoride detection	1010381
Calibration standard fluoride 1 mg/l, for calibration of the photometer during fluoride determination	1010382
3 spare cells: round cells with covers for DPD phenol red and cyanuric acid detection (DT1 and DT2B)	1007566
3 spare cells for fluoride detection (DT2A and B)	1010396
DPD reagent set, 15 ml each: 3 x DPD 1 buffer, 1 x DPD 1 reagent, 2 x DPD 3 solution	1007567
Chlorine dioxide tablets no. 1	1039732
Chlorine dioxide tablets no. 2	1039733
Chlorine HR tablets (100 off)	
ACiDiTYiNG tablets (100 off)	

Spare parts

Chlorite Photometer

	Order no.
Stirrer for purging of chlorine dioxide (DT4)	1022754
3 spare cells: round cells with covers for DPD phenol red and cyanuric acid detection (DT1 and DT2B)	1007566

H₂O₂ measurement

	Order no.
Reagent for H ₂ O ₂ (DT3), 15 ml	1023636
Spare cell, 5x , for H ₂ O ₂ (DT3)	1024072



1 DULCODOS® Pool Swimming Pool Metering Systems

1.8

Swimming Pool Metering Systems DSMa for Swimming Pools with Several Pools

Number and type of modules required for a given number of pools

Number of filtration circuits	Central unit DXCa	P module	M module	A module*	Additional N or P module (power supply unit)	Free chlorine sensor	Total chlorine sensor (optional)
1	1	1	1	1	-	1	1
2	1	1	2	2	-	2	2
3	1	1	3	3	1	3	3
4	1	1	4	4	2	4	4
5	1	1	5	5	2	5	5
6	1	1	6	6	3	6	6
7	1	1	7	7	3	7	7
8	1	1	8	8	4	8	8
9	1	1	9	9	4	9	9
10	1	1	10	10	5	10	10
11	1	1	11	11	5	11	11
12	1	1	12	12	6	12	12
13	1	1	13	13	6	13	13
14	1	1	14	14	7	14	14
15	1	1	15	15	7	15	15
16	1	1	16	16	8	16	16

* No A module if metering pumps with CANopen are used.
 The above modules include all CAN bus connecting elements (T-distributor and spur line).
 The T-distributors can also be directly coupled.
 For distributed systems, the CAN cable must be ordered by the metre with the by-the-metre connecting kit.

	Order no.
CAN bulk cable connection kit*	1026589
Connecting cable - CAN, sold by the metre*	1022160

* The CAN by-the-metre connecting kit consists of a CAN coupling M12 5P and a CAN connector M12 5P and a wiring diagram.
 The by-the-metre connecting cable can be configured into a cable of individual length using the CAN by-the-metre connecting kit.
 One CAN by-the-metre connecting kit is required for each cable to be configured.
 The connecting cables CAN M12 5P 0.5 m (pump 1 m) supplied with the sensors and modules should be used for the spur lines.

If you have any questions, please contact our sales department.

Caution:

The maximum main bus length (not including stubs) should be at most 400 m.



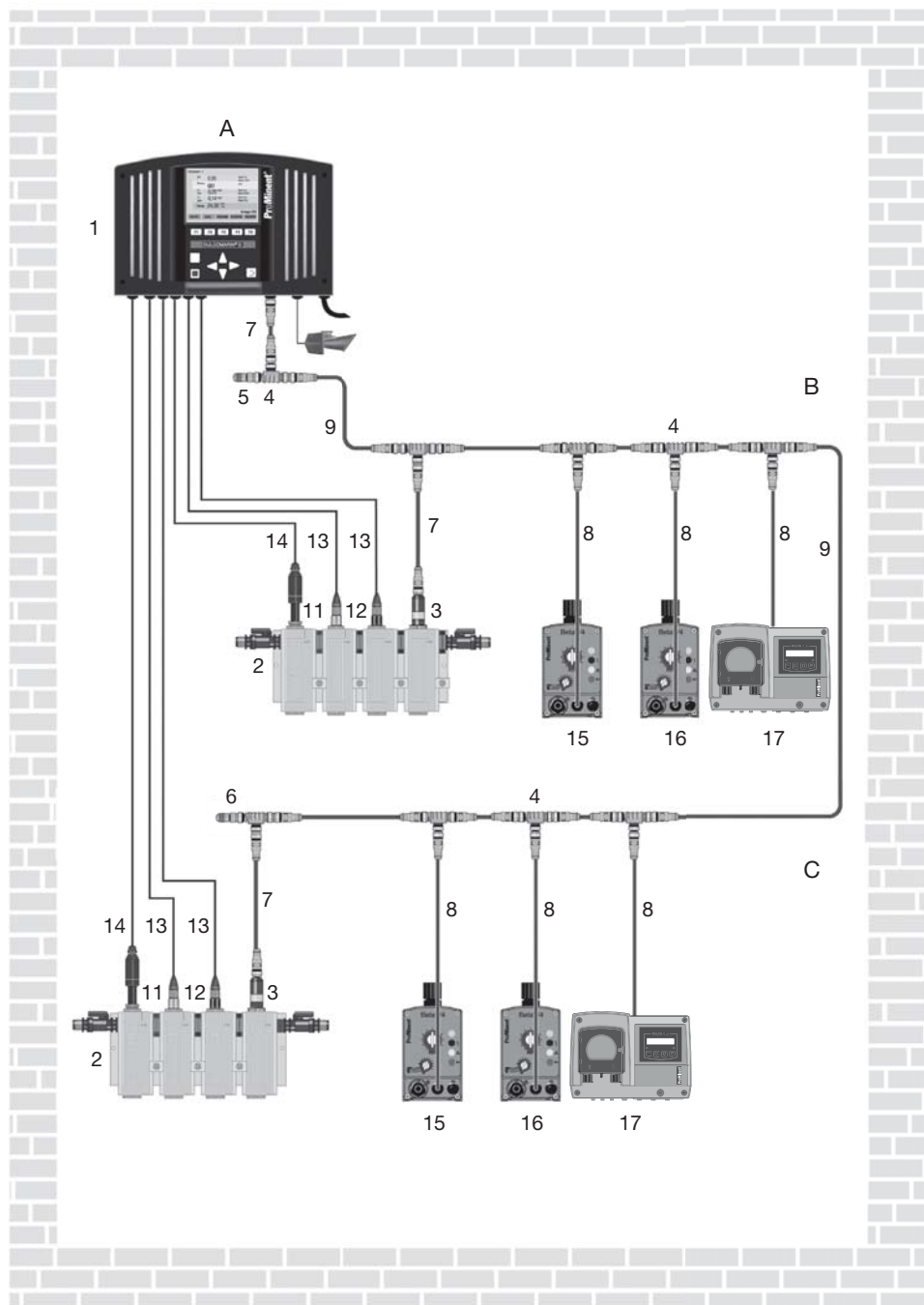
1 DULCODOS® Pool Swimming Pool Metering Systems

1.9 Configuration Examples

1.9.1 Configuration Example A: 2-Pool System

Two M modules in a central unit, use of metering pumps with CANopen bus.

- A Plant room
- B Pool 1
- C Pool 2



pk_5_022_neu

Attention:

It is very important that you adhere precisely to the principle of the design shown above because otherwise correct function is not guaranteed!

1 DULCODOS® Pool Swimming Pool Metering Systems

Measuring and control system for two filter circuits consisting of the following components:

Item	Quantity	Name	Order no.
1	1	DULCOMARIN®II central unit DXCa W 0 0 1 M M P S EN 01	–
2	2	DULCOTEST® in-line probe housing DGMa 3 2 2 T 0 0 0	–
3	2	Chlorine sensor CLE 3-CAN-10 ppm	1023425
4	9	T-distributor M12 5 pole CAN	included
5	1	Termination resistor M12 connector	included
6	1	Termination resistor M12 plug	included
7	5	Connection cable - CAN M12 5-pole 0.5 m	included
8	6	Connection cable - CAN M12 5-pole 0.3 m	included
9	–	Connecting cable - CAN M12 5-pin 10 m	1046383
11	2	pH sensor PHES 112 SE	150702
12	2	ORP sensor RHES-Pt-SE	150703
13	4	Cable combination coaxial 2 m - SN6 - pre-assembled	1024106
14	4 m	Signal cable, sold by the metre 2 x 0.25 mm ² Ø 4 mm	725122
15	2	Beta®/ 4 CANopen for pH correction BT4A0402PVT290UA000D00**	–
16	2	Beta®/ 4 CANopen for disinfectant BT4A0402PVT290UA000D00**	–
17	2	DF4a CAN for flocculant DF4aFW004015P9UA00001D10	–

* Up to 3 can be coupled from the connecting cable CAN M 12 5-pin 10 m.

** Suggested configuration

Caution:

Do not allow the maximum main bus length (without branch cables) to exceed 400 m.

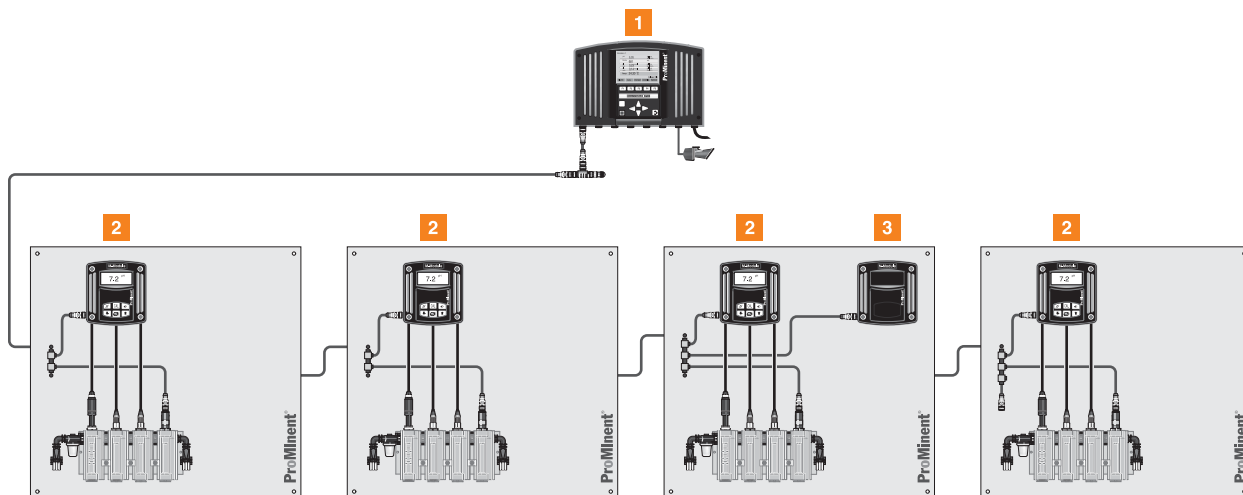


1 DULCODOS® Pool Swimming Pool Metering Systems

1.9.2

Configuration example B: Multi-pool system

Multi-pool system with measuring and control units for 4 pools.



AC_SW_0004_SW4

- 1 Central unit
- 2 M module
- 3 N module

For identity codes, please refer to pages → 1-19 and → 1-20

Components of the DSMa for a multi-pool system

- Central unit DXCa
- Measuring modules
- optionally N modules for power supply (an N module for 3 filter circuits respectively)

Central unit DXCa

a central unit is configured accordingly, depending on the application. The following versions are available:

Type	Order no.
DULCOMARIN® II central unit	DXCaW00100PSEN01 DULCOMARIN® II
Central unit with web server	DXCaW05100PSEN01 DULCOMARIN® II
Central unit with web server and OPC	DXCaW06100PSEN01 DULCOMARIN® II

Possible Configurations

Type
Central unit with web server
Measuring module DSMaPC7M00A0EN010
Measuring module DSMaPC7M0NA0EN010 with integrated N module
Measuring module DSMaPC7M0NA0EN010 with integrated N module
Measuring module DSMaPC7M00A0EN010

1 DULCODOS® Pool Swimming Pool Metering Systems

Identity code for DSMA without N module

DSMa	Measured variables											
	PC5	pH / ORP										
	PC6	pH / free chlorine										
	PC7	pH / ORP / free chlorine										
	PC8	pH / ORP / free chlorine / total chlorine										
	PC9	pH / total chlorine										
	PCA	pH / ORP / total chlorine										
	Module 1											
	M	M module										
	Module 2											
	0	none										
	G	G module										
	I	I module										
	R	R module										
	A	A module										
	Module 3 (power supply for CAN bus)											
	0	none										
	N	N module (an N module can feed a maximum of 3 filter circuits)										
	Electrical connection											
	A	230 V, 50/60 Hz, standard Euro plug										
	B	230 V, 50/60 Hz, Swiss plug										
	Sensor equipment											
	0	with sensors										
	A	Measured variable PC5 without sensors										
	E	Measured variable PC6 without sensors										
	F	Measured variable PC7 without sensors										
	G	Measured variable PC8 without sensors										
	H	Measured variable PC9 without sensors										
	I	Measured variable PCA without sensors										
	Language											
	00	none										
	DE	german										
	EN	english										
	ES	spanish										
	FR	french										
	IT	italian										
	Design											
	0	with logo										
	1	without logo										
	Assembly											
	0	supplied loose without mounting plate										
	1	assembled on a base plate										
	Certification											
0	with CE mark											
DSMa	PC7	M	0	0	A	0	EN	0	1	0		



1 DULCODOS® Pool Swimming Pool Metering Systems

Identity code for DSMa with N module

DSMa	Measured variables										
	PC5	pH / ORP									
	PC6	pH / free chlorine									
	PC7	pH / ORP / free chlorine									
	PC8	pH / ORP / free chlorine / total chlorine									
	PC9	pH / total chlorine									
	PCA	pH / ORP / total chlorine									
	Module 1										
	M	M module									
	Module 2										
	0	none									
	G	G module									
	I	I module									
	R	R module									
	A	A module									
	Module 3 (power supply for CAN bus)										
	0	none									
	N	N module (an N module can feed a maximum of 3 filter circuits)									
	Electrical connection										
	A	230 V, 50/60 Hz, standard Euro plug									
	B	230 V, 50/60 Hz, Swiss plug									
	Sensor equipment										
	0	with sensors									
	A	Measured variable PC5 without sensors									
	E	Measured variable PC6 without sensors									
	F	Measured variable PC7 without sensors									
	G	Measured variable PC8 without sensors									
	H	Measured variable PC9 without sensors									
	I	Measured variable PCA without sensors									
	Language										
	00	none									
	DE	german									
	EN	english									
	ES	spanish									
	FR	french									
	IT	italian									
	Design										
	0	with logo									
	1	without logo									
	Assembly										
0	supplied loose without mounting plate										
1	assembled on a base plate										
Certification											
0	with CE mark										
DSMa	PC7	M	0	N	A	0	EN	0	1	0	

N module configured

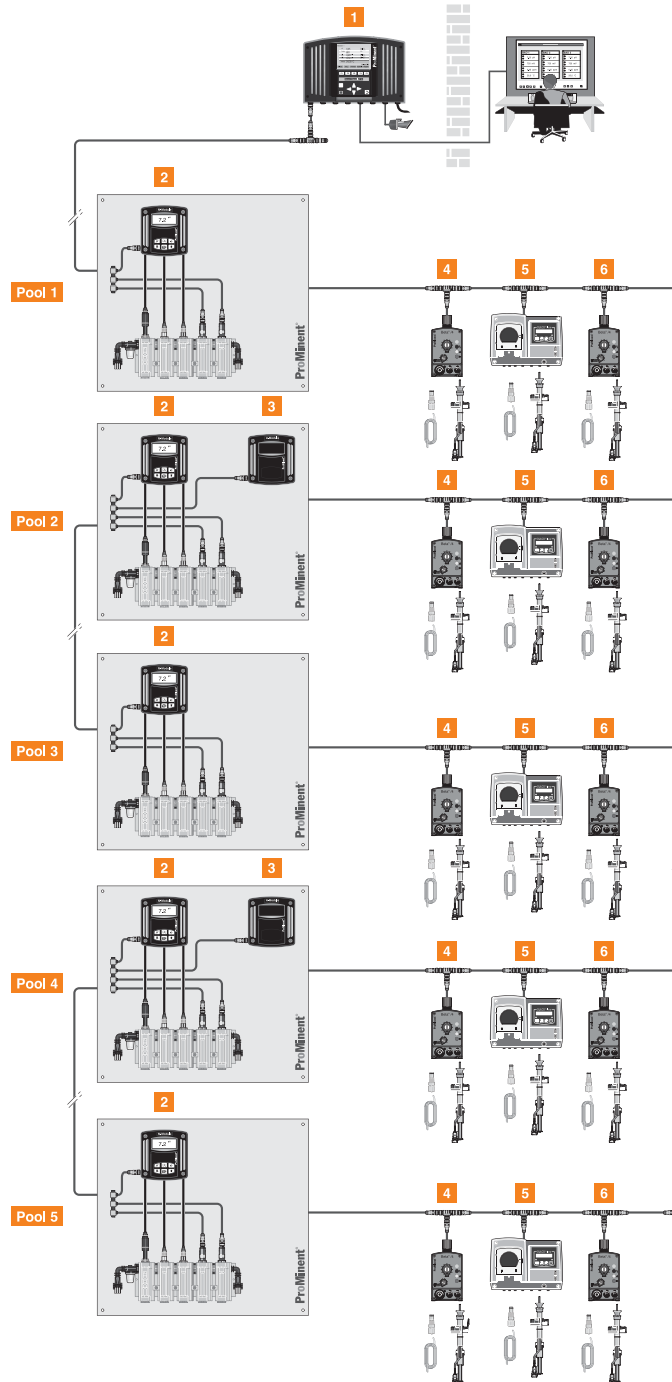


1 DULCODOS® Pool Swimming Pool Metering Systems

1.9.3

Configuration example C: Multi-pool system with metering sets for 5 pools

- 1 Central unit
- 2 M module
- 3 N module
- 4 Metering set for disinfection
- 5 Metering set for flocculation
- 6 Metering set for pH value correction



AP_SW_0003_SW

Components of the DSMa for a multi-pool system

- Central unit DXCa
- Measuring modules
- optionally N modules for power supply (an N module for 3 filter circuits respectively)
- Metering sets for disinfection, flocculation and pH value correction



1 DULCODOS® Pool Swimming Pool Metering Systems

Possible Configurations

Quantity	Type	Order no.
1	Central unit with web server	DXCaW05100PSEN01 DULCOMARIN® II
1	Measuring module DSMaPC7M00A0EN010	DSMaPC7M00A0EN010
1	Measuring module DSMaPC7M0NA0EN010 with integrated N module	DSMaPC7M0NA0EN010
1	Measuring module DSMaPC7M00A0EN010	DSMaPC7M00A0EN010
1	Measuring module DSMaPC7M0NA0EN010 with integrated N module	DSMaPC7M0NA0EN010
1	Measuring module DSMaPC7M00A0DE010	–
5	Metering set for disinfection 5.3 l with solenoid metering pump Beta®, CAN	1041560
5	Metering set for flocculation 1.5 l with peristaltic pump DF4, CAN	1043054
5	Metering set for pH value correction 2.8 l with solenoid metering pump Beta®, CAN	1041559



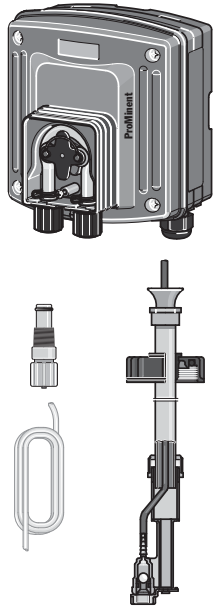
2 Metering Sets

2.1 Metering Sets for pH Value Adjustment and Disinfection

2.1.1

pH or Chlorine Metering Design with Peristaltic Pump DULCO®flex DF2a

Complete set comprising DULCO®flex hose pump with injection valve, PVC suction lance for 5 to 50 litre disposable containers (50 mm container opening), with mains cable, standard European plug for connecting to measuring and control units, as well as PVC metering line (10 m, 4/10 mm).



P_DX_0012_SW

Feed rate at max. pressure l/h	Max. pressure bar	Order no.
0.8	1.5	1041520
1.6	1.5	1041521
2.4	1.5	1041522

2.1.2

pH or Chlorine Metering Design with Peristaltic Pump DULCO®flex DF4a

The hose pump DULCO®flex DF4a has been specifically developed for metering chemicals. Typical applications include swimming pools and whirlpools.

Complete set comprising DULCO®flex hose pump DF4a with injection valve, PVC suction lance for 5 to 50 litre disposable containers (50 mm container opening), with mains cable, standard European plug for connecting to measuring and control units, as well as PVC metering line (10 m, 4/10 mm).

Feed rate at max. pressure l/h	Control	Max. pressure bar	Order no.
1.5	manual	4.0	1041568
1.5	external	4.0	1041569
6.0	manual	2.5	1041570
6.0	external	2.5	1041571
12.0	manual	2.0	1041572
12.0	external	2.0	1041573

Advantages

- Simple and convenient operation using language-neutral user guidance and programming via four buttons on the front
- Safe operation by direct input of the circulation capacity and required concentration
- Flexible to retrofit thanks to straightforward installation - also ideal for retrofitting into existing systems
- More efficient operation, for instance due to night setback
- Service-friendly design makes changing the tube easier
- Proportional adaptation of metering capacity by voltage and current input (optional = CAN bus)

Features

- Low-noise speed controllable multiphase motor with ball bearing drive shaft
- No mechanical drive, thus long operating life
- 100% switching time, operating pressure of up to 4 bar is possible
- Stepless setting of the metering capacity from 1 % to 100 %
- Metering capacities of 1.5, 6.0 or 12.0 l/h are shown in the display and can be set reproducibly
- Operating and peristaltic time counter, priming function (continuous run)
- Float switch input (contact 1 or 2 stage), hose fracture monitoring with fault indicating relay
- Switchable capacity change, for instance night setback or increased capacity



2 Metering Sets

2.1.3

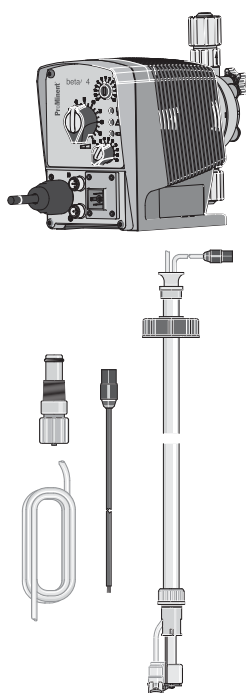
pH or Chlorine Version (DULCO®flex DF4a, CANopen)

Feed rate at max. pressure l/h	Control	Max. pressure bar	Order no.
1.5	CANopen	4.0	1041681
6.0	CANopen	2.5	1041682
12.0	CANopen	2.0	1041683

2.1.4

pH or Chlorine Metering Design with Solenoid Metering Pump Beta®

Complete set comprising a microprocessor-controlled Beta® metering pump with PVDF dosing head, pressure limitation to 4 bar 180 strokes/min., PTFE injection valve, wall brackets, PVC suction lance with pre-warning and empty signal for 5 to 50 litre disposable tanks (50 mm tank opening), 5 m signal cable for connection to measuring and control units, as well as PVC metering line (10 m, 4/10 mm) and bleed line (3 m, 4/6 mm).



P_BE_0049_SW

Feed rate at max. pressure l/h	Max. pressure bar	Order no.
1.5	4.0	1041523
2.8	4.0	1041554
4.5	4.0	1041556

2.1.5

pH or Chlorine Metering Design with Solenoid Metering Pump Beta®, CANopen

Feed rate at max. pressure l/h	Max. pressure bar	Order no.
1.5	4.0	1041558
2.8	4.0	1041559
5.3	4.0	1041560

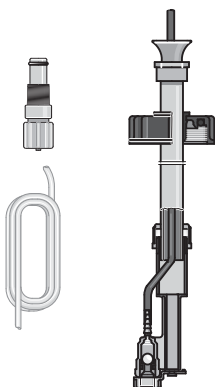
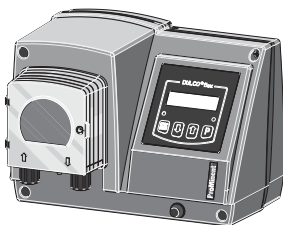


2 Metering Sets

2.2 Metering Sets for Metering of Flocculants

2.2.1

Flocculant Design With Peristaltic Pump DULCO®flex DF4a



P_DX_0011_SW

The hose pump DULCO®flex DF4a has been especially developed for metering flocculents. It is used wherever flocculents are metered constantly and precisely to the water to be treated. Typical applications include swimming pools and whirlpools.

Complete set comprising DULCO®flex peristaltic pump DF4a with metering valve, PVC suction lance for 5 to 50 litre disposable containers (50 mm container opening), with mains cable, standard European plug for connecting to measuring and control units, as well as PVC metering line (10 m, 4/10 mm).

Feed rate at max. pressure l/h	Max. pressure bar	Order no.
0.35	4.0	1044579
1.5	4.0	1041585
6	2.5	1041586
12	2.0	1041588

Advantages

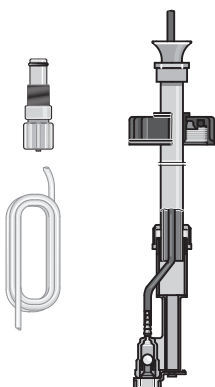
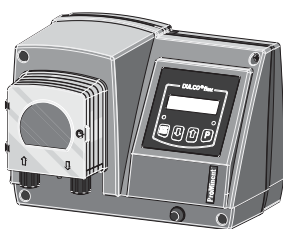
- Simple and convenient operation using language-neutral user guidance and programming via four buttons on the front
- Safe operation by direct input of the circulation capacity and required concentration
- Flexible to retrofit thanks to straightforward installation - also ideal for retrofitting into existing systems
- More efficient operation, for instance due to night setback
- Service-friendly design makes changing the tube easier
- Proportional adaptation of metering capacity by voltage and current input (optional = CAN bus)

Features

- Low-noise speed controllable multiphase motor with ball bearing drive shaft
- No mechanical drive, thus long operating life
- 100% switching time, operating pressure of up to 4 bar is possible
- Stepless setting of the metering capacity from 1 % to 100 %
- Metering capacities of 1.5, 6.0 or 12.0 l/h are shown in the display and can be set reproducibly
- Operating and peristaltic time counter, priming function (continuous run)
- Float switch input (contact 1 or 2 stage), hose fracture monitoring with fault indicating relay
- Switchable capacity change, for instance night setback or increased capacity

2.2.2

Flocculant Design with Peristaltic Pump DULCO®flex DF4a, CANopen



P_DX_0011_SW

Feed rate at max. pressure l/h	Max. pressure bar	Order no.
0.35	4.0	1044579
1.5	4.0	1043054
6	2.5	1043055
12	2.0	1043056

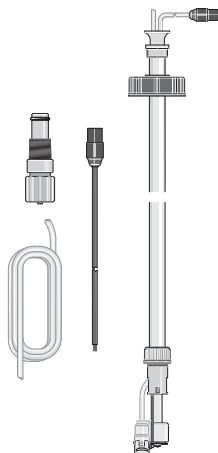
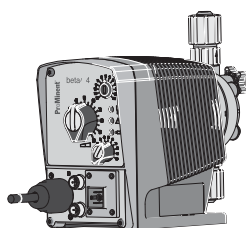


2 Metering Sets

2.2.3

Flocculant Designs With Solenoid Metering Pump Beta®

Complete system for metering flocculants, comprising a Beta® metering pump for accurate individual stroke adjustment, PVDF dosing head with bleed valve, PTFE metering valve, wall brackets, PVC suction lance with pre-warning and empty signal for 5 to 50 litre disposable tanks (50 mm tank opening), 5 m signal cable for locking to the circulation pump as well as a PVC metering line (10 m, 4/10 mm) and PVC bleed line (3 m, 4/6 mm).



P_BE_0049_SW

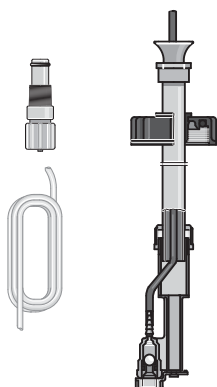
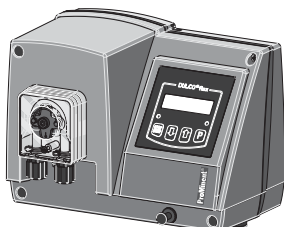
Feed rate at max. pressure l/h	Max. pressure bar	Order no.
0.74	10.0	1043051
1.1	16.0	1043052

2 Metering Sets

2.3 Metering Sets for Metering Fragrances

2.3.1

Fragrance Metering Design with Peristaltic Pump DULCO®flex DF3a



P_DX_0046_SW

- Feed rate range 0.4 - 2.4 l/h at max. 1.5 bar backpressure
- Hose material Viton®, used specifically for the metering of fragrances in wellness applications
- Control of two further peristaltic pumps for different fragrances
- Control of a solenoid valve for the diluent water
- Virtually silent operation
- Self-priming against max. 1.5 bar
- Sprung rollers for constant rolling pressure and increased service life of the hose

The DULCO®flex DF3a was specifically developed for metering fragrances in wellness facilities. This pump can be used wherever fragrances are metered in small quantities. Typical applications include the aroma infusion of douse water in saunas, steambaths, and whirlpools.

The metering pump is equipped with a process timer controlling two further peristaltic pumps for other essences. Since the essences used in saunas should not be used undiluted on the oven, the DF3a is equipped with three relays for controlling the diluent water.

To save essences when the sauna is not in use, the pump features a contact input to which e.g. a door contact or motion sensor can be connected. This ensures that fragrances are only metered when the sauna is in use.

Complete set comprising DULCO®flex hose pump DF3a with injection valve, PVC suction lance for 5 to 50 litre disposable containers (50 mm container opening), with mains cable, standard European plug and PVC metering line (10 m, 4/10 mm).

Feed rate at max. pressure l/h	Max. pressure bar	Order no.
0.8	1.5	1041589
2.4	1.5	1041591

2.4 Options

	Order no.
PVDF multifunctional valve with relief, pressure retention and bleeding function	791715
PVC metering lance with shut-off valve 10 x 4, screw thread 1/2 "	1021529
Components mounted on a wall assembly panel 500 x 600 mm with receiver tank, pump and Flow Control dosing monitor	DKFL1/200



3 Ozone Systems

3.1 Ozone for Swimming Pool Water

The use of ozone drastically solves the problems of the build-up of chloramine or THM experienced with conventional water treatment. It is the most intensive oxidising agent approved for use in water treatment processes and is usually added before the filter stage. With this system, undesirable substances in the water, such as chloramine and turbidity, are oxidised and retained in the filter.

To successfully use ozone in swimming pool water treatment, it must be added in adequate quantities and, above all, dissolved in the water. This is why a high concentration is needed when generating ozone. The addition of chlorine is also necessary when using ozone for safe disinfection, as ozone must not be present in the pool water.

The result is much clearer water without a typical indoor swimming pool odour. Moreover, the content of trihalomethanes significantly drops below the limit values.

The use of ozone in swimming pool technology is particularly valuable as

- the water does not smell, the air indoors is pleasant and healthy
- the clarity of the water improves considerably due to the flocculating effect of the ozone and
- the ozone decomposes on reacting with oxygen, a desirable factor in the water.

The use of ozone in therapy pools is a requirement in Germany.

3.2 Ozone System OZONFILT®OZVa

3.2.1 Ozone System OZONFILT®OZVa

OZONFILT® OZVa ozone plants have been designed as pressurised systems, in which the operating gas – air or oxygen – is fed into the ozone generator under pressure. The ozone is generated using medium-frequency high voltage and is primary current controlled. The introduction of PCC (primary current controlled) technology, specially developed in-house by ProMaqua, provides complete protection for the electrical components (high-voltage transformer and power stage) and also permits the correct digital display of the ozone feed rate in “grams/hour”. As a result, any required ozone volume between 3 and 100 % of the nominal capacity can be set reproducibly, and largely independently of voltage and pressure fluctuations.

The use of an integrated pressure swing dryer and a dielectric with optimum thermal conductivity makes the plant extremely compact. The unique design of the generator ensures outstanding cooling performance with low cooling water consumption and removes the heat produced quickly before the ozone produced can decompose due to excessive heat.

Operation under pressure means that the ozone generated can be introduced directly into water systems with back pressures of up to 2 bar. Additional booster pumps and injectors can therefore be dispensed with in many applications.

Combined with DULCOMETER® measuring and control technology and DULCOTEST® OZE ozone sensors, these systems are especially suitable for use where the operation is dependent on, and is controlled, by the measured data.

Features

- Simple operation
- Fully equipped
- High efficiency
- Low consumption of energy and cooling water
- High ozone concentration thanks to operation with oxygen
- PCC technology ensures complete protection of electrical components
- Correct digital display of ozone output in g/h
- Reproducible setting of the desired ozone quantity between 3 and 100 % of nominal capacity



3 Ozone Systems

3.2.2

OZONFILT® OZVa 1-4 Ozone Production Systems (Operating Gas - Air)

Under nominal conditions, the series OZVa 1 – 4 produces up to 40 g/h of ozone from oxygen in the ambient air at a concentration of 20 g/Nm³. Using the designated mixing devices, ozone concentrations of between 3 and 12 ppm can be achieved in the water to be treated, depending on the temperature (theoretical value at 30 and/or 0 °C).

Types OZVa 1 and 2 are installed in a control cabinet for wall mounting; types OZVa 3 and 4 are installed in a free-standing cabinet.

Provide an adequate supply of compressed air and a mixing unit designed for the operating conditions for operation of the ozone system.

Mixing equipment

OZVa 1 can be ordered in the following designs:

- Transparent mixing system with flow control mounted on the side of the system (see Fig. pk_7_001_1_V2)
- PVC static helical mixer mounted directly below the system, with 4 helical blades (pressure drop approx. 0.4 bar at maximum throughput) (see Fig. pk_7_042_V2)
- Without mixing system for connection of 12/10 mm stainless steel pipes or 12/9 mm PTFE pipes

OZVa 2 can be ordered in the following designs:

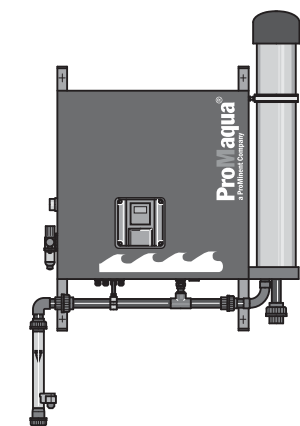
- PVC static helical mixer mounted directly below the system, with 4 helical blades (pressure drop approx. 0.4 bar at maximum throughput) (see Fig. pk_7_042_V2)
- Without mixing system for connection of 12/10 mm stainless steel pipes or 12/9 mm PTFE pipes

OZVa 3 and 4 are delivered, in principle, as designs without mixing system; order a suitable mixing system separately (see Fig. pk_07_043_V2).

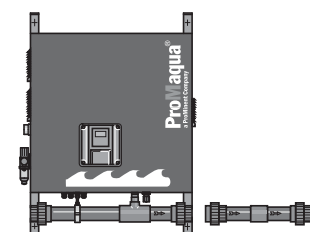
PVC or Stainless Steel Static Helical Mixer see p. → 3-33

Notes

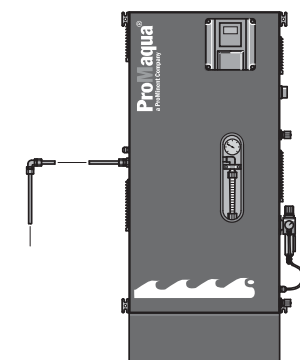
- The length of ozone gas-transporting pipes and the number of joints should be kept to a minimum. All rooms with a removable joint are to be monitored with a gas detector according to applicable German accident prevention regulations. All OZONFILT® systems are equipped for fitting a gas detector, such as a GMA 36 Ozon (see accessories).
- The ozone generator must be interlocked with the water flow into the metering point on all installations.
- A non-return valve should be installed upstream of the OVZa to prevent any return of ozonised water into the ozone-transporting pipe.



pk_7_001_1_V2
OZONFILT® OZVa 1; capacity: 5 g/h



pk_7_042_V2
OZONFILT® OZVa 2; capacity: 15 g/h



pk_7_043_V2
OZONFILT® OZVa 3; capacity: 35 g/h

3 Ozone Systems

Technical Data

OZONFILT® OZVa 1-4 ozone production systems (operating gas - air)

Ambient parameters

Max. 85 % air humidity of the ambient air, non-condensing, non-corrosive, dust-free, max. ambient temperature: 40 °C (with integrated air conditioning system: 50 °C)

		OZVa 1	OZVa 2	OZVa 3	OZVa 4
Number of generator modules		1	1	2	2
Ozone capacity, measured in accordance with DIN with air at 20 °C, cooling water at 15 °C	g/h	5	15	35	40
Air consumption (only ozone generation)	Nm³/h	0.25	0.75	1.75	2
Ozone concentration in the gas phase referenced to nominal conditions	g/Nm³ *	20	20	20	20
Specific energy requirement at nominal capacity	Wh/g	30	30	21	20
Power factor at full capacity	cos φ	0.70	0.98	0.98	0.98
Ozone connection		integrated in mixing device or G 1/4" internal	integrated in mixing device or G 1/4" internal	G 1/4" internal	G 1/4" internal

* with air at 20 °C, cooling water at 15 °C

** Nm³ = m³ under standard conditions (p = 1.013x10⁵ Pa, T = 273 K)

Electrical connection

		OZVa 1	OZVa 2	OZVa 3	OZVa 4
Connected load	V/Hz/A	230/50;60/1,2	230/50;60/3	230/50;60/6	230/50;60/6
Enclosure rating		IP 43	IP 43	IP 43	IP 43

Overall dimensions (without mixer)

		OZVa 1	OZVa 2	OZVa 3	OZVa 4
Width	mm	840	840	710	710
Height	mm	840	805	1,400	1,400
Depth	mm	310	310	310	310

Weight

		OZVa 1	OZVa 2	OZVa 3	OZVa 4
Weight	kg	70	75	121	121

Ozone mixing

		OZVa 1	OZVa 2	OZVa 3	OZVa 4
Max. raw water temperature	°C	35	35	35	35
Permissible pressure at ozone outlet	bar	0.8–2.0	0.8–2.0	0.8–2.0	0.8–1.5

Air supply

		OZVa 1	OZVa 2	OZVa 3	OZVa 4
Required air volume	l/min	6.2	17	38	42

Air quality

oil and dust-free, non-corrosive, Constant upstream pressure of 6-10 bar

Cooling water

		OZVa 1	OZVa 2	OZVa 3	OZVa 4
Cooling water requirement	l/h	10–60	20–60	50–100	70–100
Cooling water inlet pressure	bar	1–5	1–5	1–5	1–5
Cooling water inlet, PE pressure hose	mm	6 x 4	6 x 4	6 x 4	6 x 4
Cooling water outlet, open discharge	mm	6 x 4	6 x 4	6 x 4	6 x 4
Cooling water temperature at ambient temp. max. 35 °C	°C	<30	<30	<30	<30
Cooling water temperature at ambient temp. 35–40 °C	°C	<25	<25	<25	<25

Cooling water quality

No tendency to form lime scale; Removable substances: < 0.1 ml/l ; Iron: < 0.2 mg/l; Manganese: < 0.05 mg/l; no corrosive components; Conductivity: > 100 µS/cm



3 Ozone Systems

3.3 Compressors, Accessories

3.3.1 Compressors for OZONFILT® OZVa 1-4

Atlas Copco LFX compressors

The outstanding feature of this range of compressors is their outstanding value for money. They are equipped with active start unloading and automatic condensate discharge by solenoid valve. The compressors are not suitable for continuous operation and should only be used in less harsh operating conditions.

Technical Data

Type		LFX 0.7	LFX 1.5
Free air delivery rate at 7 bar	l/min	61	124
Power consumption at 7 bar	W	530	970
Number of cylinders		1	1
Sound pressure level	dB(A)	62	64
Air receiver capacity	l	20	20
Weight	kg	44	48
Suitable for OZVa Type		1 + 2	3 + 4

Type	Type	Order no.
LFX 0.7	230 V/50 Hz	1004458
LFX 0.7	230 V/60 Hz	1010719
LFX 1.5	230 V/50 Hz	1006343
LFX 1.5	230 V/60 Hz	1009638

Air filter kit

	Order no.
Air filter kit for Atlas Copco LFX compressors	1005789

Dürr ABK compressors

The outstanding feature of this continuously rated range of compressors is their extremely robust construction, making them ideally suitable for industrial use. They are equipped with active start unloading, automatic condensate discharge by solenoid valve and an hours-run meter. PTFE coated special aluminium pistons lead to the long service life and reliability of these compressor units.

Technical Data

Type		TA-080	HA-234
Free air delivery rate at 7 bar	l/min	62	152
Supply max.	VAC	230	230
Supply frequency	Hz	50/60	50
Power consumption at 7 bar	W	800	1,900
Number of cylinders		1	3
Sound pressure level	dB(A)	68	78
Air receiver capacity	l	25	55
Weight	kg	49	70
Suitable for OZVa Type		1 + 2	3 + 4

Type	Order no.
TA-080	1025398
HA-234	1025399



3 Ozone Systems

Air filter kit

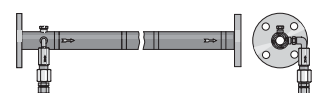
	Order no.
Air filter kit for Dürr ABK compressors*	1025400

* 1 filter kit is required per cylinder.

Compressors with refrigeration drying for operation in conditions of high humidity, and high-capacity screw compressors for connection to several ozone plants are available on request.

3.3.2

PVC or Stainless Steel Static Helical Mixer



pk_7_072
Static Helical Mixer

Designed for intensive mixing of gas with liquid flows. 4 helical blades ensure optimum mixing of the ozone with minimal pressure drop (0.1 bar per blade at maximum flow). The specified flow range of the static helical mixer should be complied with for optimum mixing results.

Version with loose flanges to DIN 2501 and integrated injection point made from stainless steel with couplings for 12 mm diam. stainless steel tube, or 12/9 mm PTFE hose, using stainless steel support inserts. In addition, the injection point is fitted with a non-return valve to protect the ozone plant from reverse flowing water. The mixers are manufactured as grease-free, so they are also suitable for Types OZVa 5-7. The stainless steel version has a G 1/4" pressure gauge tapping at the ozone mixing point.

Flow m ³ /h	Material	Overall length mm	Connector	Order no.
5 – 10	PVC-U	718	DN 40	1024324
10 – 15	PVC-U	718	DN 50	1024325
15 – 25	PVC-U	718	DN 65	1024326
25 – 35	PVC-U	1,100	DN 80	1024327
35 – 50	PVC-U	1,100	DN 100	1024328
50 – 90	PVC-U	1,300	DN 125	1034641
95 – 160	PVC-U	1,700	DN 150	1034640
5 – 10	1.4404	718	DN 40	1022503
10 – 15	1.4404	718	DN 50	1022514
15 – 25	1.4404	718	DN 65	1022515
25 – 35	1.4404	1,100	DN 80	1022516
35 – 50	1.4404	1,100	DN 100	1024154

Other sizes on request

Connecting parts for the gas pipeline

	Order no.
Stainless steel pipe 12/10 mm, Sold in metres	015743
Stainless steel pipe 12/10 mm, grease-less, 1.4 m	1022463
PTFE hose 12/9 mm, grease-less, sold in metres	037428
Stainless steel support inserts, 2 No. for 12/9 mm PTFE hose, grease-less	1025397
Stainless steel coupling 12 mm - R 1/4, grease-less	1025755
Stainless steel fitting 12 mm - R 3/8, grease-less	1034642
Stainless steel 90° elbow D 12 - D 12, grease-less	1022462
Stainless steel pressure relief valve, Adjustable pressure range 0.07 – 2 bar, Connection size: 1/4" NPT, 2 additional inputs for connecting 2 pressure gauges.	1029032
Stainless steel back pressure valve for OZMa 1 – 3 A and OZMa 4 – 6 O, adjustable pressure range 0.5 – 10 bar, connector G 3/4" external thread, grease-free	1039408
Spare parts kit for back pressure valve order no. 1039408	1039410
Stainless steel back pressure valve for OZMa 4 – 6 A, adjustable pressure range 0.5 – 10 bar, connector G 1 1/4" external thread, grease-free	1039409
Spare parts kit for back pressure valve order no. 1039409	1039411



3 Ozone Systems

3.4 Ozone System OZONFILT® Compact OMVa

3.4.1 Ozone System OZONFILT® Compact OMVa

Individually adaptable thanks to modular construction

Ozone capacity 5 to 70 g ozone/h

The OZONFILT® Compact OMVa is a complete, ready-to-use ozone system. The components are perfectly coordinated to each other.

The ozone system OZONFILT® Compact OMVa has a modular construction mounted on a stainless steel frame. It can therefore be simply adapted to and integrated in the respective application.

A sufficient quantity and constant concentration of ozonised water is produced in the system's reaction tank. From there it is pumped to where it is needed. The required ozone concentration can be adjusted and is continuously controlled and held constant by a measuring and control circuit. Depending on the application, the ozonised water is pumped by system pressure or with one or more discharge pumps to where it is needed.

With the removal and replenishment of water in the storage tank, the air, which contains ozone, is discharged out via the water phase and via a residual ozone gas destructor. Ensure that no ozone escapes into the ambient air in normal operation.

Your benefits

- Excellent process reliability through the use of a pre-assembled, complete ozone treatment stage with perfectly matched components.
- Well-thought-out installation on a stainless steel frame for plug-and-play connection
- Modular construction which can still be customised
- Compression-proof ozone generator built in compliance with DIN 19627
- Destruction of residual ozone gas for the removal of traces of ozone gas
- Room air monitoring for traces of ozone gas via a gas detector with a sensor with long-term stability
- Metering ozone, depending on the measured values, ensures a constant ozone concentration in the reaction tank

Technical details

- Ozone metering point with a downstream mixing section made of stainless steel, with a series of static mixing elements for intensive mixing of the ozone/air mix.
- Stainless steel reaction tank.
- Ozone gas is safely converted to oxygen with the integral water separator in the residual ozone gas destruction unit.
- A gas detector alerts you as soon as the limit values for ozone in the ambient air are exceeded and ozone production is stopped.
- A central electric control ensures metering of ozone depending on the measured values and the control of all connected peripheral components.
- Clear and simple operation, as well as signal exchange with higher-order control systems

Ozone generation module (1), built in accordance with DIN 19627:

The ozone is produced with an OZONFILT® OZVa in a pressure-resistant ozone generator using an electronically produced and regulated medium-frequency.

Ozone mixing module (2):

This module comprises an ozone metering point and a downstream mixing section made of stainless steel, with a series of static mixing elements for intensive mixing of the ozone/air mix with the water to be treated. The lines carrying the ozone, and the pipework from the raw water connection to the inlet to the reaction tank are made throughout in stainless steel and have been factory-pressure tested.

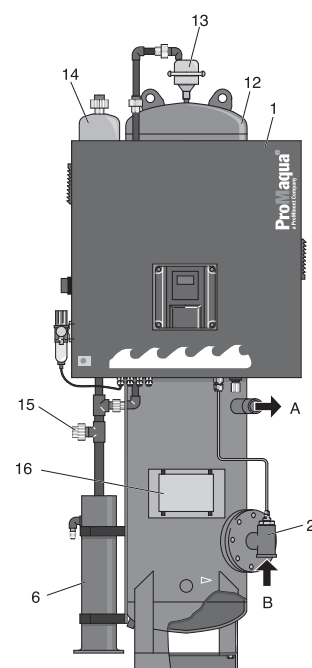
An injector for sucking out ozone is not needed with back pressures of up to 1.8 bar because the ozone is generated at positive pressure.

Reaction tank module (12):

The stainless steel reaction tank incorporates all the necessary fittings for water distribution and an automatic bleed valve (13). The ozone generation module (1), the residual ozone gas destructor (14) and room air monitor (16) are mounted on this storage tank (12).

Residual ozone gas destruction module (14):

The residual ozone gas destruction module (14) has an integrated water separator (6) to remove any traces of ozone gas in the exhaust air coming from the reaction tank (12). The connection for the exhaust air from any downstream filter system (15) that may be fitted is likewise provided.



pk_7_024_V2

A to filtration
B Raw water

3 Ozone Systems

Room air monitoring module (16):

The room air is monitored for traces of ozone gas using a calibrated gas detector with a long-term stable electrochemical sensor.

If the alarm threshold is exceeded, ozone generation is stopped and an alarm signalled. A buzzer is activated at the same time.

Field of application

- **Food and beverage industry:** Oxidation of iron and manganese, disinfection of potable water and rinsing water
- **Swimming pools:** Degradation of disinfection by-products, reliable microbiological barrier and production of crystal-clear water thanks to its microflocculating effect

Technical Data

Type		OMVa 5 – 200	OMVa 15 – 500	OMVa 35-1,000	OMVa 40-1,000	OMVa 70-2,000
Type: ozone generation system		OZVa 1	OZVa 2	OZVa 3	OZVa 4	OZMa 1A
Ozone output at 20 g/Nm ³	g/h	5	15	35	40	70
Cooling water volume (15 °C)	l/h	10–60	20–60	50–100	70–100	90
Nominal throughput	m ³ /h	1.5 – 5	5 – 15	15 – 30	30 – 45	45 – 60
Enclosure rating		IP 43	IP 43	IP 43	IP 43	IP 43
Connected load	V/Hz/A	230/50;60/1.2	230/50;60/3	230/50;60/6	230/50;60/6	230/50;60



3 Ozone Systems

3.5 Accessories and Spare Parts for Ozone Systems

3.5.1 Bleed Valves

Suitable for types	Connector	Pres- sure bar	Order no.
OZVa 1 – 7	R 3/4" internal x R 1/2" external	0 – 6.0	302525
OZMa 1 – 30/OZMa 1A	R 1" internal x R 1/2" external	0 – 2.0	302526
OZMa 2-4A / OZMa 4-6O	R 1" internal x R 3/4" external	0 – 2.0	303845

Bleed valves made of stainless steel 1.4571 in ozone-resistant version for mounting on reaction tanks.

3.5.2 Residual Ozone Gas Destructor

Residual ozone gas destruction is used to remove traces of ozone gas from the exhaust air coming from the reaction tank. Because the exhaust air from the reaction tank still contains water, the pipework should be suitably routed so as to ensure that the water is drained off at the inlet side.

A suitable drainage connection should be provided here too as the exhaust air after the residual ozone gas destructor is still up to 100 % saturated with water vapour, and because small temperature fluctuations, even on the outlet side, can lead to flowback of condensate.

The exhaust air from any downstream filter plant that may be fitted can also be routed via this ozone gas destruction unit.

PVC version

Residual ozone destructor based on active carbon granules in a PVC housing.

	Type	Ozone quantity g/h	Order no.
Residual ozone destructor 3 L	10	10	879022
Residual ozone destructor 14 L	40	40	1004267
Residual ozone destructor 30 L	100	100	879019
Residual ozone destructor 60 L	200	200	879018

Note:

The stated ozone quantities refer to quantities added to the raw water. The residual ozone destructor is designed for the normal residual ozone concentration found in swimming pool applications. It should only be used in plants with air as operating gas and a maximum added quantity of 1.5 g of ozone/m³ treated water.



3 Ozone Systems

3.6

Ozone System OZONFILT® OZMa

Powerful and yet environmentally-friendly. Disinfect and oxidise ecologically and economically.

Ozone capacity 70 to 735 g ozone/h



OZONFILT® OZMa represents maximum operational safety with minimal operating costs. The ozone generator is maintenance-free and generates up to 735 g/h of ozone from compressed air or oxygen.

The ozone systems OZONFILT® OZMa have been designed as pressurised systems, in which the operating gas – air or oxygen – is fed into the ozone generator under pressure.

Air is used as the operating gas in the ozone system Ozonfilt® OZMaA types 1 to 6

The ozone is generated from the oxygen in the ambient air and simultaneously metered. A demand-led, self-optimising pressure swing dryer reduces the consumption of compressed air to a minimum. Ozone can therefore be generated operationally safely and reliably even with a high level of ambient air humidity with ozone concentrations of up to 20 g/Nm³. Ozone concentrations of between 3 and 12 ppm can be achieved in the water to be treated with suitable mixing units, depending on the temperature.

Oxygen is used as the operating gas in the ozone system Ozonfilt® OZMaO type 1 to 6

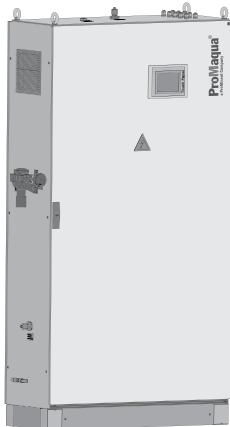
Operation with oxygen permits ozone generation with ozone concentrations of up to 150 g/Nm³. Depending on the system type, ozone is produced in 1-3 generators from oxygen provided from special oxygen generators or bottles. Using the suitable mixing equipment, ozone concentrations of up to 90 ppm can be achieved in the water to be treated, depending on the temperature.

Your benefits

- Economical: maintenance-free generator concept with virtually unlimited service life
- Up to 30% energy savings for air treatment, thanks to demand-led and self-optimising air drying compared with conventional air treatment.
- Automatic control of the operating gas depending on the ozone output, therefore reduced consumption of operating gas produced with intensive use of energy.
- High ozone concentration ensures optimum ozone solubility in water
- Direct injection without injector system at up to 2 bar back pressure
- Automatic ozone generation, virtually independent of fluctuations in main voltage and pressure
- Simple, safe and reliable operation and process visualisation thanks to large, colour and clear 7" touch panel
- Infinite adjustment and precise output control of between 3% and 100% of the nominal power with display of the ozone volume in "grammes/hour"

Technical details

- Compact mounting, ready-to-use in a painted steel cabinet or optionally in a stainless steel cabinet
- With integrated filter package for the removal of dust and small amounts of residual oil in the compressed air
- Special dielectric with excellent cooling: In spite of the low cooling water consumption, heat is quickly and efficiently discharged before the ozone produced can decompose due to excessive heat.
- PLC with integrated ozone measurement and PID control
- 7" touch panel with data logger and screen recorder
- Multiple communication interfaces (e.g. LAN, PROFIBUS® DP)
- Excellent efficiency: Over 90 % of the ozone is dissolved in the water, thanks to the special construction of the mixing unit.
- Integration of a dew point sensor to monitor the quality of compressed air
- Integration of an air conditioning unit to adjust the temperature of the ozone system
- Pause input for external switching on/off
- Contact input for locking the system, for example in the absence of flow
- Digital input for connection of a gas detector
- Digital input for control of two power stages
- 0/4-20 mA input for external output control depending on the flow or measured value with a PIC controller
- Second freely configurable 0/ 4-20 mA input
- Contact output for operating status
- Contact output for collective malfunction alert
- Contact output for limit value transgression, ozone concentration in the water too low
- One freely configurable 0/ 4-20 mA output



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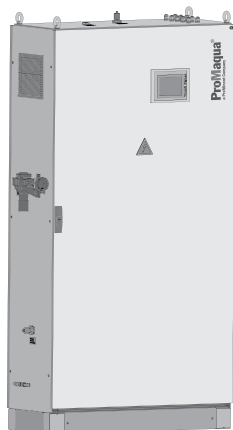
3 Ozone Systems

Field of application

- **Potable water supply:** Oxidation of iron, manganese and arsenic, refinement and taste enhancement and disinfection
- **Waste water treatment:** Degradation/reduction of COD and microcontaminants, reduction of sewage sludge
- **Food and beverage industry:** Oxidation of iron and manganese, disinfection of potable water and rinsing water
- **Swimming pools:** Degradation of disinfection by-products, reliable microbiological barrier and production of crystal-clear water thanks to its microfloculating effect
- **Industry:** Legionella prevention and disinfection of cooling water

3.6.1

Ozone Generation Systems OZONFILT® OZMa 1-6 A (Operating Gas - Air)



P_PMA_OF_0010_SW

Under nominal conditions, the OZMa 1-6 A range produces up to 420 g/h of ozone from compressed air at a concentration of 20 g/Nm³. Using the designated mixing devices, ozone concentrations of between 3 and 12 ppm can be achieved in the water to be treated, depending on the temperature (theoretical value at 30 or 0 °C).

Different feature options can be achieved by combining different Identity code characteristics.

The plants are pre-mounted ready for connection in a painted steel cabinet (optional stainless steel control cabinet) and need only be connected to a single-phase voltage supply, compressed air, cooling water/waste water and ozone metering point on the customer's site.

An adequate compressed air supply and a mixing device designed for the operating conditions should be integrated for operation of the ozone plant.

Order Information for OZONFILT® OZMa Systems see p. → 3-44, PVC or Stainless Steel Static Helical Mixer see p. → 3-33

Mixing equipment

All OZMa systems are delivered, in principle, without a mixing unit and a suitable mixing system has to be ordered separately. When selecting a suitable mixing system, please note that the mixing of ozone is more efficient the higher the water flow in the mixing system. Accordingly design the mixing system so that the flow of the water to be treated is at the upper range of the flow specification.

PVC or Stainless Steel Static Helical Mixer see p. → 3-33

Notes on installation

Keep the length of ozone gas transporting pipes and the number of joints to a minimum. All rooms with a removable joint should be monitored with a gas detector in line with the applicable German accident prevention regulations. All OZONFILT® systems are equipped for fitting a gas detector such as GMA 36 Ozon.

Ozonisation adds a large amount of gas to the water of which only a small percentage can dissolve. Adequate bleeding should therefore be integrated. Because the gases discharged in this way have a considerable residual ozone concentration, appropriate residual ozone destructors should be installed.

The ozone generator should be interlocked with the water flow into the metering point for all installations.

A non-return valve should be installed between OZMa and the ozone metering point to prevent any return of ozonised water into the ozone-transporting pipe.

Room Air Monitor see p. → 3-45, Residual Ozone Gas Destructor see p. → 3-36



3 Ozone Systems

Technical Data

Ozone Generation Systems OZONFILT® OZMa 1-3 A (Process Gas - Air)

Ambient parameters

Max. 85 % air humidity of the ambient air, non-condensing, non-corrosive, dust-free, max. ambient temperature: 40 °C (with integrated air conditioning system: 50 °C)

		OZMa 1A	OZMa 2A	OZMa 3A
Number of generator modules		1	1	1
Ozone capacity, measured in accordance with DIN with air at 20 °C, cooling water at 15 °C	g/h	70	105	140
Air consumption (only ozone generation)	Nm³/h	3.50	5.25	7.00
Ozone concentration in the gas phase referenced to nominal conditions	g/Nm³ *	20	20	20
Specific energy requirement at nominal capacity	Wh/g	16.5	16.5	16.5
Power factor at full capacity	cos φ	0.95	0.95	0.95
Ozone connection		Rp 3/8"	Rp 3/8"	Rp 3/8"

* Nm³= m³ at standard conditions (P = 1.013x10⁵Pa, T = 273 K)

Electrical connection

		OZMa 1A	OZMa 2A	OZMa 3A
Connected load	V/Hz/A	230/50;60/10	230/50;60/16	230/50;60/16
Enclosure rating		IP 43	IP 43	IP 43

Overall dimensions (without mixer)

		OZMa 1A	OZMa 2A	OZMa 3A
Width	mm	1,114	1,114	1,114
Height	mm	1,961	1,961	1,961
Depth	mm	405	405	405

Weight

		OZMa 1A	OZMa 2A	OZMa 3A
Weight	kg	270	280	300

Ozone mixing

		OZMa 1A	OZMa 2A	OZMa 3A
Max. raw water temperature	°C	35	35	35
Permissible pressure at ozone outlet	bar	0.8–2.0	0.8–2.0	0.8–2.0

Air supply

		OZMa 1A	OZMa 2A	OZMa 3A
Required air volume	l/min	73	110	147

Air quality

Oil and dust-free, Non-corrosive, Constant upstream pressure of 4.5 - 10 bar

Cooling water

		OZMa 1A	OZMa 2A	OZMa 3A
Cooling water consumption (15 °C)	l/h	90	135	180
Cooling water consumption (30 °C)	l/h	200	300	400
Cooling water inlet pressure	bar	2–5	2–5	2–5
Cooling water inlet, PE pressure hose	mm	8 x 5	8 x 5	12 x 9
Cooling water outlet, open discharge	mm	8 x 5	8 x 5	12 x 9

Cooling water quality

No tendency to form lime scale; Removable substances: < 0.1 ml/l ; Iron: < 0.2 mg/l; Manganese: < 0.05 mg/l; no corrosive components; Conductivity: > 100 µS/cm



3 Ozone Systems

Ozone Generation Systems OZONFILT® OZMa 4-6 A (Process Gas - Air)

Ambient parameters

Max. 85 % air humidity of the ambient air, non-condensing, non-corrosive, dust-free, max. ambient temperature: 40 °C (with integrated air conditioning system: 50 °C)

		OZMa 4A	OZMa 5A	OZMa 6A
Number of generator modules		2	2	3
Ozone capacity, measured in accordance with DIN with air at 20 °C, cooling water at 15 °C	g/h	210	280	420
Air consumption (only ozone generation)	Nm ³ /h	10.50	14.00	21.00
Ozone concentration in the gas phase referenced to nominal conditions	g/Nm ³ *	20	20	20
Specific energy requirement at nominal capacity	Wh/g	16.5	16.5	16.5
Power factor at full capacity	cos φ	0.95	0.95	0.95
Ozone connection		Rp 3/8"	Rp 3/8"	Rp 3/8"

* Nm³ = m³ at standard conditions (P = 1.013x10⁵Pa, T = 273 K)

Electrical connection

		OZMa 4A	OZMa 5A	OZMa 6A
Connected load	V/Hz/A	400/50;60/16	400/50;60/16	400/50;60/16
Enclosure rating		IP 43	IP 43	IP 43

Overall dimensions (without mixer)

		OZMa 4A	OZMa 5A	OZMa 6A
Width	mm	1,320	1,320	1,606
Height	mm	1,961	1,961	1,961
Depth	mm	605	605	605

Weight

		OZMa 4A	OZMa 5A	OZMa 6A
Weight	kg	420	445	589

Ozone mixing

		OZMa 4A	OZMa 5A	OZMa 6A
Max. raw water temperature	°C	35	35	35
Permissible pressure at ozone outlet	bar	0.8–2.0	0.8–2.0	0.8–2.0

Air supply

		OZMa 4A	OZMa 5A	OZMa 6A
Required air volume	Nl/min	220	293	440

Air quality

Oil and dust-free, Non-corrosive, Constant upstream pressure of 4.5 - 10 bar

Cooling water

		OZMa 4A	OZMa 5A	OZMa 6A
Cooling water consumption (15 °C)	l/h	270	360	540
Cooling water consumption (30 °C)	l/h	600	800	1,200
Cooling water inlet pressure	bar	2–5	2–5	2–5
Cooling water inlet, PE pressure hose	mm	12 x 9	12 x 9	12 x 9
Cooling water outlet, open discharge	mm	12 x 9	12 x 9	12 x 9

Cooling water quality

No tendency to form lime scale; Removable substances: < 0.1 ml/l ; Iron: < 0.2 mg/l; Manganese: < 0.05 mg/l; no corrosive components; Conductivity: > 100 µS/cm



3 Ozone Systems

3.6.2

Ozone Generation Systems OZONFILT® OZMa 1-6 O (Operating Gas - Oxygen)

Under nominal conditions, the OZMa 1-6 O range produces up to 735 g/h of ozone from oxygen at a concentration of up to 150 g/Nm³. Using the designated mixing devices, ozone concentrations in the water to be treated of up to 90 ppm can be achieved (theoretical value at 0 °C). Ozone concentration in g/Nm³ and system feed rate in g/h can be varied depending on the operating conditions and can thus be individually matched to the application conditions. Examples for various combinations are listed in the technical data table.

Different feature options can be achieved by combining different Identity code characteristics.

The systems are pre-mounted ready for connection in a painted steel cabinet (optional stainless steel control cabinet) and should only be connected to a single-phase voltage supply, oxygen, cooling water/waste water and ozone metering point on the customer's site.

Order Information for OZONFILT® OZMa Systems see p. → 3-44

Mixing equipment

All OZMa systems are delivered, in principle, without a mixing unit and a suitable mixing system has to be ordered separately. When selecting a suitable mixing system, please note that the mixing of ozone is more efficient the higher the water flow in the mixing system. Accordingly design the mixing system so that the flow of the water to be treated is at the upper range of the flow specification.

We recommend using stainless steel mixing systems because of the high ozone concentrations. Mixing systems made of PVC can have a reduced service life, depending on the operating conditions.

PVC or Stainless Steel Static Helical Mixer see p. → 3-33

Notes on installation

Keep the length of ozone gas transporting pipes and the number of joints to a minimum. All rooms with a removable joint should be monitored with a gas detector in line with the applicable German accident prevention regulations. All OZONFILT® systems are equipped for fitting a gas detector such as GMA 36 Ozon.

Depending on the operating and installation conditions, it might be necessary to also monitor the room air for excessive oxygen content. The gas detector GMA 36 Oxygen can be used for this purpose.

All gas-transporting accessories should be resistant to ozone and oxygen (e. g. fat-free).

Ozonisation adds a large amount of gas to the water of which only a small percentage can dissolve. Adequate bleeding should therefore be integrated. Because the gases discharged this way have a considerable residual ozone concentration, appropriate residual ozone destructors should be installed. Because of the high ozone concentrations, only catalytic residual ozone destructors can be used. Active carbon-based residual ozone destructors ignite spontaneously if subjected to increased ozone concentrations.

The ozone generator must be interlocked with the water flow into the metering point for all installation.

A non-return valve should be installed between OZMa and ozone metering point to prevent any return of ozonised water into the ozone-transporting pipe.

Room Air Monitor see p. → 3-45, Residual Ozone Gas Destructor see p. → 3-36



3 Ozone Systems

Technical Data

Ozone Generation Systems OZONFILT® OZMa 1-3 O (Operating Gas - Oxygen)

Ambient parameters

Max. 85 % air humidity of the ambient air, non-condensing, non-corrosive, dust-free, max. ambient temperature: 40 °C (with integrated air conditioning system: 50 °C)

		OZMa 1 O	OZMa 2 O	OZMa 3 O
Number of generator modules		1	1	1
Nominal ozone capacity at 100 g/Nm ³ ** and cooling water at 15 °C	g/h	105	158	210
Ozone capacity at 150 g/Nm ³ *	g/h	60	90	120
Ozone capacity at 80 g/Nm ³	g/h	123	184	245
Specific energy requirement at nominal capacity	Wh/g	9	9	9
Power factor at full capacity	cos φ	0.95	0.95	0.95
Ozone connection		Rp 3/8"	Rp 3/8"	Rp 3/8"

Electrical connection

		OZMa 1 O	OZMa 2 O	OZMa 3 O
Connected load	V/Hz/A	230/50;60/10	230/50;60/16	230/50;60/16
Enclosure rating		IP 43	IP 43	IP 43

Overall dimensions

		OZMa 1 O	OZMa 2 O	OZMa 3 O
Width	mm	1,114	1,114	1,114
Height	mm	1,961	1,961	1,961
Depth	mm	400	400	400

Weight

		OZMa 1 O	OZMa 2 O	OZMa 3 O
Weight	kg	220	230	250

Ozone mixing

		OZMa 1 O	OZMa 2 O	OZMa 3 O
Max. raw water temperature	°C	35	35	35
Permissible pressure at ozone outlet	bar	0.8–2.0	0.8–2.0	0.8–2.0

Specification of operating gas: oxygen

		OZMa 1 O	OZMa 2 O	OZMa 3 O
Gas volume at nominal capacity 100 g/Nm ³	l/h	1,050	1,580	2,100
Gas volume at capacity 150 g/Nm ³	l/h	400*	600*	800*
Gas volume at capacity 80 g/Nm ³	l/h	1,540	2,300	3,100
Concentration min.	vol%	90	90	90
Dew point max.	°C	-50	-50	-50
Pressure	bar	3 – 6	3 – 6	3 – 6
Max. particles	µm	5	5	5
Max. hydrocarbons	ppm	20	20	20
Max. temperature	°C	30	30	30

Cooling water

		OZMa 1 O	OZMa 2 O	OZMa 3 O
Cooling water consumption (15 °C)	l/h	120	180	240
Cooling water consumption (30 °C)	l/h	200	300	400
Cooling water inlet pressure	bar	1–5	1–5	1–5
Cooling water inlet, PE pressure hose	mm	12 x 9	12 x 9	12 x 9
Cooling water outlet, open discharge	mm	12 x 9	12 x 9	12 x 9

Cooling water quality

No tendency to form lime scale, no corrosive components; Sedimentable substances: < 0.1 ml/l; Iron: < 0.2mg/l; Manganese: < 0.05 mg/l; Conductivity: > 100 µS/cm; Chloride: < 250 mg/l

* Output 150 g/Nm³ as special version must be factory-set

** Nm³ = m³ at standard conditions (P = 1.013x10⁵Pa, T = 273 K)



3 Ozone Systems

Ozone Generation Systems OZONFILT® OZMa 4-6 O (Operating Gas - Oxygen)

Ambient parameters

Max. 85 % air humidity of the ambient air, non-condensing, non-corrosive, dust-free, max. ambient temperature: 40 °C (with integrated air conditioning system: 50 °C)

		OZMa 4 O	OZMa 5 O	OZMa 6 O
Number of generator modules		2	2	3
Nominal ozone capacity at 100 g/Nm ³ ** and cooling water at 15 °C	g/h	320	420	630
Ozone capacity at 150 g/Nm ³ *	g/h	180	240	360
Ozone capacity at 80 g/Nm ³	g/h	370	490	735
Specific energy requirement at nominal capacity	Wh/g	9	9	9
Power factor at full capacity	cos φ	0.95	0.95	0.95
Ozone connection		Rp 3/8"	Rp 3/8"	Rp 3/8"

Electrical connection

		OZMa 4 O	OZMa 5 O	OZMa 6 O
Connected load	V/Hz/A	400/50;60/16	400/50;60/16	400/50;60/16
Enclosure rating		IP 43	IP 43	IP 43

Overall dimensions

		OZMa 4 O	OZMa 5 O	OZMa 6 O
Width	mm	1,320	1,320	1,320
Height	mm	1,961	1,961	1,961
Depth	mm	605	605	605

Weight

		OZMa 4 O	OZMa 5 O	OZMa 6 O
Weight	kg	320	345	415

Ozone mixing

		OZMa 4 O	OZMa 5 O	OZMa 6 O
Max. raw water temperature	°C	35	35	35
Permissible pressure at ozone outlet	bar	0.8–2.0	0.8–2.0	0.8–2.0

Specification of operating gas: oxygen

		OZMa 4 O	OZMa 5 O	OZMa 6 O
Gas volume at nominal capacity 100 g/Nm ³	NI/h	3,200	4,200	6,300
Gas volume at capacity 150 g/Nm ³	NI/h	1,200*	1,600*	2,400*
Gas volume at capacity 80 g/Nm ³	NI/h	4,630	6,130	9,190
Concentration min.	vol%	90	90	90
Dew point max.	°C	-50	-50	-50
Pressure	bar	3 – 6	3 – 6	3 – 6
Max. particles	µm	5	5	5
Max. hydrocarbons	ppm	20	20	20
Max. temperature	°C	30	30	30

Cooling water

		OZMa 4 O	OZMa 5 O	OZMa 6 O
Cooling water consumption (15 °C)	l/h	200	280	420
Cooling water consumption (30 °C)	l/h	330	470	700
Cooling water inlet pressure	bar	1–5	1–5	1–5
Cooling water inlet, PE pressure hose	mm	12 x 9	12 x 9	12 x 9
Cooling water outlet, open discharge	mm	12 x 9	12 x 9	12 x 9

Cooling water quality

No tendency to form lime scale, no corrosive components; Sedimentable substances: < 0.1 ml/l; Iron: < 0.2mg/l; Manganese: < 0.05 mg/l; Conductivity: > 100 µS/cm; Chloride: < 250 mg/l

* Output 150 g/Nm³ as special version must be factory-set

** Nm³ = m³ at standard conditions (P = 1.013x10⁵Pa, T = 273 K)



3 Ozone Systems

3.6.3 Order Information for OZONFILT® OZMa Systems

OZMa	Type ozone generator	
	Air operation / Oxygen operation	
	g/h	g/h
01	70	105
02	105	158
03	140	210
04	210	320
05	280	420
06	420	630
	Operating gas	
A	Operating gas - air	
O	Operating gas - oxygen	
	Type	
P	ProMaqua	
S	Special version	
C	ProMaqua with air-conditioning	
	Mechanical design	
0	Standard (packaging for transport by HGV)	
1	Standard (packaging for sea/air freight)	
2	In stainless steel cabinet (packaging for transport by HGV)	
3	In stainless steel cabinet (packaging for sea/air freight)	
M	Modified	
	Operating voltage	
A	Single-phase 230 V ±10 %, 50/60 Hz (only types 01-03)	
S	Three-phase 230/400 V ±10 %, 50/60 Hz (only types 04-06)	
	Gas treatment	
0	Gas treatment not integrated (design operating gas - oxygen)	
1	Gas treatment integrated without filter package (design operating gas - air)	
2	Gas treatment integrated with filter package (design operating gas - air)	
3	Gas treatment not integrated (oxygen operating gas version), including gas control valve	
4	Gas treatment integrated without filter package (air operating gas version), including gas control valve	
5	Gas treatment integrated with filter package (air operating gas version), including gas control valve	
	Preset language	
DE	german	
EN	english	
FR	french	
IT	italian	
ES	spanish	
	Control	
0	Basic version with digital input to control two power stages	
1	External power control via 0/4-20 mA input, data logger	
2	External power control, ozone measurement and visualisation via screen recorder, 2 freely configurable 0/4-20 mA inputs, 1 freely configurable 0/4-20 mA output	
3	As 2 with additionally integrated PID controller for control of the ozone concentration independent of measured value and flow	
	Communication interfaces	
0	None	
4	PROFIBUS® DP interface	
	Additional options	
0	None	
1	Dew point sensor	
	Approvals	
01	CE-mark	
	Hardware	
0	Standard	
	Software	
0	Standard	

Explanation on the Identity code:

Mechanical design: In design 0 and 1, the plant is installed in a standard control cabinet made of powder-coated steel.

Gas treatment: Without filter package for oil-free generated or de-oiled compressed air.
With filter package for compressed air with residual oil content.



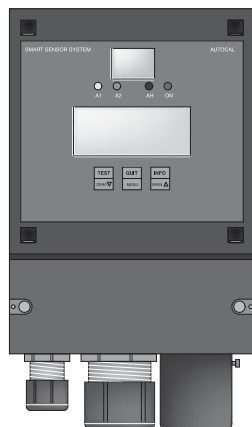
3 Ozone Systems

3.7 Safety Accessories for Ozone Systems

3.7.1

Room Air Monitor

Gas warning device GMA 36 – ozone and oxygen



pk_7_004_1
Gas warning devices GMA 36

Calibratable gas warning device with digital display of the detected gas concentration. 2 relay outputs for issue of infringements of warning and alarm thresholds, to switch external alarm sounder and for interlocking with the ozone system. The warning message relay is self-resetting, the alarm relay is a latching type and should be acknowledged at the device. 1 self-resetting relay for connection to an alarm horn is switched in the event of fault conditions and when the alarm limit is exceeded.

The ozone sensor responds to all strongly oxidising gases, hence it responds to chlorine gas or chlorine dioxide too.

The GMA 36 oxygen warning device is intended for installations where an unacceptably high oxygen enrichment of the ambient air is possible.

Technical Data

Type		Ozone	Oxygen
Warning at approx.	ppm/vol%	0.3	23.0
Alarm at approx.	ppm/vol%	0.5	25.0
Permissible ambient temperature	°C	-15...45	-15...45
Protection class housing		IP 54	IP 54
Dimensions (without PGs, without sensor) H x W x D	mm	247 x 135 x 95	247 x 135 x 95
Supply	V/Hz	85 – 264/50 – 60	85 – 264/50 – 60
Power consumption	W	5	5
Warm-up phase max.	s	150	20
"Warning" relay contact, self-resetting	V/A	230/1	230/1
"Alarm" relay contact, latching	V/A	230/1	230/1
"Horn" relay contact, latching, can be acknowledged	V/A	230/1	230/1
Sensor measuring principle		electrochemical	electrochemical
Sensor service life (depending on environmental cond.)	years	2–3	2–3

	Type	Order no.
Gas warning device GMA 36	Ozone	1023155
Gas warning device GMA 36	Oxygen	1023971

Spare Parts

	Order no.
Replacement sensor for chlorine, chlorine dioxide, ozone	1023314
Replacement sensor for oxygen	1023851
Replacement sensor for gas warning devices in the Life CGM range	1003009

Mounting kit

	Order no.
Mounting kit for direct mounting of the CGM 1060 and GMA 36 ozone warning devices on the housing of the OZVa plants	1004248
Support bracket for mounting kit for all types of OZVa except OZVa 1/2 with transparent mixing system	1005854



3 Ozone Systems

Warning light and horn

Combined horn and red warning lamp. IP 33 enclosure made of impact-resistant ABS. Dome made of clear polycarbonate. Connected load: 230 V AC, 50 mA. Supplied complete with B 15 d / 7 watt bulb.

	Order no.
Warning light and horn	1010508

Gas tracing pump

Hand operated, non-continuously working test tube pump for fast and accurate measurement of ozone gas. Complete with 10 no. ozone gas test tubes 0.05-5 ppm in carrying case.

	Order no.
Gas tracing pump	1025533

Potassium iodide starch paper

Roll with 4.8 m test strip for leak detection on pipelines carrying ozone gas.

	Order no.
Potassium iodide starch paper	1025575

3.7.2

Personal Protection Needs

Gas mask

Ozone-resistant, full-face respiratory protective mask with panoramic window shield to EN 136 Class 3. Medium size with EN 148-1 threaded pipe connection. Complete with combination filter NO-P3 and carrying case.

	Order no.
Gas mask	1025574

Warning label

Warning label in accordance with the "Guidelines for the use of ozone for water treatment" ZH 1/474, issued by the central office of the industrial safety associations. Version supplied as a combined adhesive label with markings as follows: warning sign, ozone plant room indication and prohibited activity signs.

	Order no.
Warning label	740921

Emergency stop switch

For installation near the door of the ozone plant room. IP 65 PVC enclosure.

	Order no.
Emergency stop switch	700560



4 Disinfection and Oxidation Process

4.1 CHLORINSITU® Electrolysis Systems

Chlorine and sodium hydroxide are produced in-situ in electrolysis by passing an electric current through salt water.

In **open or tubular cell electrolysis** (type CHLORINSITU® II), the electrochemical reaction takes place in a flow chamber, so that the freshly produced active chlorine immediately reacts with the sodium hydroxide to form sodium hypochlorite. A saturated brine is used as a salt solution, which is produced in a separate salt dissolving tank from salt of a defined quality. The advantage of tubular cell electrolysis lies in the simple construction of the equipment, its ease of maintenance and low investment compared with **diaphragm electrolysis systems**. The disadvantage is the relatively poor output (50%) of brine, higher entrainment of chloride into the water to be treated, higher power consumption and relatively low chlorine concentrations (5 g/l FAC) in the end product.

In **diaphragm electrolysis**, the electrochemical reaction takes place in two electrode chambers separated by a diaphragm, so that the formation of the freshly produced active chlorine and sodium hydroxide is physically separated. Systems of type CHLORINSITU® III and CHLORINSITU® III Compact bring the reaction mixtures of both electrode chambers together again after the electrochemical reaction to produce a stock solution of sodium hypochlorite (25 g/l FAC), which can be stored temporarily and metered as needed.

With systems of type CHLORINSITU® IV Compact and CHLORINSITU®, the highly pure active chlorine is fed into the water to be treated through an injector and under constant vacuum, where it dissolves as hypochlorous acid. In systems of type CHLORINSITU® V plus, any excess active chlorine gas produced is combined with the sodium hydroxide, as in the CHLORINSITU® III system, to form sodium hypochlorite and is then stored temporarily. The systems therefore only need to be designed for average chlorine demand, as peaks of capacity can be compensated for from the sodium hypochlorite temporary storage tank. In all systems of types CHLORINSITU® IV Compact, CHLORINSITU® V and CHLORINSITU® V Plus, the sodium hydroxide produced during electrolysis is stored temporarily and metered in, as required, to correct the pH value.

The advantage of diaphragm systems lies in their excellent efficiency (85% brine output) and minimal entrainment of chloride compared with open tubular cell systems. With systems of type CHLORINSITU® V and CHLORINSITU® V Plus, the entrainment of chloride and chlorate from the electrolysis cell into the water to be treated can be avoided completely. In diaphragm cell electrolysis system for producing sodium hypochlorite, the higher output results in solutions with a significantly higher chlorine content than is the case with tubular cell electrolysis.

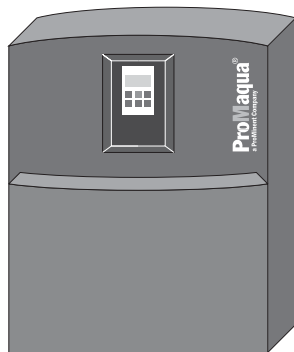
- Disinfection from natural sodium chloride
- No handling of hazardous chemicals
- Economical method thanks to low salt and energy consumption
- Ultra-pure active chlorine thanks to production in-situ and short temporary storage periods
- The fresh active chlorine is generated under a constant vacuum, absolutely reliably and with maximum operating safety, thanks to the units being designed as vacuum systems
- Chlorine generation and pH correction with one system (CHLORINSITU® IV Compact, CHLORINSITU® V and CHLORINSITU® V Plus)
- The extremely pure active chlorine is produced to meet demand and is available at atmospheric pressure
- Improved working conditions for operating personnel
- No risk of confusing dangerous chemical tanks



4 Disinfection and Oxidation Process

4.2

Membrane Electrolysis Systems CHLORINSITU® III - IV compact



P_PMA_EL_0007_SW

Electrolysis system of type CHLORINSITU® IV generate ultra-pure active chlorine in a vacuum process. A saturated solution of sodium chloride is produced in a salt dissolving tank, included in the scope of delivery for this purpose, which is then electrolysed in a diaphragm cell. Sodium hydroxide and hydrogen are produced in the cathode chamber while ultra-pure active chlorine and dilute residual brine are produced in the anode chamber, separated by the diaphragm from the cathode chamber. The resulting active chlorine is suctioned off through an injector integrated in the system and dissolved as hypochlorous acid in the water being treated. The hydrogen generated is discharged to the fresh air through a bleed line. The sodium hydroxide is disposed of or optionally used by a metering pump integrated in the system to correct the pH of the water being treated. The salt-dissolving water comes from a softener integrated in the system, thereby preventing the formation of lime deposits and ensuring the long service life of the electrolytic cell.

The microprocessor controller integrated in the system digitally indicates the current output and monitors all key functions. All operating and error messages are shown in plain text on the clear display. The output can be controlled manually, automatically (controller option) or externally.

Electrolysis systems of type CHLORINSITU® IV Compact are especially suitable for use with smaller swimming pools in residential properties and hotels (indoor pools with a total circulation capacity of up to 25 m³/hour, chlorinated in accordance with the DIN standard).

- Production and metering of ultra-pure hypochlorous acid
- Cost-effective operation thanks to the use of sodium chloride as an inexpensive raw material and no consumption of chemicals for pH correction
- Water disinfection and pH correction with a single system
- Safe vacuum system technology
- Optional integrated chlorine and pH control
- Robust, simple technology
- Compact space-saving design, ready mounted on a wall panel

CHLORINSITU® III compact (New)

produces 20-25 g/l sodium-calcium hypochlorite and stores this in a 60 l storage tank.

Type/ output g/h	Power uptake kW	Max. salt consumption g/h	Max. consumption of process water l/h	Dimensions L x W x H (mm)	Brine tank l
25	0.11	65	1.5	590 x 355 x 650	130
50	0.22	130	3	590 x 355 x 650	130

Scope of delivery:

Electrolysis systems of type CHLORINSITU® III Compact are ready mounted and wired for use on a wall panel. Chlorine electrolysis system with integrated microprocessor control and softener system. They include a diaphragm electrolysis cell, separate salt dissolving tank with level monitor and a level control for a storage tank (tank not included with the scope of delivery). A storage tank is also required as well as a metering pump for each point of injection (pump not included in the scope of delivery). Only a chlorine and pH control can be integrated as an option. The measuring and control technology then has to be offered separately for several pools.

	Order no.
CHLORINSITU® III Compact 25	1041399
CHLORINSITU® III Compact 50	1041401

The CHLORINSITU® III compact 25 is suitable for a circulation capacity of 25 m³/h, the CHLORINSITU® III compact 50 for a circulation capacity of 50 m³/h (private indoor pool).

4 Disinfection and Oxidation Process

CHLORINSITU® IV compact

Type/ output g/h	Power up- take kW	Max. salt con- sumption g/h	Max. consumption of process water l/h	Dimensions L x W x H (mm)	Brine tank l
25	0.11	65	1.5	590 x 355 x 650	130
50	0.22	130	3	590 x 355 x 650	130

Scope of delivery:

Electrolysis systems of type CHLORINSITU® IV Compact are ready mounted and wired for use on a wall panel. Chlorine electrolysis system with integrated microprocessor control and water softening system, diaphragm electrolysis cell with negative pressure monitoring, separate salt dissolving tanks with level control, integrated injector and integral feeder assembly for sodium hydroxide (optional). A booster pump is also needed (not included in the scope of delivery) for the single possible point of injection. A chlorine and pH control can be integrated as an option. Several pools cannot be fed from one Chlorinsitu® IV Compact system.

		Order no.
25 g/h	CHLORINSITU® IV Compact 25	1036461
	CHLORINSITU® IV compact 25 with pH correction	1036462
50 g/h	CHLORINSITU® IV Compact 50	1036463
	CHLORINSITU® IV Compact 50 with pH correction	1036464

<emphasis>Scope of delivery:</emphasis> without sensors, no accessories

Spare Parts

	Type	Order no.
Annual maintenance set	CHLORINSITU® IV Compact 25	1041415
3-yearly maintenance set	CHLORINSITU® IV Compact 25	1041416
Annual maintenance set	CHLORINSITU® IV Compact 25 with pH correction	1043267
3-yearly maintenance set	CHLORINSITU® IV Compact 25 with pH correction	1043268
Annual maintenance set	CHLORINSITU® IV Compact 50	1041417
3-yearly maintenance set	CHLORINSITU® IV Compact 50	1041418
Annual maintenance set	CHLORINSITU® IV Compact 50 with pH correction	1043269
3-yearly maintenance set	CHLORINSITU® IV Compact 50 with pH correction	1043270
Membrane cell	CHLORINSITU® IV Compact 25	1041419
Membrane cell	CHLORINSITU® IV Compact 50	1041420
Spare parts set	CHLORINSITU® IV Compact 25/50	1045232



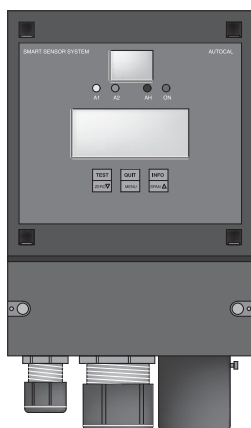
4 Disinfection and Oxidation Process

4.3 Gas Warning Device for Monitoring Chlorine Gas

The GMA 36 chlorine gas warning device is a compact measuring and switching unit designed for monitoring the surrounding air for dangerous concentrations of chlorine gas.

Gas warning device GMA 36

for monitoring chlorine gas



pk_7_004_1
Gas warning devices GMA 36

Type	Chlorine
Warning at approx.	2.0 ppm/vol%
Alarm at approx.	4.0 ppm/vol%
Permissible ambient temperature	-15...45°C
Protection class housing	IP 54
Dimensions (without PGs, without sensor) H x W x D	247 x 135 x 95 mm
Supply	85 – 264 / 50 – 60 V/Hz
Power consumption	5 W
Warm-up phase max.	150 s
"Warning" relay contact, self-resetting	230 / 1 V/A
"Alarm" relay contact, latching	230 / 1 V/A
"Horn" relay contact, latching, can be acknowledged	230 / 1 V/A
Sensor measuring principle	electrochemical
Sensor service life (depending on environmental cond.)	2–3 years

Note: The sensor reacts to all oxidising gases.

	Order no.
GMA 36 chlorine gas detector	1023157

Spare parts

		Order no.
Replacement sensor	for chlorine, chlorine dioxide, ozone	1023314
Replacement sensor	for gas warning devices in the Life CGM range	1003009

4.4 Accessories

Water hardness measuring kit

for manual determination of the overall hardness

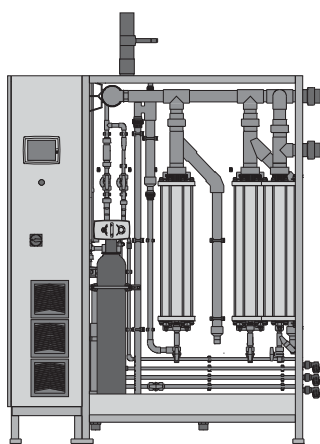
	Order no.
Water hardness measuring kit for overall hardness	505505



4 Disinfection and Oxidation Process

4.5

Electrolysis System CHLORINSITU® II



P_PMA_EL_0003_SW

Electrolysis systems of the type CHLORINSITU® II generate sodium-calcium hypochlorite with a concentration of 5 g/l. A saturated solution of sodium chloride is produced in a salt dissolving tank, included in the scope of delivery for this purpose, which, after appropriate dilution, is electrolysed in an open tubular cell. The resulting solution is collected in a storage tank and, from there, metered according to requirements using separate metering pumps. Due to its moderate pH value of 8.5 – 9, it affects the pH of the treated water significantly less than if conventional sodium-calcium hypochlorite with a pH of 12 – 13.5 were used. Much less acid is used to adjust the pH value, enabling savings of up to 80%. The hydrogen always produced during electrolysis is diluted with fresh air through an ATEX 95-approved fan and discharged safely. Both the salt-dissolving and diluting water comes from a softener unit integrated in the system, preventing the formation of lime deposits and ensuring the long service life of the electrolytic cell. There is therefore no need for acid purification. The systems are controlled using a modern PLC with a large, illuminated display and integrated Remote Control Engineer for remote diagnosis and troubleshooting. Electrolysis systems of type CHLORINSITU® II are especially suitable for applications where a robust and clearly laid out technology is required and where the entrainment of residual sodium chloride into the water being treated is not a problem.

- Robust, simple technology
- Safe system control with remote diagnosis by Remote Control Engineer
- Cost-effective operation thanks to the use of sodium chloride as an inexpensive raw material and lower chemical consumption for pH adjustment
- Compact, space-saving design
- Improved working conditions for operating personnel
- No risk of confusing dangerous chemical tanks

Technical Data

Type/ output	Fuse	Power uptake	Max. salt consumption	Max. consumption of process water	Product outlet H	Dimensions L x W x H (mm)	Brine tank	Recommended capacity storage tank
g/h	A	kW	kg/d	l/h	mm		l	l
50	16	0.78	4	10	1,188	1,050 x 600 x 1,550	130	300
100	16	1.15	8	20	1,589	1,250 x 600 x 2,000	130	500
150	16	1.53	12	30	1,589	1,250 x 600 x 2,000	200	700
200	3 x 16	1.90	16	40	1,589	1,250 x 600 x 2,000	200	1,000
300	3 x 16	2.65	24	60	1,589	1,250 x 600 x 2,000	200	1,500
400	3 x 16	3.40	32	80	1,589	1,250 x 600 x 2,000	200	2,000
500	3 x 20	4.15	40	100	1,589	1,250 x 600 x 2,000	380	2,500
600	3 x 25	4.90	48	120	1,589	1,250 x 600 x 2,000	380	3,000
800	3 x 35	6.40	65	160	1,589	1,250 x 600 x 2,000	380	3,500
1,000	3 x 35	7.90	80	200	1,589	1,250 x 600 x 2,000	520	4,500
1,200	3 x 50	9.40	95	240	1,589	1,250 x 600 x 2,000	520	5,500
1,400	3 x 50	10.90	110	280	1,589	1,250 x 600 x 2,000	520	6,000
1,600	3 x 63	12.40	130	320	1,589	1,250 x 600 x 2,000	760	7,000
1,800	3 x 63	13.90	155	360	1,589	1,650 x 600 x 2,000	760	8,000
2,000	3 x 63	15.40	175	400	1,589	1,650 x 600 x 2,000	760	9,000
2,200	3 x 80	16.90	190	440	1,589	1,650 x 600 x 2,000	760	10,000
2,400	3 x 80	18.40	210	480	1,589	1,650 x 600 x 2,000	760	11,000

Scope of delivery:

Electrolysis systems of type Chlorinsitu® II are mounted ready-wired with a PLC on a powder-coated stainless steel frame in the control cabinet. They include a Remote Control Engineer for remote diagnosis and troubleshooting, integrated water softener system, open tubular cells, ATEX 95-compliant bleed system and separate salt dissolving tanks and level monitoring unit. The system also includes liquid level sensors for monitoring the storage tank to be set up on site for sodium-calcium hypochlorite. A duplex water softener is fitted as standard for systems producing more than 1,800 g/h. Automatic monitoring of water hardness downstream of the softening system can be offered as an option.

Note:

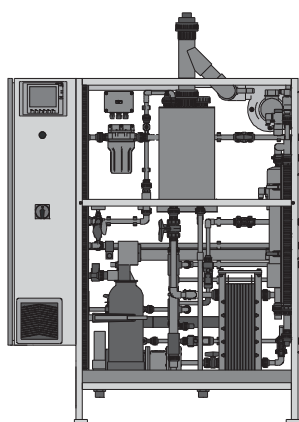
Electrolysis systems of type CHLORINSITU® II, III, V and V Plus are offered and planned to meet customer specifications. This is true both for the system documentation and the subsequent supply of spare parts and maintenance.



4 Disinfection and Oxidation Process

4.6

Electrolysis System CHLORINSITU® III



P_PMA_EL_0004_SW

Electrolysis systems of type CHLORINSITU® III generate sodium hypochlorite with a higher concentration of approximately 25 g/l with minimal entrainment of sodium chloride (85% output) from the diaphragm cell into the finished product. A saturated solution of sodium chloride is produced in a salt dissolving tank, included in the scope of delivery for this purpose, which is then electrolysed in a diaphragm cell. Sodium hydroxide and hydrogen are produced in the cathode chamber while ultra-pure active chlorine and a diluted residual brine are produced in the anode chamber, separated by the diaphragm from the cathode chamber. The resulting active chlorine is bound to the sodium hydroxide through an injector under a constant vacuum and is collected as sodium hypochlorite in a storage tank. The vacuum is kept constant by a frequency-controlled circulation pump. This creates less mechanical stress on the diaphragm in the electrolysis cell and in other parts of the system. The complete sodium hypochlorite solution can be metered, as required, by separate metering pumps. Due to its moderate pH value of 9.5 – 10, it affects the pH of the treated water significantly less than if conventional sodium-calcium hypochlorite with a pH of 12 – 13.5 were used. Much less acid is used to adjust the pH value, enabling savings of up to 70%. The hydrogen always produced during electrolysis is diluted with fresh air through an ATEX 95-approved fan and discharged safely. The salt-dissolving water comes from a softener integrated in the system, thereby preventing the formation of lime deposits and ensuring the long service life of the diaphragm cell. The efficiency of the electrolysis is constantly monitored by various flow meters, the addition of water depending on the sodium hydroxide production and a dynamic level control in the storage tanks.

The systems are controlled using a modern PLC Programmable Logic Controller with a large, illuminated display and integrated Remote Control Engineer for remote diagnosis and troubleshooting.

Electrolysis systems of type CHLORINSITU® III are especially suitable for applications in which sodium hypochlorite solution low in chloride and chlorate is required.

- Sodium hypochlorite solution low in chloride and chlorate with a high chlorine concentration (25 g/l FAC)
- Minimal acid consumption for pH correction; savings of up to 70% are possible
- Safe system control with remote diagnosis by Remote Control Engineer
- Excellent service life of the diaphragm cells, thanks to constant vacuum
- A frequency-controlled circulation pump maintains the vacuum constant in the enclosed anode area
- Maximum operating safety due to their design as negative pressure systems
- Dynamic level control in the storage tank ensures optimised chlorine production
- Active process control of production by largely internal measuring and control technology
- Cost-effective operation thanks to the use of sodium chloride as an inexpensive raw material and lower chemical consumption for pH adjustment
- Robust, simple technology
- Compact, space-saving design



4 Disinfection and Oxidation Process

Technical Data

Power supply 3 x 400 V (VAC/3P/N/PE/50 Hz)

Type/ output	Fuse	Power uptake	Max. salt consumption	Max. consump- tion of process water	Max. consump- tion of cooling water	Dimensions L x W x H (mm)	Brine tank	Recommended capacity storage tank
g/h	A	kW	kg/d	l/h	l/h		l	l
100	3 x 16	1.10	5	4	80	1,250 x 600 x 1,550	130	200
200	3 x 16	1.50	10	8	80	1,250 x 600 x 1,550	130	300
300	3 x 16	1.90	15	12	100	1,250 x 600 x 1,550	200	400
400	3 x 16	2.30	20	16	100	1,250 x 600 x 1,550	200	500
500	3 x 16	2.70	25	20	125	1,250 x 600 x 1,550	200	600
600	3 x 20	3.10	30	24	125	1,650 x 600 x 2,000	380	700
750	3 x 25	3.70	35	30	150	1,650 x 600 x 2,000	380	800
1,000	3 x 25	4.70	50	40	150	1,650 x 600 x 2,000	380	1,200
1,250	3 x 35	5.70	60	50	150	1,650 x 600 x 2,000	380	1,500
1,500	3 x 35	6.70	70	60	180	1,650 x 600 x 2,000	380	1,700
1,750	3 x 35	7.70	80	70	180	1,650 x 600 x 2,000	380	2,000
2,000	3 x 50	8.70	100	80	200	1,750 x 1,200 x 2,000	520	2,200
2,500	3 x 63	10.70	125	100	250	1,750 x 1,200 x 2,000	520	3,000
3,000	3 x 63	12.70	150	120	300	1,750 x 1,200 x 2,000	520	3,300
3,500	3 x 80	14.70	175	140	350	1,750 x 1,200 x 2,000	520	4,000
5,000	3 x 90	20.70	250	200	500	3,100 x 1,800 x 2,070	1,180	5,800
7,000	3 x 100	29.40	350	280	700	3,100 x 1,800 x 2,070	1,180	6,000
8,500	3 x 130	35.70	425	340	850	4,300 x 1,800 x 2,070	1,180	7,500
10,000	3 x 160	40.70	500	400	1,000	4,300 x 1,800 x 2,070	1,180	11,000

Scope of delivery:

Electrolysis systems of type Chlorinsitu® III are mounted ready-wired with a PLC on a powder-coated stainless steel frame in the control cabinet. They include a Remote Control Engineer for remote diagnosis and troubleshooting, integrated water softener system, diaphragm electrolysis cells, ATEX 95-compliant bleed system and separate salt dissolving tanks and level monitoring unit. Dynamic level control to monitor the storage tank to be provided on site for sodium hypochlorite. A chlorine gas warning unit and automatic monitoring of water hardness downstream of the softening system come as standard with systems producing more than 600 g/h.

Note:

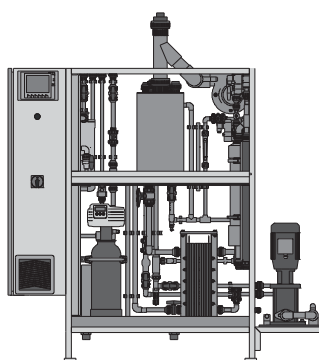
Electrolysis systems of type CHLORINSITU® II, III, V and V Plus are offered and planned to meet customer specifications. This is true both for the system documentation and the subsequent supply of spare parts and maintenance.



4 Disinfection and Oxidation Process

4.7

Electrolysis System CHLORINSITU® V



P_PMA_EL_0013_SW

Electrolysis systems of type CHLORINSITU® V generate ultra-pure active chlorine in a vacuum process. A saturated solution of sodium chloride is produced in a salt dissolving tank, included in the scope of delivery for this purpose, which is then electrolysed in a diaphragm cell. Chloride-free sodium hydroxide and hydrogen are produced in the cathode chamber, while ultra-pure chlorine gas and dilute residual brine are produced in the anode chamber, separated by the diaphragm from the cathode chamber. The active chlorine produced is suctioned off through an injector (contained in the scope of delivery) and fully dissolved as hypochlorous acid in the water being treated (through a bypass). The superchlorinated water is then distributed throughout the various pools via one of more proportionately controllable motor-driven ball valves. The vacuum is kept stable by a single frequency-controlled booster pump. This permits significant savings in terms of energy. The chloride-free sodium hydroxide is stored temporarily and can be used for pH value correction. A standard diaphragm metering pump is first used to correct the base pH (pH 6.8 – 7) of the superchlorinated water. The fine correction of the pH value is fed via additional alkali metering pumps per circuit or point of injection. To achieve this, an external pH value controller is connected directly to the system's control. The hydrogen produced is diluted with fresh air through an ATEX 95-approved fan and discharged safely. The diluted residual brine is fully discarded. To this end, the diluted brine is seriously diluted with softened water, neutralised by the addition of sodium hydroxide and disposed off in the sewer. All residual chloride and chlorate are thus disposed of and are fundamentally not mixed with the process water. Electrolysis systems of type Chlorinsitu® V can thus be compared with pure chlorine gas in terms of their oxidation strength, chloride and chlorate content in the process water. The salt-dissolving water comes from a softener integrated in the system, thereby preventing the formation of lime deposits and ensuring the long service life of the diaphragm cell. The efficiency of electrolysis is constantly monitored by various flow meters, the addition of water depending on the sodium hydroxide production and the base pH correction.

The systems are controlled using a modern PLC with a large, illuminated display and integrated Remote Control Engineer for remote diagnosis and troubleshooting. Chlorine metering control and pH value correction are provided as standard via contact inputs.

Options (always customer-specific and project-managed):

- Analogue input
- MOD-bus or PROFIBUS®
- Several points of injection
- Several booster pumps are possible with differing water qualities (e.g. brine and fresh water pools)

Electrolysis systems of type CHLORINSITU® V are suitable for all applications that require the simultaneous metering of hypochlorous acid and pH value correction.

- Production and metering of ultra-pure hypochlorous acid without temporary storage
- Chlorination and pH adjustment with a single system
- Safe system control with remote diagnosis by Remote Control Engineer
- Excellent service life of the diaphragm cells, thanks to constant vacuum
- A frequency-controlled circulation pump maintains the vacuum constant in the enclosed anode area
- Maximum operating safety due to their design as negative pressure systems
- Active process control of production by largely internal measuring and control technology
- Cost-effective operation thanks to the use of sodium chloride as an inexpensive raw material and no consumption of chemicals for pH correction
- Complete disposal of the diluted brine, nothing is fed back into the process water to be treated
- Comparable with pure chlorine gas in terms of oxidation strength, chloride and chlorate content in the process water
- Robust technology
- Compact, space-saving design



4 Disinfection and Oxidation Process

Technical Data

Power supply 3 x 400 V (VAC/3P/N/PE/50 Hz)

Type/ output g/h	Fuse A	Power uptake kW	Max. salt consumption kg/d	Max. consumption of process water l/h	(External) consumption of cooling water l/h	Dimensions L x W x H (mm)	Brine tank l
100	3 x 16	1.10	5	60	–	1,655 x 600 x 1,550	130
200	3 x 16	1.50	10	60	–	1,655 x 600 x 1,550	130
300	3 x 16	1.90	15	60	–	1,655 x 600 x 1,550	200
400	3 x 16	2.30	20	60	–	1,655 x 600 x 1,550	200
500	3 x 16	2.70	25	60	–	1,655 x 600 x 1,550	200
600	3 x 20	3.10	30	90	–	1,950 x 600 x 2,000	380
750	3 x 25	3.70	35	90	–	1,950 x 600 x 2,000	380
1,000	3 x 25	4.70	50	90	–	1,950 x 600 x 2,000	380
1,250	3 x 35	5.70	60	90	–	1,950 x 600 x 2,000	380
1,500	3 x 35	6.70	70	90	–	1,950 x 600 x 2,000	380
1,750	3 x 35	7.70	80	90	–	1,950 x 600 x 2,000	380
2,000	3 x 50	8.70	100	175	200	1,750 x 1,200 x 2,000	520
2,500	3 x 63	10.70	150	175	250	1,750 x 1,200 x 2,000	520
3,000	3 x 63	12.70	175	175	300	1,750 x 1,200 x 2,000	520
3,500	3 x 80	14.70	175	175	350	1,750 x 1,200 x 2,000	520

Capacities > 3,500 g/h upon request

Scope of delivery:

Electrolysis systems of type Chlorinsitu® V are ready mounted, wired for use, on a powder coated stainless steel frame with a Programmable Logic Controller (PLC) in the control cabinet, Remote Control Engineer for remote diagnosis and troubleshooting, integral water softening unit, diaphragm electrolysis cells, ATEX-95-compliant bleed system and separate salt dissolving tank with level monitoring. The scope of delivery also includes a frequency-controlled central injector system matched to the system to meter active chlorine and sodium hydroxide for pH correction and a single booster pump. A chlorine gas warning unit and automatic monitoring of water hardness downstream of the softening system come as standard with systems producing more than 600 g/h.

Note:

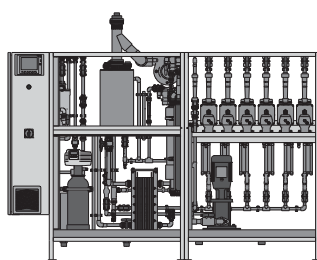
Electrolysis systems of type CHLORINSITU® II, III, V and V Plus are offered and planned to meet customer specifications. This is true both for the system documentation and the subsequent supply of spare parts and maintenance.



4 Disinfection and Oxidation Process

4.8

Electrolysis System CHLORINSITU® V Plus



P_PMA_EL_0012_SW

Electrolysis systems of type CHLORINSITU® V Plus generate ultra-pure active chlorine combined with a sodium hypochlorite solution in a vacuum process. A saturated solution of sodium chloride is produced in a salt dissolving tank, included in the scope of delivery for this purpose, which is then electrolysed in a diaphragm cell. Chloride-free sodium hydroxide and hydrogen are produced in the cathode chamber, while ultra-pure active chlorine and diluted residual brine are produced in the anode chamber, separated by the diaphragm from the cathode chamber. The resulting ultra-pure active chlorine is further processed in two ways. Firstly, as with CHLORINSITU® V systems, the gas is suctioned off through an injector that forms part of the scope of delivery (vacuum system) and dissolved as hypochlorous acid in the water being treated (via a bypass). The superchlorinated water is then distributed throughout the various pools via one of more proportionately controllable motor-driven ball valves. The vacuum is kept stable by a single frequency-controlled booster pump. This permits significant savings in terms of energy. If the complete production output is not needed, excess active chlorine can also be combined with the sodium hydroxide produced and then temporarily stored as sodium hypochlorite, as is the case with CHLORINSITU® III type systems. The system thus does not have to be dimensioned according to the maximum active chlorine demand, rather according to the average daily demand. Peaks in demand are met by the additional metering of sodium hypochlorite from the temporary storage, which, as with hypochlorous acid, is fed through a central injector system.

The chloride-free sodium hydroxide is stored temporarily and can be used for pH value correction. A standard diaphragm metering pump or the addition of sodium hypochlorite is first used to correct the base pH (pH 6.8 – 7) of the superchlorinated water. The fine correction of the pH value is fed via additional alkali metering pumps per circuit or point of injection. To achieve this, an external pH value controller is connected directly to the system's control. The hydrogen produced is diluted with fresh air through an ATEX 95-approved fan and discharged safely. The diluted residual brine is fully discarded. To this end, the diluted brine is seriously diluted with softened water, neutralised by the addition of sodium hydroxide and disposed off in the sewer. Any residual chloride and chlorate are thus disposed of and are fundamentally not mixed with the process water. Electrolysis systems of type Chlorinsitu® V Plus can thus be compared with pure chlorine gas in terms of their oxidation strength, chloride and chlorate content in the process water. The salt-dissolving water comes from a softener integrated in the system, thereby preventing the formation of lime deposits and ensuring the long service life of the diaphragm cell. The efficiency of the electrolysis is constantly monitored by various flow meters, the addition of water depending on the sodium hydroxide production and the base pH correction.

The systems are controlled using a modern PLC with a large, illuminated display and integrated Remote Control Engineer for remote diagnosis and troubleshooting. Chlorine metering control and pH value correction are provided as standard via contact inputs.

Options (always customer-specific and project-managed):

- Simultaneous production and metering of ultra-pure hypochlorous acid and sodium-calcium hypochlorite
- Analogue input
- MOD-bus or PROFIBUS®
- Several points of injection
- Several injector systems (sodium hypochlorite and active chlorine)
- Several booster pumps with differing water qualities (e.g. brine and fresh water pools)

Electrolysis systems of type CHLORINSITU® V Plus are suitable for all applications that require simultaneous hypochlorous acid metering, combined with sodium hypochlorite metering to cover peak demand and simultaneous pH value correction.

- Production and metering of ultra-pure hypochlorous acid combined with sodium hypochlorite production
- Chlorination and pH adjustment with a single system
- Safe system control with remote diagnosis by Remote Control Engineer
- Excellent service life of the diaphragm cells, thanks to constant vacuum
- A frequency-controlled circulation pump maintains the vacuum constant in the enclosed anode area
- Maximum operating safety due to their design as negative pressure systems
- Active process control of production by largely internal measuring and control technology
- Cost-effective operation thanks to the use of sodium chloride as an inexpensive raw material and no consumption of chemicals for pH adjustment
- Peak demand can be met by the presence of a daily supply of sodium hypochlorite
- Complete disposal of the diluted brine, nothing is fed back into the process water to be treated
- Comparable with pure chlorine gas in terms of oxidation strength, chloride and chlorate content in the process water
- Robust technology
- Compact, space-saving design

4 Disinfection and Oxidation Process

Technical Data

Power supply 3 x 400 V (VAC/3P/N/PE/50 Hz)

Type/ output	Fuse	Power uptake	Max. salt consumption	Max. consump- tion of process water*	(External) con- sumption of cooling water	Dimensions L x W x H (mm)	Brine tank	Recommended capacity storage tank
g/h	A	kW	kg/d	l/h	l/h		l	l
100	3 x 16	1.10	5	60	–	1,655 x 600 x 1,550	130	50
200	3 x 16	1.50	10	60	–	1,655 x 600 x 1,550	130	100
300	3 x 16	1.90	15	60	–	1,655 x 600 x 1,550	200	150
400	3 x 16	2.30	20	60	–	1,655 x 600 x 1,550	200	200
500	3 x 16	2.70	25	60	–	1,655 x 600 x 2,000	200	250
600	3 x 20	3.10	30	90	–	1,950 x 600 x 2,000	380	300
750	3 x 25	3.70	40	90	–	1,950 x 600 x 2,000	380	400
1,000	3 x 25	4.70	55	90	–	1,950 x 600 x 2,000	380	500
1,250	3 x 35	5.70	60	90	–	1,950 x 600 x 2,000	380	600
1,500	3 x 35	6.70	75	90	–	1,950 x 600 x 2,000	380	750
1,750	3 x 35	7.70	85	90	–	1,950 x 600 x 2,000	380	850
2,000	3 x 50	8.70	100	175	200	1,750 x 1,200 x 2,000	520	1,000
2,500	3 x 63	10.70	125	175	250	1,750 x 1,200 x 2,000	520	1,250
3,000	3 x 63	12.70	150	175	300	1,750 x 1,200 x 2,000	520	1,500
3,500	3 x 80	14.70	175	175	350	1,750 x 1,200 x 2,000	520	1,750

* The consumption of process water depends on the ratio of chlorine gas to stock production. The value is given here for a ratio of 70% : 30 %.

Capacities > 3,500 g/h upon request

Scope of delivery:

Electrolysis systems of type Chlorinsitu® V Plus are ready mounted, wired for use, on a powder-coated stainless steel frame with a PLC Programmable Logic Controller in the control cabinet, Remote Control Engineer for remote diagnosis and troubleshooting, integral water softening unit, diaphragm electrolysis cells, ATEX-95-compliant bleed system and separate salt dissolving tank with level monitoring. The scope of delivery also includes a frequency-controlled central injector system matched to the system to meter active chlorine and sodium hydroxide for pH correction and a single booster pump. A level control to monitor the storage tank to be provided on site for sodium hypochlorite. A chlorine gas warning unit and automatic monitoring of water hardness downstream of the softening system come as standard with systems producing more than 600 g/h.

Note:

Electrolysis systems of type CHLORINSITU® II, III, V and V Plus are offered and planned to meet customer specifications. This is true both for the system documentation and the subsequent supply of spare parts and maintenance.



4 Disinfection and Oxidation Process

4.9

Dulcodes UV Treatment Systems

Dulcodes UV treatment systems are used to remove bound chlorine (chloramine) and ozone in swimming pool water treatment. The intensive UV radiation reduces the odour intensive substances that are irritating to the eyes. The result is an improved quality of water for healthy and pleasant bathing. UV low pressure lamps, as well as UV medium pressure lamps, are used for this purpose. Whereas systems are used with low pressure lamps in private swimming pools or small pools (e.g. hotel pools), the UV treatment of the swimming pool water in larger public swimming pools is almost always carried out using medium pressure lamps due to the high circulation capacity. Although the Dulcodes UV system also improves the microbiological quality of the water, the addition of a disinfectant with depositing effect (e.g. chlorine) cannot be avoided.

Features of our Dulcodes UV system technology

- High quality radiation chambers made from stainless steel (DIN 1.4404 or 1.5671) or plastic
- Highly selective long-term and temperature-stable UVC sensors
- Uniform radiation of the entire water flow thanks to optimised system hydraulics
- Use of UV lamps with long lamp life time and high UV-C yield
- System control with comprehensive monitoring and reporting functions
- Trend display of the variation of the UV sensor signal over time
- Use of modern electronic pre-switching units with bus technology for lamp-friendly ignition and operation

The right UV system for every swimming pool

Dulcodes W standard systems for private swimming pools or small pools (e.g. hotel pools)

- Standard UV system with high-efficiency low pressure High Flux lamps with a guaranteed lamp service life of 8 - 10,000 hours

Dulcodes R systems for private and public swimming pools with manual wiper

- are particularly suitable for water which tends to form deposits on the lamp protection tubes. This range of systems is equipped with high-efficiency low pressure Opti Flux lamps and provides a guaranteed lamp service life of 14,000 hours.

Dulcodes M systems for public swimming pools with automatic wiper system

- are especially suitable for treating larger volumes of water of up to 800 m³/h, with only one lamp. Powerline medium pressure lamps have a guaranteed lamp service life of 8 - 10,000 hours.

Dulcodes K systems for swimming pools containing brine (e.g. thermal spring water, sea water)

- The radiation chamber made of UV-resistant, high-quality PE-HD plastic is equipped with high-efficiency low pressure High Flux lamps with a guaranteed lamp service life of 8 - 10,000 hours.

Enthusiastic operators

- Simple installation thanks to minimal installation, quick refitting and low space requirement
- The specified limit value for bound chlorine (0.2 mg / l) is easily achieved
- Considerable reduction in operating costs due to savings of fresh water, chemicals and heating energy.
- Low investment costs ensure short amortization periods
- Simple to operate, almost maintenance free
- Control based on measured values combined with DULCOMARIN® II
- Increased disinfection safety, destruction of chlorine-resistant germs

More enjoyable swimming

- No chlorine odour and therefore, much better air in the swimming pool
- No burning eyes, or skin irritation



4 Disinfection and Oxidation Process

UV Treatment Systems for Private Swimming Pools

Type	Max. flow	Max. flow	Lamp power	Connected load	Radiation chamber length	Ø	Connector nominal diameter
	m ³ /h	m ³ /h	W	W	mm	mm	
1x80W**	5.4*	7.2**	80	100	630	114	G 1 1/4"
1x130W	8.7*	11.6**	130	150	940	114	G 2
1x230W	20.0*	27.0**	230	250	1,468	140	DN 65
1x300R	30.0*	40.0**	1x300	320	1,562	140	DN 80

** Design for a radiation dosage of 300 J/m² with a transmission of 98%/cm

* Design for a radiation dosage of 400 J/m² with a transmission of 98%/cm

The required radiation dosage depends on the individual level of contamination of the pool. Please contact us, we look forward to advising you.

UV Treatment Systems for Public Swimming Pools

Type	Max. flow	Max. flow	Lamp power	Connected load	Radiation chamber length	Ø	Connector nominal diameter
	m ³ /h	m ³ /h	W	W	mm	mm	
1x300R	30*	40**	1x300	320	1,562	140	DN 80
2x230W	64*	85**	2x230	500	1,640	220	DN 125
1x2ML	88*	117**	2,000	2,300	850	220	DN 100
2x300R	95*	125**	2x300	650	1,633	220	DN 150
1x3ML	158*	210**	3,000	3,200	850	220	DN 150
3x300R	179*	239**	3x300	1,000	1,638	273	DN 200
1x4ML	229*	305**	4,000	4,200	1,200	270	DN 200
1x6ML	406*	540**	6,000	6,200	1,200	320	DN 250
1x8ML	541*	600**	8,000	8,200	1,500	320	DN 250
1x10ML	600*	600**	10,000	10,200	1,500	320	DN 250

** Design for a radiation dosage of 400 J/m² with a transmission of 98%/cm

* Design for a radiation dosage of 600 J/m² with a transmission of 98%/cm

The required radiation dosage depends on the individual level of contamination of the pool. Please contact us, we look forward to advising you.

UV Treatment Systems for Pools Containing Salt or Brine (PE-HD Plastic Radiation Chamber)

Type	Max. flow	Max. flow	Lamp power	Connected load	Radiation chamber length	Ø	Connector nominal diameter
	m ³ /h	m ³ /h	W	W	mm	mm	
1x290K	27*		1x290	310	1,530	138	DN 80
2x290K	94*	25**	2x290	600	1,535	188	DN 125
3x290K	193*	36**	3x290	910	1,535	268	DN 200
4x290K	250*	66**	4x290	1,200	1,535	268	DN 200

* Design for a radiation dosage of 400 J/m² with a transmission of 98 %/cm

** Design for a radiation dosage of 600 J/m² with a transmission of 98 %/cm

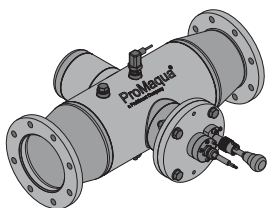
The required radiation dosage depends on the individual contamination of the pool. Further systems for higher flows are available on request. Please contact us, we look forward to advising you.



4 Disinfection and Oxidation Process

4.10

UV System Dulcodes S for Chloramine Reduction in Pool Water



P_PMA_DS_0018_SW1

Dulcodes S UV treatment systems are suitable for the photochemical degradation of combined chlorine (chloramine) in swimming pool water treatment. Special medium-pressure UV lamps generate intensive polychromatic UV radiation to reduce the odour-intensive and eye-irritating substances. The result is an improved water quality for healthy and pleasant bathing.

Features

- Flow: up to 569 m³/h (depending on transmission).
- Extremely compact inline system with small footprint.
- Simple installation due to a minimum of fitting work, quick retrofitting.
- A free choice of fitting position ensures maximum installation flexibility.
- Can be directly installed in a plastic circulation pipe because no UV radiation escapes from the radiation chamber.
- Unbeatable simple and quick maintenance: All maintenance work can be carried out quickly and conveniently from one side.
- Powerline S medium-pressure lamps with high power input of up to 3 kW.
- Special UV medium-pressure lamp with intense polychromatic UV radiation.
- Lamp service life: approximately 8,000 h.
- Long-term stable UVC sensor for monitoring the lamp output, lamp protection tube fouling and changes in water quality.
- Integral temperature sensor for monitoring the water temperature in the radiation chamber.
- Large graphic display for displaying the time dependency of the UV sensor signal with a trend indication line.
- Manual power control to optimally adapt the system to the relevant capacity needed (not for Dulcodes 1 x 0.65 and 1S).
- Automatic switching on and off based on the chloramine value, for example used in conjunction with the DULCOMARIN® II.
- Optional manual or automatic wiper system for efficient removal of deposits on the lamp protection tube. The wiper system can be easily retrofitted.
- Radiation chambers made of high-grade stainless steel 1.4404.
- Control cabinet made of painted steel.
- Optimised use of energy, thanks to large radiation chamber and uniform irradiation of the entire water flow due to optimised system hydraulics.

Application focuses

Potable water	Industrial water	Swimming pool water	Waste water	Salt water
-	✓	✓	-	-



4 Disinfection and Oxidation Process

Technical Data

Type	Max. flow	Lamp power	Connect- ed load	Radiation chamber length	Minimum clearance for mainte- nance work	Min. distance from wall	Empty weight/ Operating weight	Connection nominal diameter
	m³/h	kW	kW	mm	mm	mm	kg	mm
1x0,65S (without wiper)	20.0*	0.65	0.75	500	335	160	21/31	65
1x0.65S (without wiper)	17.0*	0.65	0.75	500	300	160	14/19	80
1x1S (without wiper)	51.0*	1.00	1.10	700	300	450	21/37	100
1x1S (without wiper)	51.0*	1.00	1.10	700	300	450	21/37	125
1x1S (with wiper)	58.0*	1.00	1.10	700	400	450	31/47	100
1x1S (with wiper)	51.0*	1.00	1.10	700	300	450	31/47	125
1x1S (automatic wiper)	51.0*	1.00	1.10	700	300	450	31/47	100
1x1S (automatic wiper)	51.0*	1.00	1.10	700	300	450	31/47	125
1x2S (without wiper)	89.0*	2.00	2.10	700	400	550	31/58	125
1x2S (without wiper)	89.0*	2.00	2.10	700	400	550	31/58	150
1x2S (with wiper)	102.0*	2.00	2.10	700	500	550	38/65	125
1x2S (with wiper)	89.0*	2.00	2.10	700	400	550	38/65	150
1x2S (automatic wiper)	89.0*	2.00	2.10	700	400	550	38/65	125
1x2S (automatic wiper)	89.0*	2.00	2.10	700	400	550	38/65	150
1x3S (without wiper)	177.0*	3.00	3.10	800	500	650	52/118	200
1x3S (without wiper)	177.0*	3.00	3.10	800	500	650	52/118	250
1x3S (with wiper)	205.0*	3.00	3.20	800	600	650	52/118	200
1x3S (with wiper)	177.0*	3.00	3.20	800	600	650	52/118	250
1x3S (automatic wiper)	177.0*	3.00	3.20	800	600	650	52/118	200
1x3S (automatic wiper)	177.0*	3.00	3.20	800	600	650	52/118	250
2x2S (automatic wiper)	240.0*	4.00	4.20	900	1,000	670	78/166	200
2x2S (automatic wiper)	240.0*	4.00	4.20	900	1,000	670	78/166	250
2x3S (automatic wiper)	330.0*	6.00	6.20	900	1,000	670	78/166	250
3x3S (automatic wiper)	500.0*	9.00	9.20	900	1,000	670	78/166	250
3x3S (automatic wiper)	500.0*	9.00	9.20	900	1,000	670	78/166	300

* 98 %/cm transmission; 600 J/m² UV dose for the breaking down of combined chlorine

Manual wipers, optionally also available as automatic wipers

Lamp type	Powerline S medium-pressure lamp
Controller type	Powerline S comfort control
Permissible operating pressure	6 bar
Permissible ambient temperature	5–40 °C
Permissible water temperature	5–40 °C

	Order no.
Powerline UV lamp 1 kW	1035179
Powerline UV lamp 2 kW	1035057
UV lamp Powerline 3 kW	1035180
Lamp protection tube for Dulcodes 1 A and 0.6 S	1035218
Lamp protection tube for Dulcodes 1 S	1035166
Lamp protection tube for Dulcodes 2 S	1035041
Lamp protection tube for Dulcodes 3 S	1035193
Wiper element (2 required per UV lamp)	1027879
Spare part set for UV S 1-3 kW motorised wiper	1037735
Spare part set for UV S 2x2 kW and 2x3 kW motorised wiper	1037756
Spare part set for UV S 3x3 kW motorised wiper	1037757
O-ring lamp protection tube/lamp cover	790410
UVC-U sensor M 1.4539	1034147
O-ring for UVC sensor	1002175
Sensor connection cable, 5 m long for systems supplied since September 2006	1021041
Replacement filter mats for control cabinet ventilation (2 off required per control cabinet)	1004212



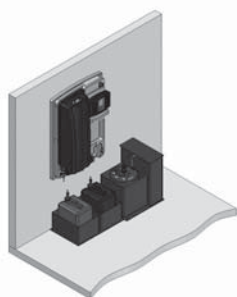
4 Disinfection and Oxidation Process

4.11

Chlorine Dioxide System Bello Zon® CDLb



P_PMA_BEZ_0077_SW



P_PMA_BEZ_0122_SW

The Bello Zon® CDLb chlorine dioxide system uses the chlorite/acid process. A chlorine-free chlorine dioxide solution is generated from a sodium chlorite solution using hydrochloric acid in a batch process. Thanks to the innovative reactor design and stepwise process flow, the production of chlorine dioxide in this manner is exceptionally safe. Depending on type, the system can produce up to 120 g chlorine dioxide per hour. This is then buffered in an integral or separate receiver module at a concentration of 1,000 or 2,000 mg/l. Thanks to the 60 g of chlorine dioxide buffered in the separate receiver module, these systems need not be dimensioned according to their peak load but rather based on their mean consumption. This drastically reduces investment costs compared with conventional systems. The ProMinent® product range includes a wide range of metering pumps and control versions from which to choose when operating several injection points using chlorine dioxide from a receiver module.

The innovative process produces a chlorine dioxide solution of exceptional long-term stability and achieves excellent efficiency of over 90 % in the chlorine dioxide reaction. No chlorine dioxide can escape from the system due to the closed gas transport system, thereby guaranteeing economical, environmentally-friendly operation with minimal use of chemicals.

The modular construction of the system means it is suitable for a range of different applications. Possible applications of the Bello Zon® CDLb chlorine dioxide system are primarily in the prevention of Legionella and in disinfection in the food and beverage industries. Other applications are in cooling and potable water treatment and in swimming pool filter disinfection. Meets the high standards stipulated in the W 224 and W 624 German Association for Gas and Water (DVGW) data sheets.

Advantages

- Safe process control
- No loss of chlorine dioxide due to closed gas transport system
- High stability of the chlorine dioxide solution
- Low investment costs
- Operation of several points of injection
- Filter disinfection
- Prevention of Legionella in hotels, hospitals, etc.
- Disinfection in the food and beverage industries (bottle rinsers, CIP, bottle washing machine, washing of fruit and vegetables)
- Market gardening (irrigation water and sprinkler irrigation water)
- Treatment of cooling water and potable water
- Swimming pools (filter backwashing and prevention of Legionella)

Technical Data

Type	Generation capacity	Solution concentration	Capacity	Dimensions (approx.) H x W x D (mm)	Weight
	g/h	mg/l	l/h	mm	kg
CDLb 06	6*	1000	8	1,236 x 878 x 306	41
CDLb 12	12*	2000	8	1,236 x 878 x 306	42
CDLb 22	22*	2000	13	1,236 x 878 x 306	46
CDLb 55	55* 1)	2000	30	1,550 x 800 x 345	73
CDLb 120	120** 1)	2000	**	1,300 x 880 x 425	55

* Option: Integrated receiver tank and integrated metering pump with suitable capacity up to 7 bar back pressure.

** with external receiver module and separate metering pump

1) without cover

Voltage supply

100 – 230 V, 50/60 Hz (16 A)

Inputs

2 freely configurable digital inputs for the functions Pause, High metering, Intermittent metering or Manual metering, as well as an external collective malfunction signal
 4 digital inputs for monitoring (warning / empty message) the chemical supply
 1 digital input for contact water meter 0.25-20 Hz
 1 frequency input for water meter 10-10,000 Hz

Outputs

1 operating signal relay
 1 alarm signal relay
 1 warning signal relay
 1 voltage output +5 V as supply voltage for a water meter with Hall sensor

Operating fluids

Sodium chlorite 7.5 %, purity according to EN 938
 Hydrochloric acid 9% purity according to EN 939
 Potable water

Enclosure rating

IP 65

4 Disinfection and Oxidation Process

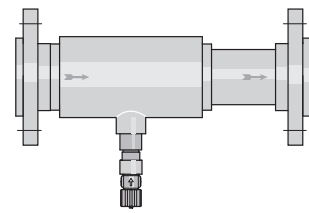
Identity code ordering system for chlorine dioxide systems Bello Zon® CDLb

CDLb	ClO ₂ production capacity					
02	CDLb 06 = 6 g/h					
04	CDLb 12 = 12 g/h					
06	CDLb 22 = 22 g/h					
08	CDLb 55 = 55 g/h, cover not included, see Accessories					
10	CDLb 120 = 120 g/h, cover not included, see Accessories					
Equipment						
0	With receiver tank, pump and multifunctional valve (not with CDLb 120) *					
1	With receiver tank and pump (not with CDLb 120) *					
2	With receiver tank, without pump (not for CDLb 120)					
3	With 30 l receiver module, without pump					
Design						
P	ProMinent					
S	Swiss, DN 15 water connection, rigid piping					
N	Neutral					
Operating voltage						
0	230 V, 50/60 Hz					
1	115 V, 50/60 Hz					
Suction lance, suction assembly						
0	None					
1	With suction lance					
2	With suction lance and collecting pan					
3	With suction lance, collecting pan, angle valve and PE hose 12x9 (10 m)					
Pre-set language						
DE	german					
EN	english					
ES	spanish					
FR	french					
IT	italian					
PL	polish					
CZ	czech					
CDLb	04	0	P	0	2	EN

4.11.1

Accessories and Maintenance Sets for Chlorine Dioxide Systems Bello Zon® CDLb

Point of injection



pk_7_066

Corrosion-resistant point of injection made of PVC-U or PVC-C for warm water applications with integrated mixer elements and maintenance-free PVDF metering valve.

	Material	installation length mm	Order no.
CDL DN 50 point of injection	PVC-U	450	1027611
CDL DN 65 point of injection	PVC-U	400	1026490
CDL DN 80 point of injection	PVC-U	400	1027612
CDL DN 100 point of injection	PVC-U	470	1034693
CDL DN 125 point of injection	PVC-U	550	1047692
CDL DN 150 point of injection	PVC-U	680	1047693
CDL DN 65 point of injection	PVC-C	400	1029326
CDL DN 80 point of injection	PVC-C	400	1029327

Temperature/pressure resistance – metering station CDL

Water temperature (°C)	maximum permissible operating pressure (bar)	
	PVC-U	PVC-C
40	12	12
50	7	9.5
60	4.5	7.5
70	–	5
80	–	3



4 Disinfection and Oxidation Process

Back pressure valve and angle valve

Back pressure valve type MFV with wall bracket and 6 x 4 mm hose connection is ideal for fitting in the chlorine dioxide metering line. Angle valve for the transition from the customer's pipeline to the 12x9 hose connector on the CDLb.

	Order no.
MFV pressure relief valve with wall mounting bracket	1027652
Angle valve D15 G 1/2" brass	1046115

Fault indicating relay for the ClO₂ pump

Fault indicating relay retrofit kit for the ClO₂ discharge pump

	Order no.
Relay 3-pin	1029309

Hood for CDLb

	Order no.
Hood for CDLb 55 PE black	1045889
Hood for CDLb 120 PE black	1045890

Safety collecting pan for chemical containers

Collecting pan with two separate compartments - 1 No. 25 l Bello Zon® acid and 1 No. 10 l Bello Zon® chlorite chemical container.

Dimensions (HxWxD): 290 x 700 x 350 mm

	Order no.
Safety collecting pan CDLa	1026744

Safety collecting pan for chemical tanks (CDLb)

Collecting pan for a 25 l Bello Zon® acid or Bello Zon® chlorite chemical canister.

Dimensions (HxWxD): 266 x 400 x 500 mm

	Order no.
Safety collecting pan CDLb	791726

Service kits for CDLa

The kits contain all parts subject to wear and tear that need to be replaced at regular service intervals. The 1-year kit should be used every year and the 3-year kit in addition every 3 years.

	Order no.
1-year service kit for pressure relief valve	1029442

For CDLa with ClO₂ pump

	Type	Order no.
1-yearly maintenance set	CDL 5	1027263
3-yearly maintenance set, 230 V	CDL 5	1049659
1-yearly maintenance set	CDL 10	1031549
3-yearly maintenance set, 230 V	CDL 10	1049665
3-yearly maintenance set, 100 V	CDLa 5	1049655
3-yearly maintenance set, 115 V	CDLa 5	1049657
3-yearly maintenance set, 100 V	CDLa 10	1049661
3-yearly maintenance set, 115 V	CDLa 10	1049663



4 Disinfection and Oxidation Process

For CDLa without ClO₂ pump

	Type	Order no.
1-yearly maintenance set	CDL5	1042829
3-yearly maintenance set, 230 V	CDL5	1049660
1-yearly maintenance set	CDL10	1042830
3-yearly maintenance set, 230 V	CDL10	1049666
3-yearly maintenance set, 100 V	CDLa 5	1049656
3-yearly maintenance set, 115 V	CDLa 5	1049658
3-yearly maintenance set, 100 V	CDLa 10	1049662
3-yearly maintenance set, 115 V	CDLa 10	1049664

Maintenance sets for Bello Zon® CDLb

For CDLb with receiver tank, pump and multifunctional valve

	Type	Order no.
Annual maintenance set	CDLb 06, CDLb 12	1044484
Annual maintenance set	CDLb 22	1044501
Annual maintenance set	CDLb 55	1044509
3-yearly maintenance set, 230 V	CDLb 06, CDLb 12	1044494
3-yearly maintenance set, 230 V	CDLb 22	1044502
3-yearly maintenance set, 230 V	CDLb 55	1044510
3-yearly maintenance set, 115 V	CDLb 06, CDLb 12	1045212
3-yearly maintenance set, 115 V	CDLb 22	1045216
3-yearly maintenance set, 115 V	CDLb 55	1045220

For CDLb with receiver tank and pump

	Type	Order no.
Annual maintenance set	CDLb 06, CDLb 12	1044495
Annual maintenance set	CDLb 22	1044503
Annual maintenance set	CDLb 55	1044511
3-yearly maintenance set, 230 V	CDLb 06, CDLb 12	1044496
3-yearly maintenance set, 230 V	CDLb 22	1044504
3-yearly maintenance set, 230 V	CDLb 55	1044512
3-yearly maintenance set, 115 V	CDLb 06, CDLb 12	1045213
3-yearly maintenance set, 115 V	CDLb 22	1045217
3-yearly maintenance set, 115 V	CDLb 55	1045221

For CDLb with receiver tank without pump

	Type	Order no.
Annual maintenance set	CDLb 06, CDLb 12	1044497
Annual maintenance set	CDLb 22	1044505
Annual maintenance set	CDLb 55	1044513
3-yearly maintenance set, 230 V	CDLb 06, CDLb 12	1044498
3-yearly maintenance set, 230 V	CDLb 22	1044506
3-yearly maintenance set, 230 V	CDLb 55	1044514
3-yearly maintenance set, 115 V	CDLb 06, CDLb 12	1045214
3-yearly maintenance set, 115 V	CDLb 22	1045218
3-yearly maintenance set, 115 V	CDLb 55	1045222



4 Disinfection and Oxidation Process

For CDLb with 30 l receiver module without pump

	Type	Order no.
Annual maintenance set	CDLb 06, CDLb 12	1044499
Annual maintenance set	CDLb 22	1044507
Annual maintenance set	CDLb 55	1044515
Annual maintenance set	CDLb 120	1044517
3-yearly maintenance set, 230 V	CDLb 06, CDLb 12	1044500
3-yearly maintenance set, 230 V	CDLb 22	1044508
3-yearly maintenance set, 230 V	CDLb 55	1044516
3-yearly maintenance set, 230 V	CDLb 120	1044519
3-yearly maintenance set, 115 V	CDLb 06, CDLb 12	1045215
3-yearly maintenance set, 115 V	CDLb 22	1045219
3-yearly maintenance set, 115 V	CDLb 55	1045223
3-yearly maintenance set, 115 V	CDLb 120	1044519



4 Disinfection and Oxidation Process

4.12

Chlorine Dioxide System Bello Zon® CDVc

Bello Zon®CDVc is the convenient system for the treatment of average to large volumes of water with chlorine dioxide.

1 to 2,000 g/h chlorine dioxide. Max. flow at 0.2 ppm ClO_2 metering is 10,000 m³/h



Chlorine dioxide system for monitoring and metering chlorine dioxide and diluted chemicals. Maximum output and safety due to special reactor concept. Bello Zon® CDVc can be easily and safely integrated into any water treatment process.

Continuous water treatment using the chlorine dioxide system Bello Zon® CDVc can be simply and safely integrated into any process. The special reactor concept generates chlorine dioxide safely and simply with maximum output.

Food-compatible PVDF is used instead of PVC generally used in the industry. This results in improved operating safety and reliability and improved purity of the chlorine dioxide generated. The central system controller manages the precise production of the chlorine dioxide. All parameters relevant for water treatment are recorded and logged.

The stroke lengths of ProMinent® metering pumps are monitored online. This rules out hazardous operating statuses arising from incorrect pump stroke length adjustments.

The precise production of chlorine dioxide is managed by the central system control. Chlorine dioxide, chlorite, pH or redox potential sensors DULCOTEST® are directly connected to the two mA inputs. The chlorine dioxide in the treated water, as well as its main by-product chlorite, is thus monitored and documented online. The chlorine dioxide concentrations in the water can be adjusted automatically depending on the measurement by the integrated PID controller.

The integrated data logger documents all status messages and measured values, which the screen writer then visualises on the clear colour display.

The systems meet all the requirements of the DVGW specifications W 224 and W 624 with regard to construction and operation and are designed for operation with diluted chemicals Bello Zon® chlorite (7.5 % NaClO_2) and acid (9 % HCl).

Your benefits

- Efficient operation, thanks to production, metering and monitoring of ClO_2 with just one system
- Maximum operating safety and purity of the ClO_2 produced through the use of PVDF reactors and stroke length-monitored pumps
- No need for external control due to integrated measuring and control technology
- Perfect quality management, thanks to integrated storage of all operating parameters and measured values
- Automatic monitoring of operating parameters and maintenance intervals
- Simple and safe operation, thanks to clear navigation in plain text

Technical details

Power supply

- 100-230 V, 50/60 Hz

Inputs

- 2 freely configurable analogue inputs (0/4-20 mA)
- 7 digital inputs for monitoring
- 1 digital input for contact water meter 0.25-20 Hz
- 1 frequency input for water meter 10-10,000 Hz

Outputs

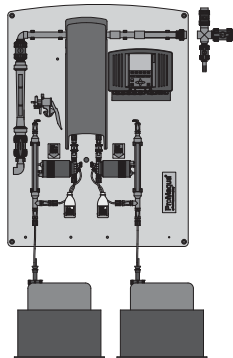
- 1 operating signal relay
- 1 alarm signal relay 1 warning signal relay
- Mains output for controlling the bypass pump
- 1 freely configurable analogue output (0/4-20 mA)
- 1 voltage output +5 V as supply voltage for water meter with Hall sensor

Operating substances

- Sodium chlorite 7.5 %, purity according to EN 938
- Hydrochloric acid 9% purity according to EN 939
- Particle-free water

Degree of protection

- IP 65



P_PMA_BEZ_0009_SW
CDVc 20-120 (figure shows optional configuration)



4 Disinfection and Oxidation Process

Field of application

- Municipal potable water and waste water treatment
- Industrial process and cooling water
- Disinfection in the food and beverage industry, above all with inlet water treatment.
- Market gardening: Germ-free irrigation water and sprinkler irrigation water

Technical Data

Type	Chlorine dioxide dosing capacity*		Max. operating pressure**	Operating temp.	Dimensions*** H x W x D (mm)	Weight*** kg	Power consumption (max.) ****	
	min.-max./hour g/h	min./day g/d					230 V A	115 V A
CDVc 20	1–20	6.4	8	10–40	1,344 x 1,002 x 200	26	2.7	0.9
CDVc 45	2–45	16.0	8	10–40	1,344 x 1,002 x 200	27	2.7	0.9
CDVc 120	6–120	40.0	8	10–40	1,344 x 1,002 x 200	28	2.7	0.9
CDVc 240	12–240	80.0	8	10–40	1,342 x 1,000 x 248	45	2.7	1.2
CDVc 600	30–600	140.0	8	15–40	1,711 x 1,200 x 273	75	2.8	1.4
CDVc 2000	100–2,000	468.0	5	15–40	1,900 x 1,400 x 370	120	4.1	3.2

* The metering figures relate to 5 bar back pressure and an ambient temperature of 20 °C. The minimum capacity/per hour is based on the fact that when the system is operating at below 5 % of the nominal power, continuous metering is no longer possible because of the correspondingly low pumping frequency of the metering pumps. When systems are not operating continuously, the reactor contents should be changed at least twice daily. The system should not, therefore, be operated below the stated minimum capacity/day.

** at 35 °C ambient temperature

*** without bypass pump, flushing valve and water supply line

**** 230 V values with bypass pump, 115 V values without bypass pump

Interfaces

Type	Chlorine dioxide dosing capacity*		Hose connection dimensions of metering pumps	Dimensions of the bypass connector DN
	min.-max./hour g/h	min./day g/d		
CDVc 20	1–20	6.4	6x4	25
CDVc 45	2–45	16.0	6x4	25
CDVc 120	6–120	40.0	6x4	25
CDVc 240	12–240	80.0	8x5	25
CDVc 600	30–600	140.0	8x5	25
CDVc 2000	100–2,000	468.0	DN 10	40



4 Disinfection and Oxidation Process

4.12.1 Identity Code Ordering System for CDVc Systems

CDVc	System type, metering output ClO ₂															
02	CDVc 20= 20 g/h															
04	CDVc 45= 45 g/h															
06	CDVc 120= 120 g/h															
08	CDVc 240= 240 g/h															
10	CDVc 600= 600 g/h															
14	CDVc 2000= 2,000 g/h															
	Type															
	P	ProMaqua														
		Power supply														
	U	100-230 V ± 10 %, 50/60 Hz														
	A	230 V ± 10 %, 50/60 Hz														
	B	100-115 V ± 10 %, 50/60 Hz (not available for version with „bypass“ 04)														
		Bypass version														
	02	PVC-U bypass with float flow meter and pump, unit l/h														
	04	Bypass PVC-U with float flow meter and bypass pump (not CDVc 2000), only selectable with operating voltage A and 50 Hz mains frequency, unit l/h														
	08	PVC-U bypass with float flow meter, unit gpm														
		Calibration device														
	0	Without calibration device, but with measuring cylinder														
	1	With calibration device														
		Suction lance, suction fitting, chemicals														
	0	none														
	1	Suction lance for 5-60 l container (only CDV 20-600)														
	2	Suction lance for 200 l container (only CDV 20-600)														
	3	Flexible suction fitting up to 5 m with two-phase level switch (only CDV 20-600 g/h)														
	4	Suction lance for 25 l tank with 2 drip pans 40 l without leakage sensor (only CDV 20-600 g/h)														
		Mechanical design														
	0	Standard														
		Preset language														
	DE	german														
	EN	english														
	FR	french														
	IT	italian														
	ES	spanish														
		Control														
	0	Basic version *)														
	1	With measuring and control properties (only in connection with version inputs and outputs 1 or 3)														
	2	With measuring and control properties, data logger and screen recorder (only in connection with version inputs and outputs 1 or 3)														
		Extended inputs and outputs														
	0	none														
	1	2 analogue inputs, freely configurable for controller output and flow rate														
	2	1 analogue output, freely configurable														
	3	2 analogue inputs and 1 analogue output, freely configurable														
		Communication interfaces														
	0	Standard														
		Approvals														
	01	CE-mark														
		Temperature monitoring														
	0	without temperature monitoring														
		Hardware														
	0	Standard														
		Software														
	0	Standard														
CDVc	04	P	U	02	1	1	0	EN	1	1	0	01	0	0	0	0

* 4 contact inputs for leakage, external fault, high dosage and pause plus 3 contact outputs for operating, warning and alarm messages.

1 digital and 1 frequency input for connection of flow meters.



4 Disinfection and Oxidation Process

4.12.2

Maintenance Sets for Bello Zon® CDV Chlorine Dioxide Systems

The maintenance kits contain all of the wear parts that may need to be replaced during regular system maintenance.

Maintenance sets for CDVc systems

	Order no.
Maintenance set, complete CDVc 20	1034758
Maintenance set, complete CDVc 45	1034759
Maintenance set, complete CDVc 120	1034760
Maintenance set, complete CDVc 240	1034761
Maintenance set, complete CDVc 600	1034762
Maintenance kit, complete CDVc 2000 up to delivery date 03/2011	1034763
Maintenance kit, complete CDVc 2000 from delivery date 04/2011	1048801

Maintenance sets for CDVb systems

	Order no.
Maintenance set, complete CDVb 15	1022252
Maintenance set, complete CDVb 35	1022253
Maintenance set, complete CDVb 60	1022264
Maintenance set, complete CDVb 120	1022265
Maintenance set, complete CDVb 220	1024614

Maintenance sets for CDVa systems

	Order no.
Maintenance set, complete 230 V CDVa 35	791842
Maintenance set, complete 230 V CDVa 60	791913
Maintenance set, complete 230 V CDVa 120	791915
Maintenance set, complete 230 V CDVa 220	740824
Maintenance set, complete 230 V CDVa 400	740765
Maintenance set, complete 230 V CDVa 600	740826
Maintenance set, complete 230 V CDVa 2000	1005333
Maintenance set, complete 115 V CDVa 35	791860
Maintenance set, complete 115 V CDVa 60	791914
Maintenance set, complete 115 V CDVa 120	791916
Maintenance set, complete 115 V CDVa 220	740825
Maintenance set, complete 115 V CDVa 400	740819
Maintenance set, complete 115 V CDVa 600	740827
Maintenance set, complete 115 V CDVa 2000	1005344

Additional spare parts are listed in the operation instructions for the systems.



5 Individual Devices and Accessories

5.1 Peristaltic Pumps

5.1.1 Peristaltic Pump DULCO®flex DF2a

The optimum pump product range for use in swimming pools, hot tubs, and spa and wellness facilities

Capacity range 0.4 - 2.4 l/h at max. 1.5 bar back pressure

The peristaltic pump DULCO®flex DF2a meters chemicals functionally, cost-effectively and quietly – ideal for use in swimming pools, hot tubs, and in spa and wellness facilities.

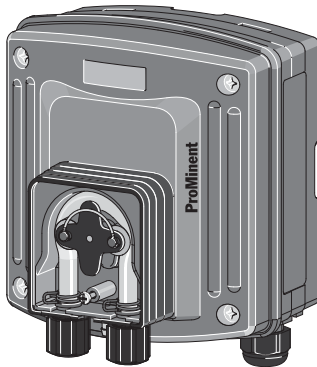
The feed chemical is transported by the rotor squeezing the hose in the direction of flow. This explains why there is no need for valves. The feed chemical is thus handled with care. Typical applications: wherever lower pump pressure is sufficient. For example when metering conditioners in private pools.

Your benefits

- Smooth inner wall reduces deposits.
- Hose materials: PharMed® or Viton®
- Virtually silent operation
- Simple handling
- Enhanced service life of the hose due to spring-loaded rollers, which keep the rolling pressure constant
- Robust and protected against spray water from all sides: Housing made of impact-resistant and chemical-resistant PPE

Technical details

- Self-priming against max. 1.5 bar
- Control or flow control via ON/OFF power supply
- Degree of protection IP 65
- OEM versions on request



pk_1_130

Technical Data

Type	Capacity		Frequency	Connector size	Suction lift	Intake head
	bar	l/h				
0204	1.5	0.4	5	6x4/10x4	4	3
0208	1.5	0.8	10	6x4/10x4	4	3
0216	1.5	1.6	20	6x4/10x4	4	3
0224	1.5	2.4	30	6x4/10x4	4	3

Admissible ambient temperature: 10-45 °C

Power consumption approx.: 5 W

Switching duration: 100 %

Enclosure rating: IP 65

	Order no.
Spare hose set, complete, PharMed®	1009480
Replacement hose compl. Viton®	1023842
Union nut 6 x 4	1009968
Union nut 10 x 4	1008908



5 Individual Devices and Accessories

5.1.2

Identity Code Ordering System

DULCO®flex product range, version DF2a

DF2a	Type	Capacity								
		bar	l/h							
	0204	1.5	0.4							
	0208	1.5	0.8							
	0216	1.5	1.6							
	0224	1.5	2.4							
		Hose material								
	P	PharMed®								
	V	Viton® for fragrances (special version)								
		Version								
		0	With ProMinent® logo							
		1	Without ProMinent® logo							
		Hydraulic connectors								
		0	Connector for hose 6/4 mm suction and discharge side							
		9	Connector for hose 10/4 mm discharge side only							
		Power supply								
		A	230 V ± 10 %, 50/60 Hz							
		Cable and plug								
		0	No mains lead							
		1	With 2 m mains lead, open ended							
		A	With mains cable, European plug							
		Drive								
		0	Mains ON/OFF							
		Installation								
		W	Wall mounted							
		Accessories								
		0	No accessories							
DF2a	0204	P	0	9	A	A	0	W	0	

Viton® and PharMed® are registered trademarks.

5 Individual Devices and Accessories

5.1.3

Peristaltic Pump DULCO®flex DF4a

The optimum pump for use in swimming pools, hot tubs and spa and wellness facilities.

Capacity range 1.5 - 12 l/h, 4 - 2 bar.



The peristaltic pump DULCO®flex DF4a for metering flocculants and activated charcoal treats water precisely and accurately. It is ideal for use in swimming pools, hot tubs or spa and wellness facilities. An operating pressure up to 4 bar is possible.

There are three designs of DULCO®flex DF4a available.

- 1 Metering chemicals
- 2 Metering activated charcoal
- 3 Metering flocculants

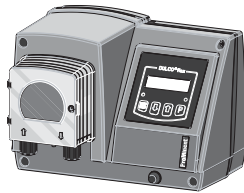
This guarantees that the operating menu, inputs and outputs are always adapted to the respective application.

Your benefits

- Language-neutral user navigation
- Continuous adjustment of capacity
- Hose materials in PharMed® and Tygon®

Technical details

- Priming function
- Night setback
- Inputs for contacts and analogue signals
- Housing degree of protection IP 65
- One or two-stage float switch input
- Operating hour counter
- CANopen interface



P_DX_0003_SW1



5 Individual Devices and Accessories

5.1.4

Identity Code Ordering System

DULCO®flex product range, version DF4a

DF4a	Application															
	0	Chemical pump														
	A	Activated charcoal metering														
	F	Flocculant metering														
	Installation															
	W	Wall mounting														
	Version															
	0	With ProMinent® logo														
	1	Without ProMinent® logo														
	Type		Capacity													
		bar	l/h													
	04004	4.0	0.35													
	04015	4.0	1.50													
	03060	2.5	6.00													
	02120	2.0	12.00													
	Hose material															
	P	PharMed®														
	T	Tygon® not for type 04004														
	Hydraulic connectors															
	0	Standard connector 6x4														
	9	Special connector 10x4 discharge side														
	Power supply															
	U	100 - 240 VAC, 50/60 Hz														
	Cable and plug															
	0	Without cable														
	1	With cable 2.0 m; open end														
	A	With cable 2.0 m; Euro connector														
	B	With cable 2.0 m; Swiss connector														
	Accessories															
	0	Without accessories														
	2	With lip-seal metering valve PCB and 10 m PE metering line														
	Hardware extension															
	0	None														
	Language default															
	00	Language-neutral														
Relay																
1	Fault signalling relay, drop-out action															
3	Fault signalling relay, pick-up action															
Control versions																
8	manual + external contact and analogue 0/4 - 20 mA + 0 - 10 V															
C	as "8" and CANopen															
D	such as "8" and CANopen and CAN connector															
Further input																
1	Pause + 2-stage level + AUX1															
2	Pause + 1-stage level + AUX1 + AUX2															
Pause/level																
0	Pause break contact + level break contact															
Approvals																
01	CE-Symbol															
DF4a	F	W	0	04015	P	9	U	A	0	0	00	1	8	1	0	01

Tygon® and PharMed® are registered trademarks.

Priming lift	3 mWS	Approx. power consumption:	24 W
Suction lift	4 mWS	Switching duration:	100 %
Speed	0 - 85 RPM	Degree of protection:	IP 65
Permissible ambient temperature:	10-45 °C		

All data refers to water at 20 °C.

Spare hoses

	Order no.
For type 04004 PharMed®	1034997
For type 04015 PharMed®	1030722
For type 04015 Tygon®	1030775
For type 03060 PharMed®	1030723
For type 03060 Tygon®	1030776
For type 02120 PharMed®	1030774
For type 02120 Tygon®	1030777

5 Individual Devices and Accessories

5.2 Metering Pumps

5.2.1

Solenoid Driven Metering Pump Beta®

Equipped with all the features and properties to guarantee superior process management.

Capacity range 0.74 - 32 l/h, 25 - 2 bar



All-purpose solenoid metering pump for metering liquid media in water treatment and chemical processes: Solenoid driven metering pump Beta®. Cost-effective, overload-proof, adaptable to existing signal transducers.

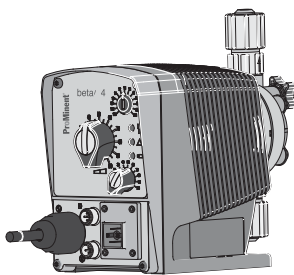
A range of different pump types and material combinations are available for virtually all metering applications. The virtually wear-free solenoid drive guarantees an exceptionally long service life even under maximum load.

Your benefits

- Simple adjustment of metering capacity via stroke rate and stroke length
- Adaptation to existing signal transducers by external control via potential-free contacts with pulse step-up and step-down
- Suitable for use with almost all liquid chemicals thanks to the available material combinations: PP, PVDF, clear acrylic, PTFE and stainless steel
- Self-bleeding dosing head design in clear acrylic/PVC and PP
- Virtually wear-free solenoid drive: economical and overload-proof
- Economical operation with up to 50% energy-savings, thanks to higher pump efficiency
- Everything in sight and under control: 3 LED display for operating, warning and error messages

Technical details

- External control via potential-free contacts with pulse step-up and step-down to adapt to existing signal transducers of 64:1 to 1:64
- Stroke rate adjustment in 10% steps of 10 - 100% corresponds to 18 - 180 strokes/minute
- Continuous stroke length adjustment between 0 - 100% (recommended 30 - 100%)
- Connector for 2-stage level switch
- Wide-ranging electrical connection: 100 - 230 V, 50/60 Hz
- Optional relay module, can also be easily and reliably retrofitted
- Design for low voltage 12-24 V DC



P_BE_0048_SW1
Beta® b



5 Individual Devices and Accessories

Technical Data

Pump type	Delivery rate at max. back pressure			Delivery rate at medium back pressure			Number of strokes	Connection size o Ø x i Ø	Suction lift	Average power consumption	Shipping weight		
	bar	l/h	ml/stroke	bar	l/h	ml/stroke					Strokes/min	mm	mWC
Beta® b													
BT4b 1000***	10	0.74	0.07	5.0	0.82	0.08	180	6 x 4	6.0**	7,2	2.9	3.6	
BT4b 1601***	16	1.10	0.10	8.0	1.40	0.13	180	6 x 4	6.0**	9,6	2.9	3.6	
BT4b 1602***	16	2.20	0.20	8.0	2.50	0.24	180	6 x 4	6.0**	11,2	2.9	3.6	
BT4b 1604***	16	3.60	0.33	8.0	4.30	0.40	180	6 x 4	6.0**	15,2	3.1	3.9	
BT4b 0708***	7	7.10	0.66	3.5	8.40	0.78	180	8 x 5	6.0**	15,2	3.1	3.9	
BT4b 0413	4	12.30	1.14	2.0	14.20	1.31	180	8 x 5	3.0**	15,2	3.1	3.9	
BT4b 0220	2	19.00	1.76	1.0	20.90	1.94	180	12 x 9	2.0**	15,2	3.3	4.4	
BT5b 2504	25	2.90	0.27	10.0	5.00	0.46	180	8 x 4****	6.0**	19,2	4.5	5.3	
BT5b 1008	10	6.80	0.63	5.0	8.30	0.76	180	8 x 5	6.0**	19,2	4.5	5.3	
BT5b 0713	7	11.00	1.02	3.5	13.10	1.21	180	8 x 5	4.0**	19,2	4.5	5.3	
BT5b 0420	4	17.10	1.58	2.0	19.10	1.77	180	12 x 9	3.0**	19,2	4.7	5.8	
BT5b 0232	2	32.00	2.96	1.0	36.20	3.35	180	12 x 9	2.0**	19,2	5.1	6.6	
Beta® b metering pumps with self-degassing liquid end													
BT4b 1601	16	0.59	0.06	8.0	0.78	0.07	180	6 x 4	1.8**	9,6	2.9	–	
BT4b 1602	16	1.40	0.13	8.0	1.70	0.16	180	6 x 4	2.1**	11,2	2.9	–	
BT4b 1604	16	2.70	0.25	8.0	3.60	0.33	180	6 x 4	2.7**	15,2	3.1	–	
BT4b 0708	7	6.60	0.61	3.5	7.50	0.69	180	8 x 5	2.0**	15,2	3.1	–	
BT4b 0413	4	10.80	1.00	2.0	12.60	1.17	180	8 x 5	2.0**	15,2	3.1	–	
BT4b 0220	2	16.20	1.50	1.0	18.00	1.67	180	12 x 9	2.0**	15,2	3.3	–	
BT5b 1008	10	6.30	0.58	5.0	7.50	0.69	180	8 x 5	3.0**	19,2	4.5	–	
BT5b 0713	7	10.50	0.97	3.5	12.30	1.14	180	8 x 5	2.5**	19,2	4.5	–	
BT5b 0420	4	15.60	1.44	2.0	17.40	1.61	180	12 x 9	2.5**	19,2	4.7	–	

* The given performance data constitutes guaranteed minimum values, calculated using water as the medium at room temperature.

** Suction lift with a filled dosing head and filled suction line, with a self-bleeding dosing head with air in the suction line.

*** Pressure-reduced pump types are available in the pressure ratings 4, 7 and 10 bar for special applications, for example in the swimming pool sector. More detailed information is available upon request.

**** With stainless steel design 6 mm connector width.

All data refers to water at 20 °C.

Materials in contact with the medium

	Dosing head	Suction/pressure connector	Ball seat	Seals	Balls
PPE	Polypropylene	Polypropylene	EPDM	EPDM	Ceramic
PPB	Polypropylene	Polypropylene	FPM	FPM	Ceramic
PPT	Polypropylene	PVDF	PVDF	PTFE	Ceramic
NPE	Clear acrylic	PVC	EPDM	EPDM	Ceramic
NPB	Clear acrylic	PVC	FPM	FPM	Ceramic
NPT	Clear acrylic	PVDF	PVDF	PTFE	Ceramic
PVT	PVDF	PVDF	PVDF	PTFE	Ceramic
TTT	PTFE with carbon	PTFE with carbon	Ceramic	PTFE	Ceramic
SST	Stainless steel material no. 1.4404	Stainless steel material no. 1.4404	Ceramic	PTFE	Ceramic

Only the self-bleeding design in material designs PPE, PPB, NPE and NPB with a valve spring made of Hastelloy C, PVDF valve insert. Diaphragm with a PTFE coating.

FKM = fluorine rubber

Metering reproducibility: ± 2% when used according to the operating instructions.

Permissible ambient temperature -10 °C to +45 °C

Degree of protection: IP 65, insulation class F

5 Individual Devices and Accessories

5.2.2 Identity Code Ordering System

BT4b	Type	Capacity													
		bar	l/h												
BT5b	1000	10	0.74												
	1601	16	1.10												
	1602	16	2.20												
	1604	16	3.60												
	0708	7	7.10												
	0413	4	12.30												
	0220	2	19.00												
	2504	25	2.90												
	1008	10	6.80												
	0713	7	11.00												
	0420	4	17.10												
	0232	2	32.00												
Liquid end/valve material															
PP		Polypropylene/PVDF, for version self-degassing Polypropylene/Polypropylene													
NP		Acrylic glass/PVDF, for version self-degassing Acrylic glass/PVC													
PV		PVDF/PVDF													
TT		PTFE/PTFE													
SS		Stainless steel 1.4404/1.4404													
Seal/diaphragm material															
E		EPDM/PTFE coated, only with PP and NP, self-bleeding													
B		FPM-B/PTFE coated, only with PP and NP, self-bleeding													
T		PTFE/PTFE coated													
S		Diaphragm additionally with FKM coating for siliceous media													
Liquid end version															
0		Non-bleed version, no valve spring, for TT, SS and type 0232 only													
1		Non-bleed version, with valve spring, for TT, SS and type 0232 only													
2		With deaerator, no valve spring, PP, PV, NP only, not type 0232													
3		With deaerator, with valve spring, PP, PV, NP only, not type 0232													
4		version for highly viscous media, only PVT, types 1604, 0708, 1008, 0413, 0713, 0220, 0420													
9		self-bleeding only with PP/NP, not for types 1000 and 0232													
Hydraulic connections															
0		Standard according to technical data													
5		Connector for 12/6 hose, delivery side only													
9		Connector for 10/4 hose, delivery side only													
Version															
0		Standard													
Logo															
0		with ProMinent® logo													
Power supply															
U		100-230 V ± 10%, 50/60 Hz													
M		12 V DC, only with BT4b													
N		24 V DC													
Cable and plug															
A		2 m European													
B		2 m Swiss													
C		2 m Australian													
D		2 m USA													
1		2 m, open-ended													
Relay															
0		No relay													
1		Fault indicating relay, normally energised, 1 x changeover contact 230 V - 2 A													
3		Fault indicating relay, normally de-energised, 1 x changeover contact 230 V - 2 A													
4		as 1 + pacing relay 2 x normally open contacts 24 V - 100 m													
5		as 3 + pacing relay 2 x normally open contacts 24 V - 100 mA													
Accessories															
0		No accessories													
1		With foot and dosing valve, 2 m PVC suction tubing, 5 m PE discharge tubing													
Control type															
0		No lock													
1		With lock: manual operation locked when external cable plugged in													
Control Variants															
0		Standard													
Options on request															
0 0		No options													
BT4b	1602	PV	T	2	9	0	0	N	A	0	0	0	0	0 0	



5 Individual Devices and Accessories

Plexiglas version for chlorine products / acid / alkali

Capacity bar	l/h	Connector size o dia. x i dia.	Order no.
4.0	0.8	6 x 4	BT4b 0400 PVT 2900UA001000
4.0	1.5	6 x 4	BT4b 0401 PVT 2900UA001000
4.0	2.8	6 x 4	BT4b 0402 PVT 2900UA001000
4.0	5.3	8 x 5	BT4b 0404 PVT 2900UA001000
4.0	8.3	8 x 5	BT4b 0408 PVT 2900UA001000

Plexiglas version for chlorine products, self-bleeding

Capacity bar	l/h	Connector size o dia. x i dia.	Order no.
4.0	0.9	6 x 4	BT4b 0401 NPB 9900UA001000
4.0	2.1	6 x 4	BT4b 0402 NPB 9900UA001000
4.0	4.2	8 x 5	BT4b 0404 NPB 9900UA001000

Spare parts kit

Type	Order no.
BT4b 0400 PVT	1023107
BT4b 0401 PVT	1023108
BT4b 0402 PVT	1023109
BT4b 0404 PVT	1035332
BT4b 0408 PVT	1023111

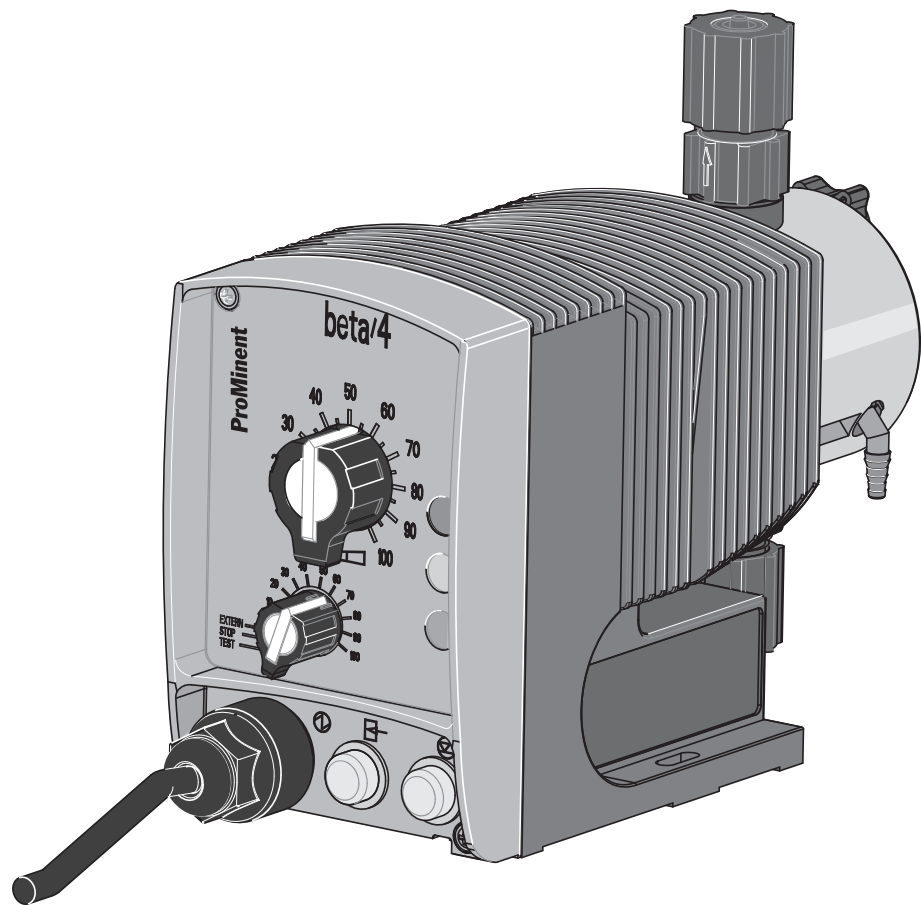
Type	Order no.
BT4b 0401 NPB	1001722
BT4b 0402 NPB	1001723
BT4b 0405 NPB	1001724



5 Individual Devices and Accessories

5.2.3

Beta® 4a Diaphragm Metering Pumps with CAN bus Interface



pk_1_004_1

- CANopen bus interface for DULCOMARIN® II
- Pump capacity 0.2-1,030 l/h
- Stroke length continuously adjustable between 0 - 100% (recommended 30 - 100%)
- Transmission of the stroke length setting from the DULCOMARIN® II
- Material designs PP, clear acrylic/PVC
- Patented coarse/fine bleed valve for PP and clear acrylic/PVC
- Self-bleeding dosing head design in PP and clear acrylic/PVC
- Connector for 2-stage level switch
- design for low voltage 12-24 V DC, 24 V AC
- 4 LED display for operation, warning and error messages
- Alarm in the event of stroke length changes of $> \pm 10\%$
- Transmission of level alarm without alarm relay via the bus



5 Individual Devices and Accessories

5.2.4

Solenoid Diaphragm Metering Pump gamma/ X

gamma/ X - the proven best-seller intelligently extended

Capacity range 1 ml/h – 45 l/h, 25 – 2 bar



The solenoid diaphragm metering pump gamma incorporates a wealth of eXcellent ingenuity! With integrated pressure measurement, it ensures the smooth running of your metering process. The gamma/X is ideal for all metering work involving liquid media.

The new solenoid diaphragm metering pump gamma/ X is user-friendly and has an outstandingly long service life, just like its predecessor. An ingenious solenoid control measures the back pressure and protects the system from overload. This technology makes a pressure sensor superfluous, meaning that operating safety can be significantly increased: no additional parts come into contact with the feed chemical, there are no additional sealing surfaces and no electronic components come into contact with the feed chemical. Whether the metering volume fluctuates or hydraulic failures affect the metering process – the gamma/ X keeps everything at your fingertips.

It independently ensures a trouble-free metering process and should the pump ever need maintenance its service module draws attention to this.

Your benefits

- Simple adjustment of the capacity directly in l/h
- Trouble-free processes by the detection of hydraulic malfunctions or blocked discharge lines
- Integrated pressure measurement and display for greater safety during commissioning and in the process
- Adaptation to existing signal transducers by external control via potential-free contacts with pulse step-up and step-down
- External control via 0/4-20 mA standard signal with adjustable assignment of signal value to stroke rate
- Integrated 7-day timer for timed metering tasks
- Guaranteed metering by means of automatic bleeding
- Connection to process control systems via bus interfaces, such as Profinet, CAN bus, from the 3rd quarter of 2015, others on request
- Organise work processes conveniently with the optional process timer. The alternative to a timer or PLC
- Virtually wear-free solenoid drive, overload-proof and economical
- Suitable for continuous micro-metering from 2 ml/h thanks to the regulated solenoid drive

Technical details

- Available material combinations: PP, PVDF, clear acrylic, PTFE and stainless steel
- Special dosing head designs for gaseous and high-viscosity media
- Illuminated LC display and 3-LED display for operating, warning and error messages, visible from all sides
- Factor with external contact control 99:1 - 1:99
- Batch operation with max. 65,536 strokes/start pulse
- Input concentration for simple adjustment with volume-proportional metering tasks
- Stroke rate adjustment in 1 stroke/hour increments from 0 to 12,000 strokes/h
- Continuous electronic stroke length adjustment from 0 - 100% (recommended 30 - 100%)
- Connector for 2-stage level switch
- External control via 0/4-20 mA standard signal with adjustable assignment of signal value to stroke rate
- Optional 4-20 mA output for remote transmission of stroke length and stroke rate
- Universal power supply unit 100 V - 230 V, 50/60 Hz
- Optional 230 V relay module, can also be easily and reliably retrofitted
- Optional 24 V combined relay, can also be easily and reliably retrofitted



P_GX_001_SW1



5 Individual Devices and Accessories

Technical Data

Pump type	Delivery rate at max. back pressure			Number of strokes	Connection size o Ø x i Ø	Suction lift	Shipping weight	
	bar	l/h	ml/stroke				Strokes/min	mm
gamma/ X								
GMXa 1602	16	2.30	0.19	200	6 x 4	6.0**	3.6	4.1
GMXa 1604	16	3.60	0.30	200	6 x 4	6.0**	3.6	4.1
GMXa 0708	7	7.60	0.63	200	8 x 5	6.0**	3.7	5.0
GMXa 0414	4	14.00	1.17	200	8 x 5	3.0**	3.7	5.0
GMXa 0220	2	19.70	1.64	200	12 x 9	2.0**	3.7	5.0
GMXa 2504	25	3.80	0.32	200	8 x 4***	6.0**	4.9	5.5
GMXa 1009	10	9.00	0.75	200	8 x 5	6.0**	5.1	6.5
GMXa 0715	7	14.50	1.21	200	8 x 5	4.0**	5.1	6.5
GMXa 0424	4	24.00	2.00	200	12 x 9	3.0**	5.1	6.5
GMXa 0245	2	45.00	3.70	200	12 x 9	2.0**	5.2	7.0
gamma/ X metering pumps with self-bleeding dosing head*								
GMXa 1602	16	1.30	0.11	200	6 x 4	2.1**	3.6	–
GMXa 1604	16	2.40	0.21	200	8 x 5	2.7**	3.6	–
GMXa 0708	7	6.80	0.57	200	8 x 5	2.0**	3.7	–
GMXa 0414	4	12.00	1.00	200	8 x 5	2.0**	3.7	–
GMXa 0220	2	18.00	1.50	200	12 x 9	2.0**	3.7	–
GMXa 1009	10	8.00	0.67	200	8 x 5	3.0**	5.1	–
GMXa 0715	7	12.00	1.00	200	8 x 5	2.5**	5.1	–
GMXa 0424	4	20.00	1.67	200	12 x 9	2.5**	5.1	–

* The given performance data represent guaranteed minimum values, calculated using water as the medium at room temperature. Bypass connector with self-bleeding dosing head (SEK): 6 x 4 mm.

** Suction lift with a filled dosing head and filled suction line, with a self-bleeding dosing head with air in the suction line

*** With stainless steel design 6 mm connector width

All data refers to water at 20 °C.

Materials in contact with the medium

	Dosing head	Suction/pressure connector	Ball seat	Seals	Balls
PPE	Polypropylene	Polypropylene	EPDM	EPDM	Ceramic
PPB	Polypropylene	Polypropylene	FKM	FKM	Ceramic
PPT	Polypropylene	Polypropylene	PVDF	PTFE	Ceramic
NPE	Clear acrylic	PVC	EPDM	EPDM	Ceramic
NPB	Clear acrylic	PVC	FKM	FKM	Ceramic
NPT	Clear acrylic	PVC	PVDF	PTFE	Ceramic
PVT	PVDF	PVDF	PVDF	PTFE	Ceramic
TTT	PTFE with carbon	PTFE with carbon	Ceramic	PTFE	Ceramic
SST	Stainless steel material no. 1.4404	Stainless steel material no. 1.4404	Ceramic	PTFE	Ceramic

Self-bleeding design only in material designs PP and NP with a valve spring made of Hastelloy C and a PVDF valve insert. Diaphragm with a PTFE coating.

FKM = fluorine rubber

Metering reproducibility: ±2% when used according to the operating instructions

Permissible ambient temperature: -10 °C to +45 °C

Mean power consumption: 24/30 W

Degree of protection: IP 65, insulation class F



5 Individual Devices and Accessories

5.2.5 Identity Code Ordering System

GMXa	Type	Capacity															
		bar l/h															
	1602	16	2.3														
	1604	16	3.6														
	0708	7	7.6														
	0414	4	14.0														
	0220	2	19.7														
	2504	25	3.8														
	1009	10	9.0														
	0715	7	14.5														
	0424	4	24.0														
	0245	2	45.0														
			Liquid end/valve material														
			PP Polypropylene/PVDF, with self-bleeding design polypropylene/polypropylene														
			NP Clear acrylic/PVDF, with self-bleeding design clear acrylic/PVC														
			PV PVDF/PVDF														
			TT PTFE/PTFE														
			SS Stainless steel 1.4404/1.4404														
			Seal/diaphragm material														
			E EPDM/PTFE coated, only with PP and NP, self-bleeding design														
			B FKM-B/PTFE coated, only with PP and NP, self-bleeding design														
			T PTFE/PTFE coated														
			S Diaphragm additionally with FKM coating for siliceous media, FKM-B seals on PP and NP, PTFE on TT, PV and SS														
			Liquid end version														
			0 Non-bleed version, no valve springonly with NP, TT and SS and type 0245														
			1 Non-bleed version, with valve springonly with NP, TT and SS and type 0245														
			2 Bleed function, no valve springonly with PP, PV, NP not for type 0245														
			3 Bleed function, with valve springonly with PP, PV, NP not for type 0245														
			4 Version for highly viscous mediaonly with PV, types 1604, 0708, 0414, 2504, 1009, 0715, 0424														
			9 self-bleeding with bypass (SEK), only with PP/NP, not for types 2504 and 0245														
			Hydraulic connections														
			0 Standard according to technical data														
			5 Discharge side connection for hose 12/6, suction side standard, only with materials PP, NP and PV														
			9 Discharge side connection for hose 10/4, suction side standard , only with materials PP, NP and PV														
			Diaphragm rupture indicator														
			0 Without diaphragm rupture indicator														
			1 With diaphragm rupture indicator, optical sensor														
			Version														
			0 Standard														
			Logo														
			0 with ProMinent® logo														
			Power supply														
			U 100-230 V, ±10 %, 50/60 Hz														
			Cable and plug														
			A 2 m European				D 2 m USA										
			B 2 m Swiss				E 2 m Great Britain										
			C 2 m Australian				1 2 m, open-ended										
			Relay, pre-set to														
			0 No relay														
			1 1 x changeover contact 230 V – 2 A, fault indicating relay N/C														
			4 2 x N/O 24 V – 100 mA, such as 1 + pacing relay														
			C 1 x N/O 24 V – 100 mA, such as 1 + 4 – 20 mA output														
			Accessories														
			0 No accessories														
			1 With foot valve and discharge valve, 2 m PVC suction tubing, 5 m PE delivery tubeonly with PP, PV and NP, not with PVT4														
			Control version														
			0 Manual + external 1:1 with pulse control														
			3 Manual + external with pulse control + analogue 0/4 - 20 mA														
			4 such as 0 + 4 week process timer														
			5 such as 3 + 4 week process timer														
			C such as 3 + CANopen														
			R as 3 + PROFIBUS® DP interface, M12														
			– no relay with PROFIBUS® version														
			Metering monitor														
			0 Pulse signal input														
			Pause/level														
			0 Pause N/C, level N/C														
			Language														
			DE German														
			EN English														
			FR French														
			ES Spanish														
GMXa	1602	PV	T	2	9	0	0	0	0	U	A	0	0	0	0	0	EN

5 Individual Devices and Accessories

5.2.6

Solenoid Driven Metering Pump delta®

Virtually an all-rounder and just the right solution for exacting requirements. A high-end product with a wide range of settings and control options.

Capacity range 7.5 - 75 l/h, 25 - 2 bar.

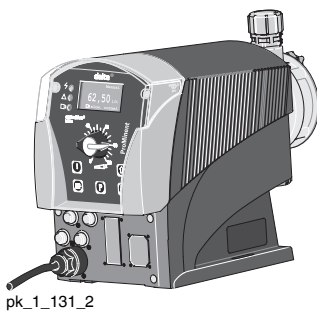


A high-end diaphragm metering pump: The solenoid driven metering pump delta® is the first of its kind to have a regulated solenoid drive. Virtually wear-free, extremely economical and with a self-bleeding dosing head design.

A range of different pump types and material combinations are available for virtually all metering applications. The optional 4-week process timer offers a variety of installation options. The pump achieves maximum precision even with fluctuating back pressure thanks to the regulated solenoid drive. This guarantees an exceptionally long service life even under maximum load. The integrated opto-Guard® monitoring function reports hydraulic fault statuses, such as excess pressure or ruptured metering line. The large illuminated LC display guarantees good readability of all displayed values. The capacity is shown directly in l/h.

Your benefits

- Adjustment of the capacity directly in l/h
- Adaptation to existing signal transducers by external control via potential-free contacts with pulse step-up and step-down
- External control via 0/4 – 20 mA standard signal with adjustable assignment of signal value to stroke rate
- Organise work processes conveniently with the optional process timer. The alternative to timers or PLCs.
- Optional PROFIBUS® interface for connection to process control systems
- Suitable for use with almost all liquid chemicals, thanks to the available material combinations PVDF, clear acrylic and stainless steel
- Virtually wear-free solenoid drive: overload-proof and economical
- Everything in sight and under control: illuminated LED display and 3-LED display for operating, warning and error messages
- Reporting of hydraulic error statuses, blocked points of injection, ruptured metering lines and air and/or gas in the dosing head, which the integrated monitoring system optoGuard® detects
- Automatic bleed function
- Maximum dosing precision by compensation of pressure fluctuations
- Also ideal for continuous micro-metering from around 6 ml/h



pk_1_131_2

Technical details

- External control via potential-free contacts with pulse step-up and step-down to adapt to existing signal transducers of 99:1 to 1:99
- Batch operation with max. 65,536 strokes/start pulse
- External control via 0/4 – 20 mA standard signal with adjustable assignment of signal value to stroke rate
- Stroke rate adjustment in 1 stroke/hour steps of 0 to 12,000 strokes/h and/or 200 strokes/min
- Stroke length continuously adjustable between 0 – 100% (recommended 30 – 100%)
- Connector for 2-stage level switch
- Dosing monitor input with adjustable number of tolerated defective strokes
- Optional optical diaphragm rupture indicator detects droplets behind the diaphragms
- Optional 4 – 20 mA output for remote transmission of stroke length and stroke rate
- "Concentration input" option for volume-proportional metering
- PROFIBUS® or CAN Open interface option
- Control module option with connecting option for chlorine, pH, ORP sensors or flow meter DFMA
- Wide-range electrical connection: 100 – 230 V, 50/60 Hz
- Optional relay module, can also be easily and reliably retrofitted



5 Individual Devices and Accessories

Technical Data

Pump type	Max. pressure bar	Delivery rate l/h	Stroke volume ml/stroke	Max. stroke rate Strokes/min	Connector size outside Ø x inside Ø	Suction lift mWC	Shipping weight NPE, NPB, PVT / SST kg
DLTa 2508	25	7.5	0.62	200	8 x 4** mm	5*	10/11
DLTa 1608	16	7.8	0.65	200	8 x 5** mm	5*	10/11
DLTa 1612	16	11.3	0.94	200	8 x 5 mm	6*	10/11
DLTa 1020	10	19.1	1.59	200	12 x 9 mm	5*	10/11
DLTa 0730	7	29.2	2.43	200	12 x 9 mm	5*	10/11
DLTa 0450	4	49.0	4.08	200	G 3/4 - DN 10	3*	10/11
DLTa 0280	2	75.0	6.25	200	G 3/4 - DN 10	2*	10/11

* Suction lift (mWS) = Suction lift with filled dosing head and filled suction line

** With stainless steel design 6 mm connector width

All data refers to water at 20 °C.

Materials in contact with the medium

Design	Dosing head	Suction/pressure connector	Ball seat	Seals	Valve balls
NPE	Clear acrylic	PVC	EPDM	EPDM	Ceramic
NPB	Clear acrylic	PVC	FKM	FKM	Ceramic
PVT	PVDF	PVDF	PVDF	PTFE	Ceramic
SST (8-12 mm)	Stainless steel 1.4404	Stainless steel 1.4404	Ceramic	PTFE	Ceramic
SST (DN 10)	Stainless steel 1.4404	Stainless steel 1.4404	PTFE with carbon	PTFE	Ceramic

Design of connectors

Plastic	8-12 mm	Hose squeeze connection
	DN 10	d16 DN 10 hose nozzle
Stainless steel	6-12 mm	Swagelok system
	DN 10	Rp 3/8 insert

Diaphragm with PTFE coating.

Repeatability of metering ± 2 % when used according to the operating instructions.

Permissible ambient temperature: -10 °C to 45 °C

Mean power consumption 78 W

Degree of protection IP 65, insulation class F

5 Individual Devices and Accessories

5.2.7 Identity Code Ordering System

DLTa	Type	Capacity														
		bar	l/h													
	2508	25	7.5													
	1608	16	7.8													
	1612	16	11.3													
	1020	10	19.1													
	0730	7	29.2													
	0450	4	49.0													
	0280	2	75.0													
		Liquid end/valve material														
	PV	PVDF/PVDF not for pump type 2508														
	NP	Acrylic glass/PVC only for pump type 2508, 1608, 1612, 1020, 0730														
	SS	Stainless steel/stainless steel														
		Seal/diaphragm material														
	T	Only with PV and SS														
	S	PTFE/diaphragm additionally with FKM coating for silica-laden media, not for types 0450 and 0280														
	B	FKM-B, only with NP														
	E	EPDM, only with NP														
		Liquid end version														
	0	Non-bleed, without valve spring, only with material TT and SS														
	1	Non-bleed, with valve spring, only with material TT and SS														
	2	Bleed version, without valve spring, only with material NP and PV														
	3	Bleed version, with valve spring, only with material NP and PV														
	4	HV version for higher-viscosity media, only for types 1608, 1612, 1020 and 0730														
		Hydraulic connections														
	0	Standard connectors as per technical data														
	5	Discharge-side connector for 12/6 hose, suction-side standard, only with material NP and PV														
	F	Connector on discharge side for 8/4 hose, standard on suction side, only with material NP and PV														
		Diaphragm rupture indicator														
	0	Without diaphragm rupture indication														
	1	With diaphragm rupture indicator, optical sensor														
	2	With dual diaphragm system and diaphragm rupture indicator, pressure sensor, only with material SS														
		Version														
	0	With ProMinent logo														
		Power supply														
	U	Universal controller 100-230 V 50/60 Hz														
		Cable and plug														
	A	2 m Europe								D	2 m USA / 115 V					
	B	2m Switzerland								1	2 m without plug					
	C	2 m Australia														
		Relay														
	0	Without relay														
	1	alarm relay normally energised 1 x C/O contact 230 V – 8 A														
	3	alarm relay normally de-energised 1 x C/O contact 230 V – 8 A														
	4	as 1 + pacing relay 2 x N.O. contacts 24 V – 100 mA														
	5	as 3 + pacing relay 2 x N.O. contacts 24 V – 100 mA														
	A	Shutdown and alarm relay normally energised 2 x N.O. contacts 24 V – 100 mA														
	C	as 1 + 4-20 mA output 1 x N.O. contact 24 V – 100 mA														
	F	with automatic bleed valve, 230 V not for pump type 2508														
	G	with automatic bleed valve, 24 V DC and relay output														
		Accessories														
	0	Without accessories														
	1	With foot and metering valve, 2m suction line and 5 m discharge line														
	2	As 0 + measuring cup (only for type 2508, 1608, 1612, 1020, and 0730)														
	3	As 1 + measuring cup (only for type 2508, 1608, 1612, 1020, and 0730)														
		Control version														
	0	Manual + external contact with pulse control														
	3	Manual + external contact with pulse control + analogue 0/4-20 mA														
	4	as 0 + 4-week process timer														
	5	as 3 + 4-week process timer														
	C	as 3 + CANopen														
	M	As 3 + pH, ORP and chlorine + DFMA control module														
	R	as 3 + PROFIBUS® interface, M12														
		Access code														
	0	Without access code														
	1	With access code														
		Language														
	DE	german														
	EN	english														
	FR	french														
	ES	spanish														
		Pause/level														
	0	Pause N.C. contact level, N.C. contact														
DLTa	1612	PV	T	2	0	0	0	0	U	A	0	0	0	0	EN	0



5 Individual Devices and Accessories

5.2.8

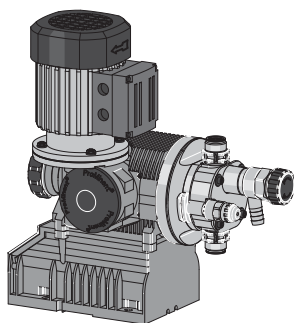
Motor Driven Metering Pump Sigma/ 1 (Basic type)

The robust pump for safe and reliable use

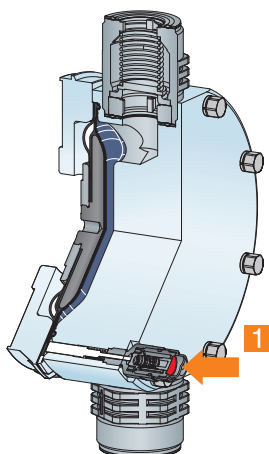
Capacity range 17 – 144 l/h, 12 – 4 bar



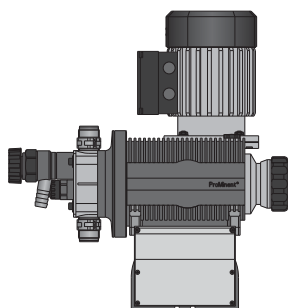
The Sigma/ 1 Basic is an extremely robust motor-driven metering pump with patented multi-layer safety diaphragm for excellent process safety. It offers a wide range of power end designs, such as three-phase or 1-phase AC motors, even for Exe and Exde areas with ATEX certification.



P_SI_0128_SW
Sigma/ 1 Basic version



P_SI_0065_C1
1: Diaphragm rupture sensor



P_SI_0152_SW
Sigma/ 1 liquid end on left

The Sigma/ 1 diaphragm metering pump together with pumps of type Sigma/ 2 and Sigma/ 3 represent an integrated product range. They cover the capacity range from 17 to 1,030 l/h, with a consistent operating concept, control concept and spare parts management. A wide range of drive versions is available, including some for use in Exe and Exde areas with ATEX certification.

Your benefits

Excellent process safety and reliability:

- In the event of an accident, the feed chemical does not escape to the outside nor into the pump's power end, thanks to the patented multi-layer safety diaphragm with optical (optionally electric) signalling
- Integrated relief valve protects the pump against overloading
- Reliable operation by bleed option during the suction process

Flexible adaptation to the process:

- The entire Sigma product range is available as standard in a "Physiologically safe in respect of wetted materials" design.
- Metering pumps with electro-polished stainless steel metering head and EHEDG certification enable them to be used in hygienically challenging applications
- Adaptation to specific installation situations, as the "Liquid end on left" is available as standard
- Wide range of power end versions, also for use in Exe and Exde areas and different flange designs for the use of customised motors
- Customised designs are available on request

Technical details

- Stroke length: 4 mm,
- Stroke length adjustment range: 0 – 100%
- Stroke length adjustment: manually by self-locking rotary dial in 1% increments (optionally with actuator or control drive)
- Metering reproducibility is better than $\pm 2\%$ within the 30-100% stroke length adjustment range under certain defined conditions and after proper installation.
- Wetted materials: PVDF, stainless steel 1.4571/1.4404, special materials on request
- Patented multi-layer safety diaphragm with optical diaphragm rupture display (optionally with diaphragm rupture warning system via a contact)
- Integrated hydraulic relief and bleed valve
- A wide range of power end versions is available: Three-phase standard motor, 1-phase AC motor, motors for use in Exe and Exde areas and different flange designs for use in customer-specific motors
- Degree of protection IP 55 (optionally II2GExeIIIT3, II2GExdIICT4)
- Fibreglass-reinforced plastic housing
- Liquid end on left is available as standard
- For reasons of safety, provide suitable overload protection mechanisms in all mechanically deflected diaphragm metering pumps

Field of application

- Volume-proportional addition of chemicals in water treatment, e.g. sodium-calcium hypochlorite for the disinfection of potable water
- Addition of chemicals depending on the measured value, e.g. metering of acid and alkali for pH neutralisation in waste water treatment
- Time-controlled addition of chemicals in the cooling water circuit
- Pulse-controlled metering in the bottling of different volumes e.g. glycerin filling of manometers



5 Individual Devices and Accessories

Sigma Basic Type Control Functions (S1Ba)

Stroke length actuator/controller

Actuator for automatic stroke length adjustment, actuating period approx. 1 sec for 1 % stroke length, 1 k Ohm response signal potentiometer, enclosure rating IP 54.

Controller consists of actuator with servomotor and integrated servo control for stroke length adjustment via a standard signal. Standard signal input 0/4-20 mA corresponds to stroke length 0 - 100 %. Automatic/manual operation selection key for manual stroke adjustment. Mechanical status display of actual stroke length value output 0/4-20 mA for remote display.

Variable speed motors with integrated frequency converter (identity code specification V)

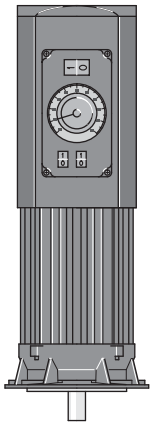
Power supply 1ph 230 V, 50/60 Hz, 0.18 kW

Externally controllable with 0/4-20 mA (see Fig. pk_2_103).

Upon request externally controllable via PROFIBUS® DP

Speed controllers with frequency converter (identity code specification Z)

The speed controller assembly consists of a frequency converter and a variable speed motor of 0.09 kW



pk_2_103

Variable speed motor with integrated frequency converter

"Physiologically safe (FDA) in respect of wetted materials" version

All wetted materials in the "Physiologically safe (FDA) in respect of wetted materials" design comply with the FDA guidelines.

FDA guidelines:

- Material PTFE: FDA No. 21 CFR § 177.1550
- Material PVDF: FDA No. 21 CFR § 177.2510

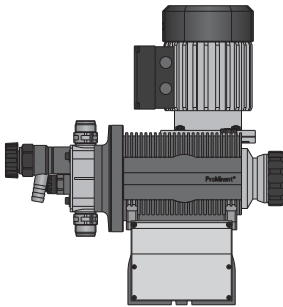
Available for material version PVT and SST.

Identity code example: S1BaH04084PVTS00 F S000

Sigma / 1 Basic Type version "left liquid end "

This version offers additional adaptability to special installation situations, e.g. in combination with storage tanks, brackets, etc.

Identity code example: S1BaH07042PVTS00 5 S000



P_SI_0152_SW

Sigma / 1 liquid end on left



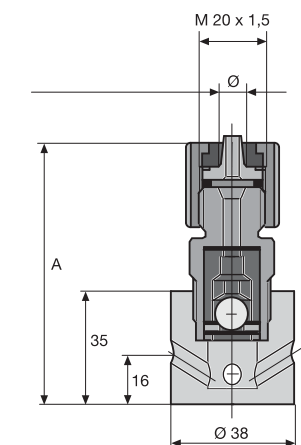
5 Individual Devices and Accessories

5.3 Accessories for Metering Pumps

5.3.1 Accessories, Suction Side

PVT Foot valve

PVDF housing, PTFE seals.

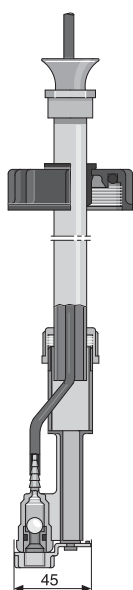


pk_1_040

Connector	oØ x iØ mm	A mm	fig.	Order no.
6/4 for hose	6 x 4	79	pk_1_040	1024705
8/5 for hose	8 x 5	79	pk_1_040	1024706
12/9 for hose	12 x 9	82	pk_1_040	1024707
DN 10 for hose	24 x 16	92	P_AC_0206_SW	1029471

Variable suction lance without level switch

680 mm long for connection to disposable 5 - 60 litre tank, consisting of foot valve, retaining tube, vertically adjustable screw cap and 2 m intake hose.



pk_1_067

PCB

Material of retaining tube and foot valve PVC
Seal material FKM
Hose material soft PVC

Material	Hose o Ø x i Ø mm		fig.	Order no.
PCB	6 x 4	For 50 mm tank opening	pk_1_067	790536
PCB	8 x 5	For 50 mm tank opening	pk_1_067	790537
PCB	12 x 9	For 50 mm tank opening	pk_1_067	790538

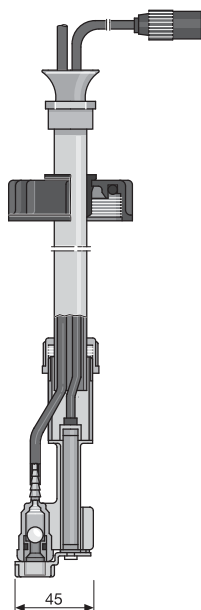
5 Individual Devices and Accessories

Variable suction lance with two-stage level switch

680 mm long for connection to disposable 5 - 60 litre tanks, comprising a foot valve, level switch with round plug and support pipe, height adjustable screw head and 2 m long suction hose.

For Beta®, gamma and delta® metering pump product ranges

Switching mode at liquid level low 2 x NC.



pk_1_075

PCB

Material of retaining tube and foot valve	PVC
Seal material	FKM
Hose material	soft PVC

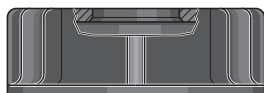
Material	Hose o Ø x i Ø mm		fig.	Order no.
PVC	6 x 4	PVC for Ø 50 tank opening, suction hose	pk_1_075	802077
PVC	8 x 5	PVC for Ø 50 tank opening, suction hose	pk_1_075	802078
PVC	12 x 9	PVC for Ø 50 tank opening, suction hose	pk_1_075	790371

Variable suction lance with two-stage level switch

PCB

Material of retaining tube and foot valve	PCB
Seal material	FPM
Hose material	Soft PVC

Material	Hose o Ø x i Ø mm		fig.	Order no.
PCB	6 x 4	PP for Ø 50 tank opening, suction hose	P_AC_02 34_SW1	790650



pk_1_066

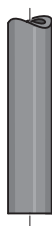
Screw cap

For tanks with opening Ø 44, customers need to order the Ø 44 screw cap as a spare part to replace Ø 50 screw cap.

	Order no.
Ø 44 screw cap	811626

Suction line, soft PVC

Material	Length m	oØ x iØ mm	Permissible pressure bar	Order no.
PVC flexible	5	6 x 4	0.5*	1004520
	5	8 x 5	0.5*	1004521
	5	12 x 9	0.5*	1004522



pk_1_013

* Admissible operating pressure at 20 °C in accordance with DIN EN ISO 7751, subject to chemical resistance and correct assembly.



5 Individual Devices and Accessories

5.3.2

Accessories for Metering End

Multifunctional valve type MFV-DK, PVDF

Multifunctional valve for assembly directly onto the liquid end of the pump with the following functions:

- Back pressure valve, opening pressure approx. 1.5 bar, with open discharge or positive pressure on the suction side (black rotary knob)
- Relief valve, opening pressure approx. 6, 10 or 16 bar (red rotary knob)
- Priming aid with pending back pressure, no need to de-pressurise pipes
- Pressure relief, e.g. prior to servicing

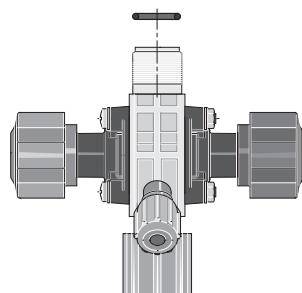
The ProMinent® multifunctional valve is simple to operate using smooth action rotary knobs, which return to their initial position on release. This ensures safe operation even under difficult access conditions. The ProMinent® multifunctional valve is made of PVDF and can be used with virtually all chemicals.

Warning: Back pressure valves are not intended as completely sealed units!

Caution: The bypass line must always be connected.

Valve body	PVDF
Diaphragm	PTFE- coated
Seal	FKM and EPDM (enclosed)

Type	Relief opening pressure*	Connection	Bypass connector	Order no.
Size I	6 bar	6-12	6/4	1005745

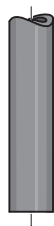


pk_1_053

Suction and discharge, PE

Material	Length m	oØ x iØ mm	Permissible pressure bar	Order no.
Polyethylene	5	6 x 4	10*	1004492
	5	8 x 5	10*	1004493

* Admissible operating pressure at 20 °C in accordance with DIN EN ISO 7751, subject to chemical resistance and correct assembly

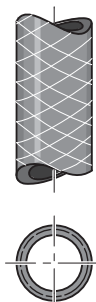


pk_1_013

Suction and discharge line, soft PVC with woven fabric core

Material	Length m	oØ x iØ mm	Permissible pressure bar	Order no.
Fabric-reinforced flexible PVC	5	10 x 4	18*	1004533
	5	12 x 6	17*	1004538

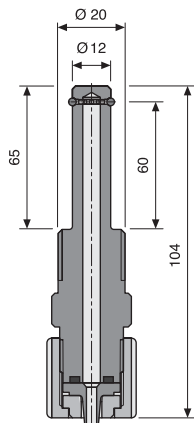
* Admissible operating pressure at 20 °C in accordance with DIN EN ISO 7751, subject to chemical resistance and correct assembly.



pk_1_060



5 Individual Devices and Accessories

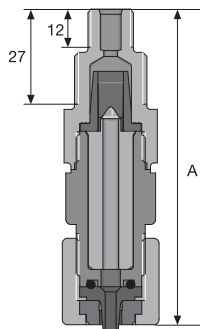


P_AC_0183_SW

PTFE Injection valves O-ring loaded

PTFE housing, FKM seals.

Connection	oØ x iØ mm	A mm	fig.	Order no.
6/4 – for PE/PTFE line	6 x 4	104	P_AC_0183_SW	809484
8/5 – for PE/PTFE line	8 x 5	104	P_AC_0183_SW	809485
10/4 – for PVC hose	10 x 4	104	P_AC_0183_SW	1002925
12/6 – for PVC hose	12 x 6	104	P_AC_0183_SW	809487
12/9 – for PE/PTFE line	12 x 9	104	P_AC_0183_SW	809486

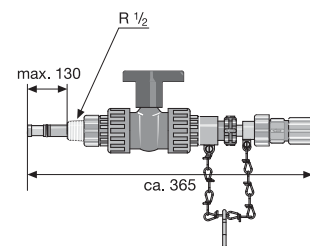


pk_1_070

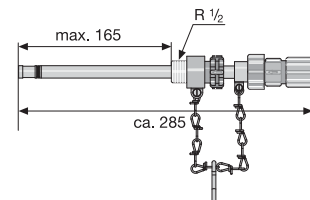
Lip seal injection valve PCB

Body PVC, seals FKM, inlet pressure approx. 0.05 bar. For metering sodium hypochlorite and for use in conjunction with the peristaltic pump DF2a.

Connection	oØ x iØ mm	A mm	fig.	Order no.
6/4 - R 1/2 - 1/4 for PE/PTFE pipe	6 x 4	90	pk_1_070	1019953
10/4 - R 1/2 - 1/4 for PE/PTFE pipe	10 x 4	90	pk_1_070	1024697



pk_1_007



pk_1_062

PPE injection lance

For immersion depths of 20 - 165 mm, in large diameter pipe to prevent chemical deposition at the point of injection. Consists of spring-loaded metering valve, Hastelloy C spring, ceramic ball, adjustable immersion rod and hose valve. With connectors for all hose sizes used with solenoid metering pumps: 6/4, 8/5, 12/9, 10/4 and 12/6.

Type	Seal material	Max. pressure at 25 °C bar	fig.	Order no.
PPE without stopcock	EPDM/silicone	6	pk_1_007	1021530
PPE with stopcock	EPDM/silicone	6	pk_1_062	1021531
PCB without stopcock	FKM/silicone*	6	pk_1_007	1021528
PCB with stopcock	FKM/silicone*	6	pk_1_062	1021529

* Caution: The product contains adhesive joints with Tangit. Please note the resistance of Tangit adhesive.



5 Individual Devices and Accessories

5.3.3

General Accessories

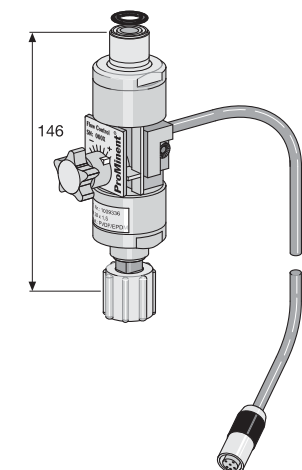
Solenoid Dosing Pump Accessories

Suitable for product range gamma/ X in material designs PP, PC, NP and TT. Complete with connector cable for assembly directly on the dosing head.

For monitoring the individual strokes based on the floating body principle. The adjustment screw is used to match the partial flow flowing past the float to the respective stroke volume so that an alarm is emitted if the level is transgressed by approx. 20%. The gamma/ L enables the permitted number of incompletely performed strokes to be selected between 1 to 127, ensuring optimum adaptation to process requirements.

Materials

Housing:	PVDF
Float:	PTFE-coated
Seals:	FKM/EPDM



pk_1_086_2

Flow Control	For pump type	Material	Order no.
Size I	1602	PVDF/EPDM	1009229
	1602	PVDF/FKM	1009335
Size II	1604, 0708, 1009, 0414, 0220, 0715, 0424	PVDF/EPDM	1009336
	1604, 0708, 1009, 0414, 0220, 0715, 0424	PVDF/FKM	1009338

External signal cable

Only for external control of Beta®, gamma/ X and delta® via contacts. With 5-pin round plug, internally bridged and 2-wire cable with open end.



pk_1_085

	Lead length	Order no.
2-core external cable, 5-pin round plug	2 m	707702
2-core external cable, 5-pin round plug	5 m	707703
2-core external cable, 5-pin round plug	10 m	707707

Solenoid Dosing Pump Accessories



pk_1_085

	Lead length	Order no.
5-core universal cable, 5-pin round plug	2 m	1001300
5-core universal cable, 5-pin round plug	5 m	1001301
5-core universal cable, 5-pin round plug	10 m	1001302

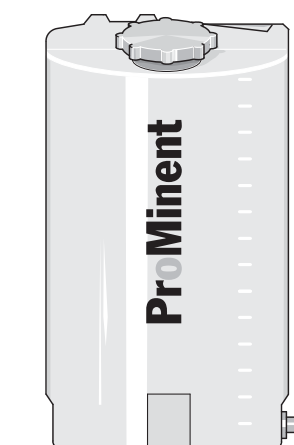
Storage tank

60 litre cylindrical tank, made of coloured PE with scale and opening for manual mixer with screw cover and threaded sleeves for mounting the ProMinent® metering pumps.

Dimensions

Diameter: 410 mm

Height: 490 mm



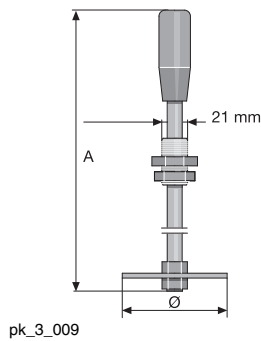
pk_3_001_1

Type	Order no.
PE black	791999
PE blue	1003813
PE yellow	1003819
PE red	1003825

5 Individual Devices and Accessories

PP Hand mixer

fully assembled.



pk_3_009

	A mm	Ø mm	Order no.
for 35 und 60 l storage tanks	515	90	741118

PPE wall mounting bracket

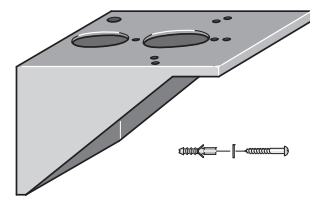
With fittings, for mounting a metering pump of size Beta®/ 4, Beta®/ 5, gamma/ X and alpha.

The Beta®/ 4, gamma/ X can either be mounted parallel or diagonally to each other.

Dimensions L x W x H: 208 x 120 x 140 mm

Material

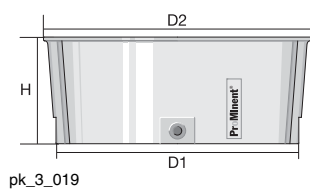
Fibreglass-reinforced plastic PPE



pk_1_092

	fig.	Order no.
for BT4, BT5, gamma/ X, G/ 4, G/ 5, D_4a	pk_1_092	810164

PE colourless/transparent stackable collecting pans



pk_3_019

Usable capacity l	D2 mm	D1 mm	H mm	Weight kg	Order no.
35	565	507	220	3.0	1010879

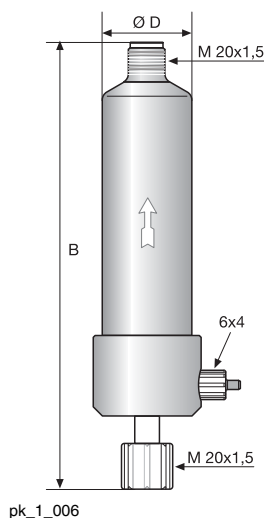
PP Pressure accumulator

Caution: Always install an overflow valve when using pressure accumulators.

Operating range

20 °C - max. operating pressure 10 bar

40 °C - max. operating pressure 6 bar



pk_1_006

	Volume l	Permissible stroke volume ml	Connection	fig.	Order no.
Size 0*	0.15	1.0	M 20 x 1.5	pk_1_006	1021157

* With bleed valve. Install directly at the pressure connector.

Blue protective tray for the metering station

to accommodate metering pumps (hose, diaphragm metering pumps) when it is necessary to separate the metering technology from the I&C technology. Can be installed freely in plant rooms, including protective tray with rear wall bracket.

	Order no.
Blue protective tray	DSS 2002



5 Individual Devices and Accessories

5.4 Measuring and Control Equipment

5.4.1

Multi-Channel Multi-Parameter Measuring and Control System DULCOMARIN® II



pk_5_045



The multi-channel measuring and control system DULCOMARIN® II stands out thanks to the following features:

- 5,7", 1/4 VGA colour display for ultra-simple operation
- Integral data logger with screen plotter: view measured data directly on the controller
- SD card and card reader for PC included: simply transfer measured data to your PC as standard
- Control of one to 16 drinking water systems or filtration circuits in swimming pools
- CAN bus system: Simple to wire, subsequently extendable
- Visualisation*: simple with embedded web server* and standard web browser
- LAN interface*: simple connection to your PC, PC network or the internet
- Intelligent sensors: save sensor data with CANopen bus and ensure you always remain within the optimum measuring range with auto ranging
- Intelligent metering pumps: use CANopen bus to obtain information on operating parameters, such as: chemical level statuses and pump capacity in the dosing range of 0.74 l/h to 1,030 l/h
- Standby metering pump for disinfectants (automatic switchover in the event of liquid level low and pump malfunction)

Drinking water applications (and general applications)

- Measure and display the following measurement parameters using 0/4...20 mA with a current input module (I-module). These values are also available on the data logger/screen plotter, the web and OPC server:
 - Flow (as feed forward control for pH and chlorine regulation)
 - UV intensity
 - Conductivity
 - Chlorine dioxide
 - Chlorite
 - Ammonia
 - Fluoride (using D1Ca)
 - Pt100 resistance thermometer via a transducer
- Display and control of free chlorine and total available chlorine
- OPC server*: Simple connection to higher-level visualisation systems

***optional**

Swimming pool applications

- Combined chlorine: reliably reduced by the control output and corresponding systems
- OPC server*: simple connection to higher-level visualisation systems
- Regulate the pool temperature using standard temperature controllers
- Superchlorination or night setback by contact or time-controlled by a second parameter set
- The decentralised modular DULCOMARIN® II system is designed for use in DIN 19643-compliant public swimming pools. The system can be configured according to requirements as the compact system DULCOMARIN® II compact or as the decentralised modular system DULCOMARIN® II DULCO®-Net .

The applications are defined in the identity code

Every drinking water system or every filtration circuit has a proprietary on-site calibration option for all measured variables.

The examples shown in the following are suitable for applications in drinking water treatment and in swimming pool technology.

What is Eco!Mode operating mode?

There is a menu in the DULCOMARIN® II for each of the measured variables controlled, in which the control parameters (setpoint, proportional coefficient etc.) are defined. Eco!Mode permits alternative control parameters to be enabled for every measured variable controlled via a digital input on the M module. The alternative control parameters can be used, for instance, in setback mode to optimise the control parameters to this operating mode or activate higher setpoints for chlorine in the event of superchlorination. Eco!Mode remains active for as long as the digital input is set, however, it can also be activated via a timer. This can be checked via the recorder function!



5 Individual Devices and Accessories

What is a web server?

A web server is a software application executed by the DULCOMARIN® II.

The web server delivers web pages with information about measurements, control, sensor calibration and control configuration to a PC with a web browser (e.g. Microsoft® Internet Explorer).

The web server enables simple and straightforward visualisation of the DULCOMARIN® II, without special visualisation software being required on the PC. The web server is independent of the PC's operating system. It can be linked by Wi-Fi to an iPad, Smart phone or laptop via an access point.

The DULCOMARIN® II is connected to a PC via a LAN/Ethernet interface and this connection can be made directly, via a network or via the internet. The cables needed for direct connection to a PC or network connection are included in the option.

Standard commercially available network components can be used as accessories for cables, routers and WLAN access points etc.

The same information can be accessed via the web server as is available on the DULCOMARIN® II itself, for instance changing setpoints for all control variables, switching off the different controllers and entering names for the pools/systems. The exceptions are the control settings and bus configuration that can only be entered directly on the controller.

What is OPC?

OPC stands for Openness, Productivity, Collaboration (formerly OLE for Process Control) and is used to describe a uniform software interface independent of specific manufacturers. OPC Data Access (OPC DA) is based on Windows technology COM (Component Object Model) and DCOM (Distributed Component Object Model). OPC XML, in contrast, is based on the internet standards XML, SOAP and HTTP.

OPC is used wherever sensors, controllers and controls supplied by different manufacturers are used to create a common, flexible network. Without OPC, two devices would require precise knowledge about the communication options of the other device to be able to exchange data and extensions and exchanges would be correspondingly difficult. With OPC it is sufficient to write an OPC-compliant driver precisely once for each device and ideally this is provided by the manufacturer. An OPC driver can be integrated without extensive adaptation into any large control and monitoring systems.

ProMinent supplies an OPC server/driver such as this for the multi-channel measuring and control system DULCOMARIN® II.

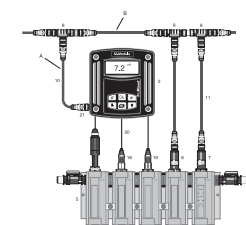


5 Individual Devices and Accessories

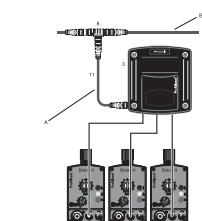
5.4.2

Multi-Channel Multi-Parameter Measuring and Control System DULCOMARIN® II

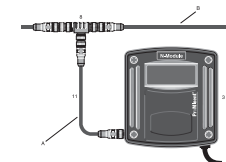
The DULCOMARIN® II multi-channel measuring and control system is suitable for 1 to 16 filtration circuits or potable water systems. The following bus modules are available for the control:



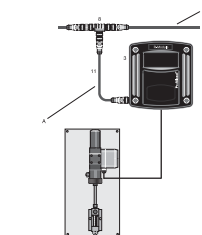
M-Module



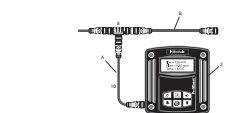
A-Module



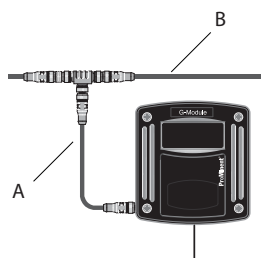
N-Module



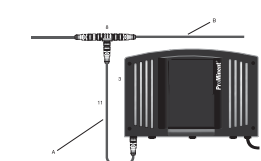
R-Module



I-Module



G-Module



M module (measurement and control):

- Measurement and control of the pH value
- Measurement and display (optional control) of the ORP
- Measurement and display of the temperature of the sample water
- Sample water monitoring
- Measurement of free chlorine
- Measurement of combined chlorine (optional, calculated from total chlorine and free chlorine)

Chlorine sensors:

- Measurement of free chlorine and temperature
- Measurement of total available chlorine and temperature
- Measurement of combined chlorine as differential chlorine measurement

A module (control of metering pumps, analogue outputs):

- 3 frequency outputs to control metering pumps for pH correction, disinfection and flocculant metering
- 3 contact inputs to process pump alarm relays or tank fill level monitoring
- 4 freely programmable analogue outputs 0/4...20 mA for pH, ORP, free chlorine, combined chlorine or temperature

P module (controlling of peristaltic pumps, power supply of bus modules):

- Power relay pulse length control for pH value (e.g. control of the peristaltic pump)
- Power relay pulse length control of disinfectant (e.g. control of the chlorine electrolysis plant)
- Power relay limit value output to minimise combined chlorine
- Alarm relay
- Power supply of bus modules

N module (power supply of bus modules):

- Power supply of bus modules with no further function

R module (control of the chlorine gas metering units):

- Control of a chlorine gas metering unit and processing of a position feedback potentiometer (0...10 kΩ) (only possible as external module)

Metering pumps with CANopen interface of type Beta®, delta®, Sigma/ 1, Sigma/ 2, and Sigma/ 3

- Direct connection to the bus
- When using Beta/4aCANopen metering pumps, the A module is not required (provided no current outputs are required).

I module (current input module)

- 2 active/passive current inputs (e.g. for the connection of 2-wire transmitters)
- 1 passive current input (e.g. for the connection of a magnetic-inductive flow meter)
- 2 digital inputs for sample water alarm and pause control
- 1 channel with controller function

G module (limit value module)

Individual parameters can be defined and different limit values activated here.

Installation of this module is necessary to comply with Austrian standards.

F module (functional module)

- Display of hydraulic functions
- Control of attractions
- Monitoring of the water circuits

5 Individual Devices and Accessories

5.4.3 Technical Data for the DULCOMARIN® II Multi-Channel Measuring and Control System

Measuring range	pH -1...15 ORP: -1,200 ... +1,200 mV Chlorine, free 0.01...10 ppm/100 ppm Chlorine, total 0.01...10 ppm Chlorine, combined 0.01... 2.00 ppm Bromine: 0.01...10 ppm Chlorine dioxide: 0.01...10 ppm Chlorite anion: 0.10...2 ppm
Temperature	-20 ... 150 °C Pt 100 or Pt 1000
Resolution	0.01 pH / 1 mV / 0.01 ppm / 0.1 °C
Accuracy	0.5% of the final value of the measuring range (at 25 °C)
Measurement input	ph and ORP via terminal mV Chlorine via CANopen bus
Control characteristic	P/PI/PID control, intelligent control
Control	Acid and/or alkali and chlorine (2 control circuits), temperature
Digital inputs	5 potential-free inputs (sample water, pause, 3 pump failures, 2nd parameter set)
Signal current output	4 x 0/4-20 mA max. load 600 Ω range adjustable. An isolating amplifier, e.g. order no. 1033536, is required for connection to units which are not electrically isolated!
Control outputs	3 reed contacts for acid, alkali or flocculants and chlorine (pulse frequency to control metering pumps) 3 relays (pulse length) contact type changeover to control solenoid valves or peristaltic pumps
Alarm relay	250 V ~3 A, 700 VA contact type, changeover
Interfaces	LAN, SD-expansion slot
Electrical connection	100...240 V~, 50/60 Hz
Permissible ambient temperature	-5...45 °C
Storage temp.	-10...70 °C
Enclosure rating	IP 65
Climate	Permissible relative humidity: 95% non-condensing DIN IEC 60068-2-30
Dimensions H x W x D	227 x 342 x 78 mm

Compliance of all devices with CANopen specifications:

On the hardware side, all devices comply with the harmonised CAN specification 2.0 (ISO99-1, ISO99-2). This includes the CAN protocol (ISO 11898-1) and details on the physical layer in compliance with ISO 11898-2 (high speed CAN up to 1 Mbit/sec) and ISO 11898-3 (low speed CAN up to 125 kBit/sec). The unit complies with the CAN-Open specification CiA-DS401 that forms the basis of the European standard EN50325-4 and also complies with the controller device profile CiA-404.



5 Individual Devices and Accessories

5.4.4

Identity Code Ordering System DULCOMARIN® II

DULCOMARIN®II DXC range

DXCa	Installation	
W	W	Wall mounting (IP 65)
	S	Control cabinet (IP 54)
0	Version	
	0	With controls
D	with operating elements for use in potable water/disinfection applications	
	Communication interfaces	
5	0	None
	5	Embedded web server, LAN including 5 m LAN patch cable 1:1, LAN coupling, 5 m crossover cable ¹⁾
6	6	OPC server + embedded web server, LAN including 5 m LAN patch cable 1:1, LAN coupling, 5 m crossover cable ¹⁾
	Option (the corresponding communications modules are required, see accessories)	
0	0	none
	1	Videographic recorder with data logger including SD card and USB card reader for PC
2	2	SoftPLC function (communication option 5 or 6 needed)
	3	KNX function (communication option 5 or 6 needed)
4	4	Alarm signalling via text, e-mail (communication option 5 or 6 needed)
	5	SoftPLC function + KNX function + alarm signalling via text, e-mail (communication option 5 or 6 needed)
6	6	SoftPLC function + alarm signalling via text, e-mail (communication option 5 or 6 needed)
	7	SoftPLC function + KNX function (communication option 5 or 6 needed)
8	8	KNX function + alarm signalling via text, e-mail (communication option 5 or 6 needed)
M	Module 1	
	M	M module, measuring module for pH, ORP, temperature
A	A	A module, control module: 3 pump and 4 analogue outputs
	I	I module, current input module, 3 mA, 2 digital inputs
M	Module 2	
	0	Not used
A	A	A module, control module: 3 pump and 4 analogue outputs
	M	M module, measuring module pH, ORP, temperature
I	I	I module, current input module, 3 mA, 2 digital inputs
	F	F module, module for filter and attraction control
P	Module 3	
	P	P module, mains power module, 1 alarm relay, 3 solenoid valve relays
N	N	N module, mains power module without relay
	1	F module occupies module position 3
S	Application	
	S	Swimming pool
D	D	Potable water/disinfection
	Language default	
00	00	no operation
	DE	German
EN	EN	English
	ES	Spanish
FR	FR	French
	IT	Italian
PL	PL	Polish
	NL	Dutch
CZ	CZ	Czech
01	Approvals	
	01	CE mark
DXCa	W	0
		0
		1
	M	F
		1
	S	EN
		01

The identity code describes the DULCOMARIN® II controller.

- ¹ The supplied cable is for connection to a hub, switch, router or an intranet.
 For direct connection of the DULCOMARIN® II to a PC/MAC, the supplied LAN coupling and category 5 cross-over cable are required.
 The maximum LAN cable length is approximately 100 m.
 To operate the web server on a PC we recommend Microsoft® Internet Explorer 5 or higher as the browser.
 The scope of supply of the DXCa includes:
 1 T-coupler, 1 CAN connection cable
 1 terminating resistance coupling and
 1 terminating resistance plug,
 1 SD card, 1 card reader suitable for PCs.

Important note when ordering multi-channel measuring and control systems for potable water and pool water applications:

Potable water applications: In the identity code, a "D" for "Potable water/disinfection" must be selected under "Version" and "Application".
 The description "System" will appear in the controller menu for the different potable water lines.

Swimming pool water applications: In the identity code, a "0" for "with operating elements" must be selected under "Version" and then an
 "S" for "Swimming pool" under "Application". The description "Tank" will appear in the controller menu for the different filter circuits.

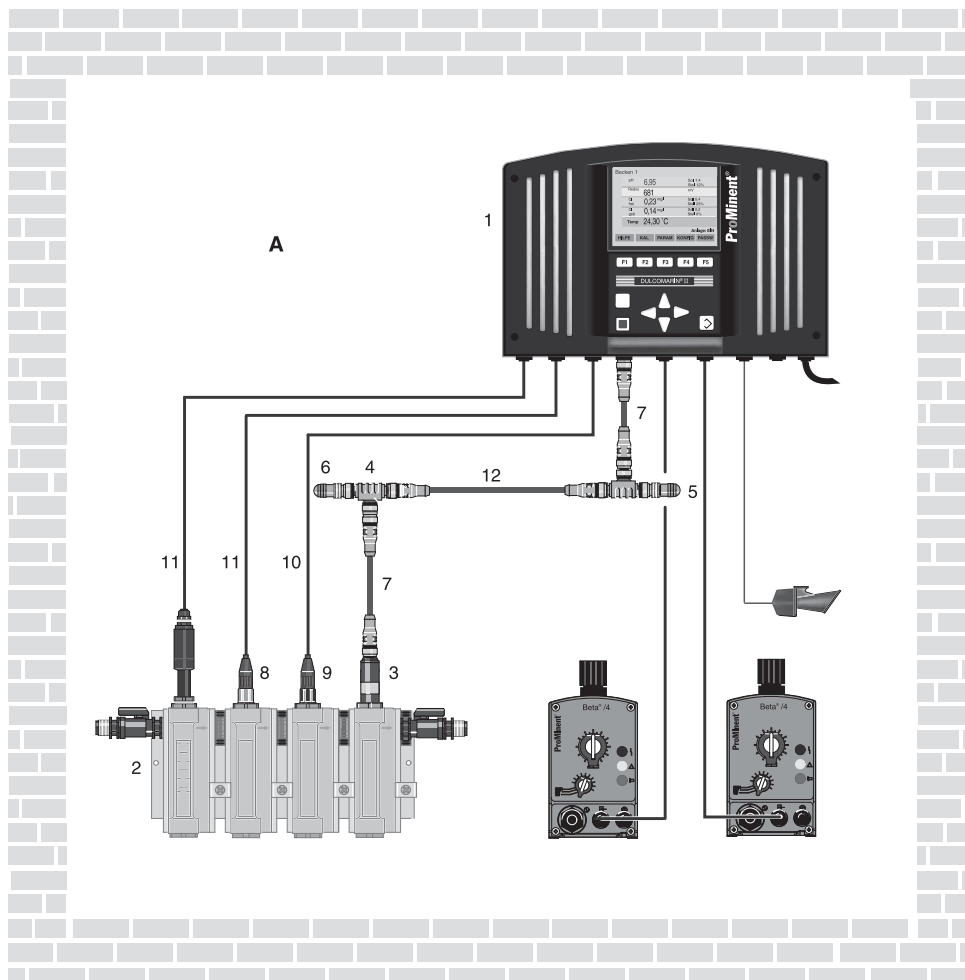
All adjustment options and the use of the different modules are identical with both applications.

5 Individual Devices and Accessories

Configuration example: 1-pool system

This example of a measuring and control system for pH, ORP, free chlorine and temperature shown for a filter circuit consists of the following components (without chemical fluid handling):

A Plant room



pk_5_020

Item	Quantity	Name	Order no.
1	1	DULCOMARIN® II central unit with measuring and control modules DXCa W 0 0 1 M A P S DE 01	–
2	1	DULCOTEST®in-line probe housing DGMa 3 2 1 T 0 0 0	–
3	1	Chlorine sensorCLE 3-CAN-10 ppm	1023425
4	3	T-distributor M12 5 pol. CAN	included in delivery
5	1	Termination resistance M12 connector	included in delivery
6	1	Termination resistance M12 plug	included in delivery
7	3	Connection cable - CAN M12 5 way 0.5 m	included in delivery
8	1	pH sensor PHES 112 SE	150702
9	1	ORP sensor RHES-Pt-SE	150703
10	2	Cable combination coaxial 2 m - SN6 - pre-assembled	1024106
11	2 m	Signal cable, sold by the metre 2 x 0.25 mm² Ø 4 mm	725122
12	–	Connecting cable - CAN, sold by the metre	1022160



5 Individual Devices and Accessories

5.4.5

Multi-Channel Multi-Parameter Measuring and Control System DULCOMARIN® II

The multi-channel multi-parameter measuring and control system DULCOMARIN® II DULCO® Net can, in its top-of-the-range optional version, control 16 potable water systems/filtration circuits, i.e. the necessary external modules for 16 tanks can be connected to and operated by the central unit. The following options are available

Measurement and control of:

Up to 16 times:

- pH value
- ORP potential
- Free chlorine
- Combined chlorine (calculated)
- Temperature of the sample water

Also in potable water applications (using the I module):

- Flow (as disturbance variable for pH and chlorine control)
- UV intensity
- Conductivity
- Chlorine dioxide
- Chlorite
- Ammonia
- Fluoride
- Pt100/Pt1000 resistance thermometer via a transducer

Other inputs and outputs:

Up to 16 times:

- 3 frequency outputs for control of metering pumps for pH-correction, disinfectant and flocculent metering
- 3 contact inputs for processing of pump fault signal relays or container level monitoring
- 4 freely programmable analogue outputs 0/4 ... 20 mA (for pH, ORP, free chlorine, combined chlorine or temperature)
- 3 output relays pulse length control of the pH value, the disinfectant and minimisation of the combined chlorine (e.g. control of a peristaltic pump and chlorine electrolysis system and UV system)
- Control of a chlorine gas metering device
- 3 Beta®/4 CANopen metering pumps
- Up to 2 F modules per filter circuit are possible

The CAN bus with CANopen protocol is used as a data transfer medium between the various bus modules. This extremely interference-proof technology was developed by Bosch and is well known from its use in automotive applications. The maximum length of the bus backbone is 400 metres.

A T-coupler is used for connection of one of each bus module (M module, A module, P module, N module, Beta® 4 CANopen metering pumps and CAN chlorine sensors), which connects the devices to the bus backbone via a branching cable.

T-coupler and branching cable are part of the scope of supply of the modules.

All bus modules are supplied via the CAN bus with 24 V operating voltage (with the exception of Beta®/4 CANopen metering pumps, P modules, N modules. These require a separate mains voltage supply).

For this reason, depending on the size of the installation (number of filtration circuits to be controlled), additional P or N modules are required that feed the operating voltage for the bus modules into the bus. The central unit always contains a power supply (N or P module).

How many additional N or P modules do you require?

Number of filtration circuits	Additional N or P modules	Number of filtration circuits	Additional N or P modules
1	–	9	4
2	–	10	5
3	1	11	5
4	2	12	6
5	2	13	6
6	3	14	7
7	3	15	7
8	4	16	8

The DULCOMARIN® II can be easily extended by the connection of bus modules.



5 Individual Devices and Accessories

Which components can a DULCOMARIN® II system comprise?

A DULCOMARIN® II DULCO® Net system comprises:

- DXCa central unit with operating elements

and a customised combination of the following components:

- M module, DXMaM (measuring and control)
- A module, DXMaA (control of metering pumps, analogue outputs)
- P module (module in the DXCa housing for power supply to the modules and alarm relays, output relays for control of, for example, peristaltic pumps)
- N module, DXMaN (power supply to external modules with no other function)
- R module, DXMaR (control of chlorine gas metering devices with response signal processing)
- I module (processing of sensor signals via 0/4...20 mA)
- F module (filter and attraction control)

The maximum bus backbone length is approximately 400 m!

5.4.6

Central Unit

The central unit can be installed anywhere, for example in the control room. It serves as an I/O unit (view measuring data, parameterise and configure individual modules). It includes the following functions: standard screen recorder/data logger function, interfaces*, embedded Web server* and power supply. As an option, the central unit can also include an M and an A module if the central unit is also located in the control room. The central unit is connected to other units via the main bus train.

For this connection, the T-distributor and the CAN connecting cable 0.5 m included in the scope of delivery are used.

The main bus train must be fitted with termination resistors at either end.

These components are included in the scope of delivery.

The central unit in the above example consists of the following components:

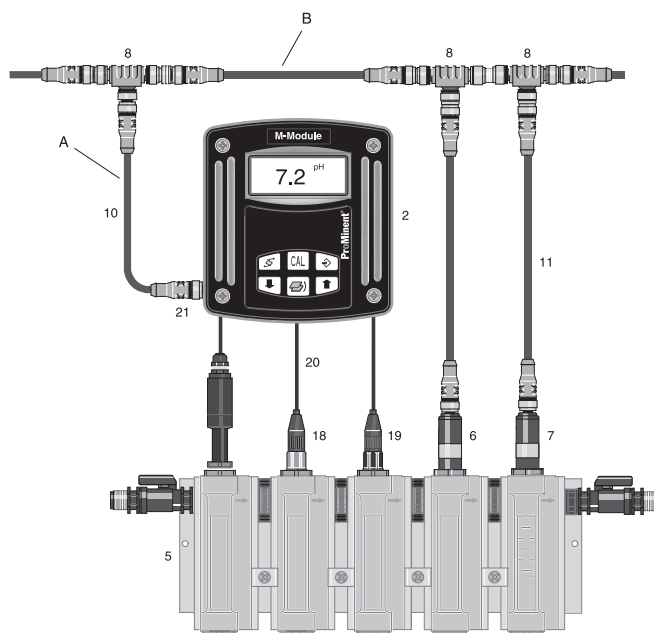
Item	Quantity	Name	Order no.
1	1	DULCOMARIN® II central unit DXCa W 0 0 1 0 0 P S EN 01	–
8	1	T-distributor M12 5 pol. CAN	included in delivery
11	1	Connection cable - CAN M12 5 way 0.5 m	included in delivery
14	1	CAN Connection cable	as required
10	1	Termination resistance M12 connector	included in delivery
12	1	Termination resistance M12 plug	included in delivery



5 Individual Devices and Accessories

5.4.7 Measuring Module (M module)

- A Stub cable
B Main BUS cable



pk_5_042

The M module with its illuminated graphic display and keypad displays the measured values and allows all sensors for the corresponding filter circuit to be calibrated on site.

The following measurements can be taken:

- pH value
- ORP potential
- free chlorine and
- total available chlorine (optional or combined chlorine is calculated) and
- sample water temperature using the temperature probe in the chlorine sensor or optionally using a separate Pt100/Pt1000 resistance thermometer

The M module has 3 digital inputs for:

- sample water monitoring
- controlling breaks in filter backwashing
- Parameter changeover for Eco!Mode

The M module is connected to the other bus modules via the main bus cable, using the T-distributor supplied and the 0.5 m CAN connection cable.

The M module in the above example consists of the following components:

Item	Quantity	Name	Order no.
2	1	M module DXMa M W 0 S EN 01	DXMa M W 0 S EN 01
5	1	In-line probe housing DGMa 3 2 2 T 0 0 0	DGMa 3 2 2 T 0 0 0
6	1	Chlorine sensor CTE 1-CAN-10 ppm	1023427
7	1	Chlorine sensor CLE 3.1-CAN-10 ppm	1023426
8	3	T-distributor M12 5 pole CAN	included in delivery
10	1	Connection cable - CAN M12 5-pole 0.5 m	included in delivery
11	2	Connection cable - CAN M12 5-pole 0.5 m	included in delivery
18	1	pH sensor PHES 112 SE	150702
19	1	ORP sensor RHES-Pt-SE	150703
20	2	Cable combination coaxial 2 m - SN6 - pre-assembled	1024106
21	2 m	Signal cable, sold by the metre 2 x 0.25 mm ² Ø 4 mm	725122

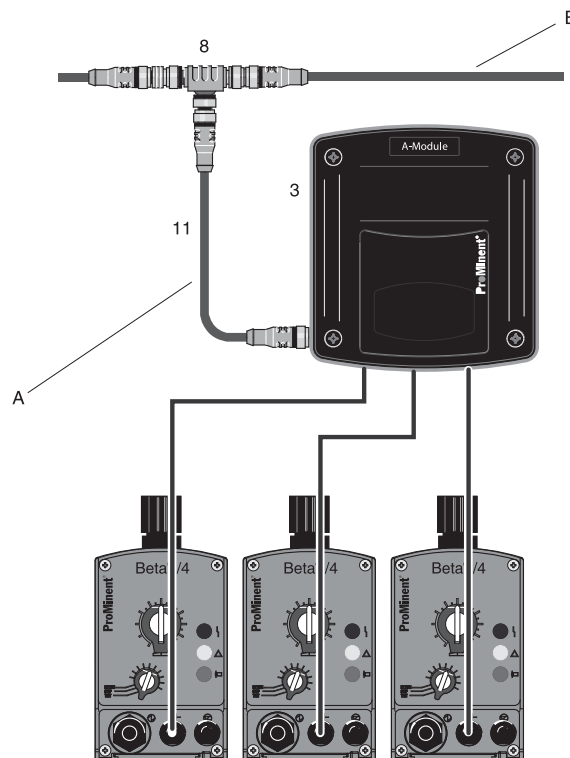
Alternative = without bound chlorine measurement

Item	Quantity	Name	Order no.
5a	1	DULCOTEST®in-line probe housing DGMa 3 2 1 T 0 0 0	-

5 Individual Devices and Accessories

5.4.8 Control Module (A module)

- A Stub cable
B Main BUS cable



pk_5_043

The A module permits the control of up to three metering pumps via pulse frequency. Possible metering combinations are:

- pH lowering and disinfectant and flocculant or
- pH raising and disinfectant and flocculant or
- pH lowering and pH raising and disinfectant

It includes 3 digital inputs to evaluate the alarm relay of metering pumps, 4 freely programmable standard signal outputs 0/4...20 mA to document measured values, or as control outputs.

The T-distributor and 0.5 m CAN connecting cable included in the scope of delivery are used for this connection.

Note: No A modules are required if Beta®/4CANopen metering pumps are used!

The A module in the above example consists of the following components (without metering technology):

Item	Quantity	Name	Order no.
3	1	A module DXMa A W 2 0 00 01	—
8	1	T-distributor M12 5 pol. CAN	included in delivery
11	1	Connection cable - CAN M12 5 way 0.5 m	included in delivery

The A module is connected to other units via the main bus train.

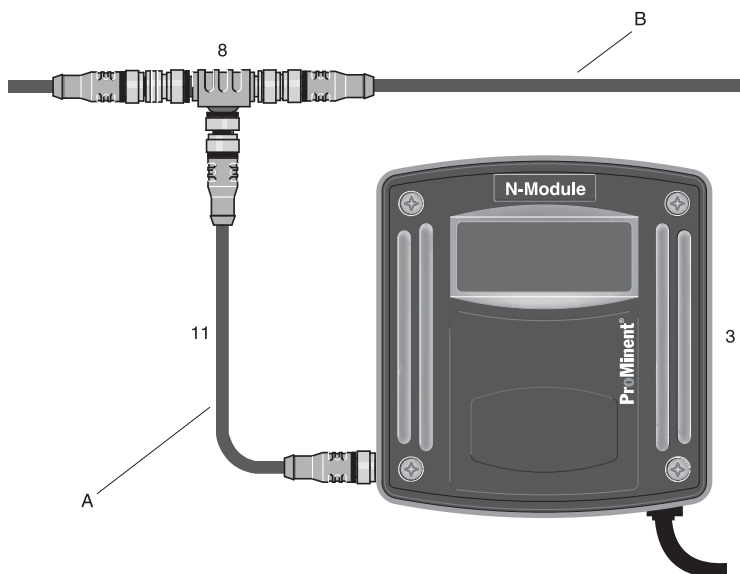
An isolating amplifier, e.g. order no. 1033536 is required for connection to units which are not electrically isolated (e.g. PLC)!



5 Individual Devices and Accessories

5.4.9 Power Supply Module (N module)

- A Stub cable
B Main BUS cable



pk_5_043_C_power

The N module (power supply) is used to supply the bus modules with power and has no further function.

The number of N modules required can be seen from the table below. If P modules are used in a system, the number of N modules is reduced accordingly. The central unit always includes a power supply unit (N or P module)

How many additional N or P modules do you require?

Number of filtration circuits	Additional N or P modules	Number of filtration circuits	Additional N or P modules
1	-	9	4
2	-	10	5
3	1	11	5
4	2	12	6
5	2	13	6
6	3	14	7
7	3	15	7
8	4	16	8

The N module requires a power supply for operation and is connected to the other bus modules via the main bus train. The T-distributor and 0.5 m CAN connecting cable included in the scope of delivery are used for this connection.

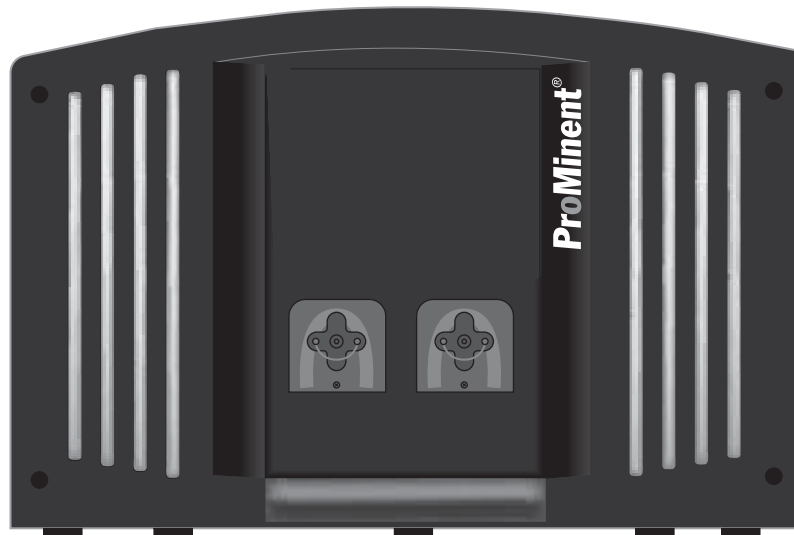
The N module in the above example consists of the following components:

Item	Quantity	Name	Order no.
3	1	N module DXMa N W 2 0 00 01	–
8	1	T-distributor M12 5 pol. CAN	included in delivery
11	1	Connection cable - CAN M12 5 way 0.5 m	included in delivery

Our Sales department would be glad to assist with any questions you may have.

5 Individual Devices and Accessories

5.4.10 Combination Module DF2a CAN SET



Combination of DF2a and P module

The combination module (DXCa without control elements) can accommodate up to two DF2 hose dosing heads that are activated via the integral P module. Metering is performed by pulse length control. Communication with the DXCa central unit is via the CAN connecting cable.

The combination module is supplied with 230 V voltage. The function of the combination module results from the function of the individual modules (see above description). The modules in the combination module are operated via the central unit DXCa.

The combination module in the above example consists of the following components (without the metering technology) (please state type DF2 with your order):

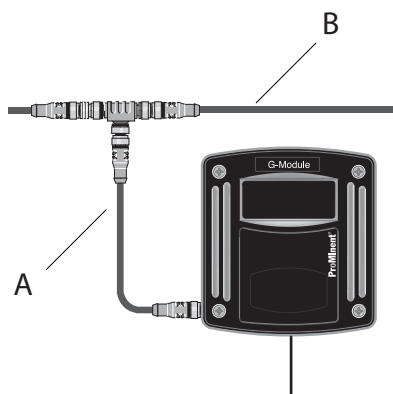
Item	Quantity	Name	Order no.
3	1	Control module DXCa W 2 0 0 0 A P S 00 01	–
8	1	T-distributor M12 5 pol. CAN	included in delivery
11	1	Connection cable - CAN M12 5 way 0.5 m	included in delivery



5 Individual Devices and Accessories

5.4.11 Limit Value and Alarm Module (G module)

- A Stub cable
B Bus main cable



P_DM_0024_SW3

The G-module is a limit value and alarm emitting module with 2 potential-free changeover relays to signal alarm states. Each of the two relays has ten different setting options to monitor measured values for minimum and maximum values and, should the values exceed or fall below these limits, this then triggers the relay. Both relays have the same setting options, thereby enabling signals for pre-warnings or shutdowns to be generated by the use of different delay periods.

The G module is connected to the other units via the main bus cable using the T-distributor and 0.5 m CAN connection cable supplied.

The G module in the above example consists of the following components:

Item	Quantity	Name	Order no.
3	1	G module DXMa G W 2 0 00 01	–
8	1	T-distributor M12 5 pol. CAN	included in delivery
11	1	Connection cable - CAN M12 5 way 0.5 m	included in delivery

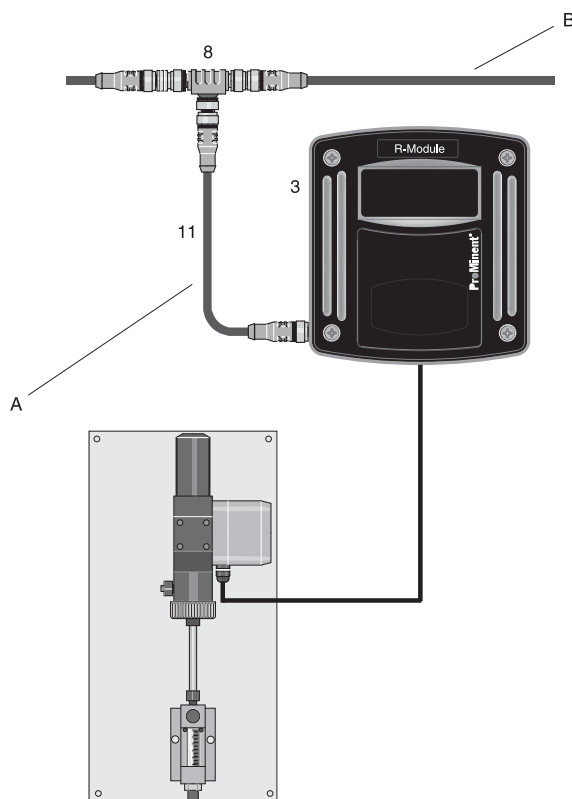
Our Sales department would be glad to assist with any questions you may have.



5 Individual Devices and Accessories

5.4.12 Control Module for Chlorine Gas Metering Devices (R module)

- A Stub cable
B Main BUS cable



pk_5_043_C

The R module permits the control of chlorine gas metering units equipped with a position feedback potentiometer.

It includes 2 power relays for opening and closing and an input for a position feedback potentiometer 1 ... 10 k Ω

The R module is connected to other units via the main bus train.

The T-distributor and 0.5 m CAN connecting cable included in the scope of delivery are used for this connection.

The R module in the above example consists of the following components (without the chlorine gas metering device):

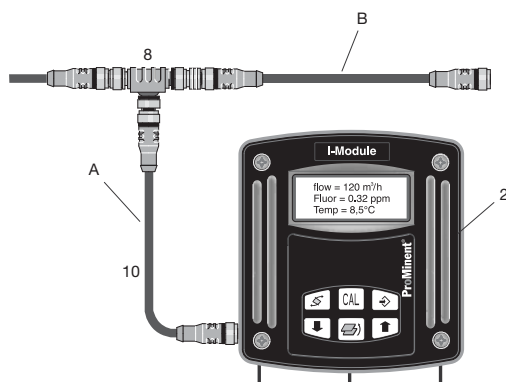
Item	Quantity	Name	Order no.
3	1	R module DXMa R W 2 0 00 01	—
8	1	T-distributor M12 5 pol. CAN	included in delivery
11	1	Connection cable - CAN M12 5 way 0.5 m	included in delivery



5 Individual Devices and Accessories

5.4.13 Current Input Module (I module)

- A Stub cable
B Main BUS cable



AP_DC_0011_SW

This I module with its illuminated graphic display and keypad is a current input module capable of processing 3 standard signals from sensors and two digital signals.

It can be used together with the multi-channel controller DULCOMARIN® II in potable water and swimming pool applications. All measured variables are available in the screen plotter and web and OPC® server.

Two analogue inputs are provided as 2-wire inputs and one as passive input. All channels have pre-selected measured variables. However the identifier and units can also be edited. Channel 1 acts as an interference variable channel for channel 2. Channel 3 acts as the temperature compensation channel for channel 2 when the measured variable is fluoride. Channel 2 has a control function.

The inputs can process the following values as 0/4... 20 mA standard signals:

- Turbidity
- Flow (can also be used as the interference variable)
- UV intensity
- Conductivity (via DMTa transmitter)
- Chlorine dioxide
- Chlorite
- Ammonia
- Fluoride
- Pt100 resistance thermometer via a transducer
- Dissolved oxygen
- Hydrogen peroxide
- Editable designation and unit for all 3 channels

The I module has 2 digital inputs for:

- Sample water monitoring and
- Pause control

The flow information can be used as an interference variable for the control of chlorine, pH correction and chlorine dioxide.

The I module is connected to other bus modules via the main bus cable using the T-distributor and 0.5 m CAN connection cable supplied as part of the delivery.

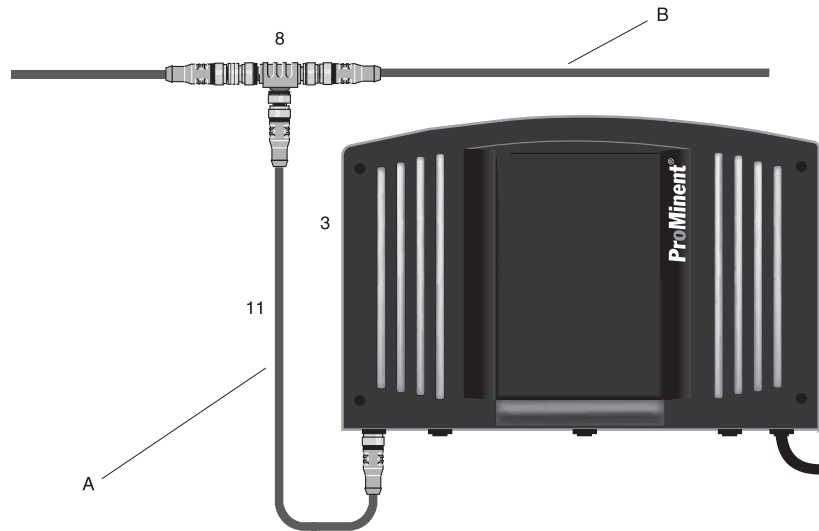
The I module in the above example consists of the following components:

Item	Quantity	Name	Order no.
2	1	I module DXMa I W 0 D EN 01	—
8	1	T-distributor M12 5P CAN	included in delivery
10	1	Connecting cable - CAN, M12, 5P, 0.5 m	Included in delivery

* Order the gateways/routers separately. Communication option 5 or 6 is always needed.

5 Individual Devices and Accessories

5.4.14 Functional Module (F Module)



P_DC_0009_SW

The F module combines functions that were until now available in the A and P module combination and also extends these functions. It includes the supply voltage (90 – 253 VAC) for the controller. The F module is selected for the 2nd module position and also occupies the 3rd module position. The following functions can be provided by the F module (you can find details for this in the individual application examples in the assembly and operating instructions). The F module also acts as an input/output module for the SoftPLC.

Hydraulic functions:

- Control of circulation operation (depending on the weekday and the time of day)
- Automatic backwashing
- Route first filtrate through the internal circuit (electrical backflow shut-off valve)
- Lowering of the water level during idle operation
- Circulation flow control
- Water level control
- Sample water valve
- Heating function
 - Heating control heat exchanger
 - Solar heating
- Gutter cleaning function

Attractions:

- Open/Close cover
- Counterflow system/JetStream
- Flood/Neck shower
- Massage nozzle
- Underwater light

Monitoring:

- Flow control
 - Current circulation flow recording
 - Fresh water top-up recording
 - Cover
 - Massage pump active
 - 1, 2 or 4-stage level functions
- Control variables for:
 - Disinfection
 - pH (+/-)
 - Flocculation
 - UV system
 - Backwashing emergency-off, if water alarm emitted



5 Individual Devices and Accessories

5.4.15

Identity Code Ordering System Multi-Channel Multi-Parameter Measuring and Control System DULCOMARIN® II (Central Unit and Combination Module)

DXCa Installation										
W	S	Wall mounting (IP 65)								
		Control cabinet (IP 54)								
		Version								
		0	With controls							
		D	With operating elements for use in potable water/disinfection applications							
		Communication interfaces								
		0	None							
		5	Embedded web server, LAN including 5 m LAN patch cable 1:1, LAN coupling, 5 m crossover cable							
		6	OPC server + embedded web server, LAN including 5 m LAN patch cable 1:1, LAN coupling, 5 m crossover cable							
		Option (the corresponding communications modules are required, see accessories)								
		0	none							
		1	Videographic recorder with data logger including SD card and USB card reader for PC							
		2	SoftPLC function (communication option 5 or 6 needed)							
		3	KNX function (communication option 5 or 6 needed)							
		4	Alarm signalling via text, e-mail (communication option 5 or 6 needed)							
		5	SoftPLC function + KNX function + alarm signalling via text, e-mail (communication option 5 or 6 needed)							
		6	SoftPLC function + alarm signalling via text, e-mail (communication option 5 or 6 needed)							
		7	SoftPLC function + KNX function (communication option 5 or 6 needed)							
		8	KNX function + alarm signalling via text, e-mail (communication option 5 or 6 needed)							
		Module 1								
		0	Not used							
		M	M module, measuring module: pH, ORP, temperature							
		A	A module, control module: 3 pump and 4 analogue outputs							
		I	I module, current input module, 3 mA inputs, 2 digital inputs							
		Module 2								
		0	Not used							
		A	A module, control module: 3 pump and 4 analogue outputs							
		M	M module, measuring module: pH, ORP, temperature							
		I	I module, current input module, 3 mA inputs, 2 digital inputs							
		F	F module, module for filter and attraction control							
		Module 3								
		P	P module, mains power module, 1 alarm relay, 3 solenoid valve relays							
		N	N module, mains power module unit without relay							
		1	F module occupies module position 3							
		Application								
		S	Swimming pool							
		D	Potable water/disinfection							
		Language default								
		DE	German							
		EN	English							
		ES	Spanish							
		FR	French							
		IT	Italian							
		PL	Polish							
		NL	Dutch							
		CZ	Czech							
		Approvals								
01	CE mark									
DXCa	W	0	0	1	M	F	1	S	EN	01

Please note the following:

Upgrade modules for existing systems require a software update for the existing system. A Software Update Kit is needed to avoid any possible incompatibility between the different modules.

The update kit is free of charge and one is also needed when ordering more than one upgrade module. The kit includes an SD memory card with the current software for the DULCOMARIN® II and a description about how to perform the software update.

Update kit/DXC and modules	Order no.
	1031284

The Identity code describes the complete **DULCOMARIN® II DULCO®-Net** central unit.

The peripheral components mentioned in the above item list, however, are not included. If modules are assigned to the central unit, the following applies:

Module 1 preferably assigned as M module

Module 2 preferably assigned as A module

Module 3 must always be assigned as P module or N module.

5 Individual Devices and Accessories

Important note when ordering multi-channel measuring and control systems for potable water and pool water applications:

Potable water applications: In the identity code, a "D" for "Potable water/disinfection" must be selected under "Version" and "Application". The description "System" will appear in the controller menu for the different potable water lines.

Swimming pool water applications: In the identity code, a "0" for "with operating elements" must be selected under "Version" and then an "S" for "Swimming pool" under "Application". The description "Tank" will appear in the controller menu for the different filter circuits.

All adjustment options and the use of the different modules are identical with both applications.

5.4.16

Identity Code Ordering System for CANopen Modules

Modules for the DULCOMARIN® II, DXM product range

DXMa	Module
M	M module, measuring module: pH, ORP, temperature
A	A module, control module: 3 pump and 4 analogue outputs
R	R module, control module: chlorine gas metering unit with feedback ^{1), 2)}
N	N module, mains power module without relay ^{1), 2)}
P	P module, mains power module with relay, only mounting type "0" ^{1), 2)}
I	I module, current input module, 3 mA inputs, 2 digital inputs
Installation	
0	No housing, only P module (IP 00)
W	Wall mounting (IP 65)
E	Retrofit module (installation module for DXCa, IP 20)
Version	
0	With controls (only M module, mounting type W) ¹
2	Without controls
3	Without controls (only mounting type "E" and "H")
Application	
0	Standard
S	Swimming pool (only M-module)
D	Potable water/disinfection (only I module)
Language default	
00	No controls ²⁾
DE	German
EN	English
ES	Spanish
FR	French
Approvals	
00	No approval, only P-module without housing
01	CE mark
DXMa	M W 0 S EN 01

Please note the following:

Upgrade modules for existing systems require a software update for the existing system. A Software Update Kit is needed to avoid any possible incompatibility between the different modules.

The update kit is free of charge and one is also needed when ordering more than one upgrade module. The kit includes an SD memory card with the current software for the DULCOMARIN® II and a description about how to perform the software update.

Order no.

Update kit/DXC and modules

1031284



5 Individual Devices and Accessories

5.4.17

Accessories for the DULCOMARIN® II Measuring and Control System

	Order no.
CLE 3-CAN-10 ppm	1023425
CLE 3.1-CAN-10 ppm	1023426
CTE 1-CAN-10 ppm	1023427
BRE 3-CAN-10 ppm	1029660
T-distributor M12 5 pole CAN	1022155
Termination resistor M12 coupling	1022154
Termination resistor M12 plug	1022592
Connecting cable - CAN M12 5 pole 0.3 m	1024568
Connecting cable - CAN M12 5-pole 0.5 m	1022137
Connecting cable - CAN M12 5-pole 1 m	1022139
Connecting cable - CAN M12 5-pole 2 m	1022140
Connecting cable - CAN M12 5-pole 5 m	1022141
Connecting cable - CAN M12 5-pin 10 m	1046383
Connecting cable - CAN, sold by the metre	1022160
CAN bulk cable connection kit	1026589
PHES 112 SE	150702
RHES-Pt-SE	150703
Cable combination coaxial 0.8 m - SN6 - pre-assembled	1024105
Cable combination coaxial 2 m - SN6 - pre-assembled	1024106
Cable combination coaxial 5 m - SN6 - pre-assembled	1024107
Signal cable, sold by the metre 2 x 0.25 mm ² Ø 4 mm	725122
Connecting cable LAN M12 - RJ45 5.0 m	1026715
Cross-over patch cable 2x RJ45 connector 5 m	1027859
LAN coupling 2x RJ45 socket 1:1	1027860
USB 2.0 SD card reader	732981
SD memory card/DXC measuring data archiving	1027470
Isolating amplifier 4-channel for mA outputs of the A module	1033536

*** up to 3 cables, each 10 m, can be coupled**

The CAN bulk cable connection kit comprises a 5-pin M12 CAN coupling and a 5-pin M12 CAN plug and a wiring diagram.

The CAN bulk cable connection kit can be used to configure the connecting cable to form a cable of any required length.

One CAN bulk cable connection kit is required for each cable to be assembled.

The 0.5 m (1 m pump) 5-pin M 12 CAN connecting cables supplied with the sensors and modules have to be used as branch cables.

Caution:

Do not allow the maximum main bus length (excluding branch cables) to exceed 400 m!

Sensor selection table (swimming pool)

Sensor	Measurement task	free chlorine for a small percentage of combined chlorine. Calibration method DPD 1	freechlorine for a large percentage of combined chlorine. Calibration method DPD 1	combined chlorine and free chlorine (differential chlorine measurement) Calibration method DPD 1+3	total availablechlorine (e.g. trichlorinated isocyanuric acid) Calibration method DPD 1	Bromine BCDMH, DBDMH Calibration method DPD1 or DPD1+3
CLE3-CAN-10 ppm (Order no.: 1023425)	X					
CLE3.1-CAN-10 ppm (Order no.: 1023426)			X	X		
CTE1-CAN-10 ppm * (Order no.: 1023427)				X		
CGE2-CAN-10 ppm (Order no.: 1024420)					X	
BRE3-CAN-10 ppm (Order no. 1029660)						X

* the CTE1-CAN-10 ppm sensor only works together with the CLE3.1-CAN-10 ppm sensor



5 Individual Devices and Accessories



P_MSRZ_0014_SW

PROFIBUS®-DP V1 gateway

The CANopen - PROFIBUS®-DP V1 gateway is an interface based on CANopen, which connects the DULCOMARIN® II swimming pool controller or disinfection controller to a PROFIBUS® DP network. Here the DULCOMARIN® II is configured as the slave and the PLC is the master. Data traffic can be cyclic or acyclic. The measured values are transmitted cyclically. Setpoints can be changed, the system can be set to pause control and Eco!Mode operation can be activated in acyclic traffic. The corresponding GSD file can be loaded from the ProMinent homepage and is also contained on the enclosed data carrier.

The module is intended for installation in a control cabinet (top hat rail) and is connected to the CAN bus in the same way as other modules. The DULCOMARIN® II Controller must have software version 3022 or greater. No specific identity code is needed.

A separate 24 VDC power supply is required.

Voltage supply	24 V DC
Typical power consumption approx.	500 mA
Max. number of measured values	116
Weight	250 g
Dimensions L x W x H (mm)	117.2 x 45 x 113.5 mm
RoHS (Restriction of Hazardous Substances)	yes
CE conformity	yes
Enclosure rating	IP 20

Order no.

CANopen - PROFIBUS®-DP V1 gateway complete

1044462

Modbus RTU gateway



P_MSRZ_0014_SW

The CANopen - Modbus RTU gateway is an interface based on CANopen, which connects the DULCOMARIN® II swimming pool controller or disinfection controller to a Modbus RTU network. Here the DULCOMARIN® II is configured as the slave and the PLC is the master. Data traffic can be cyclic or acyclic. The measured values are transmitted cyclically. Setpoints can be changed, the system can be set to pause control and Eco!Mode operation can be activated in acyclic traffic. The corresponding description table can be found in the operating instructions. It can be downloaded from the ProMinent homepage and is also contained on the enclosed data carrier.

The module is intended for installation in a control cabinet (top hat rail) and is connected to the CAN bus in the same way as other modules. No specific identity code is needed.

A separate 24 VDC power supply is required.

Voltage supply	24 V DC
Typical power consumption approx.	500 mA
Max. number of measured values	116
Weight	250 g
Dimensions L x W x H (mm)	117.2 x 45 x 113.5 mm
RoHS (Restriction of Hazardous Substances)	yes
CE conformity	yes
Enclosure rating	IP 20

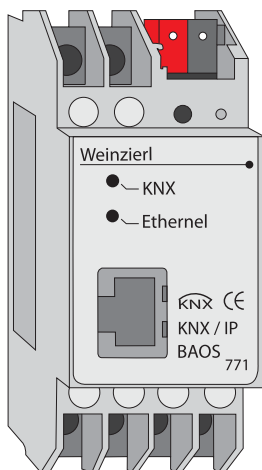
Order no.

Gateway CANopen - Modbus RTU

1047247



5 Individual Devices and Accessories



P_MSRZ_0017_SW1

Ethernet KNX gateway

The Ethernet – KNX gateway is an Ethernet based interface that connects the DULCOMARIN® II swimming pool controller or disinfection controller to a KNX building control system. It can transmit the measured values and status messages from up to 2 systems/pools. No feedback effect from the KNX network is possible.

The module is intended for installation in a control cabinet (top hat rail) and is connected to the LAN/ Ethernet connector of the DXCa. The DXCa needs to have communication option 5 = web server or 8 = web server + OPC server for this.

A separate 24 VDC power supply is required.

Voltage supply	12 – 24 V DC
Typical power consumption approx.	500 mA
Max. number of measured values (max. 2-pool system)	20
Weight	100 g
Dimensions L x W x H (mm)	117.2 x 60 x 113.5 mm
RoHS (Restriction of Hazardous Substances)	yes
CE conformity	yes
Enclosure rating	IP 20

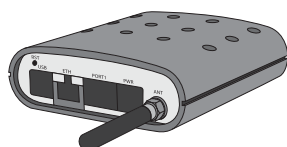
Order no.

Gateway Ethernet-KNX
1047326

ER75i mobile phone router (GSM/GPRS/EDGE)

Important for operation of the mobile communications router:

- The products do not include a mobile communications data contract, which has to be concluded separately with a mobile communications provider.
- Please check in advance the network coverage of your mobile communications provider
- Make sure that the installation can be installed in a place whether the received signal has sufficient strength and there is also a power supply



P_MSRZ_0018_SW1

You can connect to your DULCOMARIN® II Disinfection Controller using the mobile phone router ER75i irrespective of the separation.

Mobile Ethernet makes it possible to use the available infrastructure for location-independent Ethernet communication.

In addition to GSM and GPRS, EDGE technology can also be used for data transfer. Stable and permanent connections are monitored and maintained through continuous control. An integrated DHCP server makes possible simple installation and fast Internet access. The ideal device for alarm signalling, remote control and remote service.

The mobile phone router ER75i is specially configured for the DULCOMARIN® II / Disinfection Controller.

The DULCOMARIN® II Disinfection Controller must have at least communications option 5 = web server. This DXCa option does not include the mobile phone router.

Scope of supply: Router, CD, patch cable, magnetic foot aerial, plug-in power pack.

GPRS/EDGE (class 10) mobile phone router for industrial applications (max. download 236 Kbit/s, max. upload 118.4 Kbit/s)

Single web-interface, DHCP, DynDNS, VRRP, NTP, dial-in router control via SMS

Data volume / roaming control via SMS

Status Information via SNMP and SMS

LED status display

Frequency bands: 850/900/1800/1900 MHz

Interfaces: Ethernet 10/100

External GSM aerial: FME - 50 Ω

Power supply: 10 ... 30 V DC

Working temperature range: -30 °C ... +60 °C

Dimensions: 30 x 90 x 102 mm, plastic housing, also for wall mounting

Weight: 190 g (without aerial and plug-in power pack)

Degree of protection: IP 44, for use in dry rooms or offices

Order no.

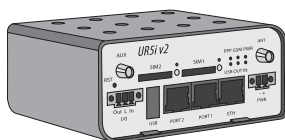
GSM/GPRS/EDGE mobile phone router ER75i
1047329

5 Individual Devices and Accessories

UR5i mobile phone router (UMTS/HSPA+)

Important for operation of the mobile communications router:

- The products do not include a mobile communications data contract, which has to be concluded separately with a mobile communications provider.
- Please check in advance the network coverage of your mobile communications provider
- Make sure that the installation can be installed in a place whether the received signal has sufficient strength and there is also a power supply



P_MSRZ_0019_SW1

You can connect to your DULCOMARIN® II Disinfection Controller using the mobile phone router UR5i via UMTS/HSPA+ irrespective of the separation.

Mobile Ethernet makes it possible to use the available infrastructure for location-independent Ethernet communication.

UMTS/HSPA+ technology can be used for data transfer. Stable and permanent connections are monitored and maintained through continuous control. An integrated DHCP server makes possible simple installation and fast Internet access. The ideal device for alarm signalling, remote control and remote service.

With WLAN access. The WLAN access has no bridge function for connection of another WLAN network.

The mobile phone router is specially configured for the DULCOMARIN® II / Disinfection Controller.

The DULCOMARIN® II Disinfection Controller must have at least communications option 4 = alarm signalling via SMS / email, or greater. This DXCa option does not include the mobile phone router.

Scope of supply: Router, CD, patch cable, magnetic foot aerial, plug-in power pack. Degree of protection: IP 44, for use in dry rooms or offices.

UMTS/HSPA+ Tri-Band (max. download 14.4 Mbit/s, max. upload 5.7 Mbit/s)

WLAN supported NAT/PAT and X.509

Integrated firewall (SPI)

Single web-interface, DHCP, DynDNS, VRRP, dial-in router control via SMS

Data volume / roaming control via SMS

Status information via SNMP and SMS

Extensive mobile connection statistics options

LED status display

Frequency bands: GSM/GPRS/EDGE: 850/900/1800/1900 MHz
UMTS: 850/900/1900/2100 MHz

External GSM aerial: SMA - 50 Ω

Power supply: 10 ... 30 V DC

Working temperature range: -30 °C ... +60 °C

Dimensions: 50 x 84 x 117 mm, DIN top hat rail 35 mm

Weight: 207 g

Degree of protection: IP 44

Order no.

UMTS/HSPA+ mobile phone router UR5i v2F

1047330



5 Individual Devices and Accessories

5.5

Sensors

5.5.1

Sensors for Chlorine, Bromine, Chlorine dioxide and Chlorite

Sensor for Free Chlorine CLE 3-CAN

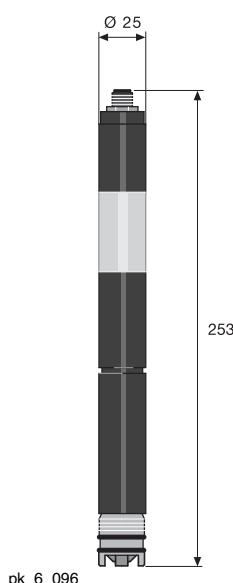


Standard sensor for measuring free chlorine in clear water. For use on controllers with CAN-bus connection

Your benefits

- Measured variable: free chlorine, no significant cross sensitivity to combined chlorine (chloramines)
- Diaphragm-covered sensor (encapsulated) minimises faults caused by changing flow or ingredients in the water
- Operation on the CAN-bus with all the associated benefits

Measured variable	free chlorine (hypochlorous acid HOCl)
Reference method	DPD1
pH range	5.5 ... 8.0
Temperature	5 ... 45 °C
Max. pressure	1.0 bar
Intake flow	30...60 l/h (in the DGM or DLG III)
Supply voltage	Via CAN interface (11 - 30 V)
Output signal	uncalibrated, temperature compensated, electrically isolated
Selectivity	Free chlorine as against combined chlorine, even if there is not an excess of it
Disinfection process	Chlorine gas, hypochlorite, electrolysis with diaphragm. Disinfectants with organic chlorine, e.g. based on cyanuric acid, are unsuitable
Installation	Bypass: open sample water outlet
Sensor fitting	DGM, DLG III
Measuring and control equipment	DULCOMARIN® II
Typical applications	CLE 3-mA-0,5 ppm: potable water; CLE 3-mA-2.0/10 ppm: swimming pools (surfactant-free)
Resistance to	Salts, acids, lyes. Not surfactants
Measuring principle, technology	amperometric, 2 electrodes, diaphragm-covered



	Measuring range	Order no.
CLE 3-CAN-10 ppm	0.01...10.0 mg/l	1023425

Chlorine sensors complete with 100 ml of electrolyte

A mounting kit, order no. 815079, is required for initial fitting of the chlorine sensors in the in-line probe housing DLG III.

Sensor for Free Chlorine CLE 3.1-CAN



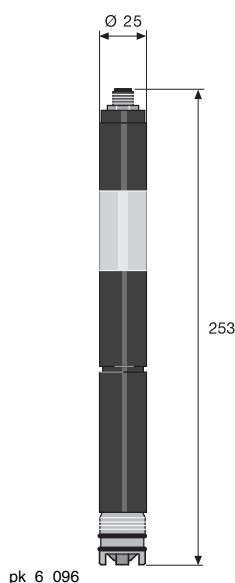
Sensor for the measurement of free chlorine in clear water with higher selectivity towards combined chlorine. For use on controllers with CAN-bus connection

Your benefits

- Measured variable: free chlorine, no cross sensitivity to combined chlorine (chloramines) even if there is an excess of it
- Diaphragm-covered sensor (encapsulated) minimises faults caused by changing flow or ingredients in the water
- Operation on the CAN-bus with all the associated benefits



5 Individual Devices and Accessories



Measured variable	free chlorine (hypochlorous acid HOCl) with large proportions of bound chlorine; to detect bound chlorine using DULCOMARIN® II and Sensor for Total Chlorine type CTE 1-CAN
Reference method	DPD1
pH range	5.5 ... 8.0
Temperature	5 ... 45 °C
Max. pressure	1.0 bar
Intake flow	30...60 l/h (in DGMa or DLG III)
Supply voltage	Via CAN interface (11 – 30 V)
Output signal	uncalibrated, temperature compensated, electrically isolated
Selectivity	Free chlorine
Disinfection process	Chlorine gas, hypochlorite, electrolysis with diaphragm. Disinfectants with organic chlorine, e.g. based on cyanuric acid, are unsuitable
Installation	Bypass: open sample water outlet
Sensor fitting	DGM, DLG III
Measuring and control equipment	DULCOMARIN® II
Typical applications	Potable water with higher volumes of combined chlorineSwimming pools, To determine the combined chlorine from the difference: Total chlorine minus free chlorine in the controller DULCOMARIN® II
Resistance to	Salts, acids, lyes. Not surfactants
Measuring principle, technology	amperometric, 2 electrodes, diaphragm-covered

	Measuring range	Order no.
CLE 3.1-CAN-10 ppm	0.01 ... 10.0 mg/l	1023426

Chlorine sensors complete with 100 ml of electrolyte

A mounting kit, order no. 815079, is required for initial fitting of the chlorine sensors in the in-line probe housing DLG III.

Sensor for Total Chlorine CTE 1-CAN



Sensor for total chlorine, including, for example, free chlorine, chloramines etc. even with high pH values in different kinds of water. For use on controllers with CAN-bus connection

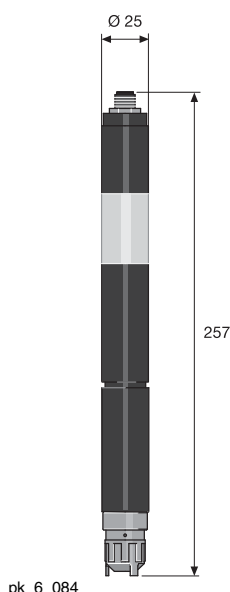
Your benefits

- Measured variable: Total chlorine, chlorine compounds, in which chlorine acts as an oxidising agent, e.g. free chlorine (HOCl and OCl⁻), chloramines etc.
- Diaphragm-covered sensor (encapsulated) prevents faults caused by changing flow or ingredients in the water
- Hydrophilic diaphragm guarantees permeability for different water-soluble oxidising agents towards the measuring electrodes
- The special reaction system of the electrolyte allows components containing oxidising chlorine to be determined and used at a high pH of up to 9.5
- Operation on the CAN-bus with all the associated benefits

Sensor for connection to a CAN interface (e.g. DULCOMARIN® II swimming pool controller)



5 Individual Devices and Accessories



Measured variable	total chlorine
Reference method	DPD4
pH range	5.5 ... 9.5 (up to pH 8.5 with D1C pH correction)
Temperature	5 ... 45 °C
Max. pressure	3.0 bar
Intake flow	30...60 l/h (in DGMa or DLG III)
Supply voltage	Via CAN interface (11 - 30 V)
Output signal	uncalibrated, temperature-compensated, electrically isolated
Selectivity	not selective, cross-sensitive towards many oxidation agents
Disinfection process	Chlorine gas, hypochlorite, electrolysis with diaphragm, Monochloramine
Installation	Bypass: open sample water outlet
Sensor fitting	DGM, DLG III
Measuring and control equipment	DULCOMARIN® II
Typical applications	CTE 1-mA-0.5 ppm: Potable water; CTE 1-mA-2/5/10 ppm: Potable, industrial, process, waste water. In swimming pools combined with CLE 3.1 to detect combined chlorine
Resistance to	Surfactants
Measuring principle, technology	amperometric, 2 electrodes, membrane-covered

	Measuring range	Order no.
CTE 1-CAN-10 ppm	0.01...10.0 mg/l	1023427

Chlorine sensors complete with 100 ml of electrolyte

A mounting kit, order no. 815079, is required for initial fitting of the chlorine sensors in the in-line probe housing DLG III.

Sensor for Total Available Chlorine CGE 2-CAN



Sensor for total available chlorine, such as derivatives of chloro(iso)cyanuric acid when used in swimming pools. For use on controllers with CAN-bus connection

Your benefits

- Measured variable: total available chlorine, for instance disinfectant with organic chlorine, such as derivatives of chloro(iso)cyanuric acid
- Diaphragm-covered sensor (encapsulated) minimises faults caused by changing flow or ingredients in the water
- Hydrophilic diaphragm guarantees the permeability of chloro(iso)cyanuric acid derivatives towards the measuring electrodes
- The special reaction system of the electrolyte allows the total available chlorine to be determined and use at a high pH of up to 9.5
- Operation on the CAN-bus with all the associated benefits

Measured variable	Total available chlorine: Total of organic combined chlorine (e.g. bound to cyanuric acid) and free chlorine
Reference method	DPD1
pH range	5.5 ... 9.5
Temperature	5 ... 45 °C
Max. pressure	3.0 bar
Intake flow	30...60 l/h (in the DGM or DLG III)
Supply voltage	via CAN interface (11 – 30 V DC)
Output signal	uncalibrated, temperature-compensated, electrically isolated
Selectivity	only limited against combined chlorine (chloramines)
Disinfection process	Disinfectants with organic chlorine, e.g. based on cyanuric acid
Installation	Bypass: open sample water outlet
Sensor fitting	DGM, DLG III



5 Individual Devices and Accessories

Measuring and control equipment	DULCOMARIN® II
Typical applications	Swimming pool water, Disinfection processes with chloro(iso)cyanuric acid derivatives
Resistance to	Surfactants
Measuring principle, technology	amperometric, 2 electrodes, membrane-covered

	Measuring range	Order no.
CGE 2-CAN-10 ppm	0.01...10.0 mg/l	1024420

A mounting kit, order no. 815079, is required for initial fitting of the chlorine sensors in the in-line probe housing DLG III.

Sensor for Total Available Bromine BRE 3-CAN



Sensor for free and combined bromine, also for use with slightly contaminated water. For use on controllers with CAN-bus connection

Your benefits

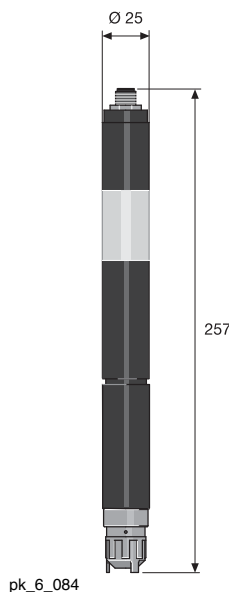
- Measured variable: total available bromine from BCDMH and other oxidative-acting bromine organic disinfectants
- Diaphragm-covered sensor minimises faults caused by changing flow or ingredients in the water
- Use with high pH values by optimisation of the electrolyte diaphragm system
- Operation on the CAN-bus with all the associated benefits

Sensor for connection to a CAN interface (e.g. DULCOMARIN® II swimming pool controller)

Measured variable	Total available bromine
Reference method	For DBDMH, free bromine: DPD1. For BCDMH: DPD4
pH dependence	If the pH changes from pH 7 to pH 8, the sensor sensitivity is reduced a) in the case of DBDMH and free bromine by approx. 10 % b) in the case of BCDMH by approx. 25 %
Temperature	5 ... 45 °C
Max. pressure	3.0 bar
Intake flow	30...60 l/h (in DGM or DLG III)
Supply voltage	Via CAN interface (11 – 30 V)
Output signal	uncalibrated, temperature-compensated, electrically isolated
Selectivity	not selective, cross-sensitive towards many oxidation agents
Disinfection process	DBDMH (1,3-dibromo-5,5-dimethyl-hydantoin), BCDMH (1-bromo-3-chloro-5,5-dimethyl-hydantoin), Free bromine (HOBr, OBr)
Installation	Bypass: open sample water outlet
Sensorarmatur	DGM, DLG III
Measuring and control equipment	DULCOMARIN® II
Typical applications	Swimming pools/whirlpools and cooling water; can also be used in sea water
Resistance to	Surfactants
Measuring principle, technology	amperometric, 2 electrodes, membrane-covered

	Measuring range	Order no.
BRE 3-CAN-10 ppm	0.02...10.0 mg/l	1029660

Note: a mounting kit (order no. 815079) is required for initial fitting of the bromine sensors in the in-line probe housing DLG III.



pk_6_084



5 Individual Devices and Accessories



Chlorine Dioxide Sensor CDR 1-CAN

Sensor for the measurement of chlorine dioxide for all kinds of water, including hot and contaminated water. Without cross-sensitivity by free chlorine. For operation on controllers with 4-20 mA input

Your benefits

- Measured variable: Chlorine dioxide, without cross sensitivity to free chlorine
- Diaphragm-covered sensor minimises faults caused by changing flow or ingredients in the water
- Resistance to films of dirt by pore-free diaphragm
- Operating temperature up to 60 °C (short term) by appropriate sensor materials
- Operation on the CAN-bus with all the associated benefits

Sensors for connection to a CAN interface (e.g. Disinfection Controller)

Measured variable	Chlorine dioxide (ClO ₂)
Reference method	DPD1
pH range	1.0 ... 10.0
Cross sensibility	Ozone
Temperature	5 ... 45 °C
Max. pressure	1.0 bar
Intake flow	30...60 l/h (in DGM or DLG III)
Supply voltage	via CAN interface (11-30 V)
Output signal	uncalibrated, temperature-compensated, electrically isolated
Response time sensor	t ₉₀ ~ 3 min.
Selectivity	Chlorite, Chlorate, Free chlorine
Installation	Bypass: open sample water outlet
Sensor fitting	DGMa/DLGIII

Measuring and control equipment DULCOMARIN® II

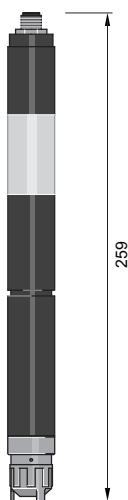
Typical applications

Contaminated industrial, process water, containing surfactants, cooling water, irrigation water, slightly contaminated waste water, warm water

Resistance to

Measuring principle, technology

Surfactants, water-soluble pollutants, Solids/dirt, Biofilms
amperometric, 2 electrodes, membrane-covered



P_DT_0071_SW1

	Measuring range	Order no.
CDR 1-CAN-10 ppm	0.01...10.0 mg/l	1041155

* Complete with 100 ml of electrolyte, connecting cable - CAN M12 5-pin 0.5 m, T-distributor M12 5-pin CAN

Chlorite Sensor CLT 1-CAN



Sensor for monitoring the disinfection by-product chlorite in compliance with potable water regulations. Without cross-sensitivity towards chlorine dioxide, chlorate and chlorine. For use on controllers with CAN-bus connection

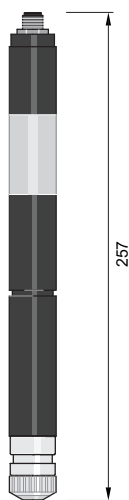
Your benefits

- Online monitoring of the disinfection by-product chlorite
- Diaphragm-covered sensor minimises faults caused by changing flow or ingredients in the water
- No interference by chlorine dioxide/chlorine/chlorate
- Online monitoring improves process reliability
- Online monitoring replaces expensive laboratory analysis
- Operation on the CAN-bus with all the associated benefits

Sensors for connection to a CAN interface (e.g. Disinfection Controller)



5 Individual Devices and Accessories



P_DT_0070_SW1

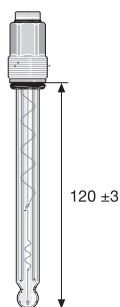
Measured variable	Chlorite anion (ClO_2^-)
Reference method	DPD method, chlorite together with chlorine dioxide
pH range	6.5 ... 9.5
Cross sensibility	Ozone
Temperature	1 ... 40 °C
Max. pressure	1.0 bar
Intake flow	30...60 l/h (in DGM or DLG III)
Supply voltage	via CAN interface (11-30 V)
Output signal	uncalibrated, temperature-compensated, electrically isolated
Response time sensor	3 min.
Selectivity	Chlorite selective towards chlorine dioxide, chlorate and free chlorine
Installation	Bypass: open sample water outlet
Parts number/identity code	DGM, DLG III
Measuring and control equipment	DULCOMARIN® II
Typical applications	Monitoring of potable water or similar water treated with chlorine dioxide. Selective measurement of chlorite and chlorine dioxide, chlorine and chlorate is also possible.
Resistance to	Surfactants
Measuring principle, technology	amperometric, 2 electrodes, membrane-covered

	Measuring range	Order no.
CLT 1-CAN-2 ppm	0.05...2.0 mg/l	1041156

* Complete with 100 ml of electrolyte, connecting cable - CAN M12 5-pin 0.5 m, T-distributor M12 5-pin CAN

5.5.2

Sensors for pH Value

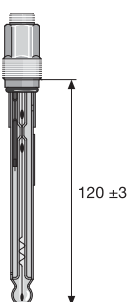


pk_6_016

PHES 112 SE

pH range	1 ... 12
Temperature	0 ... 60 °C
Max. pressure	3.0 bar
Min. conductivity	150 µS/cm
Diaphragm	Ceramic
Sensor shaft	Glass
Thread	PG 13.5
Typical applications	Swimming pools, whirlpools, potable water

	Fitting length	Order no.
PHES 112 SE	120 ± 3 mm	150702
PHES 112 SE	225 ± 3 mm	150092



pk_6_019

pH-Sensor PHEP 112 SE

pH range	1 ... 12
Temperature	0 ... 80 °C
Max. pressure	6.0 bar
Min. conductivity	150 µS/cm
Diaphragm	Ceramic
Sensor shaft	Glass
Fitting length	120 ± 3 mm
Thread	PG 13.5
Mounting hole Ø min.	—



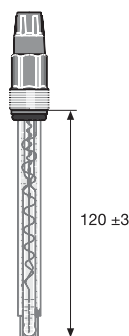
5 Individual Devices and Accessories

Typical applications

Swimming pools during pressurisation for higher temperatures and pressures, potable and industrial water, electroplating, chemical industries

Order no.

PHEP 112 SE	150041
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pk_6_090

PHEK 112 SE

pH range	1 ... 12
Temperature	0 ... 60 °C
Max. pressure	3.0 bar
Min. conductivity	150 µS/cm
Diaphragm	Ceramic
Sensor shaft	Polycarbonate
Fitting length	120 ± 3 mm
Thread	PG 13.5
Stem diameter min.	–
Typical applications	Swimming pools, potable water, aquaria

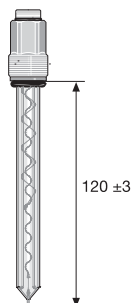
Order no.

PHEK 112 SE	1028457
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ex HD works

5.5.3

Sensors for ORP Value



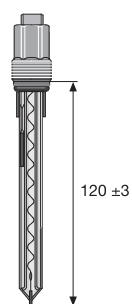
pk_6_031

RHES-Pt-SE

Temperature	0 ... 60 °C
Max. pressure	3 bar
Min. conductivity	150 µS/cm
Diaphragm	Ceramic
Sensor shaft	Glass
Fitting length	120 ± 3 mm
Thread	PG 13.5
Typical applications	Swimming pools, whirlpools, potable water

Order no.

RHES-Pt-SE	150703
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pk_6_035

RHEP-Pt-SE

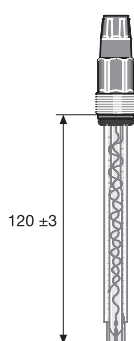
Temperature	0 ... 80 °C
Max. pressure	6.0 bar
Min. conductivity	150 µS/cm
Diaphragm	Ceramic
Sensor shaft	Glass
Fitting length	120 ± 3 mm
Thread	PG 13.5
Mounting hole Ø min.	–
Typical applications	Swimming pools during pressurisation for higher temperatures and pressures, potable and industrial water, electroplating,

Order no.

RHEP-Pt-SE	150094
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5 Individual Devices and Accessories



pk_6_091

RHEK-Pt-SE

Temperature	0 ... 60 °C
Max. pressure	3.0 bar
Min. conductivity	150 µS/cm
Diaphragm	Ceramic
Sensor shaft	Polycarbonate
Thread	PG 13.5
Fitting length	120 ± 3 mm
Typical applications	Swimming pool, Potable water, Aquariums,

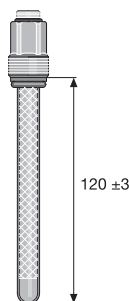
Order no.

RHEK-Pt-SE

1028459

5.5.4

Sensors for Temperature



pk_6_026

Temperature	0 ... 100 °C
Max. pressure	10.0 bar
Thread	PG 13.5
Electrical connection	SN6
Typical applications	Temperature measurement and pH temperature correction

Order no.

Pt 100 SE

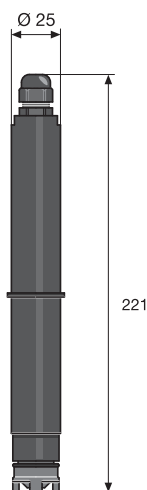
305063

Pt 1000 SE

1002856

5.5.5

Sensors for Inorganic Chlorine



pk_6_039

CLE 3-mA

Measured variable	free chlorine (hypochlorous acid HOCl)
Reference method	DPD1
pH range	5.5 ... 8.0
Temperature	5 ... 45 °C
Max. pressure	1.0 bar
Intake flow	30...60 l/h (in DGM or DLG III)
Supply voltage	16...24 V DC (two-wire technology)
Output signal	4...20 mA ≈ measuring range, temperature-compensated, uncalibrated, not electrically isolated
Selectivity	Free chlorine as against combined chlorine, even if there is not an excess of it
Disinfection process	Chlorine gas, hypochlorite, electrolysis with diaphragm. Disinfectants with organic chlorine, e.g. based on cyanuric acid, are unsuitable
Installation	Bypass: open sample water outlet
Sensor fitting	DGM, DLG III

Measuring range Order no.

CLE 3-mA-0.5 ppm	0.01...0.5 mg/l	792927
CLE 3-mA-2 ppm	0.02...2.0 mg/l	792920
CLE 3-mA-5 ppm	0.01...5.0 mg/l	1033392
CLE 3-mA-10 ppm	0.10...10.0 mg/l	792919
CLE 3-mA-20 ppm	0.20...20.0 mg/l	1002964
CLE 3-mA-50 ppm	0.50...50.0 mg/l	1020531
CLE 3-mA-100 ppm	1.00...100.0 mg/l	1022786

Chlorine sensors complete with 100 ml of electrolyte



5 Individual Devices and Accessories

	Capacity ml	Order no.
Electrolyte for CLO 2 type chlorine sensor	100	1035480
Electrolyte for CLO 1 type chlorine sensor	100	1035191
Electrolyte for CBR 1 type chlorine/bromine sensor	100	1038017
Electrolyte for BCR 1 type bromine sensor	50	1044843
Electrolyte for all chlorine sensors type CLE, CLR 1	100	506270

	Capacity ml	Order no.
Sensor cap for CLO 1	–	1035197
Accessory set for CLO 1 (electrolyte, grinding disc, plug)	100	1035482
Accessory set for CLO 2 (electrolyte, grinding disc, plug)	100	1035483
Accessory set for CLE (2 membrane caps + electrolyte)	100	1024611
Accessory set for CBR 1 (2 membrane caps + electrolyte)	100	1038984
Accessory set BCR 1 (2 membrane caps + electrolyte)	50	1044844
Membrane cap for types: CLE 2.2, CLE 3, CDE 1.2, CDE 2, OZE 2 and OZE 3	–	790488

A mounting kit, order no. 815079, is required for initial fitting of the chlorine sensors in the in-line probe housing DLG III.

5.5.6

Organic Chlorine (Total Available Chlorine)

	Capacity ml	Order no.
Electrolyte for CLO 2 type chlorine sensor	100	1035480
Electrolyte for CLO 1 type chlorine sensor	100	1035191
Electrolyte for CBR 1 type chlorine/bromine sensor	100	1038017
Electrolyte for BCR 1 type bromine sensor	50	1044843
Electrolyte for CGE/CTE/BRE type sensors	50	792892

	Capacity ml	Order no.
Accessory set for CBR 1 (2 membrane caps + electrolyte)	100	1038984
Accessory set BCR 1 (2 membrane caps + electrolyte)	50	1044844
Membrane cap for CGE/CTE 1 (2/5/10 ppm) and BRE 1 (10 ppm), BRE 2	–	792862

5.5.7

Assembly Set

Mounting kit for sensor/DGM

for mounting amperometric sensors with R 10 connection

	Order no.
Mounting kit for sensor/DGM	791818



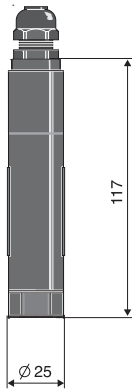
5 Individual Devices and Accessories

5.6 Accessories for Sensors

5.6.1

Measuring Transducer 4...20 mA (Two Wire)

pH measuring transducer 4 ... 20 mA type pH V1



pk_5_064

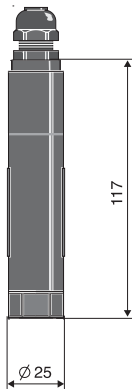
Measuring range	pH 0 ... 14
Measuring error	better than 0.1 pH (typical ± 0.07 pH)
Socket	SN6
Input resistance	$> 5 \times 10^{11} \Omega$
Signal current output	4 ... 20 mA $\approx$ -500 ... +500 mV $\approx$ pH 15.45 ... -1.45 not calibrated, not electrically isolated
Power supply DC	18...24 V DC
Ambient temperature	-5...50 °C, non-condensing
Enclosure rating	IP 65
Dimensions	141 mm (length), 25 mm ($\varnothing$)

Order no.

pH measuring transducer 4 ... 20 mA type pH V1

809126

ORP measuring transducer 4 ... 20 mA type RH V1



pk_5_064

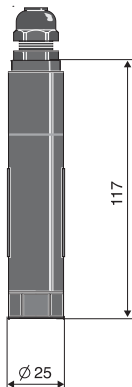
Measuring range	0 ... 1000 mV
Measuring error	better than ± 5 mV (typical ± 3 mV)
Socket	SN6
Input resistance	$> 5 \times 10^{11} \Omega$
Signal current output	4 ... 20 mA $\approx$ 0 ... +1000 mV not electrically isolated
Power supply DC	18...24 V DC
Ambient temperature	-5...50 °C, non-condensing
Enclosure rating	IP 65
Dimensions	141 mm (length), 25 mm ($\varnothing$)

Order no.

ORP measuring transducer 4 ... 20 mA type RH V1

809127

Temperature measuring transducer 4 ... 20 mA type Pt100 V1



pk_5_064

Measuring range	0 ... 100 °C
Measuring error	better than ± 0.5 °C (typical ± 0.3 °C)
Socket	SN6
Input resistance	$\sim 0 \Omega$
Signal current output	4 ... 20 mA $\approx$ 0 ... +100 °C not electrically isolated
Power supply DC	18...24 V DC
Ambient temperature	-5...50 °C, non-condensing
Enclosure rating	IP 65
Dimensions	141 mm (length), 25 mm ($\varnothing$)

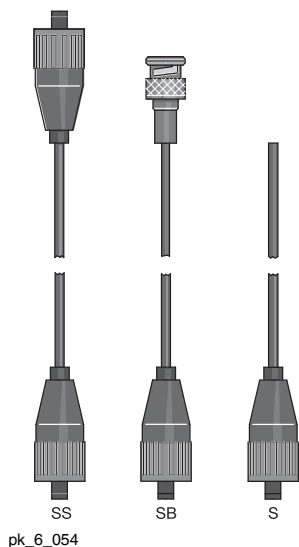
Order no.

Temperature measuring transducer 4 ... 20 mA type Pt 100 V1

809128



5 Individual Devices and Accessories



Signal leads for pH/ORP measurement

- Pre-assembled to facilitate installation
- Factory tested to ensure function reliability
- IP 65

Type	Description	Order no.
2 x SN6	Coaxial cable, Ø 5 mm, 0.8 m – SS	305077
2 x SN6	Coaxial cable, Ø 5 mm, 2 m – SS	304955
2 x SN6	Coaxial cable, Ø 5 mm, 5 m – SS	304956
2 x SN6	Coaxial cable Ø 5 mm, 10.0 m – SS	304957
SN6 - open end	Cable combination coaxial 0.8 m - SN6 - pre-assembled	1024105
SN6 - open end	Cable combination coaxial 2 m - SN6 - pre-assembled	1024106
SN6 - open end	Cable combination coaxial 5 m - SN6 - pre-assembled	1024107
SN6 - open end	Coaxial cable Ø 5 mm, 10.0 m - S	305040
SN6 - BNC	Coaxial cable Ø 3 mm, 10.0 m - SB	305099
SN6 - BNC	Coaxial cable, Ø 5 mm, 0.8 m – SB	1033988
SN6 - BNC	Coaxial cable, Ø 5 mm, 2.0 m – SB	1033011
SN6 - DIN	Coaxial cable Ø 5 mm, 0.8 m - SD	305098
SN6 - DIN	Coaxial cable Ø 5 mm, 2.0 m - SD	304810
SN6 - open end d5 (DSR)	Cable combination coax 2.0 m - S	1005672

2-wire measuring line

2-core, conductor: 0.25 mm², cable diameter: 4 mm

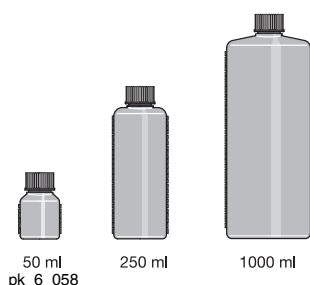
For mA type chlorine / bromine / chlorine dioxide / ozone sensors and pH, ORP, Pt 100, conductivity, hydrogen peroxide (PEROX) transducers.

	Order no.
Signal cable, sold by the metre 2 x 0.25 mm ² Ø 4 mm	725122

pH quality buffer solutions

Accuracy ±pH 0.02 (±0.05 at pH 10). The shelf life depends upon frequency of use and the amount of chemical drag-in.

Alkaline buffer solutions can react with CO₂ if left open. This will affect their values, therefore close after use. Buffer solutions should be replaced a maximum of three months after opening. The solution contains a biocide to prevent bacteria forming.



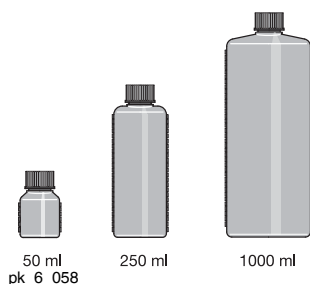
	Capacity ml	Order no.
Buffer pH 4.0 – red	50	506251
Buffer pH 4.0 – red	250	791436
Buffer pH 5.0 – red	50	506252
Buffer pH 7.0 – green	50	506253
Buffer pH 7.0 – green	250	791437
Buffer pH 9.0 –	50	506254
Buffer pH 10.0 – blue	50	506255
Buffer pH 10.0 – blue	250	791438

ORP quality buffer solutions

Accuracy to ±5 mV. Shelf life depends upon frequency of use and the strength of the chemicals in sample solutions.

Buffer solutions should be replaced a maximum of three months after opening.

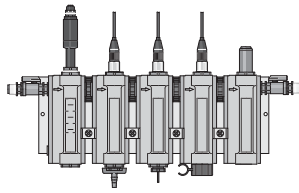
Warning: The 465 mV ORP buffer solution is an irritant!



	Capacity ml	Order no.
ORP buffer 465 mV	50	506240
ORP buffer 465 mV	250	791439
ORP buffer 220 mV	50	506244

DPD-reagents for calibration of amperometric sensors s. p. → 1-14

5 Individual Devices and Accessories



pk_6_066

DGM modular in-line probe housing

To hold conductivity, Pt 100, pH or ORP sensors with PG 13.5 screw-in thread or amperometric sensors with R 1" screw-in thread.

Advantages:

- Simple installation (completely ready-mounted on a panel); max. 7 modules on a panel
- Simple retrofit extension option (see extension modules)
- Module for sample water flow control
- Quick measurement recording due to low volume of sample water
- Each completely assembled DGM is equipped with a simple sampling tap

Ball valves on both sides for shutting off the flow and for flow adjustment

Material	All modules: Transparent PVC Seals: FKM Calibration cup: PP Mounting panel: PVC white
Max. temperature	60 °C
Max. pressure	6.0 bar up to 30 °C, 1.0 bar up to 60 °C
Max. flow rate	80 l/h
Recommended Flow volume	40 l/h
Flow sensor	Reed contact max. switch power 3 W max. switch voltage 175 V max. switch current 0.25 A max. operating current 1.2 A max. contact resistance 150 mΩ
Switching hysteresis	20 %
Enclosure rating	IP 65
Typical applications	Potable water, swimming pool water or water of similar quality with no suspended solids
Assembly	Max. 5 modules pre-assembled onto baseboard: more than 5 modules, pre-assembled onto baseboard as custom version, priced accordingly.

FKM = fluoro rubber

for pH sensors	for ORP sensors	for chlorine sensors	for temperature sensors	Flow sensor	Identity code / Order no.
x	x	-	-	x	DGM A 320 T 000
x	-	x	-	x	DGM A 311 T 000
x	x	x	-	x	DGM A 321 T 000
x	x	x	x	x	DGM A 331 T 000

x = available

- = not available

Accessories

	Order no.
Sample water filter	1021338
Nylon screen	1021337
Stainless steel screen	1038867
Laboratory ball valve	1010380
Sample water valve	1018874



5 Individual Devices and Accessories

Identity Code Ordering System for In-Line Probe Housing Modules

DGM	Series		Series Version						
	A								
			Flow monitor module						
		1	with l/h scale						
		2	with gph scale (US)						
		3	With flow monitor, l/h scale						
		4	with flow monitor, gph scale (US)						
			Number of PG 13.5 modules						
		0	without PG 13.5 modules						
		1	one PG 13.5 module						
		2	two PG 13.5 modules						
		3	three PG 13.5 modules						
		4	four PG 13.5 modules						
			Number of 25 mm modules						
		0	No 25 mm modules						
		1	One 25 mm module						
		2	Two 25 mm modules						
			Main material						
		T	Transparent PVC						
			Sealing material						
		0	FKM A						
			Hydraulic connectors						
		0	8 x 5 hose						
		1	PVC DN 10 threaded connector						
		4	Hose 12 x 6						
			Version						
		0	With ProMinent® logo						
		1	Without ProMinent® logo						
		2	With ProMinent® logo, without mounting plate						
		3	Without ProMinent® logo, without mounting plate						
DGM	A	3	2	1	T	0	0	0	

Accessories supplied:

- Wall fastenings for PG 13.5 modules: Calibration plate, mounting kits for PG 13.5 probes

The identity code DGM A 3 2 1 T 0 0 0 describes, for example, a fully assembled configuration of a flow module with sensor, two PG 13.5 modules (e.g. for pH and ORP probes) and a 25 mm module (e.g. for CLE 3 chlorine sensor) 8 x 5 tube connectors are ready mounted.

Recommended accessories

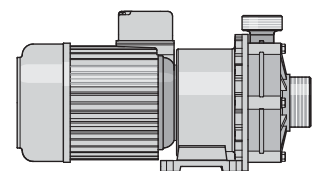
		Order no.
for potential equalizer plug	–	791663
Flow sensor for flow expansion module (optional)	–	791635
additional calibration cup	–	791229
PG 13.5 sampling tap	for 13.5 module	1004737
25 mm sampling tap	for 25 mm module	1004739
Expansion module for PG 13.5 sensors	–	1023975
Expansion module for 25 mm sensors	–	1023976
Non-return valve	–	809435

- max. 7 modules possible on a mounting plate
- more on request

5 Individual Devices and Accessories

5.6.2

von Taine® Solenoid-coupled Rotary Pumps for Application = Water Sampling



pk_3_026

Feed pumps for liquid media

von Taine® pumps are solenoid-coupled centrifugal pumps. Thanks to the solenoid coupling, the pumps transport the liquid medium from storage tank to storage tank without any leaks or even from a tank to a discharge line. von Taine® centrifugal pumps transport media at up to 22,500 l/h and up to a discharge lift of 23.5 metres. As the pump capacity is highly dependent on the back pressure, the performance curve must always be observed. When selecting the pumps check the material compatibility and consider the density, viscosity and temperature of the transport medium. The transported media must not contain any solid fractions. The pump is not self-priming and requires a feed.

The following material versions are available:

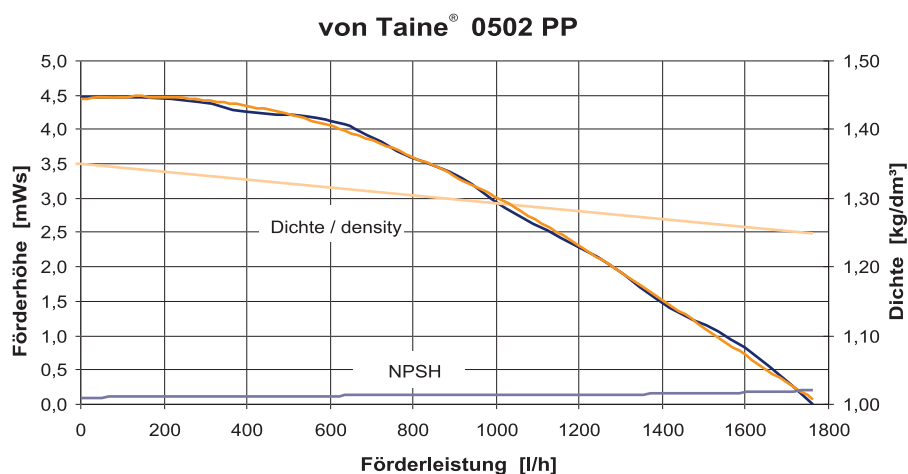
- Pump head: PP or PVDF
- Seal: FKM or EPDM

The pump bearings are made from "oxide ceramics" and should not be allowed to run dry. Protect the pump from running dry. The hydraulic connectors have pipe threadings according to DIN ISO 228-1 (cylindrical external thread).

Max. system pressure at 20° C	1.0
Suction connector (S)	G 1 1/4"
Discharge connector (D)	G 1"
Power uptake	60
Enclosure rating	IP 55
Voltage/frequency	1/230/50
Feed lift max.	4.5
Feed rate at max. pressure	1,800

	Order no.
von Taine® 0502 PP/FKM	1023089
von Taine® 0502 PVDF/FKM	1023095

Performance curve of von Taine® 0502 PP



P_vT_0006_SW



5 Individual Devices and Accessories

5.7

Test Devices

5.7.1

Portamess Portable Meters Measured Variable, Conductivity

Robust measuring instrument to withstand the most severe mechanical and chemical loading.

Measuring range 0.01 $\mu\text{S}/\text{cm}$ to 1,000 mS/cm



The measuring instrument Portamess® conductivity is a robust, leak-tight and battery-operated hand-held measuring instrument with a large measuring range and automatic or manual temperature compensation, which can be used in the industrial, environmental, food and waste water sectors.



pk_5_098

The Portamess® conductivity is used to measure conductivity and temperature in the industrial, environmental, food and waste water sectors. The unit complies with the requirements of the EMC Act and the NAMUR NE 21 recommendations. Calibration can be done with buffer solutions made of different, pre-selectable buffer sets.

Your benefits

- Robust and leak-tight
- Long lifespan: Over 1,000 h operating time with only 3 x AA batteries
- Always in sight: Large LC display

Technical details

Measuring ranges:

- Conductivity instrument: 0.01 $\mu\text{S}/\text{cm}$... 1,000 mS/cm , with sensor LF204: 1 $\mu\text{S}/\text{cm}$... 500 mS/cm
- Temperature: -20 ... 120 °C
- Salinity: 0.0 ... 45.0 g/kg (0 ... 30 °C)
- TDS: 0 ... 1,999 mg/l (10 ... 40 °C)

Measuring error:

- Conductivity < 0.5% of the measured value (with conductivities of > 500 mS/cm < 1% of the measured value) ± 1 digit
- Temperature < 0.3 K ± 1 digit

Sensor adaptation: Direct input of the cell constants, automatic establishment of the cell constants with KCl solution 0.01 or 0.1 mol/l, cell adaptation with any known solutions

Cell constant k: 0.010 ... 199.9 cm^{-1} (adjustable)

Temperature compensation: configurable, manual or measured

Degree of protection: IP 66

Operating time: approx. 1,000 hours with 3 no. AA cells

Dimensions: 160 x 133 x 30 mm (H x W x D)

Weight: 560 g with batteries

Scope of delivery: Measuring instrument, field case, conductivity sensor LF 204, operating instructions in German, English and French

Field of application

- Industry
- Environmental protection
- Food production
- Water or waste water inspection

Order no.

Portamess® 911 Cond

1008713

Note:

The scope of delivery does include the conductivity sensor LF 204.



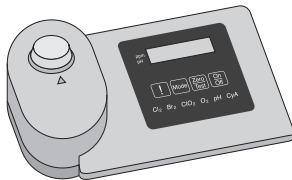
5 Individual Devices and Accessories

5.7.2

Photometer

Precise measurement results through high-quality interference filters

Photometers measure nearly all disinfectants and the pH value based on the photometric principle. They are portable, compact and make safe, simple measurement possible.



P_DT_0074_SW
Photometer

The photometers DT1B, DT2C, DT3B and DT4B are used, among other things, as a reference method for calibrating the electrochemical sensors for chlorine, chlorine dioxide, fluoride, chlorite, H_2O_2 , bromine and ozone. They have been adapted to today's requirements and can be used in almost all areas of water analysis. High-quality interference filters and long-term stable LEDs are used as the light source in the high-precision optics. The entire measuring unit is maintenance-free. Precise and reproducible analysis results are achieved with minimum time and effort. The units are winning customers over with their excellent operating convenience, ergonomic design, compact dimensions and ease of use.

Your benefits

- Portable and compact
- Simple to operate with text support
- Safe, simple measurement of chlorine, chlorine dioxide, fluoride, chlorite, H_2O_2 , bromine, ozone, pH and trichloroisocyanuric acid
- Can be calibrated
- Memory function for the last measurements
- Backlit display
- Real-time clock
- Countdown
- Watertight, degree of protection IP 68

Technical details

Measuring ranges of the DT1B:

- 0.05... 6.0 mg/l free chlorine (DPD1) + total chlorine (DPD1+3)
- 5 ... 200 mg/l free chlorine (high range)
- 0.1 ... 13.0 mg/l bromine (DPD1)
- 0.05 ... 11 mg/l chlorine dioxide (DPD1)
- 0.03 ... 4.0 mg/l ozone (DPD4)
- 6.5 ... 8.4 pH (phenol red)
- 1 ... 80 mg/l cyanuric acid

Measuring ranges of the DT2C:

- 0.05 ... 2.0 mg/l fluoride
- 0.05... 6.0 mg/l free chlorine and total chlorine
- 0.05 ... 11.0 mg/l chlorine dioxide

Measuring ranges of the DT3B:

- 1 ... 50 / 40 ... 500 mg/l hydrogen peroxide (H_2O_2)

Measuring ranges of the DT4B:

- 0.03 ... 2.5 mg/l chlorite
- 0.05 ... 11 mg/l chlorine dioxide
- 0.05 ... 6 mg/l chlorine

Measuring tolerance: depending on the measured value and measuring method

Battery: 4 no. AA/LR6 batteries

Permissible ambient temperature range: 5...40 °C

Relative humidity: 30 ... 90 % (non-condensing)

Degree of protection: IP 68

Housing material: ABS

Keypad: polycarbonate film

Dimensions: 190 x 110 x 55 mm (L x W x H)

Weight: 0.4 kg



5 Individual Devices and Accessories

Field of application

- Swimming pools
- Potable water
- Process water

	Order no.
Photometer DT1B	1039315
Photometer DT2C	1039316
Photometer DT3B	1039317
Photometer DT4B	1039318

Photometers supplied with accessories, container vessels and reagents.

Consumable items

	Order no.
DPD 1 buffer, 15 ml	1002857
DPD 1 reagent, 15 ml	1002858
DPD 3 solution, 15 ml	1002859
Phenol red tablets R 175 (100 in each)	305532
Cyanuric acid tablets (100 in each)	1039744
SPADNS reagent, 250 ml for fluoride detection	1010381
Calibration standard fluoride 1 mg/l, for calibration of the photometer during fluoride determination	1010382
3 spare cells: round cells with covers for DPD phenol red and cyanuric acid detection (DT1 and DT2B)	1007566
3 spare cells for fluoride detection (DT2A and B)	1010396
DPD reagent set, 15 ml each: 3 x DPD 1 buffer, 1 x DPD 1 reagent, 2 x DPD 3 solution	1007567
Chlorine dioxide tablets no. 1	1039732
Chlorine dioxide tablets no. 2	1039733
Chlorine HR tablets (100 off)	
ACIDITYING tablets (100 off)	

Spare parts

Chlorite Photometer

	Order no.
Stirrer for purging of chlorine dioxide (DT4)	1022754
3 spare cells: round cells with covers for DPD phenol red and cyanuric acid detection (DT1 and DT2B)	1007566

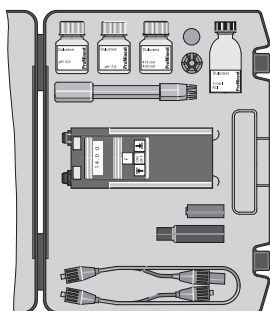
H₂O₂ measurement

	Order no.
Reagent for H ₂ O ₂ (DT3), 15 ml	1023636
Spare cell, 5x , for H ₂ O ₂ (DT3)	1024072



5 Individual Devices and Accessories

5.7.3 Manual Measuring Device for pH / ORP / Temperature



P_AC_0219_SW

Variant 1

for pH, ORP measurement and temperature, as well as simulation for functional testing of pH / ORP sensors and measuring control devices in a plastic case

	Order no.
Manual measuring device type pH/mV/T	1010984

Variant 2

as variant 1, but with additional pH sensor type PHEK-112-S, 0.8 m coaxial cable with 2 x SN 6-pin, pH 4 and pH 7 buffer solution (50 ml each)

	Order no.
Manual measuring device (variant 2)	1010985

Variant 3

as variant 2, but with additional ORP sensor type RHEK-PS, 0.8 m coaxial cable with 2 x SN 6 pin, ORP 465 mV buffer solution (50 ml each)

	Order no.
Manual measuring device (variant 3)	1010986

5.8 Compensating Tank (Surge Water Tank)

Compensating tank for overflow pool

ProMinent Dosierttechnik produces compensating tanks made of PP or PE to meet your requirements, if required also with level monitoring.



6 General information on Water Treatment

6.1 Swimming Pool Water Treatment

The German standard DIN 19643 must be complied with when treating swimming pool water, with the exception of swimming pools in the home (private pools). It defines the microbiological, chemical and physical requirements (e.g. pH value, chlorine content, etc.) regarding the composition of the water. This includes the measurement, operation and inspection processes, with the objective of providing clear, clean and especially hygienically safe water.

Swimming pool water treatment basically consists of the following components:

- Circulation / flow through the swimming pool (hydraulic)
- Flocculation
- Filtration
- pH correction
- Disinfection

Circulation / flow through the swimming pool

Together with the water treatment system and the circulation lines, the swimming pool represents a circulation system in which the following must be fulfilled:

- optimum water exchange, that is removal of impurities
- optimum distribution of the disinfection agent

Hydraulic systems that work with floor returns have proved to be the most effective and possibly drain 100% of the water via the overflow gutter or wet decks.

Flocculation

By adding flocculents, colloidal pollutants that cannot be filtered or pollutants that have been dissolved are transformed into a form that can be filtered. They can then be extracted from the water in the subsequent filtering process. Flocculation must only be performed for public swimming pools / hotel pools / therapy pools according to DIN 19643.

Filtration

Filtration has the aim of retaining any pollutants found in the water. As this represents a fundamental component of the water treatment system, filter systems should always be purchased from specialist suppliers.

pH value adjustment

The pH value is essentially affected by the outgassing of carbon dioxide (CO_2), from the disinfectant used (e.g. sodium hypochlorite) as well as by the make up water. Too low a value, i.e. acid pH values, can lead to attacks on the materials of construction, a reduction of flocculation, increased corrosion and the formation of nitrogen trichloride. The latter causes irritation to the mucous membranes. Too high a value, i.e. alkali pH values, reduce the disinfecting effect of chlorine, impair flocculation, stimulate the formation of lime scale and have an extremely negative effect on the well-being of swimming pool users. The chlorine measurement is additionally influenced negatively. This is why the pH should lie between 6.5 and 7.6 pH. The continual measurement of the pH value is carried out using appropriate pH sensors and controllers.

Disinfection

In accordance with DIN 19643, the disinfection of swimming pool water must only be carried out using chlorine in the form of chlorine gas, sodium hypochlorite or calcium hypochlorite solution. Organic chlorine products, as well as active oxygen, must not be used. Ozone, which is a mandatory specification for therapy pools, must only be used for oxidation and must be removed completely before the water is reintroduced into the swimming pool.

Disinfection using chlorine: The process of chlorination that has proved itself throughout the world for many years is characterised by a safe and quick degermination speed with a high level of oxidation. It is therefore the disinfection process of choice in swimming pool water treatment.

A decisive factor in disinfection is the concentration of free active chlorine (hypochlorous acid = HOCl). According to DIN 19643, the concentration of chlorine in a pool should lie between 0.3 and 0.6 mg per litre. With warm whirlpools, the concentration should lie between 0.7 and 1.0 mg / litre. As the concentration of chlorine is continually reduced by substances that enter the water, such as dust, cosmetics, skin particles, bacteria and algae, a reactivation is prescribed in DIN 19643 by an automatic measuring, control and metering system. The metering systems must be designed for metering feed rates of 2 g/m³ (indoor swimming pools) or 10 g/m³ (outdoor swimming pools). Systems, such as this, are also recommended for private pools. The chlorine measurement must be carried out directly via a chlorine sensor and not via the redox voltage auxiliary parameter. This produces the following benefits:

- quick response behaviour
- detection of the actual concentration of chlorine present (prevents over-chlorination)



6 General information on Water Treatment

The redox voltage is an auxiliary hygiene parameter and is used to monitor the water quality or hygienic safety. Redox voltages of > 700 mV guarantee the quick destruction of any germs in the water.

Use of ozone: The use of chlorine in conventional water treatment can lead to undesired reaction products, such as trihalomethanes and chloramines.

The chloramines (bound chlorine) are responsible for the typical indoor swimming pool odour and can lead to irritation of the skin and mucous membrane. Trihalomethanes are suspected as being carcinogenic. These problems can be drastically reduced by the use of ozone. Ozone is the strongest oxidising agent approved for use in water treatment. It is usually added before the filtration stage and then any undesirable substances in the water, chloramines and turbidity, are oxidised and retained in the filter. Moreover, the content of trihalomethanes are reduced significantly below the limit values.

The use of ozone in swimming pool water treatment provides the following advantages:

- the water does not have an unpleasant odour
- the indoor air is pleasant and healthy
- the clarity of the water is increased considerably by the flocculating effect of the ozone
- no chemical by-products, as the ozone quickly decomposes to oxygen after doing its job, a desirable factor in the water.

As ozone may have a negative impact on people's health due to outgassing, it is only allowed to be used in water treatment systems in accordance with DIN 19643. It must be removed completely before the water is re-introduced into the swimming pool.

Disinfection in the pool itself must be provided with chlorine. This is also recommended for domestic pools. However, active oxygen products can also be used instead of chlorine.

Disinfection using active oxygen: Disinfection using active oxygen is only permitted in domestic pools. The recommendations of the chemical supplier must be observed here in all cases.

Automatic measuring, control and metering units

Measuring and control units record the measured values of the sensors, process them and, for example, control the metering pump according to the actual water parameters. Such systems offer the operator safety and can relieve them of a lot of work at the same time.



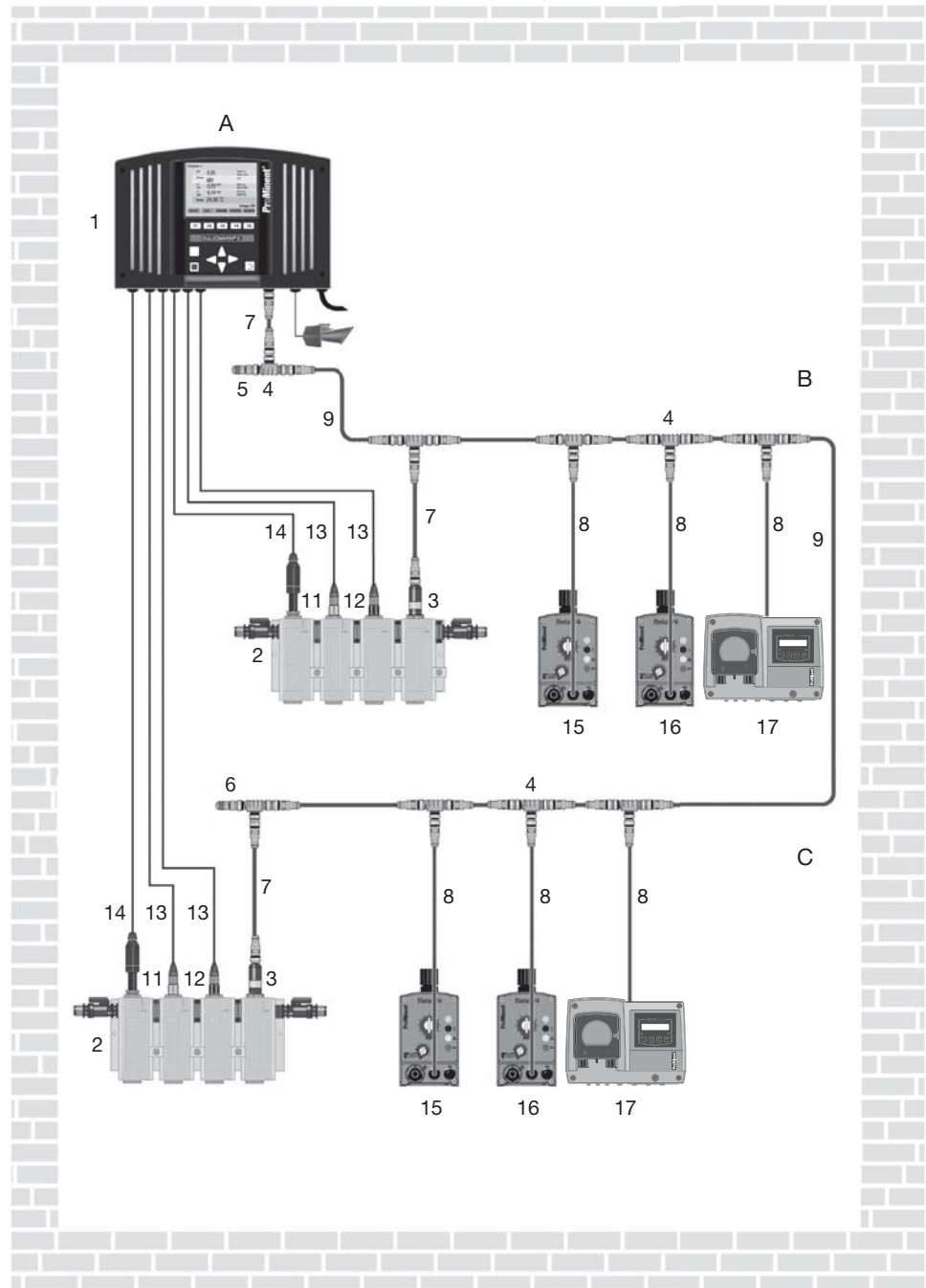
6 General information on Water Treatment

6.2 Measuring and Control System Diagram

6.2.1 Configuration Example: 2-Pool System

Two M modules in a central unit, use of metering pumps with CANopen bus.

- A Plant room
- B Pool 1
- C Pool 2



pk_5_022_neu

Attention:

It is very important that you adhere precisely to the principle of the design shown above because otherwise correct function is not guaranteed!



6 General information on Water Treatment

Measuring and control system for two filter circuits consisting of the following components:

Item	Quantity	Name	Order no.
1	1	DULCOMARIN®II central unit DXCa W 0 0 1 M M P S EN 01	–
2	2	DULCOTEST® in-line probe housing DGMa 3 2 2 T 0 0 0	–
3	2	Chlorine sensor CLE 3-CAN-10 ppm	1023425
4	9	T-distributor M12 5 pole CAN	included
5	1	Termination resistor M12 connector	included
6	1	Termination resistor M12 plug	included
7	5	Connection cable - CAN M12 5-pole 0.5 m	included
8	6	Connection cable - CAN M12 5-pole 0.3 m	included
9	–	Connecting cable - CAN M12 5-pin 10 m	1046383
11	2	pH sensor PHES 112 SE	150702
12	2	ORP sensor RHES-Pt-SE	150703
13	4	Cable combination coaxial 2 m - SN6 - pre-assembled	1024106
14	4 m	Signal cable, sold by the metre 2 x 0.25 mm ² Ø 4 mm	725122
15	2	Beta®/ 4 CANopen for pH correction BT4A0402PVT290UA000D00**	–
16	2	Beta®/ 4 CANopen for disinfectant BT4A0402PVT290UA000D00**	–
17	2	DF4a CAN for flocculant DF4aFW004015P9UA00001D10	–

* Up to 3 can be coupled from the connecting cable CAN M 12 5-pin 10 m.

** Suggested configuration

Caution:

Do not allow the maximum main bus length (without branch cables) to exceed 400 m.



6 General information on Water Treatment

6.2.2

DXCa Function Extensions for Hotel and Private Pools (F Module)

The following functions are available with the basic version:



P_DC_0010_SW1



pk_5_045

Temperature control

P-module control, for instance of a heat pump Accessory components required: Pt 100 or Pt 1000

Superchlorination

Eco mode utilises a 2nd set of control parameters and when Eco mode is activated, it is this 2nd set of parameters which determine the control output. One way to activate Eco mode is via an inbuilt timer and if the 2nd set of parameters are set to give a higher chlorine dose, then the Eco mode function could be used to give timed superchlorination.

Circulation pump speed reduction (Frequency inverter by others required)

Once all DIN 19643 or individually set parameter values have been reached, the speed of the circulation pump can be reduced. The speed reduction can be controlled in 2 ways - by timed step reduction or in relation to the ORP value. This reduces the electric power consumption.

Timer control 0/4 - 20 mA

This enables the direct control of a frequency converter for precise adjustment of the circulation to the current hygiene conditions. Freely programmable events: Circulation mode, backwashing, setback.

Backwashing

Gutter cleaning

Second circuit

Sample water monitoring

Solar heating

Chlorine reduction

Internal circuit

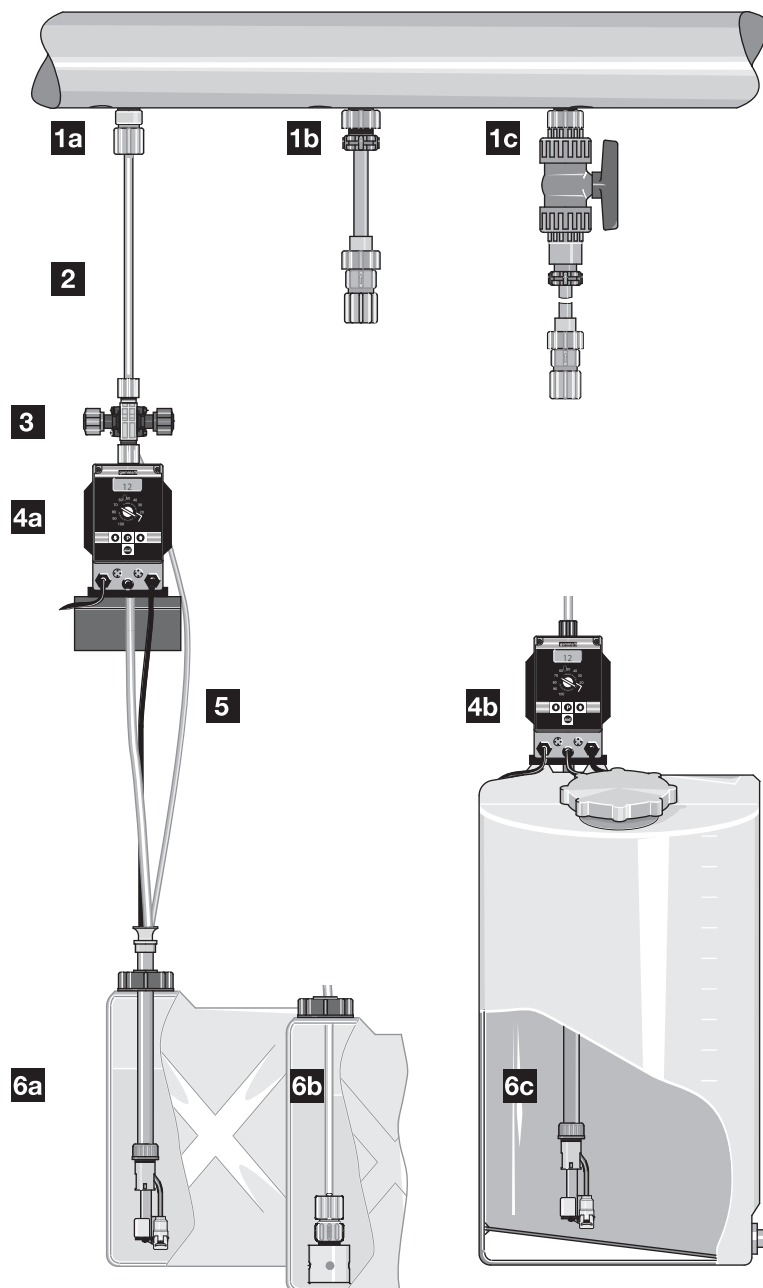
Control of attractions



6 General information on Water Treatment

6.3 Metering Station

- 1a Dosing valve PTFE
- 1b Injection lance PVC
- 1c Injection lance PVC with shut-off cock valve
- 2 Discharge tubing PE
- 3 Multifunction valve
- 4a Metering pump mounted on a wall bracket
- 4b Metering pump mounted on a storage tank
- 5 Return line
- 6a PVC suction lance for 5 - 50 litre one-way valve I
- 6b Suction line with PVC foot valve
- 6c Suction Kit



P_DST_0003_SW

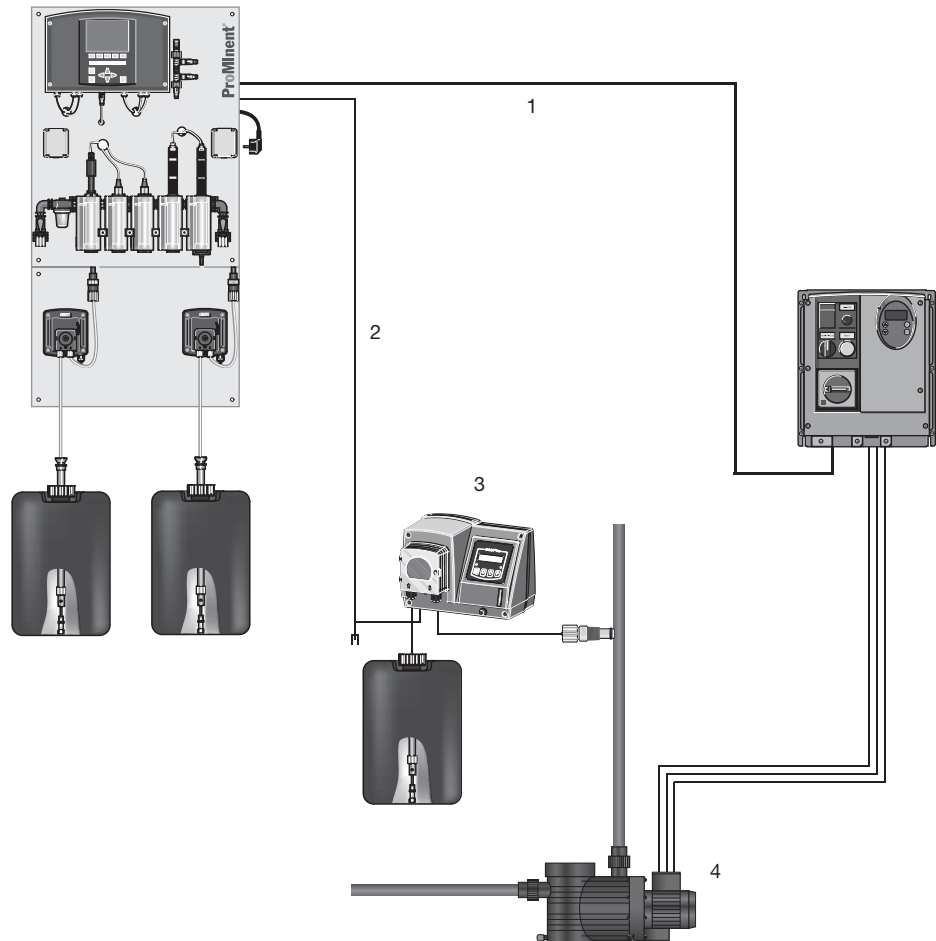
6 General information on Water Treatment

6.4 "Reduction/Energy Saving" Application Solution

	Order no.
DSPaPC7A15A00D303010	–
DF4A flocculant metering	1041585
Frequency converter 2.2 kW	1030687

DULCODOS® Pool PC7 with:

- 1 Analogue 4-20 mA signal
- 2 CAN bus line
- 3 DF4 CAN bus
- 4 Circulating pump for pools



SBK_Applikationsloesung_Absenkung



6 General information on Water Treatment

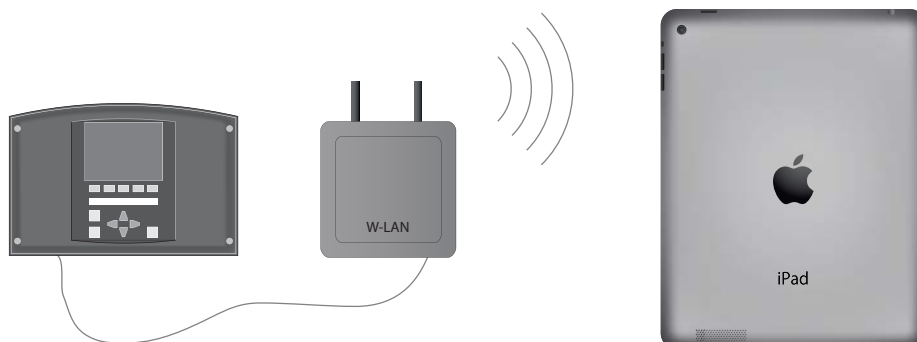
6.5

"WLAN Remote Display" Application Solution

You can use the "Remote display" package to check the status of all of your pools via a wireless connection to your iPad at any time. You also have the option of calibrating sensors for pH, redox, temperature and chlorine from your iPad, thereby significantly simplifying calibration. 1-point calibration can also be performed at the pool edge in the event of favourable pool hydraulics or if your plant room is difficult to access. The last 5 test results are saved and the display of the possible potential savings with setback is also retained.

The "PoolDiary" can be downloaded from the Apple Store to record your data and operating statuses, enabling you to detect all key parameters and evaluate them using the program.

	Order no.
DSPaPC7015A00D303010	–
DF4A flocculant metering	1041585
iPad package (incl. access point set-up and documentation in pdf format) 16 GB + 3G	DKFA

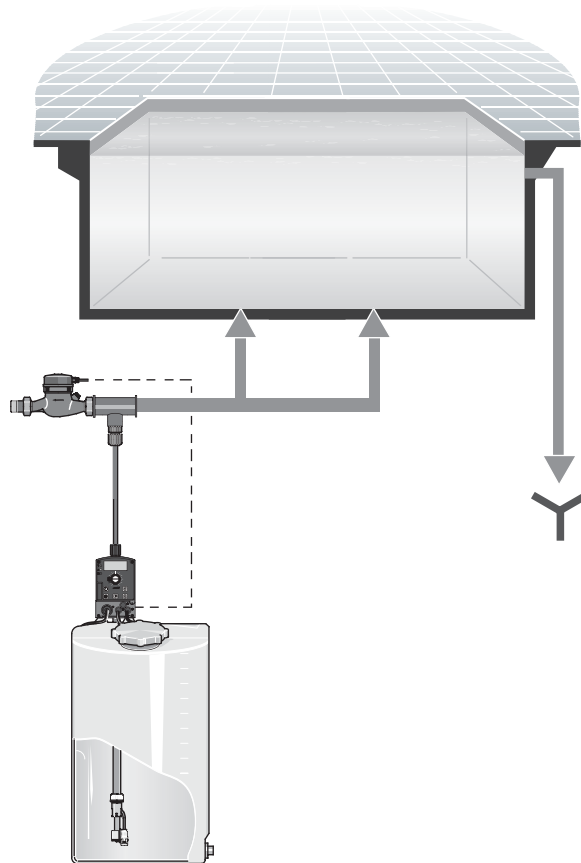


SBK_Applikationsloesung_iPad

6 General information on Water Treatment

6.6 "Immersion Tank" Application Solution

Metering system for the volume-proportional metering of sodium hypochlorite for immersion tank applications. The metering systems ensure that disinfectant is continually fed to the system when it is operational.



AP_SW_0002_SW

The system comprises:

Quantity		Order no.
1	Solenoid metering pump gamma/ L GALA1602NPB900UA001000	–
1	Contact water meter MNR-K, Q _n 2.5 m³/h, R 3/4, pulse interval 0.1	304434
1	Threaded connector complete with seal for water meter, brass, R 1 G 1 1/4	801322
1	Threaded connector complete with seal for water meter, brass, R 1 G 1 1/4 connection for injection valve G 1/4	359032
1	5-core universal cable, 5-pin round plug	1001301
1	Suction lance for connection to 5-60 litre non-reusable tank with 2 m long suction hose (6/4 mm)	802077
5 m	PTFE hose 6x4 mm	037426
1	PVC injection valve, connector 6 mm - G 1/4" for PE/PTFE line	914559
1	Wall brackets, complete, for ProMinent metering pumps gamma/ L, Beta®	810164



6 General information on Water Treatment

Calculation example for designing immersion tank metering using a "contact water meter":

Requirement	Value
Water throughput	1 m ³ /h
Pulse interval	1 l/contact
Metering concentration	1.2 ‰
Stroke volume	0.15 ml/stroke
Stroke length	30 ‰ (corresponds to 0.05 ml/stroke)
Capacity	0.05 l/h (effective)
ACTUAL stroke rate	16.67 strokes / min. (max. permissible 180 strokes/min.)
Metering proportion	0.54 mg/l

Setting/Display data:

Metering proportion	0.54 mg/l
Concentration	1.2 ‰
Stroke volume	0.05 ml/stroke
Capacity	0.05 l/h
Pulse interval	1 l
Service life (storage tank) with 24 h operation	32.41 days with a 35 l storage tank



7 ProMinent® Chemical Resistance List

Resistance of Materials Used in Liquid Ends to the Chemicals Most Frequently Used

The data apply to standard conditions (20 °C, 1,013 mbar).

s	=	saturated solution in water
+	=	resistant
+/o	=	largely resistant
o	=	conditionally resistant
-	=	not resistant
n	=	resistance not known
=>	=	see
*	=	For bonded connections, the resistance of the adhesive (e.g. Tangit) is to be considered. (Materials of the types 'o' and '-' are not recommended !)
**	=	does not apply to glass fibre reinforced material

Concentration data are stated in weight percent, referred to aqueous solutions. If percentages are stated for the level of resistance, this level of resistance is only valid up to this concentration.

NOTE:

The elastomers **CSM (Hypalon®)** and **IIR (butyl rubber)** used as diaphragm materials in pulsation dampers have properties similar to **EPDM**.

PTFE is resistant to all chemicals in this list.

PTFE filled with carbon, however, is attacked by strong oxidants such as bromine (anhydrous) or concentrated acids (phosphoric acid, sulphuric acid, chromic acid).

The resistance of PVC-U adhesive joints with Tangit deviates from the list below with regard to the following chemicals:

Medium	Concentration range
Sulfochromic acid	≥ 70 % H ₂ SO ₄ + 5 % K ₂ Cr ₂ O ₇ /Na ₂ Cr ₂ O ₇
Chromic acid	≥ 10 % CrO ₃
Hydrochloric acid	≥ 25 % HCl
Hydrogen peroxide	≥ 5 % H ₂ O ₂
Hydrofluoric acid	≥ 0 % HF

Explanation of abbreviations used as column headings:

Acrylic:	Acrylic resistance
PVC:	PVC, rigid, (PVC-U) resistance
PP:	Polypropylene resistance
PVDF:	PVDF resistance
1.4404:	Stainless steel 1.4404 & 1.4571 resistance
FKM:	Fluorine Rubber (e.g. Viton® A & B) resistance
EPDM:	Ethylene-Propylene-Dien-rubber resistance
Tygon:	Tygon® R-3603 resistance
Pharmed:	Pharmed® resistance
PE:	Polyethylene resistance
2.4819:	Hastelloy C-276 resistance
WGK:	water endangering class

Viton® is a registered trademark of DuPont Dow Elastomers

Water endangering classes (WGK):

1	=	slightly hazardous to water
2	=	hazardous to water
3	=	severely hazardous to water
(X)	=	No classification. Classification according to conclusion by analogy. To be used under reserve.

Safety data sheets

Safety data sheets on our products in a number of different languages are provided on our website.

www.prominent.com/MSDS



7 ProMinent® Chemical Resistance List

The data has been taken from relevant manufacturer's documentation and our own tests. Resistance of materials is also dependant on other factors, e.g. operating conditions, conditions of surfaces etc, and so this list must be treated as an initial guide only. It cannot claim to offer any guarantees. It should be taken into consideration in particular that usual dosing media are compounds, and their corrosiveness cannot be deducted simply by adding the corrosiveness of each single component. In such cases the chemical producers' data of the material compatibility are to be considered as a matter of prime importance for the material choice. A safety data sheet does not give this data and therefore cannot take the place of the technical documentation on the application.

Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Acetaldehyde	CH ₃ CHO	100%	-	-	o	-	+	-	+/-	-	-	+	+	2
Acetamide	CH ₃ CONH ₂	s	+	+	+	+	+	o	+	-	+/-	+	+	1
Acetic Acid	CH ₃ COOH	100%	-	50%	+	+	+	-	o	60%	60%	70%	+	1
Acetic Anhydride	(CH ₃ CO) ₂ O	100%	-	-	o	-	+	-	+/-	-	+	o	+	1
Acetic Ether => Ethyl Acetate														
Acetone	CH ₃ COCH ₃	100%	-	-	+	-	+	-	+	-	-	+	+	1
Acetophenone	C ₆ H ₅ COCH ₃	100%	-	n	+	-	+	-	+	n	n	+	+	
Acetyl Chloride	CH ₃ COCl	100%	-	+	n	-	o	+	-	-	o	n	+	1
Acetylacetone	CH ₃ COCH ₂ COCH ₃	100%	-	-	+	-	+	-	+	n	n	+	+	1
Acetylene Dichloride => Dichloro Ethylene														
Acetylene Tetrachloride => Tetrachloro Ethane														
Acrylonitril	CH ₂ =CH-CN	100%	-	-	+	+	+	-	-	-	-	+	+	3
Adipic Acid	HOOC(CH ₂) ₄ COOH	s	+	+	+	+	+	+	+	-	+/-	+	+	1
Allyl Alcohol	CH ₂ CHCH ₂ OH	96%	-	o	+	+	+	-	+	-	o	+	+/-	2
Aluminium Acetate	Al(CH ₃ COO) ₃	s	+	+	+	+	+	+	+	+	+	+	+/-	1
Aluminium Bromide	AlBr ₃	s	+	+	+	+	n	+	+	+	+	+	+	2
Aluminium Chloride	AlCl ₃	s	+	+	+	+	-	+	+	+	+	+	+	1
Aluminium Fluoride	AlF ₃	10%	+	+	+	+	-	+	+	+	+	+	+/-	1
Aluminium Hydroxide	Al(OH) ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Aluminium Nitrate	Al(NO ₃) ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Aluminium Phosphate	AlPO ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Aluminium Sulphate	Al ₂ (SO ₄) ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Ammonium Acetate	CH ₃ COONH ₄	s	+	+/-	+	+	+	+	+	+	+	+	+	1
Ammonium Bicarbonate	NH ₄ HCO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Ammonium Carbonate	(NH ₄) ₂ CO ₃	40%	+	+	+	+	+	+	+	+	+	+	+	1
Ammonium Chloride	NH ₄ Cl	s	+	+	+	+	-	+	+	+	+	+	+/-	1
Ammonium Fluoride	NH ₄ F	s	+	o	+	+	o	+	+	+	+	+	+	1
Ammonium Hydroxide	"NH ₄ OH"	30%	+	+	+	+	+	-	+	+	+	+	+	2
(25 °C)														
Ammonium Nitrate	NH ₄ NO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Ammonium Oxalate	(COONH ₄) ₂ * H ₂ O	s	+	+	+	+	+	+	+	+	+	+	+	1
Ammonium Perchlorate	NH ₄ ClO ₄	10%	+	+	+	+	+	+	+	+	+	+	+	1
Ammonium Peroxodisulphate	(NH ₄) ₂ S ₂ O ₈	s	+	+	+	+	5%	+	+	+	+	+	5%	2
Ammonium Phosphate	(NH ₄) ₃ PO ₄	s	+	+	+	+	10%	+	+	+	+	+	10%	1
Ammonium Sulphate	(NH ₄) ₂ SO ₄	s	+	+	+	+	10%	+	+	+	+	+	10%	1
Ammonium Sulphide	(NH ₄) ₂ S	s	+	+	+	+	n	+	+	n	n	+	n	2
Ammoniumaluminium Sulphate	NH ₄ Al(SO ₄) ₂	s	+	+	+	+	+	+	+	+	+	+	+	1
Amyl Alcohol	C ₅ H ₁₁ OH	100%	+	+	+	+	+	-	+	-	-	+	+	1
Aniline	C ₆ H ₅ NH ₂	100%	-	-	+	+	+	-	+/-	-	o	+	+	2
Aniline Hydrochloride	C ₆ H ₅ NH ₂ * HCl	s	n	+	+	+	-	+/-	+/-	-	o	+	+	2
Antimony Trichloride	SbCl ₃	s	+	+	+	+	-	+	+	+	+	+	n	2
Aqua Regia	3 HCl + HNO ₃	100%	-	+	-	+	-	-	o	-	-	-	-	2
Arsenic Acid	H ₃ AsO ₄	s	+	+	+	+	+	+	+	20%	o	+	+	3
Barium Carbonate	BaCO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Barium Chloride	BaCl ₂	s	+	+	+	+	-	+	+	+	+	+	+	1
Barium Hydroxide	Ba(OH) ₂	s	+	+	+	+	+	+	+	+	+	+	+	1
Barium Nitrate	Ba(NO ₃) ₂	s	+	+	+	+	+	+	+	+	+	+	+	1
Barium Sulphate	BaSO ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Barium Sulphide	BaS	s	+	+	+	+	+	+	+	+	+	+	+	(1)
Benzaldehyde	C ₆ H ₅ CHO	100%	-	-	+	-	+	+	+	-	-	o	+	1
Benzene	C ₆ H ₆	100%	-	-	o	+	+	o	-	-	-	o	+	3
Benzene Sulphonic Acid	C ₆ H ₅ SO ₃ H	10%	n	n	+	+	+	+	-	-	-	n	+	2
Benzoic Acid	C ₆ H ₅ COOH	s	+	+	+	+	+	+	+	-	+/-	+	+	1
Benzoyl Chloride	C ₆ H ₅ COCl	100%	-	n	o	n	o	+	+	n	n	o	+	2



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Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Benzyl Alcohol	$C_6H_5CH_2OH$	100%	-	-	+	+	+	+	-	-	+	+	+	1
Benzyl Benzoate	$C_6H_5COOC_6H_5$	100%	-	-	+	o	+	+	-	-	-	+	+	2
Benzyl Chloride	$C_6H_5CH_2Cl$	90%	-	n	o	+	+	+	-	-	-	o	+	2
Bitter Salt => Magnesium Sulphate														
Bleach => Sodium Hypochlorite														
Blue Vitriol => Copper Sulphate														
Borax => Sodium Tetraborate														
Boric Acid	H_3BO_3	s	+	+	+	+	+	+	+	+	+	+	+	1
Brine		s	+	+/o	+	+	+/o	+	+	+	+	+	+	1
Bromine (dry)	Br_2	100%	-	-	-	+	-	-	-	-	-	-	+	2
Bromine Water	$Br_2 + H_2O$	s	-	+	-	+	-	-	n	n	-	n	(2)	
Bromo Benzene	C_6H_5Br	100%	n	n	o	+	+	o	-	-	-	o	+	2
Bromochloro Methane	CH_2BrCl	100%	-	-	-	+	+	n	+/o	-	-	o	+	2
Bromochlorotrifluoro Ethane	$HCClBrCF_3$	100%	-	-	o	+	+	+	-	+	+	o	+	(3)
Butanediol	HOC_4H_8OH	10%	n	+	+	+	+	o	+	+	+	+	+	1
Butanetriol	$C_4H_{10}O_3$	s	+	+	+	+	+	o	+	+	+	+	+	1
Butanol	C_4H_9OH	100%	-	+	+	+	+	o	+/o	-	-	+	+	1
Butyl Acetate	$C_7H_{13}O_2$	100%	-	-	+	+	+	-	-	-	+/o	+	+	1
Butyl Acetate	$CH_3COOC_4H_9$	100%	-	-	o	+	+	-	+/o	-	+/o	-	+	1
Butyl Alcohol => Butanol														
Butyl Amine	$C_4H_9NH_2$	100%	n	n	n	-	+	-	-	n	n	+	+	1
Butyl Benzoate	$C_6H_5COOC_4H_9$	100%	-	-	o	n	+	+	+	-	-	o	+	2
Butyl Mercaptane	C_4H_9SH	100%	n	n	n	+	n	+	-	n	n	n	n	3
Butyl Oleate	$C_{22}H_{42}O_2$	100%	n	n	n	+	+	+	+/o	n	n	n	+	1
Butyl Stearate	$C_{22}H_{44}O_2$	100%	o	n	n	+	+	+	-	n	n	n	+	1
Butyraldehyde	C_3H_7CHO	100%	-	n	+	n	+	-	+/o	-	-	+	+	1
Butyric Acid	C_3H_7COOH	100%	5%	20%	+	+	+	+	+	-	+/o	+	+	1
Calcium Acetate	$(CH_3COO)_2Ca$	s	+	+	+	+	+	+	+	+	+	+	+	1
Calcium Bisulphite	$Ca(HSO_3)_2$	s	+	+	+	+	+	+	+	+	+	+	+	(1)
Calcium Carbonate	$CaCO_3$	s	+	+	+	+	+	+	+	+	+	+	+	1
Calcium Chloride	$CaCl_2$	s	+	+	+	+	-	+	+	+	+	+	+	1
Calcium Cyanide	$Ca(CN)_2$	s	+	+	+	+	n	+	+	+	+	+	n	3
Calcium Hydroxide	$Ca(OH)_2$	s	+	+	+	+	+	+	+	+	+	+	+	1
Calcium Hypochlorite	$Ca(OCl)_2$	s	+	+	o	+	-	o	+	+	+	+	+	2
Calcium Nitrate	$Ca(NO_3)_2$	s	+	50%	50%	+	+	+	+	+	+	+	+	1
Calcium Phosphate	$Ca_3(PO_4)_2$	s	+	+	+	+	+	+	+	+	+	+	+	1
Calcium Sulphate	$CaSO_4$	s	+	+	+	+	+	+	+	+	+	+	+	1
Calcium Sulphide	CaS	s	+	+	+	+	n	+	+	+	+	+	+	(2)
Calcium Sulphite	$CaSO_3$	s	+	+	+	+	+	+	+	+	+	+	+	(1)
Calcium Thiosulphate	CaS_2O_3	s	+	+	+	+	-	+	+	+	+	+	+	1
Carbolic Acid => Phenole														
Carbon Disulphide	CS_2	100%	-	-	o	+	+	+	-	-	-	o	+	2
Carbon Tetrachloride	CCl_4	100%	-	-	-	+	+	+	-	-	-	o	+	3
Carbonic Acid	" H_2CO_3 "	s	+	+	+	+	+	+	+	+	+	+	+	1
Caustic Potash => Potassium Hydroxide														
Caustic Soda => Sodium Hydroxide														
Chloric Acid	$HClO_3$	20%	+	+	-	+	-	o	o	+	+	10%	+	2
Chlorinated Lime => Calcium Hypochlorite														
Chlorine Dioxide Solution	$ClO_2 + H_2O$	0.5%	o	+	o	+	-	o	-	o	-	o	+	
Chlorine Water	$Cl_2 + H_2O$	s	+	+	o	+	-	+	+	o	-	o	+	
Chloro Benzene	C_6H_5Cl	100%	-	-	+	+	+	+	-	-	-	o	+	2
Chloro Ethanol	$ClCH_2CH_2OH$	100%	-	-	+	o	+	-	o	-	+	+	+	3
Chloro Ethylbenzene	$C_6H_4ClC_2H_5$	100%	-	-	o	n	+	o	-	-	-	o	+	(2)
Chloro Phenole	C_6H_4OHCl	100%	-	n	+	+	+	n	-	-	-	+	+	2
Chloro Toluene	C_7H_8Cl	100%	-	-	n	+	+	+	-	-	-	n	+	2
Chloroacetone	$ClCH_2COCH_3$	100%	-	-	n	n	+	-	+	-	-	n	+	3
Chlorobutadiene	C_4H_5Cl	100%	-	-	n	n	+	+	-	-	-	n	+	1
Chloroform	$CHCl_3$	100%	-	-	o	+	+	+	-	-	o	-	+	2
Chlorohydrin	C_3H_5OCl	100%	-	n	+	-	+	+	o	-	+	+	+	3
Chloroprene => Chlorobutadiene														
Chlorosulphonic Acid	$SO_2(OH)Cl$	100%	-	o	-	+	-	-	-	-	-	-	o	1
Chrome-alum => Potassium Chrome Sulphate														
Chromic Acid	H_2CrO_4	50%	-	+	o	+	10%	+	-	o	o	+	10%	3



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Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Chromic-Sulphuric Acid	$K_2CrO_4 + H_2SO_4$	s	-	+	-	+	n	n	n	-	-	-	n	3
Chromium Sulphate	$Cr_2(SO_4)_3$	s	+	+	+	+	+	+	+	+	+	+	+	1
Citric Acid	$C_6H_8O_7$	s	+	+	+	+	+	+	+	+	+	+	+	1
Cobalt Chloride	$CoCl_2$	s	+	+	+	+	-	+	+	+	+	+	+	2
Copper-II-Acetate	$Cu(CH_3COO)_2$	s	+	+	+	+	+	+	+	+	+	+	+	3
Copper-II-Arsenite	$Cu_3(AsO_3)_2$	s	+	+	+	+	+	+	+	+	+	+	+	3
Copper-II-Carbonate	$CuCO_3$	s	+	+	+	+	+	+	+	+	+	+	+	2
Copper-II-Chloride	$CuCl_2$	s	+	+	+	+	1%	+	+	+	+	+	+	2
Copper-II-Cyanide	$Cu(CN)_2$	s	+	+	+	+	+	+	+	+	+	+	+	(3)
Copper-II-Fluoride	CuF_2	s	+	+	+	+	+	+	+	+	+	+	+	(2)
Copper-II-Nitrate	$Cu(NO_3)_2$	s	+	+	+	+	+	+	+	+	+	+	+/o	2
Copper-II-Sulphate	$CuSO_4$	s	+	+	+	+	+	+	+	+	+	+	+	2
Cresols	$C_6H_4CH_3OH$	100%	o	o	+	+	+	+	-	-	-	+	+	2
Crotonaldehyde	$CH_3C_2H_2CHO$	100%	n	-	+	+	+	-	+	-	-	+	+	3
Cubic Nitre => Sodium Nitrate														
Cumene => Isopropyl Benzene														
Cyclo Hexane	C_6H_{12}	100%	+	-	+	+	+	+	-	-	-	+	o	1
Cyclohexanole	$C_6H_{11}OH$	100%	o	+/o	+	+	+	+	-	-	-	+	+	1
Cyclohexanone	$C_6H_{10}O$	100%	-	-	+	-	+	-	+/o	-	-	+	+	1
Cyclohexyl Alcohol => Cyclohexanol														
Cyclohexylamine	$C_6H_{11}NH_2$	100%	n	n	n	n	+	-	n	n	n	n	+	2
Decahydronaphthalene	$C_{10}H_{18}$	100%	-	+/o	o	+	n	o	-	-	-	o	+	2
Decaline => Decahydronaphthalene														
Dextrose => Glucose														
Diacetonalcohol	$C_6H_{12}O_2$	100%	-	-	+	o	+	-	+	-	-	+	+	1
Dibromoethane	$C_2H_4Br_2$	100%	-	-	n	+	+	+	-	-	-	-	+	3
Dibutyl Ether	$C_4H_9OC_4H_9$	100%	-	-	+	+	+	-	o	-	-	+	+	2
Dibutyl Phthalate	$C_{16}H_{22}O_4$	100%	-	-	+	+	+	+	+/o	o	+	o	+	2
Dibutylamine	$(C_4H_9)_2NH$	100%	n	n	+	+	+	-	-	n	n	+	+	1
Dichloro Acetic Acid	$Cl_2CHCOOH$	100%	-	+	+	+	+	-	+	-	o	+	+	1
Dichloro Benzene	$C_6H_4Cl_2$	100%	-	-	o	+	+	+	-	-	-	o	+	2
Dichloro Butan	$C_4H_8Cl_2$	100%	-	-	o	+	+	+	-	-	-	o	+	3
Dichloro Butene	$C_4H_6Cl_2$	100%	-	-	o	+	+	o	-	-	-	o	+	3
Dichloro Ethane	$C_2H_4Cl_2$	100%	-	-	o	+	+	+	-	-	o	-	+	3
Dichloro Ethylene	$C_2H_2Cl_2$	100%	-	-	o	+	+	o	-	-	o	-	+	2
Dichloro Methane	CH_2Cl_2	100%	-	-	o	o	o	+	-	-	o	-	+	2
Dichloroisopropyl Ether	$(C_3H_6Cl)_2O$	100%	-	-	o	n	+	o	o	-	-	o	+	(2)
Dicyclohexylamine	$(C_6H_{12})_2NH$	100%	-	-	o	n	+	-	-	-	-	o	+	2
Diethyleneglycol	$C_4H_{10}O_3$	s	+	+	+	+	+	+	+	+	+	+	+	1
Diethyleneglycolethyl Ether	$C_8H_{18}O_3$	100%	n	n	+	+	+	n	+/o	-	o	+	+	1
Diethylether	$C_2H_5OC_2H_5$	100%	-	-	o	+	+	-	-	-	o	o	+	1
Diglycolic Acid	$C_4H_6O_5$	30%	+	+	+	+	+	+	n	+	+/o	+	+	3
Dihexyl Phthalate	$C_{20}H_{26}O_4$	100%	-	-	+	+	+	-	n	o	+	+	+	(1)
Diisobutylketone	$C_9H_{18}O$	100%	-	-	+	+	+	-	+	-	-	+	+	1
Di-iso-nonyl Phthalate	$C_{26}H_{42}O_4$	100%	-	-	+	+	+	n	n	o	+	+	+	1
Diisopropylketone	$C_7H_{14}O$	100%	-	-	+	+	+	-	+	-	-	+	+	1
Dimethyl Carbonate	$(CH_3O)_2CO$	100%	n	n	+	+	+	+	-	n	n	+	+	1
Dimethyl Ketone => Acetone														
Dimethyl Phthalate	$C_{10}H_{10}O_4$	100%	-	-	+	+	+	-	+/o	o	+	+	+	1
Dimethylformamide	$HCON(CH_3)_2$	100%	-	-	+	-	+	-	+	-	+/o	+	+	1
Dimethylhydrazine	$H_2NN(CH_3)_2$	100%	n	n	+	n	+	-	+	n	n	+	+	3
Diocetyl Phthalate	$C_{44}H_{88}O_4$	100%	-	-	+	+	+	-	+/o	o	+	+	+	1
Dioxane	$C_4H_8O_2$	100%	-	-	o	-	+	-	+/o	-	-	+	+	1
Disodium Hydrogenphosphate	Na_2HPO_4	s	+	+	+	+	+	+	+	+	+	+	+	1
Disulfur Acid -- Oleum														
Disulphur Dichloride	S_2Cl_2	100%	n	n	n	+	n	+	-	-	-	n	n	
DMF => Dimethylformamide														
Engine Oils		100 %	n	+/o	+	+	+	+	-	-	-	+	+	2
Epsom salts => Magnesium Sulphate														
Ethanol	C_2H_5OH	100%	-	+	+	+	+	-	+	-	+	+	+	1
Ethanol Amine	$HOC_2H_4NH_2$	100%	o	n	+	-	+	-	+/o	-	o	+	+	1
Ethyl Acetate	$CH_3COOC_2H_5$	100%	-	-	35%	+	+	-	+/o	-	+/o	+	+	1
Ethyl Acrylate	$C_2H_3COOC_2H_5$	100%	-	-	+	o	+	-	+/o	-	-	+	+	2



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Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Ethyl Benzene	$C_6H_5-C_2H_5$	100%	-	-	o	+	+	o	-	-	-	o	+	1
Ethyl Benzoate	$C_6H_5COOC_2H_5$	100%	n	-	+	o	+	+	-	-	-	+	+	1
Ethyl Bromide	C_2H_5Br	100%	-	n	+	+	n	+	-	-	o	+	+	2
Ethyl Chloroacetate	$ClCH_2COOC_2H_5$	100%	-	o	+	+	+	+	-	-	-	+	+	2
Ethyl Chlorocarbonate	$ClCO_2C_2H_5$	100%	n	n	n	n	n	+	-	n	n	n	n	(2)
Ethyl Cyclopentane	$C_5H_4C_2H_5$	100%	+	+	+	+	+	+	-	-	-	+	+	(1)
Ethylacetacetate	$C_6H_{10}O_3$	100%	n	-	+	+	+	-	+/o	-	+/o	+	+	1
Ethylacrylic Acid	C_4H_7COOH	100%	n	n	+	+	+	n	+/o	n	n	+	+	(1)
Ethylene Diamine	$(CH_2NH_2)_2$	100%	o	o	+	-	o	-	+	n	n	+	o	2
Ethylene Dibromide => Dibromoethane														
Ethylene Dichloride => Dichloro Ethane														
Ethylene Glycol => Glycol														
Ethylenglycol Ethylether	$HOC_2H_4OC_2H_5$	100%	n	n	+	+	+	n	+/o	-	o	+	+	1
Ethylhexanol	$C_8H_{16}O$	100%	n	+/o	+	+	+	+	+	-	-	+	+	2
Fatty Acids	$R-COOH$	100%	+	+	+	+	+	+	o	-	o	+	+	1
Ferric Chloride	$FeCl_3$	s	+	+	+	+	-	+	+	+	+	+	+/o	1
Ferric Nitrate	$Fe(NO_3)_3$	s	+	+	+	+	+	+	+	+	+	+	+	1
Ferric Phosphate	$FePO_4$	s	+	+	+	+	+	+	+	+	+	+	+	1
Ferric Sulphate	$Fe_2(SO_4)_3$	s	+	+	+	+	o	+	+	+	+	+	+	1
Ferrous Chloride	$FeCl_2$	s	+	+	+	+	-	+	+	+	+	+	+/o	1
Ferrous Sulphate	$FeSO_4$	s	+	+	+	+	+	+	+	+	+	+	+	1
Fixing Salt => Sodium Thiosulphate														
Fluoro Benzene	C_6H_5F	100%	-	-	+	+	+	o	-	-	-	o	+	2
Fluoroboric Acid	HBF_4	35%	+	+	+	+	o	+	+	+	-	+	+	1
Fluorosilicic Acid	H_2SiF_6	100%	+	30%	30%	+	o	+	+	25%	o	40%	+/o	2
Formaldehyde	CH_2O	40%	+	+	+	+	+	-	+/o	-	-	+	+	2
Formalin => Formaldehyde														
Formamide	$HCONH_2$	100%	+	-	+	+	+	+	+	n	n	+	+	1
Formic Acid	$HCOOH$	s	-	+/o	+	+	+	-	-	+/o	+/o	+	+	1
Furane	C_4H_4O	100%	-	-	+	-	+	-	n	-	-	+	+	3
Furane Aldehyde	$C_5H_4O_2$	100%	n	n	n	o	+	-	+/o	-	-	n	n	2
Furfuryl Alcohol	$OC_4H_3CH_2OH$	100%	-	-	+	o	+	n	+/o	-	-	+	+	1
Gallic Acid	$C_6H_2(OH)_3COOH$	5%	+	+	+	+	+	+	+/o	+	+	+	+	1
Gasoline		100 %	-	-	+	+	+	+	-	-	-	+	+	2
Glauber's Salt => Sodium Sulphate														
Glucose	$C_6H_{12}O_6$	s	+	+	+	+	+	+	+	+	+	+	+	1
Glycerol	$C_3H_5(OH)_3$	100%	+	+	+	+	+	+	+	+	+	+	+	1
Glycerol Triacetate	$C_3H_5(CH_3COO)_3$	100%	n	n	+	+	+	-	+	n	n	+	+	1
Glycine	NH_2CH_2COOH	10%	+	+	+	+	+	+	+	+	+	+	+	1
Glycol	$C_2H_4(OH)_2$	100%	+	+	+	+	+	+	+	+	+	+	+	1
Glycolic Acid	$CH_2OHCOOH$	70%	+	37%	+	+	+	+	+	+	+/o	+	+	1
Gypsum => Calcium Sulphate														
Heptane	C_7H_{16}	100%	+	+	+	+	+	+	-	-	-	+	+	1
Hexachloroplatinic Acid	H_2PtCl_6	s	n	+	+	+	-	n	+	n	n	+	-	
Hexanal	$C_5H_{11}CHO$	100%	n	n	+	+	+	-	+/o	-	-	+	+	1
Hexane	C_6H_{14}	100%	+	+	+	+	+	+	-	-	-	+	+	1
Hexanol	$C_6H_{13}OH$	100%	-	-	+	+	+	n	+	-	o	+	+	1
Hexantriol	$C_6H_9(OH)_3$	100%	n	n	+	+	+	+	+	n	n	+	+	1
Hexene	C_6H_{12}	100%	n	+	+	+	+	+	-	-	-	+	+	1
Hydrazine Hydrate	$N_2H_4 \cdot H_2O$	s	+	+	+	+	+	n	+	-	o	+	+	3
Hydrobromic Acid	HBr	50%	+	+	+	+	-	-	+	+	-	+	o	1
Hydrochloric Acid	HCl	38%	32%	+	+	+	-	+	o	+	o	+	o	1
Hydrofluoric Acid	HF	80%	-	40% *	40% **	+	-	+	o	40%	-	40%	+/o	1
Hydrogen Cyanide	HCN	s	+	+	+	+	+	+	+	+	+	+	+	3
Hydrogen Peroxide	H_2O_2	90%	40%	40%*	30%	+	+	30%	30%	30%	+	+	+	1
Hydroiodic Acid	HI	s	+	+	+	+	-	-	n	+	-	+	n	1
Hydroquinone	$C_6H_4(OH)_2$	s	o	+	+	+	+	+	-	+	+/o	+	+	2
Hydroxylamine Sulphate	$(NH_2OH)_2 \cdot H_2SO_4$	10%	+	+	+	+	+	+	+	+	+	+	+	2
Hypochlorous Acid	$HOCl$	s	+	+	o	+	-	+	+/o	+	+	o	+	(1)
Iodine	I_2	s	o	-	+	+	-	+	+/o	+	+	o	+/o	
Iron Vitriol => Ferrous Sulphate														
Isobutanol => Isobutyl Alcohol														
Isobutyl Alcohol	$C_2H_5CH(OH)CH_3$	100%	-	+	+	+	+	+	+	-	o	+	+	1



7 ProMinent® Chemical Resistance List

Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Isopropanol => Isopropyl Alcohol														
Isopropyl Acetate	$\text{CH}_3\text{COOCH}(\text{CH}_3)_2$	100%	-	-	+	+	+	-	+/-	-	+/-	+	+	1
Isopropyl Alcohol	$(\text{CH}_3)_2\text{CHOH}$	100%	-	+/-	+	+	+	+	+	-	o	+	+	1
Isopropyl Benzene	$\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)_2$	100%	-	-	o	+	+	+	-	-	-	o	+	1
Isopropyl Chloride	$\text{CH}_3\text{CHClCH}_3$	80%	-	-	o	+	+	+	-	-	o	o	+/-	2
Isopropyl Ether	$\text{C}_6\text{H}_{14}\text{O}$	100%	-	-	o	+	+	-	-	-	o	o	+	1
Kitchen Salt => Sodium Chloride														
Lactic Acid	$\text{C}_3\text{H}_6\text{O}_3$	100%	-	+	+	+	+/-	+	10%	-	+/-	+	+	1
Lead Acetate	$\text{Pb}(\text{CH}_3\text{COO})_2$	s	+	+	+	+	+	+	+	+	+	+	+	2
Lead Nitrate	$\text{Pb}(\text{NO}_3)_2$	50%	+	+	+	+	+	+	+	+	+	+	+	2
Lead Sugar => Lead Acetate														
Lead Sulphate	PbSO_4	s	+	+	+	+	+	+	+	+	+	+	+	(2)
Lead Tetraethyl	$\text{Pb}(\text{C}_2\text{H}_5)_4$	100%	+	+	+	+	+	+	-	n	n	+	+	3
Lime Milk => Calcium Hydroxide														
Liquid Ammonia => Ammonium Hydroxide														
Lithium Bromide	LiBr	s	+	+	+	+	+	+	+	+	+	+	+	1
Lithium Chloride	LiCl	s	+	+	+	+	-	+	+	+	+	+	n	1
Lunar Caustic => Silver Nitrate														
Magnesium Carbonate	MgCO_3	s	+	+	+	+	+	+	+	+	+	+	+/-	1
Magnesium Chloride	MgCl_2	s	+	+	+	+	o	+	+	+	+	+	+	1
Magnesium Hydroxide	$\text{Mg}(\text{OH})_2$	s	+	+	+	+	+	+	+	+	+	+	+	1
Magnesium Nitrate	$\text{Mg}(\text{NO}_3)_2$	s	+	+	+	+	+	+	+	+	+	+	+	1
Magnesium Sulphate	MgSO_4	s	+	+	+	+	+	+	+	+	+	+	+/-	1
Maleic Acid	$\text{C}_4\text{H}_4\text{O}_4$	s	+	+	+	+	+	+	+	-	o	+	+	1
Malic Acid	$\text{C}_4\text{H}_6\text{O}_5$	s	+	+	+	+	+	+	+	+	+	+	+	1
Manganese-II-Chloride	MnCl_2	s	+	+	+	+	-	+	+	+	+	+	+	1
Manganese-II-Sulphate	MnSO_4	s	+	+	+	+	+	+	+	+	+	+	+	1
MEK => Methyl Ethyl Ketone														
Mercury	Hg	100%	+	+	+	+	+	+	+	+	+	+	+	3
Mercury-II-Chloride	HgCl_2	s	+	+	+	+	-	+	+	+	+	+	+	3
Mercury-II-Cyanide	$\text{Hg}(\text{CN})_2$	s	+	+	+	+	+	+	+	+	+	+	+	3
Mercury-II-Nitrate	$\text{Hg}(\text{NO}_3)_2$	s	+	+	+	+	+	+	+	+	+	+	+	3
Mesityl Oxide	$\text{C}_6\text{H}_{10}\text{O}$	100%	-	-	n	n	+	-	+/-	-	-	n	+	1
Methacrylic Acid	$\text{C}_3\text{H}_5\text{COOH}$	100%	n	n	+	+	+	o	+/-	-	+/-	+	+	1
Methanol	CH_3OH	100%	-	-	+	+	+	o	+	-	+/-	+	+	1
Methoxybutanol	$\text{CH}_3\text{O}(\text{CH}_2)_4\text{OH}$	100%	-	-	+	+	+	+	o	-	o	+	+	(1)
Methyl Acetate	$\text{CH}_3\text{COOCH}_3$	60%	-	-	+	+	+	-	+/-	-	+/-	+	+	2
Methyl Acrylate	$\text{C}_2\text{H}_3\text{COOCH}_3$	100%	-	-	+	+	+	-	+/-	-	o	+	+	2
Methyl Benzoate	$\text{C}_6\text{H}_5\text{COOCH}_3$	100%	-	-	+	o	+	+	-	-	-	+	+	2
Methyl Catechol	$\text{C}_6\text{H}_3(\text{OH})_2\text{CH}_3$	s	+	+	+	+	+	+	-	+	+/-	+	+	(1)
Methyl Cellulose		s	+	+	+	+	+	+	+	+	+	+	+	1
Methyl Chloroacetate	$\text{ClCH}_2\text{COOCH}_3$	100%	-	o	+	+	+	o	-	-	-	+	+	2
Methyl Cyclopentane	$\text{C}_5\text{H}_9\text{CH}_3$	100%	+	+	+	+	+	+	-	-	-	+	+	(1)
Methyl Dichloroacetate	$\text{Cl}_2\text{CHCOOCH}_3$	100%	-	-	+	n	+	-	n	-	-	+	+	2
Methyl Ethyl Ketone	$\text{CH}_3\text{COC}_2\text{H}_5$	100%	-	-	+	-	+	-	+	-	-	+	+	1
Methyl Glycol	$\text{C}_3\text{H}_8\text{O}_2$	100%	+	+	+	+	+	-	+/-	+	+	+	+	1
Methyl Isobutyl Ketone	$\text{CH}_3\text{COC}_4\text{H}_9$	100%	-	-	+	-	+	-	o	-	-	+	+	1
Methyl Isopropyl Ketone	$\text{CH}_3\text{COC}_3\text{H}_7$	100%	-	-	+	-	+	-	+/-	-	-	+	+	1
Methyl Methacrylate	$\text{C}_3\text{H}_5\text{COOCH}_3$	100%	-	-	+	+	+	-	-	-	-	+	+	1
Methyl Oleate	$\text{C}_{17}\text{H}_{33}\text{COOCH}_3$	100%	n	n	+	+	+	+	+/-	n	n	+	+	1
Methyl Salicylate	$\text{HOC}_6\text{H}_4\text{COOCH}_3$	100%	-	-	+	+	+	n	+/-	-	-	+	+	1
Methylacetyl Acetate	$\text{C}_5\text{H}_8\text{O}_3$	100%	-	-	+	+	+	-	+/-	-	o	+	+	2
Methylamine	CH_3NH_2	32%	+	o	+	o	+	-	+	+	+	+	+	2
Methylene Chloride => Dichloro Methane														
Mirabilite => Sodium Sulphate														
Morpholine	$\text{C}_4\text{H}_9\text{ON}$	100%	-	-	+	-	+	n	n	-	-	+	+	2
Muriatic Acid => Hydrochloric Acid														
Natron => Sodium Bicarbonate														
Nickel-II-Acetate	$(\text{CH}_3\text{COO})_2\text{Ni}$	s	+	+	+	+	+	-	+	+	+	+	+	(2)
Nickel-II-Chloride	NiCl_2	s	+	+	+	+	-	+	+	+	+	+	+	2
Nickel-II-Nitrate	$\text{Ni}(\text{NO}_3)_2$	s	+	+	+	+	+	+	+	+	+	+	+/-	2
Nickel-II-Sulphate	NiSO_4	s	+	+	+	+	+	+	+	+	+	+	+/-	2
Nitrate of Lime => Calcium Nitrate														



7 ProMinent® Chemical Resistance List

Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Nitric Acid	HNO ₃	99%	10%	10%*	50%	65%	50%	65%	10%	35%	35%	50%	65%	1
Nitro Methane	CH ₃ NO ₂	100%	-	-	+	o	+	-	+/-	-	-	+	+	2
Nitro Propane	(CH ₃) ₂ CHNO ₂	100%	-	-	+	n	+	-	+/-	-	-	+	+	2
Nitro Toluene	C ₆ H ₄ NO ₂ CH ₃	100%	-	-	+	+	+	o	-	-	-	+	+	2
Octane	C ₈ H ₁₈	100%	o	+	+	+	+	+	-	-	-	+	+	1
Octanol	C ₈ H ₁₇ OH	100%	-	-	+	+	+	+	+	-	-	+	+	1
Octyl Cresol	C ₁₅ H ₂₄ O	100%	-	-	+	+	+	o	n	-	-	+	+	(1)
Oil => Engine Oils														
Oleum	H ₂ SO ₄ + SO ₃	s	n	-	-	-	+	+	-	+	+	-	+	2
Orthophosphoric Acid => Phosphoric Acid														
Oxalic Acid	(COOH) ₂	s	+	+	+	+	10%	+	+	+/-	+/-	+	+/-	1
Pentane	C ₅ H ₁₂	100%	+	+	+	+	+	+	-	-	-	+	+	1
Pentanol => Amyl Alcohol														
Perchloric Acid	HClO ₄	70%	n	10%	10%	+	-	+	+/-	o	+	+	n	1
Perchloroethylene => Tetrachloro Ethylene														
Perhydrol => Hydrogen Peroxide														
Petroleum Ether	C _n H _{2n+2}	100%	+	+/-	+	+	+	+	-	-	-	+	+	1
Phenole	C ₆ H ₅ OH	100%	-	-	+	+	+	+	-	10%	+	+	+	2
Phenyl Ethyl Ether	C ₆ H ₅ OC ₂ H ₅	100%	-	-	+	n	+	-	-	-	-	+	+	2
Phenyl Hydrazine	C ₆ H ₅ NNH ₂	100%	-	-	o	+	+	o	-	-	-	o	+	2
Phosphoric Acid	H ₃ PO ₄	85%	50%	+	+	+	+	+	+	+	+	+	+	1
Phosphorous Oxychloride	POCl ₃	100%	-	-	+	+	n	+	+	n	n	+	+	1
Phosphorous Trichloride	PCl ₃	100%	-	-	+	+	+	o	+	+	+/-	+	+	1
Phthalic Acid	C ₆ H ₄ (COOH) ₂	s	+	+	+	+	+	+	+	-	+	+	+	1
Picric Acid	C ₆ H ₂ (NO ₃) ₃ OH	s	+	+	+	+	+	+	+	+	-	+	+	2
Piperidine	C ₅ H ₁₁ N	100%	-	-	n	n	+	-	-	-	-	n	+	2
Potash Alum => Potassium Aluminium Sulphate														
Potassium Acetate	CH ₃ COOK	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Aluminium Sulphate	KAl(SO ₄) ₂	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Bicarbonate	KHCO ₃	40%	+	+	+	+	+	+	+	+	+	+	+/-	1
Potassium Bifluoride	KHF ₂	s	n	+	+	+	+	+	+	+	+	+	+	1
Potassium Bisulphate	KHSO ₄	5%	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Bitartrate	KC ₄ H ₅ O ₆	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Borate	KBO ₂	s	+	+	+	+	+	+	+	+	+	+	+	(1)
Potassium Bromate	KBrO ₃	s	+	+	+	+	+	+	+	+	+	+	+	2
Potassium Bromide	KBr	s	+	+	+	+	10%	+	+	+	+	+	0,1	1
Potassium Carbonate	K ₂ CO ₃	s	+	+	+	+	+	+	+	55%	55%	+	+	1
Potassium Chlorate	KClO ₃	s	+	+	+	+	+	+	+	+	+	+	+	2
Potassium Chloride	KCl	s	+	+	+	+	-	+	+	+	+	+	+/-	1
Potassium Chromate	K ₂ CrO ₄	10%	+	+	+	+	+	+	+	+	+	+	+	3
Potassium Chrome Sulphate	KCr(SO ₄) ₂	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Cyanate	KOCN	s	+	+	+	+	+	+	+	+	+	+	+	2
Potassium Cyanide	KCN	s	+	+	+	+	5%	+	+	+	+	+	5%	3
Potassium Cyanoferate II	K ₄ Fe(CN) ₆	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Cyanoferate III	K ₃ Fe(CN) ₆	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Dichromate	K ₂ Cr ₂ O ₇	s	+	+	+	+	25%	+	+	+	+	+	10%	3
Potassium Fluoride	KF	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Hydroxyde	KOH	50%	+	+	+	+	+	-	+	10%	10%	+	+	1
(25 °C)														
Potassium Iodide	KI	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Nitrate	KNO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Perchlorate	KClO ₄	s	+	+	+	+	n	+	+	+	+	+	+	1
Potassium Permanganate	KMnO ₄	s	+	+	+	+	+	+	+	6%	6%	+	+	2
Potassium Persulphate	K ₂ S ₂ O ₈	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Phosphate	KH ₂ PO ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Pyrochromate => Potassium Dichromate														
Potassium Sulphate	K ₂ SO ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Potassium Sulphite	K ₂ SO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Propionic Acid	C ₂ H ₅ COOH	100%	o	+	+	+	+	+	+	-	+/-	+	+	1
Propionitrile	CH ₃ CH ₂ CN	100%	n	n	+	+	+	+	-	-	-	+	+	2
Propyl Acetate	CH ₃ COOC ₃ H ₇	100%	-	-	+	+	+	-	+/-	-	-	+	+	1
Propylene Glycol	CH ₃ CHOHCH ₂ OH	100%	+	+	+	+	+	+	+	+	+	+	+	1
Prussic Acid => Hydrogen Cyanide														
Pyridine	C ₅ H ₅ N	100%	-	-	o	-	+	-	-	-	o	+	+	2



7 ProMinent® Chemical Resistance List

Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Pyrrole	C ₄ H ₄ NH	100%	n	n	+	n	+	-	-	-	-	+	+	2
Roman Vitriol => Copper Sulphate														
Salicylic Acid	HOC ₆ H ₄ COOH	s	+	+	+	+	+	+	+	+	+	+	+/o	1
Salmiac => Ammonium Chloride														
Saltpeter => Potassium Nitrate														
Silic Acid	SiO ₂ * x H ₂ O	s	+	+	+	+	+	+	+	+	+	+	+	1
Silver Bromide	AgBr	s	+	+	+	+	+/o	+	+	+	+	+	+	1
Silver Chloride	AgCl	s	+	+	+	+	-	+	+	+	+	+	+/o	1
Silver Nitrate	AgNO ₃	s	+	+	+	+	+	+	+	+	+	+	+/o	3
Slaked Lime => Calcium Hydroxide														
Soda => Sodium Carbonate														
Sodium Acetate	NaCH ₃ COO	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Benzoate	C ₆ H ₅ COONa	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Bicarbonate	NaHCO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Bisulphate	NaHSO ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Bisulphite	NaHSO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Borate	NaBO ₂	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Bromate	NaBrO ₃	s	+	+	+	+	+	+	+	+	+	+	+	3
Sodium Bromide	NaBr	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Carbonate	Na ₂ CO ₃	s	+	+	+	+	+/o	+	+	+	+	+	+	1
Sodium Chlorate	NaClO ₃	s	+	+	+	+	+	+	+	+	+	+	+	2
Sodium Chloride	NaCl	s	+	+	+	+	-	+	+	+	+	+	+	1
Sodium Chlorite	NaClO ₂	24%	+	+	+	+	10%	+	+	+	+	+	10%	2
Sodium Chromate	Na ₂ CrO ₄	s	+	+	+	+	+	+	+	+	+	+	+	3
Sodium Cyanide	NaCN	s	+	+	+	+	+	+	+	+	+	+	+	3
Sodium Dichromate	Na ₂ Cr ₂ O ₇	s	+	+	+	+	+	+	+	+	+	+	+	3
Sodium Dithionite	Na ₂ S ₂ O ₄	s	+	10%	10%	+	+	n	n	+	+	10%	+/o	1
Sodium Fluoride	NaF	s	+	+	+	+	10%	+	+	+	+	+	+	1
Sodium Hydrogen Sulphate => Sodium Bisulphate														
Sodium Hydroxide	NaOH	50%	+	+	+	+	+	-	+	10%	30%	+	+	1
(60%/ 25 °C)														
Sodium Hypochlorite	NaOCl + NaCl	12%	+	+	o	+	-	+	+	+	+	o	> 10%	2
Sodium Iodide	NaI	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Metaphosphate	(NaPO ₃) _n	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Nitrate	NaNO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Nitrite	NaNO ₂	s	+	+	+	+	+	+	+	+	+	+	+	2
Sodium Oxalate	Na ₂ C ₂ O ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Perborate	NaBO ₂ *H ₂ O ₂	s	+	+/o	+	+	+	+	+	+	+	+	+/o	1
Sodium Perchlorate	NaClO ₄	s	+	+	+	+	10%	+	+	+	+	+	10%	1
Sodium Peroxide	Na ₂ O ₂	s	+	+	+	+	+	+	+	n	n	-	+	1
Sodium Persulphate	Na ₂ S ₂ O ₈	s	n	+	+	+	+	+	+	+	+	+	+	1
Sodium Pyrosulphite	Na ₂ S ₂ O ₅	s	+	+	+	+	+	n	n	+	+	+	+	1
Sodium Salicylate	C ₆ H ₄ (OH)COONa	s	+	+/o	+	+	+	+	+	+	+	+	+	1
Sodium Silicate	Na ₂ SiO ₃	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Sulphate	Na ₂ SO ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Sulphide	Na ₂ S	s	+	+	+	+	+	+	+	+	+	+	+	2
Sodium Sulphite	Na ₂ SO ₃	s	+	+	+	+	50%	+	+	+	+	+	50%	1
Sodium Tetraborate	Na ₂ B ₄ O ₇ * 10 H ₂ O	s	+	+	+	+	+	+	+	+	+	+	+	1
Sodium Thiosulphate	Na ₂ S ₂ O ₃	s	+	+	+	+	25%	+	+	+	+	+	25%	1
Sodium Tripolyphosphate	Na ₅ P ₃ O ₁₀	s	+	+	+	+	+	+/o	+	+	+	+	+	1
Starch	(C ₆ H ₁₀ O ₅) _n	s	+	+	+	+	+	+	n	+	+	+	+	1
Starch Gum		s	+	+	+	+	+	+	+	+	+	+	+	1
Styrene	C ₆ H ₅ CHCH ₂	100%	-	-	o	+	+	o	-	-	-	o	+	2
Sublimate => Mercury-II-Chloride														
Succinic Acid	C ₄ H ₆ O ₄	s	+	+	+	+	+	+	+	+	+	+	+	1
Sugar Syrup		s	+	+	+	+	+	+	+	+	+	+	+	1
Sulphur Chloride => Disulphur Dichloride														
Sulphuric Acid	H ₂ SO ₄	98%	30%	50%	85%	+	20%	+	+	30%	30%	80%	+	1
Sulphuric Acid, fuming --> Oleum														
Sulphurous Acid	H ₂ SO ₃	s	+	+	+	+	10%	+	+	+	+	+	+	(1)
Sulphuryl Chloride	SO ₂ Cl ₂	100%	-	-	-	o	n	+	o	-	-	-	n	1
Tannic Acid	C ₇₆ H ₅₂ O ₄₆	50%	+	+	+	+	+	+	+	+	+	+	+	1
Tartaric Acid	C ₄ H ₆ O ₆	s	50%	+	+	+	+	+	+/o	+	+	+	+	1



7 ProMinent® Chemical Resistance List

Chemical	Formula	Conc	Acryl	PVC	PP	PVDF	1.4404	FPM	EPDM	Tygon	PharMed	PE	HastelloyC	WPC
Tetrachloro Ethane	$C_2H_2Cl_4$	100%	-	-	o	+	+	o	-	-	o	o	+	3
Tetrachloro Ethylene	C_2Cl_4	100%	-	-	o	+	+	o	-	-	o	o	+	3
Tetrachloromethane => Carbon Tetrachloride														
Tetrahydro Furane	C_4H_8O	100%	-	-	o	-	+	-	-	-	-	o	+	1
Tetrahydro Naphthalene	$C_{10}H_{12}$	100%	-	-	-	+	+	+	-	-	-	o	+	3
Tetralin => Tetrahydro Naphthalene														
THF => Tetrahydrofuran														
Thionyl Chloride	$SOCl_2$	100%	-	-	-	+	n	+	+	+	+	-	n	1
Thiophene	C_4H_4S	100%	n	-	o	n	+	-	-	-	-	o	+	3
Tin-II-Chloride	$SnCl_2$	s	+	o	+	+	-	+	+	+	+	+	+/o	1
Tin-II-Sulphate	$SnSO_4$	s	n	+	+	+	+	+	+	+	+	+	+/o	(1)
Tin-IV-Chloride	$SnCl_4$	s	n	+	+	+	-	+	+	+	+	+	+	1
Titanium Tetrachloride	$TiCl_4$	100%	n	n	n	+	n	o	-	n	n	n	n	1
Toluene	$C_6H_5CH_3$	100%	-	-	o	+	+	o	-	-	-	o	+	2
Toluene Diisocyanate	$C_7H_3(NCO)_2$	100%	n	n	+	+	+	-	+/o	n	n	+	+	2
Tributyl Phosphate	$(C_4H_9)_3PO_4$	100%	n	-	+	+	+	-	+	o	+	+	+	1
Trichloro Ethane	CCl_3CH_3	100%	-	-	o	+	+	+	-	-	o	o	+	3
Trichloro Ethylene	C_2HCl_3	100%	-	-	o	+	+/o	o	-	-	o	o	+	3
Trichloro Methane => Chloroform														
Trichloroacetaldehyde Hydrate	$CCl_3CH(OH)_2$	s	-	-	o	-	+	o	o	n	n	+	+	2
Trichloroacetic Acid	CCl_3COOH	50%	-	+	+	+	-	-	o	+	+/o	+	+	1
Tricresyl Phosphate	$(C_7H_7)_3PO_4$	90%	-	-	+	n	+	o	+	o	+	+	+	2
Triethanol Amine	$N(C_2H_4OH)_3$	100%	+	o	+	n	+	-	+/o	-	o	+	+	1
Trilene => Trichloro Ethane														
Trioctyl Phosphate	$(C_8H_{17})_3PO_4$	100%	n	-	+	+	+	o	+	o	+	+	+	2
Trisodium Phosphate	Na_3PO_4	s	+	+	+	+	+	+	+	+	+	+	+	1
Urea	$CO(NH_2)_2$	s	+	+/o	+	+	+	+	+	20%	20%	+	+	1
Vinyl Acetate	$CH_2=CHOOCCH_3$	100%	-	-	+	+	+	n	n	-	+/o	+	+	2
Water Glass => Sodium Silicate														
Xylene	$C_6H_4(CH_3)_2$	100%	-	-	-	+	+	o	-	-	-	o	+	2
Zinc Acetate	$(CH_3COO)_2Zn$	s	+	+	+	+	+	-	+	+	+	+	+	1
Zinc Chloride	$ZnCl_2$	s	+	+	+	+	-	+	+	+	+	+	n	1
Zinc Sulphate	$ZnSO_4$	s	+	+	+	+	+	+	+	+	+	+	+/o	1



7 ProMinent® Chemical Resistance List

Pump Specification Data

Min./max. required feed rate l/h _____
 Available power supply _____ V, _____ Hz
 Min./max. operating temperature °C _____
 Properties of process chemical _____
 Name, concentration % _____
 Solids content % _____
 Dynamic viscosity mPa (= cP) _____
 Vapour pressure at operating temperature bar _____
 Remarks, e.g. abrasive, _____
 gaseous, flammable, _____
 corrosive towards _____

Suction conditions:

Min./max. suction lift m _____
 Min./max. positive suction head m _____
 Pressure in chemical tank bar _____
 Suction line length m _____
 Suction line diameter mm _____

Discharge conditions:

Min./max. back pressure bar _____
 Min./max. discharge head m _____
 Min./max. negative discharge head m _____
 Discharge line length m _____
 Discharge line diameter mm _____
 Number of valves and fittings in
 suction and discharge line _____

Data required for proportional dosing:

Water flow Q min./max. m³/h _____
 Required final concentration g/m³, ppm _____

Example:

A required dose in mg/l = g/m³ = ppm

(Water flow Q max. 50 m³/h)

Pulse spacing (flow volume per pulse) of water meter 5 l.

Process fluid = sodium hypochlorite solution Na OCl with 12 % chlorine (by weight) = 120 g/kg = 150 g/l = 150 mg/ml

Selected dosing pump GALa 1005 NPB2 with 0.41 ml/per stroke volume, at max. 10800 strokes/h.

Variables: pump type, pulse spacing and concentration. The stroke rate (max. throughput l/h: pulse spacing l/pulse = 50,000 l/h : 5 l/pulse = 10000 pulses/h) must not exceed the max. stroke frequency (10800 strokes/h) of the dosing pump.

$$\text{Feed quantity} = \frac{\text{water throughput Q max. (l/h)} \times \text{stroke volume (l)}}{\text{pulse spacing (l)}} = \frac{50,000 \text{ l} \times 0.00041 \text{ l}}{\text{h} \times 5 \text{ l}} = 4.1 \text{ l/h}$$

$$\begin{aligned} \text{Final dose} &= \frac{\text{concentration (mg/ml)} \times \text{stroke volume (l)}}{\text{pulse spacing (l)}} = \frac{150 \text{ mg} \times 0.41 \text{ ml}}{\text{ml} \times 5 \text{ l}} = 12.3 \text{ mg/l} \\ &= 12.3 \text{ g/m}^3 \\ &= 12.3 \text{ ppm chlorine Cl}_2 \end{aligned}$$

pk_0_002





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